An Interpretation of Relativistic Spin Entanglement Using Geometric Algebra

by

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Author's Declaration of Originality

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Abstract

Entangled states are often given as one of the most bizarre examples of "weirdness" described as inherent to quantum mechanics. The present work reinterprets entanglement as not being a property of states at all, but rather as a relationship between the reference frames in which the states reside, which proposes to reduce "weirdness" of interpretation.

Using the geometric Algebra of Physical Space, it has been shown that a classical form of the Dirac equation can be satisfied by any eigenspinor, which is a Lorentz transformation operator describing the relative velocity and relative orientation of the rest frame of a system as seen from a particular lab frame from which it is described. The real linear nature of the Dirac equation means any real linear superposition of such eigenspinors are also solutions. Thus, with entanglement modelled as an operator consisting of a linear superposition of rotation operators describing the possible relative orientations of a particular particle frame and the frame from which it is observed, it too can satisfy a bipartite form of the Dirac equation.

To investigate this model, the present work applies relativistic boost transformations to the entangling operator in various ways, including as an identical boost of both parts in the same direction, and also as equal and oppositely-directed boosts. The resulting "entangling eigenspinors" are then analyzed in various ways, including the application to specific spin states — only to discover that doing this results in a reduction of the information, which can be interpreted as a reduction in the amount of entanglement. By comparing this to the treatment of the Dirac equation in APS, it may be concluded that the application of the entangling eigenspinor to a state — which models the typical approach of simply boosting an entangled state — gives an incomplete account of what is happening. The full information is thus contained within the entangling eigenspinor, justifying the interpretation of the entanglement in terms of geometric information relating the reference frames, rather than as a property of the state.

This, I dedicate
to everyone I have taught
or will teach someday.

I try to inspire,
but in truth, you inspire me
to be inspiring.

Through learning, I hope
humankind will save itself
from its own follies.

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much of my attention. I am looking forward to seeing them all again without the distracting shadow of "needing to get back to work" looming over me — in fact, I am looking forward to joyfully explaining my work, now that it is finished and finally clear to me.

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All in all, I am grateful to everyone who has supported me throughout this experience. Without the abovementioned people (as well as others not mentioned by name but who I also appreciate greatly), I guarantee that this dissertation would not have been possible — after all, it was barely possible even *with* all this encouragement. I am very pleased that I was able to bring this to completion, not just to validate all their faith in me, but also to grow into the accomplished person that I finally feel I am.

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Chapter 1

The Prelude

In the search for a greater understanding of physical reality, geometric algebras show much promise. The insights revealed by these algebras tend to emphasize geometric relationships which may not be obvious in other representations. The primary purpose of the present work at the outset was to seek geometric insights from an investigation of spin entanglement within the relativistic framework provided by a specific geometric algebra known as the Algebra of Physical Space (APS).

The investigation has revealed an unusual insight into a possible interpretation of entanglement, one which is more closely related to the transformations of special relativity instead of the states of quantum mechanics. Specifically, the interpretation claims that entanglement can be usefully represented in the relationships between pairs of reference frames, including superpositions of relative orientations and relative velocities. It is this interpretation that is examined within the present work.

This first chapter begins by separately introducing each of the subtopics that are involved in this dissertation, as well as the limitations of the scope. Section 1.2 provides reasons for studying the topics, and most importantly, some evidence that a new interpretation may be useful. Section 1.3 gives an overview of the ways this dissertation contributes to the knowledge base of humankind.

1.1 What this Dissertation Examines

This dissertation examines the intersection of several physical concepts, each of which can be defined separately. The following brief descriptions provide an overview of the ideas discussed in the present work, with attention to the context provided by each of the others. Also described here are some specific restrictions to the scope, with some explanation of why these restrictions are reasonable and even ideal.

1.1.1 Quantum-Mechanical Interpretation

Quantum mechanics is a mathematical model of physics at very small scales. The predictions of quantum mechanics have repeatedly been verified experimentally [5, 131, 132, 154, 162], and its representations are straightforward and self-consistent.

However, there is much disagreement in how to interpret the mathematics [59, 73, 106, 119, 138, 145]. What is the nature of the physical reality described by the mathematical model? Is the wavefunction representing the quantum state a physically "real" entity [127, 159], or only a representation of possibilities such that only the measurement results are real [6]? Are there some unknown but nonlocal [34] "hidden variables" [39, 40] which, if known, would complete the description and make it deterministic, or is reality inherently probabilistic? Does a wavefunction "collapse" upon measurement [85], or does something else explain why the post-measurement state coincides with one of the eigenstates of the observable that is measured?

The interpretation developed in the present work is closely related to **relational quantum mechanics** (**RQM**) [103, 130, 152], which is based on the idea that only relative information is physically relevant. In the RQM interpretation, any quantum state information must be described relative to some specific observer, just as any velocity information must be described relative to some specific observer in the theory of relativity — and in both cases, the "observer" need not be some conscious entity or device capable of computing or measuring, but rather the "observer" refers simply to the reference frame in which an observer may be.

In RQM, if a state as seen from a particular reference frame is described as a superposition of measurement eigenstates, it is interpreted as probabilistic. If a state as seen from a particular reference frame is in an eigenstate of a particular measurement operator, it still may be in a superposition of eigenstates of that same operator (perhaps relativistically modified) as seen from another reference frame. As such, the state itself may not be considered "objectively real" since it is not "universal" or independent of all observers. Instead, the state describes the **information** available to a particular reference frame.

The present work does not examine the process of measurement, but it does make reference to the idea that the post-measurement state as seen from a measuring observer's reference frame is one of the eigenstates of the observable representing the property that is measured. It does this by expressing a state in a "pre-measured" form, as a linear superposition of the eigenstates that correspond to the property of interest (which, here, is spin).

1.1.2 Relativistic Bipartite Entanglement

Entanglement is one of the key features that sets quantum mechanics apart from the classical description of reality. It is typically defined [92, 124] in contrast to *separability* of quantum states: if

a multiparticle state may be created using only operations local to each of the parts, and possibly also some classical communications between the holders of each of the parts [36], the state is **separable**; if it cannot be created in this way, it is **entangled** [47]. Entanglement is a form of **correlation** beyond what is possible classically. It is typically modelled as a superposition of correlated possibilities, such that the result of measuring one part narrows down the possibilities for the other parts.

Bipartite entanglement involves quantum correlations between only two parts. The parts in question may be described as **substates** or as existing within **subspaces** even if the information within those subspaces cannot be described individually. Bipartite entanglement must be well-understood before extending to multipartite entanglement; thus, since the present work is providing a new interpretation of entanglement, it is important to begin with the simplest case before building upon it.

Relativistic entanglement takes into account relative motion between the entangled state and an observer, and/or relative motion between the entangled substates. Specifically, relativity [142] can involve velocities great enough to require taking into account the constancy of the speed of light, which may also be considered the maximal speed of information transfer. The investigation in this dissertation only treats special relativity [67], and thus only considers inertial reference frames, i.e. ones of constant velocity.

1.1.3 Spin-one-half Particle States

One of the simplest quantum systems to model and discuss is one that has only two discrete **states**. It was Pauli [121] who first recognized that a "two-valuedness not describable classically" [123] may be ascribed to electrons, and Uhlenbeck and Goudsmit [151] who first interpreted this as an intrinsic angular momentum, or **spin**, with "spin up" and "spin down" as the two values possible as measurement outcomes. The magnitude of one-half was realized as necessary to explain why the Stern-Gerlach experiment [76] involved a beam of silver atoms splitting into two beams, instead of three which would be the case if the spin angular momentum was an integer.

Particles having half-integer spin are known as **fermions**, and the most elementary fermions — *quarks* and *leptons* — all have spin of one-half. Thus, an understanding of spin of this magnitude is of fundamental importance. Particles of spin other than one-half are not discussed, although pairs of spin-$\frac{1}{2}$ particles are mentioned as collectively having spin of 0 or 1.

Finally, by specifying that the treatment here involves **states**, what is implied is that the treatment is not dealing with *ensembles* of particles. Ensembles may be in a **mixture** of states, rather than a **pure state**, and thus must be represented by a **density matrix** (see Section 2.3.5) instead of a state vector or wavefunction.

1.1.4 Distinguishable Massive Particles

The assembly of the abovementioned ideas is tempered by two restrictions that keep the discussion as straightforward as possible, so as to appropriately begin the geometric investigation intended in this dissertation.

First, the main body of this work is limited to discussing **particles**, rather than also including antiparticles. The formalism applied here does include a connection to a description of antiparticles [11], and they are discussed briefly in Section 2.3.2. However, there is an extension to the mathematical formalism that seems more appropriate for modelling antiparticles [31], and which is being considered for a future extension of the present work.

A further restriction in this dissertation is to discuss only **massive** particles, as opposed to massless particles which necessarily travel at the speed of light in all reference frames. This is done primarily to ensure it is possible to model the particles under discussion from within their own rest frames, before transforming their description to include speeds relative to some observer.

Finally, this dissertation investigates only massive particles that are **distinguishable**. This is in contrast to the situation where particles are identical [94] and thus their properties may be interchanged, with only either a symmetric or antisymmetric effect on the wavefunction describing the situation. "Since the major conceptual problems posed by quantum mechanics—nonlocality and the measurement problem—are present even in the quantum mechanics of distinguishable particles, this simplification does not appear to evade essential features." [109, fn. 7]

1.1.5 Geometric Algebra

A geometric algebra is an algebra of vectors. It is defined on a vector space, and it involves products of vectors that are interpreted geometrically. Since the physical world is intuitively described in terms of vectors and geometry, a geometric algebra is most useful in describing physics.

The specific geometric algebra applied in this dissertation — the Algebra of Physical Space, or APS — is described and demonstrated in Chapter 2.

1.1.6 Additional Limitations to the Scope

Several important investigations fall just outside the scope of this work, but would be interesting to apply towards extending the present approach. One such exclusion is that of **measurement** — although the squared coefficients in a superposition are associated with measurement probabilities, measurement itself is not explicitly handled within the present work. Another exclusion is the

comparison of descriptions from multiple inertial reference frames, which would be most interesting in conjunction with an application of measurement. Also, this work does not examine density matrices, and thus cannot consider mixed states, but the ingredients to do so are all here.

There are also some restrictions to the **boosts** that are investigated in Chapter 3, because it was realized that too general a treatment of boosts made it difficult to uncover any useful geometric relationships. Thus, the only boosts investigated are ones where the two reference frames are described as having equal velocities relative to the observer (for reasons explained in Section 3.2.4), and they are either parallel or antiparallel. Furthermore, the directions investigated are either aligned with the initial spin direction that is used to orient one of the coordinate axes, or perpendicular to the spin direction.

Finally, note that forces are not considered at all: the particles are in field-free space, and thus the velocities involved are constant. The interactions preparing the initial state are presumed to have taken place well before this model begins its description. The "application" of a relativistic boost is meant to be interpreted in a mathematical sense: the model is formed by a series of algebraic steps that each add additional relevant information, until the situation is described to the desired extent. After all the "internal" relative information is implemented in the model, only relative information "external" to the system may be added, which changes the relative information between the system and the observer.

1.2 Why this Dissertation Examines What it Does

There are several reasons why this dissertation covers the subjects that it does, using the method that it does. In general, the subject matter for this dissertation was chosen based on the pure intrigue about the nature of spin entanglement, and also the fairly recent innovations involving its applications, both of which are explained in Section 1.2.1. The inclusion of relativistic considerations in the present work was an obvious choice to make, in light of the inherently relativistic nature of APS, but there are also interesting applications of relativistic spin entanglement, as mentioned in Section 1.2.2.

The main reason this dissertation can discuss a new interpretation is because an interesting one was discovered in the process of investigating the subject matter. It is, of course, necessary to justify why it is worth considering a new interpretation of spin entanglement, and this is done in Section 1.2.3. Finally, Section 1.2.4 wraps up by noting some of the reasons for the specific geometric algebra used in the present work, including an acknowledgement of how it contributes to the justification of the new interpretation.

1.2.1 Why Investigate Spin Entanglement?

Since spin and entanglement are both individually intriguing subjects, their combination is intriguing as well, which is part of what inspired the investigation of spin entanglement in the present work. Some of what makes these subjects intriguing is reviewed below.

1.2.1.1 Early entanglement and its interpretation

The idea of entanglement was first introduced by Einstein, Podolsky, and Rosen [64] as a conundrum, highlighted in a gedankenexperiment as a way to demonstrate that quantum mechanics does not provide a complete description of physical reality. The concerns introduced in the original "EPR" paper [64] were nonetheless realized as reasonably depicting the complementarity [42] of position and momentum, showing how the measurement probabilities are indeed related to one another [140].

However, even though the mathematical model was accepted as accurate, the *interpretation* continued to pose difficulties. Questions remained, such as *how* it is possible that the measurement probabilities can be correlated even over a significant distance, especially after it was realized [34, 46] that local hidden variables could not be employed to explain the correlation. Thus more investigation and contemplation followed — and is still ongoing [58, 105, 153].

1.2.1.2 Entangled spins, and spin itself

A simpler system than the EPR experiment that can also model entanglement [41] involves discrete quantum properties instead of continuous ones. Specifically, a pair of particles with a total spin of zero is described as having entangled spin states, and the mathematically non-separable parts can be spatially separated over a significant distance in a way similar to that described by the EPR [64] gedankenexperiment — this type of situation is described in this dissertation as "**EPR-like**". The result of a measurement of either of the two parts along any chosen direction would give a result of either spin up or spin down along the direction of measurement. What makes entangled spins interesting is that a subsequent measurement of the other part would have its result correlated to have opposite spin, *no matter the choice of measurement direction.*

Such a system of entangled spins can be useful not only for investigating entanglement, but also the nature of spin itself, which is also curiosity-provoking. Although spin is described as an intrinsic angular momentum, there is some debate about whether it is appropriate to treat spin as an actual self-rotation, although such a treatment has been done [8, 135, 137, 150]. The debate is over whether or not an elementary entity should be modelled as a point particle: if it is, it has no structure that can be described as "spinning"; if it is not, its surface rotation speed would exceed the speed of light. Electron spin could have even more complicated interpretations of its motion [7, 88], or — as

described in Section 2.3.4 — it may be treated as a rest-frame oscillation which, in a moving frame, may be associated with the de Broglie waves associated with particles of matter [22].

1.2.1.3 Applications of entanglement, especially spin entanglement

No matter how spin and its entanglement are interpreted, there are valuable applications [160] pressing the development forward. In particular, quantum information processing [62, 116, 117] harnesses the quantum powers of entanglement and superposition for computational purposes.

Spin is particularly useful for computation since its two measurable values can clearly be associated with the binary 0 and 1 required for quantum bits or **qubits**. Other two-valued quantum systems may be usefully entangled and applied computationally as well, but the present treatment specifically involves states that are physically rotated in order to invert the qubit value.

All in all, increased understanding of both spin and entanglement may inspire new ways of implementing quantum computing devices, thus supporting the continued investigation of both subjects, separately and together.

1.2.1.4 Continuing studies of entanglement

Overall, despite the numerous in-depth review papers [1, 46, 47, 92, 113, 124] that have been published on the subject of entanglement, there is still much value to be found in the continuing investigation of the subject. In fact, papers continue to be published [52, 61, 66, 112, 143] that examine entanglement within a restricted scope, similar to that of the present work.

Evidently, there is not yet enough clarity provided by the existing work to quell any further investigations, and there are sufficient unanswered questions to prompt continuing efforts.

1.2.2 Why Investigate Relativistic Entanglement?

Early models of quantum computers described only the qubits and manipulations thereof. Velocity was not treated, either because it would not be involved in the actualization of the physical model (as in a static array of qubits), or because it was being treated as negligible because the spin degrees of freedom were represented independently from momentum degrees of freedom (as is discussed in more detail in Section 1.2.3). However, the applications of quantum systems have been extended to cases involving motion — specifically, quantum teleportation [37] and cryptography [60, 65] both involve the transmission of quantum states, with velocities that may be quite significant. Thus, it is crucial that an appropriate mathematical model of entanglement incorporates relativity.

It is also possible to expand the overall understanding of reality by studying combinations of physical systems. For example, an examination of entangled particles in the vicinity of a black hole may bring illumination to the nature of entanglement, black holes, or both [44, 98, 104, 107, 114]. While the present work does not include such considerations of general relativity, it is a step towards doing so. The cumulative nature of knowledge means that any step towards increased understanding, on any topic, may be considered valuable for its own sake, in addition to the practical applications that may result from its investigation.

1.2.3 Why Offer a New Interpretation of Spin Entanglement?

This section outlines some reasons to consider a new interpretation of spin entanglement. It begins by considering the multitude of interpretations of quantum mechanics as a whole, then discusses a concern regarding some of the common interpretations of spin, followed by a geometric way to address the problem.

1.2.3.1 Why are there so many interpretations of quantum mechanics?

Before answering the question of why this dissertation offers a new interpretation of spin entanglement, there is a more fundamental question worth asking first. Why are there are so many different interpretations of quantum mechanics? The answer is at least threefold:

1. because the predictions of calculable results made by the existing interpretations are equivalent [147];
2. because interpretation itself is regarded by some as less important than obtaining results (colloquially known as the "shut-up-and-calculate" [110] interpretation of quantum mechanics); and
3. because none of the interpretations have sufficiently convinced the scientific community at large [139].

These three reasons are listed in a likely order of influence: if the interpretations give equivalent results, and only the results are important, then why would anyone bother with investigating the possible interpretations, other than those with more philosophical leanings? Also, if newer interpretations are not getting much attention because of these influences, how is anyone to know that there are convincing arguments in their favour?

Fortunately, although the perception of interpretation may be considered "a matter of personal philosophical prejudice" [139, p. 226], there seems to be some agreement [139, p. 225] that a

reconstruction of quantum theory will prove useful. The continuing investigations of interpretations of quantum mechanics [55, 58, 69, 91, 127, 152, 153] bring hope that the multitude of interpretations may someday be reduced to one satisfactory explanation.

The present work does not attempt to tackle all the details that would make up an interpretation of quantum mechanics as a whole, but it does involve an interpretation that is related to the larger picture. Specifically, the interpretation here is related to relational quantum mechanics (RQM) [130, 152], in that it describes quantum states as *relative information*. The way this is done provides some improvements on the typical interpretations of quantum states.

1.2.3.2 What is wrong with typical interpretations?

A very general working definition of "an interpretation" in any field of physics may be as *a description of the physical meaning that corresponds to a mathematical representation*. In this sense, any mathematical representation of a physical concept involves some degree of interpretation, in the definitions of its symbols and the manner in which they combine and give results. A particular physical concept may have several appropriate mathematical representations, with interpretations that may either agree or give differing perspectives of the same thing. Any given interpretation might be helpful in enhancing understanding, or it might hinder the understanding or expansion of an idea by being incomplete, misleading, more complicated than necessary, or even outright wrong.

In many treatments of spin entanglement [1, 52, 70, 77, 112, 148], spin and momentum states are represented by a tensor product of state kets, giving an abstract representation of the physical information:

$$|p, s\rangle \equiv |p\rangle \otimes |s\rangle \tag{1.1}$$

where the spin s is typically meant to be interpreted as being in the reference frame with momentum p relative to the observer. The observed spin direction in such a state may not be the same as it would be in the particle rest frame [82, 97], since the transformation of the state from rest frame to moving frame involves a Lorentz transformation which has the effect of rotating a spin state [124]. Further Lorentz transformations are then typically applied, with the resulting effect including not only a boost but also a Wigner rotation (C.1) acting on the spin in the rest frame. In the end, it is presumed that as long as the momentum is known fairly precisely, it can be taken into account [1, p. 29][148] by the rotation of a measurement device to account for the effect that the boost transformation has on the spin direction. (However, see Appendix D for evidence that the relationship between observed spin and relative motion is not quite that straightforward.)

The realization that the relative spin direction of massive particles can be shifted by a velocity transformation has resulted in some questionable interpretations. Some papers [56, 97, 134] have described this relationship as a *dependence*, even claiming that spin and linear momentum cannot be considered independent degrees of freedom [61]. Some [124, 125] go so far as classifying spin as

a *secondary variable*, saying it "depends" both on the Lorentz transformation and the momentum, where momentum is a *primary variable* since it "depends" only on the Lorentz transformation.

While the abovementioned descriptions may be useful in some contexts, they risk introducing some confusion: after all, rest-frame spin and linear momentum are in fact *independent*, since each can be set independently of the other. Furthermore, it is important to realize that the effect of a Lorentz transformation on a property does not change the property itself; for example, it is no longer common to speak of "relativistic mass", instead treating mass as its constant rest-frame value which may have a relativistic factor multiplying it, where necessary. Similarly, it is likely only reasonable to refer to describe spin in its rest frame — but what of entanglement? That is what the present work investigates.

1.2.3.3 How can a new interpretation help?

A means of resolving the problem with the above interpretations is to describe both the relative orientation and velocity as *information* that may be described from a particular reference frame. In fact, the Lorentz transformation itself describes both the velocity and orientation relative to the rest frame of the particle, but the relative orientation requires one additional piece of information: an initial spin direction, which is used to orient the coordinate system for the description. The momentum, for the massive particles under discussion here, is then defined as the Lorentz-transformed mass.

If a mathematical treatment of spin or entanglement does not explicitly specify reference frames, their implicit existence may be responsible for some of the confusions in interpretation. This problem may be related to the disconnect between the abstract representation and the physical nature of the subject under investigation. No matter how these problems are diagnosed, what is shown in the present work is that a promising cure may be found in geometric algebra.

1.2.4 Why Use Geometric Algebra?

There are many reasons for using geometric algebra to describe physical systems. First and foremost, the specific geometric algebra — the Algebra of Physical Space (APS) — that is used in the present work has provided such clarity in previous insights that it is of great interest to consider what further insights it may provide. The main reasons for its use in the present work are outlined in Section 2.1. Following this are demonstrations of particular APS insights relevant to the present work: relativity is covered in Section 2.2, and quantum mechanics in 2.3.

To address one of the concerns mentioned above, note that the geometric algebra approach avoids the interpretational problems that result from artificially splitting the information into parts attempting

to separably describe momentum and spin. Instead, a geometric entity known as a spinor is used to represent the spin in the rest frame, and the direction of its vector part is used to orient the coordinate system. Other states are obtained from this by the application of physically-realizable transformations, including spatial rotations as well as boost transformations to reference frames of other velocities. The resulting expressions may be linear superpositions, with probability amplitudes that take into account the relative velocity.

Overall, one of the main benefits of a geometric model for any physical system is the fact that it is based on physical geometry. This fact can make it easier to visualize and interpret the topics under examination, due to everyday spatial intuition. At the same time, the geometric model can help extend humankind's understanding of entanglement, spin, quantum information, and even spacetime itself, especially when using a geometric algebra that is based on the everyday three-dimensional Euclidean space, as APS is. All of this may also make it easier to design experiments to test the revelations that may arise from using such a model.

1.3 How this Dissertation Examines the Subject

The present study of relativistic spin entanglement involves several novel and noteworthy approaches, which are listed below. After this, there is a brief overview of this entire dissertation, including the appendices (several of which include some supplementary material that is original, yet did not fit within the flow of the rest of this dissertation).

1.3.1 What Is Novel or Noteworthy Here?

This section gives an overview of the methods applied within the present work towards examining the subject of relativistic spin entanglement, with emphasis on what aspects were developed for this dissertation.

In general, the present work extends the existing preliminary APS approach to spin and entanglement (which is described in Section 2.3) to describe entangled bipartite states in relativistic motion (where relativity in APS is described in Section 2.2). However, there are specific and interpretations and tactics that were employed which go beyond the mere application of a mathematical method.

1.3.1.1 A geometric rather than abstract representation

Part of what the present work does is make use of a geometric alternative to the abstract representation for spin states, which is not original to this work [20] but it not a common approach. More

typical is a representation of spin as a linear superposition of discrete measurement outcomes. However, such a linear superposition may be obtained from a continuous transformation of spin (2.39), and thus spin itself may be considered continuous [155, 161].

In the present work, the state information that is obtained from a physically-realizable transformation of a geometric spinor not only contributes to the probability amplitudes, but also gives a geometric representation of spin itself. The geometric interpretation of spin as a rest-frame oscillation becomes a plane wave under a Lorentz boost [22]; see the brief description of this in Section 2.3.4. By continuing to use this model [20], and by discovering no discrepancies from its use, the geometric interpretation of spin [22] becomes even more justifiable.

The geometric state representation is distinct from another common representation for spin and entanglement: the density matrix formalism. While density matrices may be useful due to their ability to represent mixed ensembles of states, they have the disadvantage of obscuring probability amplitudes. In the present work, probability amplitudes are treated as important because they contain geometric information, and they imply the possibility of interference with other state wavefunctions.

1.3.1.2 Entanglement as a transformation

Representing entanglement in this particular geometric algebra is also not original to the present work [23]. What is new here is an interpretation of entanglement that is not found elsewhere: as residing within a transformation operator, rather than within the bipartite spin state. With this interpretation, the entanglement of spin states is modelled as a superposition of rotations acting upon a separable state, with coefficients representing the probability amplitudes that can be associated with the probability of each transformation taking place.

The justification for this interpretation is introduced at the beginning of Chapter 3 and evaluated in more detail in Section 3.4. The details of the derivation of the transformation operators — the **entanglors** — is in Section 3.1, where they are also investigated relative to each other and relative to the entangled bipartite basis states in order to establish their utility without involving the states themselves.

1.3.1.3 Boosts as information

The next interpretation applied within this dissertation is that of treating Lorentz velocity transformations — **boosts** — as *information* relating reference frames. Specifically, the velocity information "exists" only in a relative sense, as described from a particular reference frame deemed the **lab frame** or observer's frame, regarding a particular observed frame (which may be an object or system's **rest frame**).

In Section 3.2, it is noted that there are two ways of implementing bipartite boosts: the boosts can be describing relative information between the pair of subsystems — as described from a common vantage point — or between an observer and the combined bipartite system. The relative information between subsystems is a transformation that can only be applied in the initial mathematical setup; any further transformations must only shift the observer's reference frame relative to the combined bipartite system, or equivalently, the system relative to the observer.

This approach to modelling the relative velocities within a bipartite state was not seen elsewhere in the literature. Instead, the momentum of a state is typically included as information within a ket, as described above (1.1), and a bipartite state is described by a tensor product of such kets, with entangled states being a linear superposition of such bipartite states. The present approach differs from these others in that it consistently models all such information as physically-realizable transformations, which are described relatively between reference frames. In fact, even the entanglement transformation may be interpreted as a superposition of rotations described relatively between the reference frames of the substates.

1.3.1.4 Combination of entangling and boosting information

Since the interpretation of entanglement as a transformation is unique to this work, and the method of modelling bipartite state velocities as relative information between reference frames is also new, clearly the *combination* of such approaches is original as well.

What is not new is the idea that combinations of boosts and rotations make up the group of restricted Lorentz transformations, and that the description of a particle is often assisted by describing a specific restricted Lorentz transformation — known as the **eigenspinor** [11] — which relates the velocity and orientation of rest frame of the particle to the lab frame from which the particle is observed or described. However, with the present work describing entanglement as a superposition of relative rotations, the application of relative boosts to such a description enables an investigation of the new idea of a superposition of bipartite eigenspinors, which is done in Section 3.3.

The reason this is interesting is because a classical form of the Dirac equation in APS 2.20 has been shown to be satisfied by an eigenspinor [11], and since the Dirac equation is a real linear equation, a real linear superposition of solutions is also a solution. Thus, the **entangling eigenspinors** that arise from the combination of bipartite boosts with entanglors can be interpreted (as described in Section 3.4) as satisfying a **bipartite Dirac equation**.

There are other studies that investigate entanglement using the Dirac equation [3, 48, 56, 75, 143], and/or the algebra of Dirac gamma matrices [84, 128], and/or using Dirac spinors [100], and even using the Algebra of Physical Space [20, 23, 25]. However, in none of these works is entanglement examined in the way it is in the present work, let alone relativistic entanglement.

1.3.1.5 Explanation of Wigner rotations, and a derivation using APS

The Wigner rotation [155] is a rotation that relates a product of non-collinear boosts to an overall boost that is nearly equivalent to the initial product (C.1). However, there are examples in the literature[3, 77, 134, 133, 144, 148] of the Wigner rotation being invoked as an explanation for the effective rotation of a spin direction due to a boost, even if there is only one boost applied.

The APS can clarify the confusion between the two ideas, as is done in Appendix C. The Wigner rotation itself is investigated first, because it is interesting and worth noting for cases where multiple boosts may be applied. Then the rest-frame rotation of a spin as a result of a boost is examined, and it can be explained in APS as a result of the spinorial nature of the rest-frame spin state, as is demonstrated in Appendix C.5.

Although neither of these ideas is new, the fact that they have been confused in the literature gave reason to investigate them both using APS and point out their inequivalence, as was done for the first time in the present work.

1.3.1.6 A revised measure of concurrence, and a derivation using APS

In any investigation of entanglement, questions arise about the degree or extent of the entanglement: is a state fully or partly entangled, or is it separable? If an entangled state is manipulated or viewed in another way does the amount of entanglement change, and if so, by how much?

There are many different measures of entanglement [35, Ch. 15][47, 92], and not all are appropriate for every situation. In the present work, the quantifying of entanglement is used only to give a general idea of the effect of the processes being explored, so there is no need to use a complicated measure involving quantum information or entropy. Furthermore, since only pure states (not mixtures) are treated, a fairly simple measure is sufficient. The selected measure for the present work is the **concurrence** [156].

For the purposes of the present work, it is useful to make two modifications to the basic formula for the concurrence of a pure state (F.9):

- The first modification acknowledged that a relativistic boost of a spin state is equivalent to a dilated rotation, which results in a state that is no longer normalized. To take this into account, a normalization factor can be added to the formula for the concurrence (F.10). With this modification, maximal entanglement is still represented by a concurrence of 1, which makes it straightforward to compare the entanglement in different situations.

- The second modification expressed the concurrence in terms of the coefficients of the entangled basis states (F.11), instead of the coefficients of the separable basis states. This is done to ease the application of this formula to the results found in Section 4.3.2, which have a simple form in terms of the entangled basis states.

Both of these modifications are original ideas presented within this work, and the derivation using APS is also an original reinterpretation of the source material [156]. However, since this work plays only a supporting role to the main purpose of this dissertation, the efforts are presented as a supplement, in Appendix F.4.

1.3.2 Where Is What?

The upcoming chapters of this dissertation are organized as follows:

- Chapter 2 covers all the relevant **background details** of the mathematical method applied in the present work. It begins by describing the Algebra of Physical Space (APS), emphasizing its overall benefits. This is followed by a review of the relativistic and quantum-mechanical approaches and corresponding geometrical interpretations that have been developed using APS, focusing on those that are applicable to the rest of this dissertation.
- Chapter 3 contains the **original interpretation** that is the main feature of the present work, specifically that entanglement can be modelled as a transformation operator rather than a part of the state. Bipartite boosts are examined and interpreted as well, with attention to what cases could be applied to the entangling operators to gain the most useful geometric insights. Finally, the boosts are applied to obtain a variety of entangling eigenspinors, i.e. transformation operators containing both relative rotations and relative boosts.
- Chapter 4 begins with a summary of the method applied and the **results** obtained. Next, the entangling eigenspinors that were the results of the previous chapter are **analyzed**, feature by feature, before being **applied** to spin states in order to interpret them physically.
- Chapter 5 discusses the **significance** of the interpretation and results, and gives several leads into how the present work may be expanded in the **future**.

The appendices include the following:

- Appendix A outlines the **notation**, **symbols**, **conventions**, and gives some **definitions** of terms used in the Algebra of Physical Space.

- Appendix B defines **projectors** and lists their **properties**. It also demonstrates some of the most useful ways projectors are applied to general elements of the APS, and the resulting interpretations of these: as two-component column **spinors**, also known as elements of a particular **minimal left ideal**. There are also some comments on the many ways projectors are **applied**. Finally, pairs of projectors are combined into what are defined here as **biprojectors**, and various relationships involving these are summarized.
- Appendix C discusses and derives the **Wigner rotation**, and also examines a rotation that has occasionally been **confused** with Wigner rotations.
- Appendix D investigates single-particle **relativistic spin** in a way similar to the bipartite approach taken in Section 4.3,
- Appendix E defines and gives rules for applying **tensor products**.
- Appendix F summarizes the ways a **bipartite spin state** can be represented in terms of various **basis** states, including separable basis states, entangled basis states, and in terms of singlet and triplet states. These representations are then related to one another via their coefficients. This appendix wraps up with a derivation of **concurrence** using APS, with some revisions to assist with its implementation in the present work, particularly in light of ensuring that it can handle the effects of the relativistic boosts.

To summarize what was discussed above: Section 1.1 introduced and described *what* concepts are crucial to this work; Section 1.2 explained *why* these concepts are interesting; and Section 1.3 overviewed *how* the topic is investigated, including *what* new work is presented here, *how* it is limited, and *where* to find it all.

Thus concludes Chapter 1 and the introduction it provides to this dissertation.

Chapter 2

The Algebra of Physical Space: Motivations and Methods

This chapter introduces the Algebra of Physical Space (APS), beginning in Section 2.1 with a general introduction followed by a comparison to other geometric algebras. This section then describes the motivations behind choosing this particular geometric algebra over the others.

Further understanding of the motivation for choosing APS can be found in the next two sections, which explain how APS has been usefully applied to describe relativity (in Section 2.2) and quantum spin and entanglement (in Section 2.3). The focus is on the features that are applied and extended in the new material presented in Chapter 3, where relativistic boosts are applied to entangled spin transformation amplitudes.

Key revelations throughout the chronicle in this chapter serve to demonstrate how this algebra provides geometric clarity through its descriptions of physical processes, thus justifying its continued extension to further topics that also promise to benefit from the greater understanding it provides.

2.1 Geometric Algebras

This section starts by defining the Algebra of Physical Space (APS), first in terms of its elements and interpretation, and then in comparison to other geometric algebras. This section wraps up by outlining the motivations for the choice of APS for use in this dissertation.

Further details about the specific applicability of APS to the topics of this dissertation will follow this section.

2.1.1 Elements of The Algebra of Physical Space

The Algebra of Physical Space (APS) refers to a real geometric algebra [57] generated by products of vectors in three-dimensional Euclidean space. An element of APS involves real linear combinations of **geometric products** of the orthonormal basis vectors $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$, and thus can have the form of any of the following:

- a scalar element, where for example the unit scalar is obtained by squaring any of the basis vectors: $\mathbf{e}_j^2 = 1$;
- a vector element, which can be any linear combination of the basis vectors:

$$\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + v_3\mathbf{e}_3;$$

- a **bivector**, which is an oriented area element described as the product of vectors in two different directions: e.g. $\mathbf{e}_1\mathbf{e}_2$ is a unit-magnitude area element coplanar with vectors of the form $v_1\mathbf{e}_1 + v_2\mathbf{e}_2$, with orientation given by the axis direction about which $\mathbf{e}_1$ would rotate towards $\mathbf{e}_2$;
- a **trivector**, which is the oriented volume element of the space: $\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3$;
- or a real linear combination of any of the above.

The structure [16, pp. 8–9] of the algebra is defined by an **associative vector product**,

$$\mathbf{vv} = \mathbf{v} \cdot \mathbf{v} = |v|^2 \tag{2.1}$$

where the dot indicates the usual dot product (inner product) of vectors, and v is the scalar magnitude of the vector $\mathbf{v}$. Since a vector may represent the sum of a pair of other vectors, $\mathbf{v} = \mathbf{u} + \mathbf{w}$, the associative vector product may be expanded and the common terms on both sides may be subtracted away to leave

$$\mathbf{uw} + \mathbf{wu} = 2\mathbf{u} \cdot \mathbf{w} \tag{2.2}$$

which indicates that vectors generally do not commute in this algebra. In fact, if nonzero vectors $\mathbf{u}$ and $\mathbf{w}$ are perpendicular (such as the basis vectors), then $\mathbf{u} \cdot \mathbf{w} = 0$ and they anticommute: $\mathbf{uw} = -\mathbf{wu}$. The anticommuting of the basis vectors,

$$\mathbf{e}_j\mathbf{e}_k = -\mathbf{e}_k\mathbf{e}_j \text{ for } j \neq k \tag{2.3}$$

gives rise to the trivector being oriented: inverting any two factors introduces a minus sign, so for example, $\mathbf{e}_1\mathbf{e}_3\mathbf{e}_2 = -\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3$.

From this simple yet elegant beginning arises a wealth of geometric understanding, which enables APS to be clear enough to serve as a pedagogical tool [13, 16, 21] and interesting enough to serve as an exploratory one [22, 25, 33], even if the subjects under investigation are already well-developed using other notations.

2.1.2 A Complex Interpretation of the Geometry of Physical Space

One particularly compelling feature of APS is in its treatment of the unit trivector. By the anticommutation of the basis vectors, it is easy to see that the unit trivector squares to -1, and commutes with all elements of APS. These properties allow the unit trivector to be associated with the **unit imaginary** i.

As a commuting element that changes sign under a parity inversion, the unit trivector $i \equiv \mathbf{e}_1\mathbf{e}_2\mathbf{e}_3$ may be considered a **pseudoscalar**. Furthermore, this geometric interpretation of the unit imaginary allows a relation between bivectors (spatial planes) and axial vectors (**pseudovectors**, or imaginary vectors):

$$\mathbf{e}_j\mathbf{e}_k = i\epsilon_{jkl}\mathbf{e}_l$$

where ϵ_{jkl} is the fully antisymmetric Levi-Civita symbol. Furthermore, any *real* element of APS expressed in terms of scalars, vectors, bivectors, and trivectors, i.e. in the eight-element basis $\{1,\ \mathbf{e}_j,\ \mathbf{e}_j\mathbf{e}_k\ \mathbf{e}_1\mathbf{e}_2\mathbf{e}_3\}$ for $1 \leq j < k \leq 3$, can instead be expressed as a *complex* linear combination of scalar and vector elements, i.e. in the basis $\{1,\ \mathbf{e}_1,\ \mathbf{e}_2,\ \mathbf{e}_3\}$.

Since any appearance of the unit imaginary in an expression can be replaced by the unit trivector, the algebra itself is still considered real. It is a matter of convenience to associate i with the unit trivector, and doing so is not considered a "complexification" of the algebra (though this has also been done [30, 99]). However, the treatment of the unit imaginary as a geometric object gives great insight into its ubiquity in mathematical physics [26].

2.1.3 Alternative Geometric Algebras

In terms of comparing APS to other geometric algebras, note that APS is referred to as the Clifford algebra $C\ell_3$, where the subscript gives the number of basis vectors that square to $+1$. In general, a Clifford algebra denoted by $C\ell_{p,q}$ has p basis vectors that square to $+1$ and q that square to -1.

Note that APS is not the only Clifford algebra that has been used to describe physical reality. Interestingly, there have been so many discoveries of Clifford algebras by physicists attempting to represent various physical phenomena that it has been proposed that Clifford algebras in general are "somehow intrinsic to the physical world" [136].

Specific other geometric algebras found to be useful in describing physical phenomena have included Hamilton's quaternions [81] $\mathbb{H} \cong C\ell^{+}_{3,0}$ (where + indicates the real part), the biquaternions $\mathbb{H} \oplus \mathbb{H}$ or complex quaternions $\mathbb{C} \otimes \mathbb{H}$ [80], as well as Hestenes' **space-time algebra** (STA) [89, 90] which is closely related to the Clifford algebra $C\ell_{1,3}$ of Dirac's gamma matrices [63]. In particular, since the first STA textbook [86] was released in 1966, researchers have used STA to uncover many valuable insights into the underlying geometry of various physical phenomena, including several investigations [84, 87, 88, 102, 143] of topics particularly relevant to this present work.

2.1.4 Arguing Against the Alternative Algebras

The alternative geometric algebras tend to define new quantities to use as a basis, which contrasts the APS approach of using the basis vectors of familiar three-dimensional Euclidean space. The geometric product of the familiar basis vectors gives results that are interpreted in a straightforward and illuminating manner, with "fewer new rules and operators" [21].

The alternative options described above all have the ability to describe the same physical features, since there are isomorphisms between these algebras (or parts thereof, since $C\ell_{1,3}$ is twice the size of the others), but the ease of use and clarity of description differ greatly. In comparison to APS, the geometric interpretation of the complex quaternions "is rather disguised, and it does not lend itself easily to a covariant formulation" [21, p. 872]. Similarly, the space-time algebra (STA) has been found to contain "abstractions and potential hazards" [21, p. 872] such as the fact that the unit pseudoscalar element γ_5 does not commute with all elements. The commuting unit pseudoscalar element from APS, as well as the valuable geometric interpretation of the imaginary unit, can be argued [50] to make APS preferable over STA.

One putative advantage of STA is that its "conventions have been chosen to differ as little as possible from those in formalisms with which physicists are already familiar" [86, p. vii], in particular, using the gamma matrices γ_μ — familiar from Dirac theory — as basis vectors. However, this familiarity is nonexistent to someone who has not yet studied the Dirac equation. Thus, it stands to reason that an introduction to geometric algebras would be easier using a notation that is closer to what is truly familiar at the earlier pedagogical levels. Furthermore, it has been shown that all of the relationships that are given in STA can be represented equally well in APS [32], yet since the algebra is half the size (specifically, APS is isomorphic to the even subalgebra of STA), the representations cannot help but be more concise in APS.

So although many revelations have been drawn from the alternative Clifford algebras described above (and from other geometric algebras), there are sufficient advantages to the particular choice of APS to select it for use in the present work, as will be frequently emphasized throughout this chapter.

There does exist a further complexification of APS [31, 99], which defines the volume element $I = \mathbf{e}_1\mathbf{e}_2\mathbf{e}_3$ as an imaginary element that exists *in addition* to the imaginary unit i, with both as commut-

ing pseudoscalars. The two imaginary elements are related by a new scalarlike element $h = i/I = I/i$, where $h^2 = 1$, and the effect is to double the size of the algebra (in a way that is distinct from STA). This Complexified APS (or **CAPS**) approach may have great value in representing certain physical relationships, including in regards to the subject of this dissertation, but it was of this author's opinion that it would be useful to withhold the additional complexification in order to determine whether a clear need for CAPS might arise during the investigation of relativistic entanglement. (See Section 5.2 for further thoughts on using CAPS to describe relativistic entanglement.)

2.1.5 Use of APS in This Dissertation

Thus, the reasons for choosing to use APS in this dissertation follow directly from the points mentioned above:

- First and foremost, the clarity of the geometric representation is well-suited to providing insight into the interface between the quantum and classical relativistic descriptions of physical systems.
- Second, the alternative geometric approaches have enough disadvantages to dissuade the author of this dissertation against pursuing their study, especially after becoming accustomed to the APS and the advantages it embodies.
- Finally, the potential exists for the future use of APS as a means of teaching physics concepts, in which case further development of APS is required so that such an education could continue through increasingly advanced topics. An increase in the number of resources that use APS would furthermore provide a foothold for APS-trained students entering the field to apply their existing geometric comprehension towards expanding the understanding within the field itself.

Thus concludes the reasoning behind the choice of using the Algebra of Physical Space for this dissertation.

2.2 APS and Relativity

The earliest papers [10, 24, 27, 28, 29] introducing the ideas behind the Algebra of Physical Space (APS) had revealed remarkable relativistic insights by using this algebra, well before the name "Algebra of Physical Space" and corresponding acronym "APS" were established in about 2004 [18, 19, 20, 21, 32]. The specific features of APS that make it ideal for relativistic treatments of physical topics are outlined below.

2.2.1 Time Component

One of the particularly noteworthy advantages of using this algebra in relativity is in how simply it represents temporal information alongside spatial information: as scalar parts of APS elements [21], where the vector parts represent spatial properties. A sum of scalar and vector parts is deemed a **paravector**.

An arbitrary real paravector such as $p = p^0 + \mathbf{p} = p^0 + p^k \mathbf{e}_k$ can represent a **spacetime vector**. In such a case, the scalar component p^0 is interpreted as the temporal component, while the spatial part is $p^k \mathbf{e}_k$.

The **spacetime interval** (or "square length") of a paravector p is given by $p\bar{p}$, where the bar conjugate (A.6) negates the vector part:

$$p\bar{p} \;=\; \left(p^0 + \mathbf{p}\right)\left(p^0 - \mathbf{p}\right) \;=\; \left(p^0\right)^2 - |\mathbf{p}|^2 \tag{2.4}$$

This definition of the spacetime interval gives a quadratic form to the algebra that enables it to be associated with the **Minkowski metric** $\{1, -1, -1, -1\}$ of spacetime.[1]

The square length of a paravector determines its spacetime classification:

$$\begin{aligned} x\bar{x} > 0 \qquad & x \text{ is } \textbf{timelike} \\ x\bar{x} = 0 \qquad & x \text{ is } \textbf{lightlike} \text{ or } \textbf{null} \\ x\bar{x} < 0 \qquad & x \text{ is } \textbf{spacelike} \end{aligned}$$

A real paravector orthogonal (A.13) to a real (A.10) spacelike paravector is timelike, and vice versa. A lightlike paravector is orthogonal to itself, because of its null norm.

Since APS has such a clear association with Minkowski spacetime, the "four-vectors" that are traditionally used in relativistic contexts are easily represented by real paravectors in APS. In the traditional approach, the spacetime metric is *imposed* on the mathematical model in order to ensure spacetime intervals remain invariant under Lorentz transformations. In contrast, in APS (or in its matrix algebra predecessor [71]) this invariance happens quite naturally, since the spacetime interval (square length) is a Lorentz scalar, and such scalars remain unchanged by Lorentz transformations.

2.2.2 Lorentz Transformations

A second clear advantage of APS is how simply a Lorentz transformation is represented, applied, and interpreted, especially as compared to any explicitly matrix-based representation [21]. The

[1] The metric with opposite sign, $\{-1, +1, +1, +1\}$, is also an option. However, using this would require the definition of the square length given above to be negated, which seems a less intuitive choice, given this mathematical framework.

relativistic effect of "mixing" space and time in inertial frames moving with relativistic velocity is modelled in APS as a rotation in spacetime, and evaluated through calculations of simple geometric products.

The mathematical representation of a restricted Lorentz transformation (i.e. one that is restricted to be physically realizable, thus excluding reflections, spatial inversion, and time inversion) may involve:

- a **boost** B which is a real operator ($B^\dagger = B$) that relates inertial reference frames in relative motion and can be expressed as

$$B = \exp\left(\frac{\mathbf{w}}{2}\right) = \cosh\frac{w}{2} + \hat{\mathbf{w}}\sinh\frac{w}{2} \tag{2.5}$$

  where $\mathbf{w} = w\hat{\mathbf{w}}$ and $w = \operatorname{artanh}\frac{v}{c}$ is the rapidity corresponding to speed v in direction $\hat{\mathbf{w}}$;

- a **rotor** R, or spatial rotation, which is even ($\overline{R}^\dagger = R$) and unitary ($R^\dagger = \overline{R}$), and can be expressed as

$$R = \exp\left(-i\boldsymbol{\theta}/2\right) = \cos\frac{\theta}{2} - i\hat{\boldsymbol{\theta}}\sin\frac{\theta}{2} \tag{2.6}$$

  where $\boldsymbol{\theta} \equiv \theta\hat{\boldsymbol{\theta}}$ and θ is the rotation angle about the $\hat{\boldsymbol{\theta}}$ axis;

- or some product thereof.

Any combination of rotations and boosts may be equivalently expressed as the product of a single boost and a single rotor, $L = BR$, where the rotor acts in the rest frame before the boost is applied. (Note that the conjugations $x^\dagger$ and $\bar{x}$ in the above are defined in Appendix A.2.)

Boosts and rotors may each be interpreted as a rotation, but in a *spacetime plane* which is *spacelike* for rotors and *timelike* for boosts. A geometric interpretation of a timelike plane includes the time axis, $\mathbf{e}_0$ ($= 1$), so the boost $B = \exp\left(\mathbf{w}/2\right)$ in fact applies a spacetime rotation in the $\mathbf{e}_0\hat{\mathbf{w}}$ plane. Furthermore, since the argument to the exponential function is real, the rotation is *hyperbolic* in nature (see Figure 2.1); also, the greater the rapidity, the more closely the result is tilted towards the light cone, with infinite rapidity representing the speed of light.

The traditional matrix- or tensor-based approach to Lorentz transformations may be well-established due to frequent use, and while it is certainly powerful and effective, it is rather unwieldy[2]. Contrast this with APS, where the geometry of spacetime is emphasized by use of geometric basis elements, rather than buried within tensor components.

[2]Even frequent users of tensors for relativity are aware of how cumbersome the notation is. For example, in a 2013 textbook focusing on the tensorial approach to spacetime, it is mentioned [67, Section 5.1] that "A fundamental addition to the general baggage of tensors is the causal character of 4-vectors in Minkowski spacetime." This not only describes tensors in a far from complimentary manner, but it also implies that the imposition of the Minkowski metric is a further complication; perhaps the author would be surprised at how the Minkowski metric arises naturally within the APS approach!

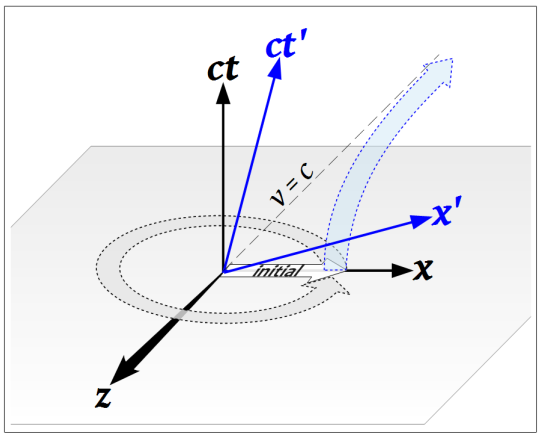


Figure 2.1: The two types of physically-realizable Lorentz transformations are shown here as spacetime rotations. Purely spatial transformations would rotate the vector labelled "initial" along the circular path in the **xz** plane. A boost transformation of the object would hyperbolically rotate the vector in the plane of **x** and **t**. The axes marked $\mathbf{x}'$ and $\mathbf{ct}'$ show that an arbitrary boost shifts the axes towards the light cone at $v = c$.

2.2.3 Focus on Physical Significance

Another significant feature of APS is the fact that it only treats as significant the **relative** orientations and motions of an observer and the observed system. In other words, it does not matter if a Lorentz transformation is applied as an "active" change of the observed reference frame, or as a "passive" change of the observer's reference frame, or a combination of the two; the results are all represented in the same way in APS. (Note that some authors reverse these definitions.)

This is notably [19, pp. 376–378] unlike the space-time algebra, STA, which emphasizes descriptions in terms of the frames themselves rather than the relations between them. The relationship between the reference frames is all that is physically significant, and this is reflected in APS (where, indeed, there is rarely a reason to mention whether a transformation is active or passive).

2.2.4 Covariance and Invariance

Elements or expressions that are simply rotated and/or reflected by Lorentz transformations are said to be **covariant** [19, p. 15]. Such quantities generally have meaningful geometric interpretations in any reference frame. Components of APS elements (that is, the coefficients of the basis elements) are not generally covariant, but it is generally easy enough to manipulate the covariant quantities and then extract components as necessary.

Properties that are unchanged by Lorentz transformations are said to be **invariant** [19, p. 15] (or *Lorentz-invariant* when emphasizing that the invariance is with respect to Lorentz transformations). Algebraic examples of such properties include a particular scalar product of paravectors (A.12) and

the square of a biparavector. A geometric example of a Lorentz-invariant property is the square length of an APS element (2.4), though in fact the invariance of the square length is part of the definition of a Lorentz transformation.

Both *covariance* and *invariance* are useful properties to note within any relativistic treatment. In APS, covariance is used as a guiding principle towards determining the most useful representations of expressions, and invariance is used to simplify calculations (since it is often true that performing a calculation in a particular inertial reference frame — usually the rest frame — is easier than in any other frame, and invariance means the result holds true in any other frame).

2.2.5 The Eigenspinor

A particular Lorentz transformation that arises frequently in this dissertation (and most other APS resources) is the **eigenspinor** [11, p. 4295], denoted by

$$\Lambda = BR \tag{2.7}$$

which describes the relative motion (via the boost B) and relative orientation (via the rotor R) of the rest frame of a particle as observed from a specific lab frame.

The "**spinor**" part of the name "eigenspinor" is indicative of the distinct **unilateral** form of its transformations. For example, an additional transformation $L = B_2R_2$ would be applied to an eigenspinor $\Lambda = B_1R_1$ to obtain

$$\begin{aligned} \Lambda' &= L\Lambda \\ &= B_2R_2B_1R_1 \\ &= B'R' \end{aligned}$$

with a new overall boost B' and rotor R' factor.[3] The "**eigen**" part of the name (from the German for "own" or "innate") indicates that this spinor is a *characteristic* relationship between the system in question and the frame from which it is being described.

The eigenspinor can be used to transform any property that is easily described in the rest frame to its equivalent in the lab frame. A typical transformation of an APS element is enacted by a **bilateral** application of the eigenspinor and one of its conjugates. For example, using units where $c = 1$, the transformation of the rest-frame velocity $u_{rest} = \mathbf{e}_0$ (which may also be described as a *unit*

[3] This unilateral transformation is distinct from typical transformations of paravectors, vectors, and bivectors in APS, which are performed **bilaterally**: an operator acts from the left, and some conjugate operator acts from the right: e.g. $R\mathbf{v}R^\dagger$ rotating a vector $\mathbf{v}$ or $B\mathbf{F}\bar{B}$ boosting a bivector $\mathbf{F}$. See Section 2.2.2 for details about rotations and boosts in APS. Note that defining a spinor [51] by its transformational properties is not the only possibility [11, Note 12], but it is useful here.

displacement along the time axis) gives the following **proper velocity** of a particle as seen in the lab frame:

$$u = \Lambda u_{rest} \Lambda^\dagger = \Lambda \mathbf{e}_0 \Lambda^\dagger = \Lambda\Lambda^\dagger \tag{2.8}$$

where $\Lambda^\dagger$ is the Hermitian conjugate (A.7) of the eigenspinor, and the unit scalar $\mathbf{e}_0 = 1$ commutes with everything. Here, it may be concluded that the proper velocity u is a unimodular paravector, since Lorentz transformations leave square length $u\bar{u}$ unchanged [16, p. 36] and the rest-frame velocity is of unit magnitude (in units of c).

Since rotors are unitary $\left(R^\dagger = \bar{R}\right)$ and boosts are real $\left(B^\dagger = B\right)$, the proper velocity is seen to be $u = \Lambda\Lambda^\dagger = B^2$, and thus the boost may be expressed [16, p. 55] in terms of the proper velocity u as

$$B = u^{1/2} = \frac{1+u}{\sqrt{2\left(\gamma+1\right)}}$$

where

$$\gamma \equiv \langle u \rangle_S = \left(1 - \mathbf{v}^2/c^2\right)^{-1/2} = \cosh w = \frac{dt}{d\tau} \tag{2.9}$$

is the relativistic **Lorentz factor** for velocity $\mathbf{v}$ (rapidity $w = \operatorname{artanh}\frac{v}{c}$), τ is the **proper time** (the time as seen in the rest frame of the particle), and $\langle u \rangle_S$ is the scalar part (A.9) of the proper velocity.

The proper velocity may also be expressed as $u = \gamma\left(1 + \mathbf{v}\right)$, which, along with noting its unimodularity,

$$u\bar{u} = \gamma^2\left(1 - v^2\right) \ , \tag{2.10}$$

is useful in expressing the boost (2.5) in terms of the Lorentz factor γ and boost direction,

$$B = \sqrt{\frac{\gamma+1}{2}} + \hat{\mathbf{v}}\sqrt{\frac{\gamma-1}{2}}, \tag{2.11}$$

which is the form that will be used in Section 3.2 of this dissertation.

Since the particle and/or observer may be in non-uniform relative motion, the eigenspinor relating them may be changing in time, so $\Lambda = \Lambda\left(\tau\right)$. The time evolution of the eigenspinor is given by the following equation of motion [20, p. 4]:

$$\dot{\Lambda} = \frac{1}{2}\mathbf{\Omega}\Lambda = \frac{1}{2}\Lambda\mathbf{\Omega}_{rest} \tag{2.12}$$

where

$$\mathbf{\Omega}_{rest} \equiv 2\bar{\Lambda}\dot{\Lambda} \tag{2.13}$$

is a biparavector that represents the instantaneous **spacetime rotation rate** of the system in its own rest frame (co-moving, but not co-rotating). In the lab frame, the spacetime rotation rate is

$$\Omega = \Lambda\Omega_{rest}\bar{\Lambda} \tag{2.14}$$

where the appropriate conjugation involved in the bilateral application of the eigenspinor to a biparavector is the bar conjugate (A.6) instead of dagger conjugate (A.7), as is used for transformations of real paravectors. It is also useful to note that the spacetime rotation rate in the lab frame can be expressed as

$$\Omega = 2\dot{\Lambda}\bar{\Lambda} \tag{2.15}$$

where the unimodularity of the eigenspinor, $\Lambda\bar{\Lambda} = 1$, interrelates the four expressions above.

2.2.6 Gauge Transformations

A **gauge transformation** is one which leaves measured values unchanged [20, p. 131]. One example is a transformation of the form

$$\Lambda \to \Lambda R_0 \tag{2.16}$$

where R_0 is a rotation in the rest frame, which leaves the proper velocity $u = \Lambda\mathbf{e}_0\Lambda^\dagger$ unchanged. (The invariance can be proven by noting that R_0 commutes with $\mathbf{e}_0$ to simplify with $R_0^\dagger$, since $R_0 R_0^\dagger = R_0\bar{R}_0 = 1$.) This transformation effectively reorients the rest frame, and thus it is permissible to select any orientation for the coordinates representing a system, without loss of generality.

Specifically relevant to this dissertation, when analyzing the spin of a particle it is mathematically convenient to orient the coordinates so that the $\mathbf{e}_3$ direction is along the rest-frame spin direction. Once the gauge freedom has been used to orient the coordinate system in this way, any further gauge rotations may only be about the $\mathbf{e}_3$ direction:

$$\Lambda \to \Lambda \exp\left(-i\mathbf{e}_3\phi\right) \tag{2.17}$$

where the scalar parameter ϕ sets the angular position from which any further rotations about $\mathbf{e}_3$ are measured [20, p. 131].

It is possible for the rest-frame orientation to be time dependent, described by a rotor $R_0(\tau)$. If so, then the spacetime rotation rate (2.13) that appears in the eigenspinor equation of motion (2.12) is transformed as follows:

$$\boldsymbol{\Omega}_{rest} \to \bar{R}_0\left(\boldsymbol{\Omega}_{rest} - i\omega_0\mathbf{e}_3\right)R_0$$

where $-i\omega_0\mathbf{e}_3 = 2\dot{R}_0\bar{R}_0$ is the spatial rotation rate that generates the rest-frame rotation $R_0(\tau)$. This transformation leaves the proper acceleration as well as the proper velocity of a massive particle unchanged [20, p. 130].

The subject of time-dependent rest-frame rotations about $\mathbf{e}_3$ will be discussed again in Section 2.3.4.

2.2.7 Relativistic Applications of APS

The arguments in favour of using APS for relativity have inspired its continued use for explorations of physical theories. This dissertation continues the trend of applying APS to a subject befitting a relativistic treatment and which benefits from a geometric interpretation.

For an example of how APS has provided valuable perspective on other subjects, note that electromagnetism has been treated in detail using APS [14, 17, 28], with results that have united other known approaches [24], enhanced understanding of known results [33], and had considerable educational value [16, 21].

As another example more closely related to the present dissertation, **de Broglie waves** have been examined from an APS standpoint [22], from which they are seen to be the result of a relativistic desynchronizing of phase oscillations at points throughout a boosted extended system, which had been synchronized in the rest frame. The rest-frame phase oscillations that give rise to de Broglie waves will be discussed further in Section 2.3.4, after an introduction to how APS has been applied to various topics in quantum mechanics.

2.3 APS and Quantum Mechanics

The Algebra of Physical Space (APS) has historical connections to, and useful descriptions of, several aspects of quantum mechanics. The relationship of this algebra to descriptions involving a single spin-$\frac{1}{2}$ particle are outlined below in approximate historical order, working towards the recent developments leading up to the topic of the present work. The subsection titles below give a general indication of the noteworthy feature or property of APS introduced or defined within.

2.3.1 Basis of APS

The early papers [10, 71] foreshadowing the development of APS had applied the **Pauli spin matrices** in ways beyond their initial application by Pauli [122] where they were introduced to help describe the interaction between spin-$\frac{1}{2}$ particles and an external electromagnetic field. Specifically, these pre-APS papers highlighted that the three Pauli matrices (A.2),

$$\sigma_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

plus the 2×2 identity matrix, $I_2 \equiv \sigma_0$, together can serve as a basis for relativistic 4-vectors.[4]

[4]Note that the earliest papers used the Pauli matrices directly while emphasizing features of the underlying algebra, while others used the algebra of the matrices, initially calling it the Pauli algebra. Later [12, Ch. 9], it was noted that the term "Pauli algebra" would be a more appropriate name for the algebra of the Pauli matrices, which are but one example of the algebras that adhere to the same rules as the vector-based one discussed here.

It was eventually acknowledged [27] that the Clifford algebra underlying the Pauli matrices was remarkably useful without explicitly requiring the matrix representation. Thus, it was possible to identify the following correspondence between the Pauli matrices and APS basis elements:

$$\mathbf{e}_0 \approx \sigma_0, \quad \mathbf{e}_1 \approx \sigma_1, \quad \mathbf{e}_2 \approx \sigma_2, \text{ and } \mathbf{e}_3 \approx \sigma_3 \tag{2.18}$$

although there are in fact infinitely many matrix representations that may be used to represent the basis elements while still adhering to the same algebraic rules — in particular, the anticommuting products of basis vectors (2.3). Although the use of the algebra itself is the prime focus of this dissertation, it is useful to acknowledge this specific matrix representation when comparing to existing research. This is especially useful when discussing spin, since spin is often represented in terms of matrix components, or as elements in a column, as a spinor.

The emphasis in those pre-APS papers was primarily on the novelty of using these matrices on topics other than spin, but spin itself was not forgotten, as will be described next.

2.3.2 The Dirac Equation

The relativistic nature of the early form of APS, along with its roots in the Pauli spin matrices, highlighted a particular subject as a prime candidate for studying using that newly-discovered algebraic approach [10, 72]: the **Dirac equation**, which describes massive spin-$\frac{1}{2}$ particles in relativistic motion [63]. The Dirac equation has remained a subject of continuing interest throughout the development of APS [11, 20, 23, 31], and the essential details that apply to the present work are outlined here.

2.3.2.1 Classical Dirac equation in APS

The relationship between the paravector momentum $p = mcu$ for a particle and the eigenspinor $\Lambda = BR$, which is the transformation amplitude that is used to obtain the **proper velocity** of the particle via $u = \Lambda\Lambda^\dagger$, gives rise to an equation that may be deemed the **"classical Dirac equation"** in reference to how it relates to the quantum Dirac equation when the momentum is replaced by its quantum operator.

First, note that the eigenspinor is generally taken to be unimodular, $\Lambda\bar{\Lambda} = 1 = \Lambda^\dagger\bar{\Lambda}^\dagger$, which allows a rearrangement of the expression for momentum $p = mc\Lambda\Lambda^\dagger$ by multiplying through by $\bar{\Lambda}^\dagger$ on the right. The result is

$$p\bar{\Lambda}^\dagger = mc\Lambda \tag{2.19}$$

which is an equation satisfied [11, p. 4296] by any eigenspinor Λ up to an arbitrary initial rotation (represented by ΛR_0, where R_0 acts in the rest frame and Λ converts the rest-frame description to

the lab frame) and a real scalar multiplying factor. Any real linear combination of solutions is also a solution [20, p. 11], because of the real linearity of the Dirac equation. (The reality of the solutions may be understood by recalling that a general expression in APS may be treated as a real linear combination of scalar, vector, bivector, and trivector parts, without necessarily involving the unit imaginary.)

2.3.2.2 Antiparticles in the classical Dirac equation

It is also reasonable to consider an eigenspinor that is antiunimodular, $\Lambda\bar{\Lambda} = -1$, since the usual bilateral application of Lorentz transformations (e.g. $x \to x' = \Lambda x \Lambda^\dagger$) only requires the square length of the transformed entity to remain unchanged (e.g. $x\bar{x} = x'\bar{x}'$); thus it is only the *magnitude* $\left|\Lambda\bar{\Lambda}\right|$ of the square length of the transformation operator that must equal one [16, pp. 42–43]. We typically take the square length to be positive, since bilateral applications of such a transformation would ignore any additional phase factor; however, in the case of the Dirac equation, there is valid reason to consider both possible real values satisfying $\left|\Lambda\bar{\Lambda}\right| = 1$. Thus, the Dirac equation can also be satisfied [23, p. 404] by a *dual* eigenspinor:

$$\Lambda_- = \star\Lambda = -i\Lambda$$

where $\star$ represents the **Hodge dual** (A.15), or any real scalar multiple of this.

With this antiunimodular eigenspinor, the classical Dirac equation (2.19) can be rearranged to obtain an expression for momentum that is different from the one originally used to derive the Dirac equation:

$$p = -m\Lambda_-\Lambda_-^\dagger$$

which can be interpreted as describing a particle having energy of sign opposite its momentum, i.e. an **antiparticle**. Thus, the classical Dirac equation is satisfied by an eigenspinor that can be a real linear combination of particle and antiparticle solutions.

2.3.2.3 A more general form of the classical Dirac equation

In general, the unimodularity of the particle eigenspinor Λ_+ and antiunimodularity of the antiparticle eigenspinor Λ_- may be jointly represented [146, 158] using the Yvon-Takabayasi angle $\beta_\pm$, such that $\Lambda_\pm\bar{\Lambda}_\pm = e^{i\beta_\pm}$ and $\Lambda_\pm = e^{i\beta_\pm/2}\Lambda$ where $\Lambda = BR$ is just the Lorentz transformation without any consideration of the particle being acted upon. Thus:

- $\beta_+ = 0$ represents positive-energy solutions, i.e. particles, and gives $e^{i\beta_+} = 1$ and modulus $\Lambda_+\bar{\Lambda}_+ = 1$ and thus $\Lambda_+ = \Lambda = BR$; and

- $\beta_- = \pm\pi$ represents negative-energy solutions, i.e. antiparticles, and gives $e^{i\beta_-} = -1$, $e^{i\beta_-/2} = \pm i$, $\Lambda_-\bar{\Lambda}_- = -1$, and $\Lambda_- = \pm i\Lambda_+$).

In this dissertation, the sign relating the antiunimodular eigenspinor Λ_- to the unimodular Lorentz transformation $\Lambda = BR$ is taken to be negative (as if taking $\beta_- = -\pi$, so $e^{i\beta_-/2} = -i$) as a matter of convenience, in order for the Hodge dual operator to be used to describe the antiparticle eigenspinor relative to the particle one. Of course, since any real scalar multiplying the eigenspinor does not change its validity as a solution to the Dirac equation, this choice makes no difference to the interpretation, only the way it is represented.

It is convenient to represent a more general arbitrary solution to the Dirac equation as Ψ, and the equation itself as

$$p\bar{\Psi}^\dagger = mc\Psi \tag{2.20}$$

so as to compare more directly to the traditional form (2.23). A general solution can represent any real linear combination of particle or antiparticle eigenspinors (but not both), and may take into account a position-dependent scalar function $\rho(x)$ representing a rest-frame distribution [20, p 12]. Furthermore, the orientation and trajectory at various positions x may be described by an eigenspinor field $\Lambda(x)$ rather than a constant eigenspinor. The most general solution $\Psi(x)$, then, can then be deemed a *density-normalized eigenspinor field*, with corresponding *current density* $j(x) = \Psi\Psi^\dagger$, as in the Born interpretation [45] of quantum states being described in terms of probability amplitudes and probability current [25, p. 26] .

Note that even the most general solution $\Psi(x)$ describes an *elementary* particle (or antiparticle); even a real linear superposition of solutions may be considered a single eigenspinor [11, Sec III]. A particle that is not elementary is instead made up of parts that can be described relatively to each other by different eigenspinors (i.e. different orientations or relative motions).

2.3.2.4 Quantum Dirac equation in APS

The differential form of the standard representation of the field-free Dirac equation replaces the momentum p^μ, with an operator, $i\hbar\partial_\mu$. In APS, this is achieved similarly, by replacing the term containing the paravector momentum p in the classical Dirac equation (2.19) with the following:

$$p\bar{\Psi}^\dagger \rightarrow i\hbar\partial\bar{\Psi}^\dagger\mathbf{e}_3 \tag{2.21}$$

where ∂ is the APS gradient operator, $\partial \equiv \mathbf{e}_\mu\partial^\mu = \frac{1}{c}\partial_0 - \nabla$ (A.19).

With the replacement above, and using units where $c = 1$, the classical Dirac equation for particles in APS (2.19) can be expressed in quantum form [31, p. 550]:

$$i\hbar\partial\bar{\Psi}^\dagger\mathbf{e}_3 = m\Psi \tag{2.22}$$

which is comparable to the traditional form [111, p. 598],

$$i\hbar\gamma^\mu \frac{\partial \Psi}{\partial x^\mu} = m\Psi \tag{2.23}$$

where γ^μ are the *Dirac gamma matrices* (A.3).

Note that where the traditional Dirac treatment uses i to represent a quantity that squares to -1, an appropriate geometric version [20, 89] of the Dirac equation uses the bivector $i\mathbf{e}_3$ instead[5]. This bivector not only squares to -1 but also represents the spin plane in the rest frame of the particle — or antiparticle, if negative.

2.3.2.5 Traditional solutions to the Dirac equation

Note that the traditional form of the Dirac equation involves the gamma matrices (A.3), which have dimension 4×4. Its solutions are typically [79] expressed in the form of a four-component column matrix, known as a four-spinor or **bispinor** since it consists of two two-component entities known as spinors:

$$\Psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix} = \begin{pmatrix} \varphi \\ \chi \end{pmatrix} \text{ where} \tag{2.24}$$

$$\varphi = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} \quad \text{and} \quad \chi = \begin{pmatrix} \psi_3 \\ \psi_4 \end{pmatrix}. \tag{2.25}$$

The existence of the two components for each of φ and χ is because there are two eigenstates for a **spin operator**: spin-up or spin-down. The existence of a pair of such two-component parts has two common interpretations; either:

- φ and χ describe the positive and negative energy states, if the gamma matrices are defined using the standard (Dirac-Pauli) representation; or
- φ and χ describe the left- and right-chiral components of states, if the gamma matrices are defined using the chiral (Weyl) representation.[6]

Note, however, that the gamma matrix representation involves matrices are relatively sparse — with no more than half the entries nonzero — and have quantities duplicated throughout, so it is reasonable to consider that a more concise representation is possible.

[5] Recall that the rest-frame rotational gauge freedom (2.16) had enabled selecting the coordinate direction $\mathbf{e}_3$ to align with the spin axis.

[6] Although the Weyl representation is often reserved for massless particles, massive particles can be described in terms of their chiral parts. In other words, since even neutrinos have been realized to have nonzero mass, "Although no known (or even conjectured) particle can possibly be a Weyl fermion, [...] the concept is very useful because Weyl fermions can be used as building blocks of any fermion field." [120].

The APS representation is indeed more concise, as its elements can be represented by 2×2 matrices. Furthermore, in APS, the eigenspinor Ψ can be expressed as a square matrix containing four elements, the same number of elements as the column spinors in the traditional representation. The reconciliation of the two representations will be examined in the following subsections.

2.3.2.6 Decomposition of the eigenspinor

To compare the information content of the traditional and APS representations of the solutions to the Dirac equation [20, 23], we can re-express the APS representation of Ψ in terms of complementary **projectors** (B.1) $\mathrm{P}_3 = \frac{1}{2}(1 + \mathbf{e}_3)$ and $\overline{\mathrm{P}}_3 = \mathrm{P}_{-3} = \frac{1}{2}(1 - \mathbf{e}_3)$ that are associated with the initial spin direction $\mathbf{e}_3$. Since projectors that are **complementary** (B.4) sum to one, the eigenspinor can be multiplied and separated out as follows:

$$\Psi \;=\; \Psi\left(\mathrm{P}_3 + \overline{\mathrm{P}}_3\right) \;=\; \Psi\mathrm{P}_3 + \Psi\overline{\mathrm{P}}_3$$

where each term is expressed in a different *minimal left ideal*, which can be denoted [30] $(\mathrm{APS})\,\mathrm{P}_3$ and $(\mathrm{APS})\,\overline{\mathrm{P}}_3$. In these ideals (see Appendix B.4), each term can be simplified to an even element — scalar plus bivector — multiplying the projector. This is useful because each even element (A.11) can be expressed as a dilated rotor, and thus the entire eigenspinor can be decomposed into ideals (B.12) (B.14) as:

$$\Psi\mathrm{P}_3 + \Psi\overline{\mathrm{P}}_3 \quad = \quad k_+ R_+ \mathrm{P}_3 \;+\; k_- R_- \overline{\mathrm{P}}_3 \tag{2.26}$$

where $k_\pm$ are the scalar dilation factors, and $R_\pm$ represent the rotations, and where the subscripts indicate the corresponding projectors.

The effect of expressing the eigenspinor in terms of projectors is to transform the boost operators so they become **unitary** ($U^\dagger U = 1$), which is essential for describing a transformation of one quantum state to another, preserving the sum of probabilities.

2.3.2.7 From projected eigenspinors to column spinors

If the APS basis elements are replaced by Pauli matrices (A.2) to obtain a matrix representation, the effect of multiplying an APS element from the right by each of these particular projectors individually can be interpreted as extracting one of the columns of the matrix, leaving zeros in the other column (see Appendix B.3). The information in the nonzero columns can be seen to be comparable to the two-component spinors from the traditional representation.

The Dirac equation (2.22) can thus be split into two parts, each of which describes one column from its 2×2 Pauli matrix representation:

$$\begin{aligned} & p\bar{\Psi}^\dagger \mathrm{P}_{+3} = i\hbar\partial\bar{\Psi}^\dagger \mathbf{e}_3 \mathrm{P}_{+3} = m\Psi\mathrm{P}_{+3} \\ \rightarrow & \boxed{p\bar{\Psi}^\dagger \mathrm{P}_{+3} = i\hbar\partial\bar{\Psi}^\dagger \mathrm{P}_{+3} = m\Psi\mathrm{P}_{+3}} \end{aligned} \tag{2.27}$$

$$\begin{aligned} & p\bar{\Psi}^\dagger \mathrm{P}_{-3} = i\hbar\partial\bar{\Psi}^\dagger \mathbf{e}_3 \mathrm{P}_{-3} = m\Psi\mathrm{P}_{-3} \\ \rightarrow & \boxed{p\bar{\Psi}^\dagger \mathrm{P}_{-3} = -i\hbar\partial\bar{\Psi}^\dagger \mathrm{P}_{-3} = m\Psi\mathrm{P}_{-3}} \end{aligned} \tag{2.28}$$

where the projectors have each "gobbled" the unit vector $\mathbf{e}_3$ in a different way via the Pacwoman property (B.7): $\mathbf{e}_3\mathrm{P}_{\pm 3} = \pm\mathrm{P}_{\pm 3}$.

The second of the projected forms above can be expressed in the same minimal left ideal (APS) P_3 as the first, by taking its bar-dagger conjugate:

$$\bar{p}\Psi\mathrm{P}_{+3} = i\hbar\bar{\partial}\Psi\mathrm{P}_{+3} = m\bar{\Psi}^\dagger\mathrm{P}_{+3} \ . \tag{2.29}$$

This way, both projections (2.27) and (2.29) of the Dirac equation (2.22) are in a form that have matrix representations whose first column is equivalent to a two-component spinor, as seen in Equations (B.9) and (B.11).

2.3.2.8 Weyl bispinor

The projected solutions $\Psi\mathrm{P}_3$ and $\bar{\Psi}^\dagger\mathrm{P}_3$, which may be treated as independent of each other [11, Sec. IV], can be combined [31, p. 550] into a single four-component spinor Ψ_4:

$$\Psi_4 = \frac{1}{\sqrt{2}} \begin{pmatrix} \bar{\Psi}^\dagger\mathrm{P}_3 \\ \Psi\mathrm{P}_3 \end{pmatrix}$$

where the order of elements has been selected so that Ψ_4 can satisfy the following single matrix equation formed from the pair of coupled equations that are satisfied by the projected solutions:

$$\begin{aligned} & \begin{bmatrix} 0 & i\hbar\bar{\partial} \\ i\hbar\partial & 0 \end{bmatrix} \Psi_4 = m\Psi_4 \\ \text{i.e.} \quad & \left(\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} i\hbar\partial^0 + \begin{bmatrix} 0 & -\mathbf{e}_k \\ \mathbf{e}_k & 0 \end{bmatrix} i\hbar\partial^k \right) \Psi_4 = m\Psi_4 \end{aligned}$$

which, in terms of the Weyl representation for the gamma matrices (A.3), gives us the Dirac equation in its traditional form: $\gamma_\mu i\hbar\partial^\mu\Psi_4 = m\Psi_4$. This means the spinor Ψ_4 can be associated with the

Weyl bispinor, and can be denoted

$$\Psi_W \equiv \begin{pmatrix} \Psi_R \\ \Psi_L \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \bar{\Psi}^\dagger \mathrm{P}_3 \\ \Psi \mathrm{P}_3 \end{pmatrix} \tag{2.30}$$

so that the right-chiral part of the Weyl bispinor is given by $\frac{1}{\sqrt{2}}\bar{\Psi}^\dagger \mathrm{P}_3$ and the left-chiral part by $\frac{1}{\sqrt{2}}\Psi \mathrm{P}_3$. Thus, the projectors have a similar effect as the traditional *chirality operators* $\frac{1}{2}(1 \pm \gamma_5)$, where $\gamma_5 = -i\gamma_0\gamma_1\gamma_2\gamma_3$, in that they extract the left-chiral and right-chiral parts from the bispinor.

2.3.2.9 Dirac bispinor

If the desired information about the dynamics is in terms of the particle-antiparticle split, rather than the chirality, we can take the even and odd parts of the eigenspinor, $\langle\Psi\rangle_\pm = \frac{1}{2}\left(\Psi \pm \bar{\Psi}^\dagger\right)$, and combine their P_3 projections to form the four-component spinor in Dirac-Pauli (or "standard") representation:

$$\Psi_{DP} = \begin{pmatrix} \langle\Psi\rangle_+ \mathrm{P}_3 \\ \langle\Psi\rangle_- \mathrm{P}_3 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \Psi_W = \frac{1}{\sqrt{2}} \begin{pmatrix} \Psi_L + \Psi_R \\ \Psi_L - \Psi_R \end{pmatrix} . \tag{2.31}$$

where Ψ_W is the Weyl bispinor (2.30).

Note that the even $\langle\Psi\rangle_+$ and odd $\langle\Psi\rangle_-$ parts are not only even and odd in terms of their vector grades in APS, but also in their response to parity inversion. For positive-energy states, the projected even and odd parts correspond to the large (major) and small (minor) components respectively, with the odd component reducing to zero in the rest frame; thus, particles have even intrinsic parity. The opposite is true for negative-energy states: the even and odd parts of Ψ are small and large respectively, with the even part reducing to zero in the rest frame, and thus antiparticles have odd intrinsic parity [15].

2.3.2.10 Interpretation of Dirac spinors

The descriptions above have shown that the APS eigenspinor representation of solutions to the Dirac equation contains the same information as traditional representations, though some manipulation is required to compare the results. This is because the geometrical significance is "hidden" in the traditional representations [25, p. 533] and only by projecting the APS eigenspinor in particular ways is the traditional information obtained.

The fact that the APS solutions can be expressed entirely as eigenspinors suggests that the Dirac equation primarily describes *the dynamics of transformation amplitudes*. However, mathematically projecting the solutions is also essential, since this extracts the information about the effect of the

transformation on the **spin** of the fundamental entity being examined (whether it is a particle or antiparticle). It is interesting to consider that the Dirac equation itself might not describe quantum states at all, not until the projectors are applied.

This dissertation will be continuing with the idea of a separation between the transformation and the state, applying it to the notion of spin entanglement. The idea (described in Chapter 3) is to apply relativistic boosts to the eigenspinors describing stationary entangled states, and examine the resulting transformation amplitudes *before* applying the projectors to extract state information. By doing so, it is the intention of this author to further demystify the quantum nature of spin and entanglement.

2.3.3 Spin States

As noted in Section 2.3.2, the Dirac equation is traditionally known for describing spin. The very form of the solution to the Dirac equation is as a set of two-component spinors (2.25), where the two components represent spin up and spin down basis states along the selected quantization axis, where these basis states are the eigenstates of a spin operator.

However, the APS description makes available a solution that describes the dynamics of transformation amplitudes, which must be *projected* in order to compare to traditional descriptions involving spin. Thus, it may be interpreted that it is the *projectors* that provide the key to describing the spin states themselves. This interpretation is verified below.

2.3.3.1 Spin-up basis state spinor

As mentioned in Section 2.2.6, the orientation of the coordinate system is selected (without loss of generality) so that the $\mathbf{e}_3$ direction aligns with the initial rest-frame spin direction. This direction is also associated with the Pauli matrix that is diagonal, σ_3, and therefore the two-component column spinors that represent the eigenstates each have only one nonzero component in the traditional representation. The **spin operator** for this direction is traditionally given by $S_z = \frac{\hbar}{2}\sigma_3$.

In APS, the projector associated with the initial spin direction $\mathbf{e}_3$ has an important role in relating the APS Dirac equation to the traditional representations, in that it projects out the first column of the matrix representation so that the column is comparable (B.9) to a two-component spinor. Examining the matrix representation of the projector itself,

$$\mathrm{P}_3 = \frac{1}{2}\left(1 + \mathbf{e}_3\right) \approx \frac{1}{2}\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} . \tag{2.32}$$

it is seen that its own first column can be associated with the two-component spinor that typically represents a spin-up state.

Furthermore, the P_3 projector is easily seen via the Pacwoman property (B.7) to behave as an eigenstate of the APS version of the spin operator, $S_z = \frac{\hbar}{2}\mathbf{e}_3$:

$$S_z \mathrm{P}_3 = \frac{\hbar}{2}\mathbf{e}_3 \mathrm{P}_3 = \frac{\hbar}{2}\mathrm{P}_3$$

with eigenvalue $\frac{\hbar}{2}$. Thus, it is convenient and appropriate to define the **spin-up state spinor** as

$$\psi_\uparrow \equiv \mathrm{P}_3 \sim \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{2.33}$$

where the two-component representation typically associated with this state is shown here only for clarity.

2.3.3.2 Spin-down basis state spinor

With the spin-up state $\psi_\uparrow$ defined as a spinor, transformations of it must be applied unilaterally[7]. Thus, the **spin-down state spinor** $\psi_\downarrow$ can be defined through the unilateral application of a rotor, $R_\pi = \exp\left(\mathbf{e}_1\mathbf{e}_3\frac{\pi}{2}\right) = \mathbf{e}_1\mathbf{e}_3$, which enacts a rotation by π in the $\mathbf{e}_1\mathbf{e}_3$ plane.

Note that the $\mathbf{e}_1$ coordinate direction is *defined* as aligning with the rotation plane used to turn spin-up to spin-down, without loss of generality because of the rotational gauge freedom (2.17) about the spin direction. Choosing a different direction perpendicular to $\mathbf{e}_3$ to form the plane of rotation would affect only the phase of the resulting spin-down state; hence, selecting a particular direction fixes a particular representation as a zero-phase reference spin-down state. A geometric interpretation of selecting such a direction must wait until there is additional geometry described by the system, such as that of a direction of motion.

Thus, the spin-down state can be expressed as

$$\psi_\downarrow \equiv \mathbf{e}_1\mathbf{e}_3\mathrm{P}_3 = \mathbf{e}_1\mathrm{P}_3 \sim \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{2.34}$$

where the Pacwoman property (B.7) is used to simplify the expression to a less cumbersome way of denoting the state. Furthermore, $\psi_\downarrow$ is easily seen via the Pacwoman property (B.7) and a related property for unit vectors perpendicular to the projector, $\mathbf{e}_1\mathrm{P}_3 = \overline{\mathrm{P}}_3\mathbf{e}_1$ (B.8), to behave as

[7] See Footnote 3 on page 25.

an eigenstate of the spin operator, $S_z = \frac{\hbar}{2}\mathbf{e}_3$:

$$\begin{aligned} S_z\psi_\downarrow &= \frac{\hbar}{2}\mathbf{e}_3\mathbf{e}_1\mathrm{P}_3 = \frac{\hbar}{2}\mathbf{e}_3\overline{\mathrm{P}}_3\mathbf{e}_1 \\ &= -\frac{\hbar}{2}\overline{\mathrm{P}}_3\mathbf{e}_1 = -\frac{\hbar}{2}\mathbf{e}_1\mathrm{P}_3 \\ \therefore S_z\psi_\downarrow &= -\frac{\hbar}{2}\psi_\downarrow \end{aligned}$$

with eigenvalue $-\frac{\hbar}{2}$.

The Pauli matrix representation of the spin-down state is

$$\begin{aligned} \psi_\downarrow = \mathbf{e}_1\mathrm{P}_3 = \frac{1}{2}\left(\mathbf{e}_1 + \mathbf{e}_1\mathbf{e}_3\right) &= \frac{1}{2}\left(\mathbf{e}_1 - i\mathbf{e}_2\right) \\ &\approx \frac{1}{2}\left(\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} - i\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}\right) = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \end{aligned} \tag{2.35}$$

which again can have its second column dropped, so that the correspondence between the APS representation and the traditional column representation is established.

These representations for the spin-up and spin-down states may be used as a basis for the two-dimensional abstract Hilbert space that typically describes spin states as a simple two-level system. The underlying geometric nature of these states can serve well in illuminating physical effects and in possibly guiding development of physically-realizable experiments.

2.3.3.3 General spin states as rotated spin-up states

A general spin state may be modelled as a rotated spin-up state spinor, given by $R\psi_\uparrow = R\mathrm{P}_3$, using the **Bloch sphere** interpretation of spin states as depicted in Figure 2.2. The rotor R here may be acknowledged as a simpler case of the eigenspinor introduced in Section 2.2.5, one not containing any velocity transformation, so it describes a particle in a rotated lab frame that remains at rest relative to the particle.

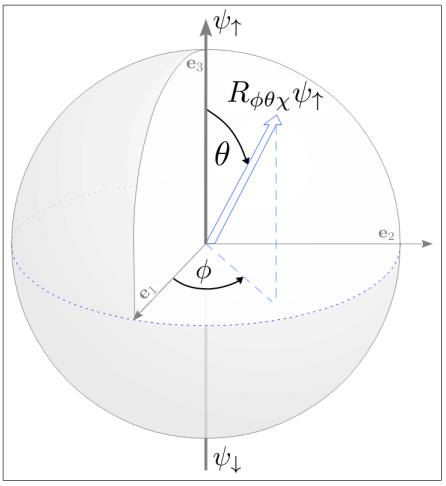


Figure 2.2: The Bloch sphere interpretation models a pure spin state ψ as a point on a sphere, where the north and south poles represent spin-up and spin-down respectively. All other complex linear superpositions of spin-up and spin-down are given by points on the sphere. An arbitrary spin state given by a Euler rotation of spin-up (2.37) is rotated by θ about the $\mathbf{e}_2$ direction and ϕ about the $\mathbf{e}_3$ direction. A phase rotation of χ is modelled as an initial rotation about the spin-up direction before the other rotations are applied.

Consider an arbitrary rotation expressed in terms of Euler angles [20, Eq. 38],

$$R_{\phi\theta\chi} = R_\phi R_\theta R_\chi = \exp\left(-i\mathbf{e}_3\frac{\phi}{2}\right)\exp\left(-i\mathbf{e}_2\frac{\theta}{2}\right)\exp\left(-i\mathbf{e}_3\frac{\chi}{2}\right) \tag{2.36}$$

which, when applied to the spin-up projector, becomes

$$R_{\phi\theta\chi}\psi_\uparrow = \exp\left(-i\mathbf{e}_3\frac{\phi}{2}\right)\exp\left(-i\mathbf{e}_2\frac{\theta}{2}\right)\exp\left(-i\frac{\chi}{2}\right)\mathrm{P}_3 \tag{2.37}$$

where the rotor R_χ adjacent to the projector P_3 was simplified by the Pacwoman property (B.7) to become a phase factor (see also Section 2.3.3.5).

An interpretation of the arbitrarily rotated spinor $R\mathrm{P}_3$ may be as a rotor R that is "projected" onto the minimal left ideal (APS) P_3, so that its matrix representation (B.9) is equatable to a traditional two-component spinor.

2.3.3.4 Linear combination spinor states

To see how an arbitrarily rotated state $R\mathrm{P}_3$ can be expressed in terms of spin-up and spin-down components, we multiply it from the left by the sum of two complementary projectors $\mathrm{P}_3 + \overline{\mathrm{P}}_3 = 1$ (B.4):

$$\begin{aligned}\left(\mathrm{P}_3 + \overline{\mathrm{P}}_3\right)(R\mathrm{P}_3) &= \mathrm{P}_3\,(R\mathrm{P}_3) + \overline{\mathrm{P}}_3\,(R\mathrm{P}_3)\\ &= a\mathrm{P}_3 \quad + b\mathbf{e}_1\mathrm{P}_3\\ &= a\psi_\uparrow \quad + b\psi_\downarrow\,.\end{aligned} \tag{2.38}$$

The resulting scalar coefficients a and b are known as **probability amplitudes**[8], and they may be complex. Their absolute squares give the measurement probabilities for each basis state, and thus the state must be **normalized** so that these probabilities sum to one: $|a|^2 + |b|^2 = 1$.

For the arbitrarily rotated state expressed in terms of Euler angles (2.37), the application of complementary projectors gives

$$\left(\mathrm{P}_3 + \overline{\mathrm{P}}_3\right)\left(R_{\phi\theta\chi}\mathrm{P}_3\right) = e^{-i\chi/2}\left[e^{-i\phi/2}\cos\frac{\theta}{2}\psi_\uparrow + e^{i\phi/2}\sin\frac{\theta}{2}\psi_\downarrow\right] \tag{2.39}$$

such that $|a|^2 = \cos^2\frac{\theta}{2}$ is the probability of measuring spin up and $|b|^2 = \sin^2\frac{\theta}{2}$ is the probability of measuring spin down. Note that $e^{i\phi}$ is the relative phase between the components, while $e^{-i\chi/2}$ is a global phase which may be measurable in an ensemble of identical states (see Section 2.3.4 for an interpretation).

The linear combination state may be easily expressed in Pauli matrix representation using the matrix representations of the basis states (2.32) and (2.35), where again the column of zeros may be dropped to obtain the usual two-component spinor. The two coefficients of the APS basis states may also be understood to correspond to the two components of the spinor, so the matrix representation need not be given explicitly here. Indeed, there is no further need to refer to the matrix representation at all.

2.3.3.5 Phase factors

One of the features of the Hilbert space that underlies the description of spin states is that a state is unchanged by multiplication by a complex scalar. Multiplying factors may have useful interpretations when they arise in a calculations, but in general, an overall unit-magnitude phase factor is generally ignored unless it is involved in a comparison relative to another state.

A phase factor in APS can be given a geometric interpretation. Since a phase factor such as $\exp(i\alpha)$ is a complex scalar, it commutes with everything in the algebra and may be moved adjacent to the P_3 projector. In this position, the phase factor may be equivalently represented as a rest-frame rotation about the $\mathbf{e}_3$ direction:

$$\exp(i\alpha)\,\mathrm{P}_3 = \exp(i\alpha\mathbf{e}_3)\,\mathrm{P}_3 \tag{2.40}$$

since the Pacwoman property (B.7) "ungobbles" the unit vector corresponding to the projector. While this rotation has no effect on the relative measurement probabilities of the state, its interpretation as a rotation about the spin direction is geometrically more interesting than an overall multiplicative factor. Further interesting effects of such a rotation are discussed in Section 2.3.4.

[8] The coefficients are referred to as **amplitudes** because spin states can interfere, like waves, such as in a Stern-Gerlach measurement device before the two beams are completely clear of each other's influence [25, pp20-22].

It is worth mentioning that a rotation by 2π about any direction introduces an overall phase factor of -1, which has been touted as a particularly odd quantum property of spin. With spin being represented by a spinor, the mathematical reason for the phase factor follows from the unilateral application of the rotor describing the rotation, and the fact that a rotor has the form $R = \cos\frac{\theta}{2} - i\hat{\theta}\sin\frac{\theta}{2}$ (where the factor of $\frac{1}{2}$ arises from the usual bilateral form for vectorial rotations; see Footnote 3 on page 25). Thus, a rotation angle of $\theta = 2\pi$ always gives $R = -1$, no matter the rotation axis $\hat{\theta}$. The geometric interpretation of this overall factor of -1 may still be puzzling, but at least its presence reinforces the idea that spin states are correctly modelled as spinors in the APS representation, since this factor is present in traditional representations as well.

2.3.4 Geometric Interpretation of Spin

If a phase factor applied to a rest-frame state projector P_3 — or a global phase factor of an arbitrarily rotated state (2.39) — is time-dependent, it can be interpreted as a self-rotation about the spin axis $\mathbf{e}_3$. Furthermore, as mentioned in Section 2.2.6, any rest-frame rotation of the eigenspinor about the $\mathbf{e}_3$ direction leaves calculations of velocity and acceleration unchanged, and is thus considered a gauge transformation.

Thus, the self-rotation of the rest-frame state may be considered a gauge transformation, and the geometric interpretation of this may be as the **intrinsic spin** of the particle [20, pp. 5-6]. This does not necessarily claim that the particle itself is spinning, but this is evidence that spin angular momentum may be modelled geometrically and dynamically.

It may be shown [23] that the geometric treatment of spin as a physical (yet intrinsic) spinning motion may have observable consequences in cases allowing superposition and interference. Specifically, such a rest-frame oscillation may be detected as **de Broglie waves** in a moving reference frame [22], and interpreted as an intrinsic rotation rate at the **Zitterbewegung** frequency [25, p. 13] of $2\omega_0 = 2E_0/\hbar = 2mc^2/\hbar$. Although an intriguing relationship between spin and Zitterbewegung has been acknowledged in several instances [88, 93, 141], there are also examples of research [9, 101] claiming Zitterbewegung is unphysical or undesirable. Continuing to investigate spin using a geometric interpretation could reveal new ways to investigate Zitterbewegung and set the matter to rest.

The notion of an intrinsic rotation may suggest renaming the "rest frame" to something else, such as *reference frame* [23], but the author of this present work prefers to simply define "rest frame" as the frame instantaneously co-moving with the system in question, but not co-rotating with any aspect of it.

2.3.5 Spin Density Operators

The spinor state representation falls short when there is a need to describe spin systems having some unavailable information, such as when there is decoherence, or when examining a subsystem without access to the whole system or when ignoring part of a system (performing a partial trace) for the sake of making a calculation. Such cases are beyond the scope of this dissertation, however the representation that is used to describe such states has some geometric value that is worth mentioning.

The **state density operator** has the form of a product of states ψ_i with their Hermitian conjugates $\psi_i^\dagger$:

$$\rho = \sum_i \psi_i \psi_i^\dagger \ .$$

Its value will become more clear with some examples.

2.3.5.1 Pure spin states

A system is said to be a **pure state** if there is complete knowledge about the state it is in, or if that knowledge is theoretically possible[9]. Such a system can be described as being in a superposition of basis eigenstates, or if described in an appropriate basis, the system can be said to be in one of the basis eigenstates.

In APS, a pure spin state can be described as a rotated spin-up state $\psi_R = R\psi_\uparrow$. The density operator for this pure spin state [20, p. 15] is:

$$\rho = \psi_R \psi_R^\dagger = (R\mathrm{P}_3)\left(\mathrm{P}_3^\dagger R^\dagger\right) = R\mathrm{P}_3 R^\dagger = \mathrm{P}_s \tag{2.41}$$

and thus the density operator for a pure state is a projector, $\mathrm{P}_s = \frac{1}{2}(1 + \hat{\mathbf{s}})$, with all the powers that such a description entails (see Appendix B). This projector is an eigenstate of the spin operator $S_s = \frac{\hbar}{2}\hat{\mathbf{s}}$, and the spin itself can be modelled as being in the direction $\hat{\mathbf{s}}$ in the Bloch sphere representation (see Figure 2.2).

Since the spin-up state in the $\mathbf{e}_3$ direction involves no rotation from the standard basis, the density operator for it is clearly seen to be $\rho_\uparrow = \mathrm{P}_3$, which gives the P_3 projector another[10] interesting application within the APS repertoire. The corresponding spin-down density operator is

$$\begin{aligned} \rho_\downarrow &= \psi_\downarrow \psi_\downarrow^\dagger = \mathbf{e}_1 \mathrm{P}_3 \left(\mathbf{e}_1 \mathrm{P}_3\right)^\dagger \\ &= \mathbf{e}_1 \mathrm{P}_3 \mathrm{P}_3 \mathbf{e}_1 = \mathbf{e}_1 \mathrm{P}_3 \mathbf{e}_1 = \mathbf{e}_1 \mathbf{e}_1 \overline{\mathrm{P}}_3 \\ \therefore \rho_\downarrow &= \overline{\mathrm{P}}_3 = \frac{1}{2}(1 - \mathbf{e}_3) \end{aligned}$$

[9] This "theoretical possibility of certainty" can be demonstrated using an ensemble of identically-prepared particles in an unknown yet pure spin state, where it is possible to take enough measurements to enable calculating the spin probabilities in all directions, and thus determine the unknown spin direction with certainty.

[10] See Appendix B.5.

so that the spin-down state is in the direction $-\mathbf{e}_3$. Note that the spin-up and spin-down density operators are complementary: $\mathrm{P}_3 + \overline{\mathrm{P}}_3 = 1$ (B.4).

Furthermore, it is worth mentioning that traditional bra-ket notation represents a pure state density operator as a projector; for example, the spin-up density operator can be given by $\rho_\uparrow = |\uparrow\rangle \langle\uparrow|$. This is equivalent to the APS product of a state (akin to the ket) and its Hermitian conjugate (akin to the bra).

2.3.5.2 Mixed states

A state that is not pure is called a **mixed state**, so named because it often arises as the description of an ensemble of states containing a *statistical mixture* of subsystems that are each in a pure state [111, p. 366]. Such a mixed state cannot be expressed as a complex linear combination of eigenstates (which would make it a pure state); it must instead be represented as a real linear combination of density operators for the pure states that make up the mixture, with coefficients representing probabilities.

For example, if an ensemble of spin-$\frac{1}{2}$ particles known to contain a fraction a of states prepared in a spin-up state and b in a spin-down state, the density operator is

$$\rho_{a,b} = a\rho_\uparrow + b\rho_\downarrow$$

such that $a + b = 1$. However, the same density operator may be obtained from some other mixture of states, say, prepared in a different basis, so in general it is not possible to extract information about the actual composition of a mixture from the density operator. Indeed, the mixed state may be a result of incomplete information rather than a statistical mixture of known information, and thus the full extent of the details of its composition is even more assuredly unavailable.

For the case of a spin-$\frac{1}{2}$ particle in APS, an arbitrary density matrix may be expressed in the form $\rho_\alpha = \frac{1}{2}(1 + \alpha\hat{\mathbf{s}})$ where $0 \leq \alpha < 1$ [49, Appx. H] gives a mixed state. The particular mixed state with $\alpha = 0$ indicates a **fully unpolarized state** of half spin-up and half spin-down:

$$\begin{aligned}
\rho_{\alpha=0} &= \frac{1}{2}\rho_\uparrow + \frac{1}{2}\rho_\downarrow \quad = \quad \frac{1}{2}\mathrm{P}_3 + \frac{1}{2}\overline{\mathrm{P}}_3 \\
&= \frac{1}{2}\left[\frac{1}{2}(1 + \mathbf{e}_3)\right] + \frac{1}{2}\left[\frac{1}{2}(1 - \mathbf{e}_3)\right] \\
&= \frac{1}{4}(1 + \mathbf{e}_3 + 1 - \mathbf{e}_3) \\
&= \frac{1}{2}
\end{aligned}$$

so α may be described as a percentage of purity for the state: a completely mixed state has $\alpha = 0$ and is 0% pure, while a pure state has $\alpha = 1$ and is 100% pure and thus may be represented as a projector $\rho_{\mathbf{s}} = \frac{1}{2}(1 + \hat{\mathbf{s}})$.

While mixed states are not featured any further in this dissertation, they were important to introduce in order to define the scope of this work through demonstrating by counterexample what is meant by a state that is *not* pure. To reiterate the difference: a pure state is one that can be described as a linear combination of *eigenstates*; a mixed state is one that must be described as a linear combination of *density matrices*.

2.3.6 Bipartite States and Spin Entanglement

As mentioned in Chapter 1, entanglement is one of the key features that sets quantum mechanics apart from the classical description of reality. It is defined as a superposition of states that cannot have its parts described independently. In the case of entangled spin states, the spin measurement results have some degree of correlation or anticorrelation in their measurement results, beyond what can be explained classically.

This section outlines some details about how entanglement has been investigated so far using the Algebra of Physical Space [20, 23, 49], and also includes a few other important aspects of entanglement that will be used in Chapter 3. Many of the features of the treatment below are common to most of the typical mathematical treatments of entanglement [35, 117], but they are described using the APS formalism in terms of the pure spin states given in Section 2.3.3.

2.3.6.1 Tensor products for bipartite states

The APS approach to describing pairs of spin-$\frac{1}{2}$ particles follows the common approach [83] of using direct products of the vector spaces used to describe the individual parts of the pair. In such a direct product, the vector spaces are independent of each other, even though manipulations may be performed on the set of both spaces. **Bipartite spin states** are described by tensor products of single-particle spin states. (See Appendix E for some notes on tensor products of elements from a direct product of vector spaces.)

For the distinguishable[11] particles that are treated in this dissertation, the tensor product notation implies that it is understood which particle is described by the first factor in each tensor product, and which is described second. The nature of the distinguishability depends on the physical implementation, and is not addressed within the algebraic implementation.

2.3.6.2 Separable bipartite states

The simplest bipartite states are ones made from entities that can be described separately, and are combined together into a simple tensor product of single particle states. That is, if one system is

[11] Indistinguishable particles can be accommodated by including additional terms with the roles of the particles reversed.

in a state A, and another in a state B, their combined system can be said to be in a **separable bipartite state** $A \otimes B$.

For example, a pair of spin-up states can be represented by the tensor product of spin-up states, $\psi_\uparrow \otimes \psi_\uparrow$, and a pair of spin-down states by $\psi_\downarrow \otimes \psi_\downarrow$. It is also possible that a pair of separable spins has one in a spin up state and in one down, and the two ways to represent this — $\psi_\uparrow \otimes \psi_\downarrow$ or $\psi_\downarrow \otimes \psi_\uparrow$ — have distinct meanings due to the distinguishability of the particles and the implication by the direct product that each space represents a specific particle.

An arbitrary separable state at rest can be described in by separate rotations each acting on a spin-up state:

$$\begin{aligned} \Psi_{AB} &\equiv \psi_A \otimes \psi_B \\ &= R_A \psi_\uparrow \otimes R_B \psi_\uparrow \\ &= (R_A \otimes R_B)\,[\psi_\uparrow \otimes \psi_\uparrow] \\ &= \mathbf{R}_{AB}\,[\psi_\uparrow \otimes \psi_\uparrow] \end{aligned}$$

where the rotation operators have been factored away from the spin-up states using the multiplication rule of Kronecker products (E.2), $(A \otimes B)\,(C \otimes D) = AC \otimes BD$, which emphasizes the separate vector spaces for each particle. Note the notation: the boldfaced $\mathbf{R}$ indicates a bipartite version of the single-state rotor R, and the subscript AB indicates separability into "particle A" and "particle B", so that the bipartite state spinor Ψ_{AB} describes a separable state containing single-state spinors ψ_A and ψ_B.

2.3.6.3 General bipartite pure states

In general, a bipartite state may not be separable, even if it is pure. **Entangled states** are defined as those which cannot be expressed as a product of single-particle states. Such states can be expressed as linear combinations (superpositions) of separable bipartite eigenstates, so they can still be considered pure states. The information in such a state, however, cannot be associated with the individual particles, as it could in the separable case.

For example, a general pure state may be expressed as a superposition of the **separable basis states** consisting of all four pairings of spin-up and spin-down single-particle states:

$$\Upsilon = a\,[\psi_\uparrow \otimes \psi_\uparrow] + b\,[\psi_\uparrow \otimes \psi_\downarrow] + c\,[\psi_\downarrow \otimes \psi_\uparrow] + d\,[\psi_\downarrow \otimes \psi_\downarrow] \tag{2.42}$$

$$\equiv a\Upsilon_{\uparrow\uparrow} + b\Upsilon_{\downarrow\downarrow} + c\Upsilon_{\downarrow\uparrow} + d\Upsilon_{\uparrow\downarrow} \tag{2.43}$$

where state normalization requires $|a|^2 + |b|^2 + |c|^2 + |d|^2 = 1$. As usual, the coefficients of the bipartite eigenstates are probability amplitudes, which implies the possibility of interference with

other amplitudes, and they square to the probabilities associated with the possible measurement results for the pair of states.

Note that the separable bipartite basis states are represented by $\Upsilon_{jk} \equiv \psi_j \otimes \psi_k$, where $j, k \in \{\uparrow, \downarrow\}$. The unusual choice of the Greek letter Υ has been selected here so as to enable Ψ to stand out more clearly when it is used later.

If the expanded general state (2.42) can be factored into a product of the form $A \otimes B$, the state is separable. In all other cases, there is some degree of **entanglement** in the state. If the states are separable, their measurement results have no correlation other than what is possible classically (i.e. both particles may be aligned without being entangled). If there *is* any entanglement, then there is the possibility for correlation in the measurement results beyond what is possible classically, which can be demonstrated experimentally if the state represents a collection of identical pairs of particles upon which repeated measurements can be made.

2.3.6.4 Concurrence as a measure of entanglement

A simple and appropriate measure of entanglement for the pure states that are the focus of this dissertation is the **concurrence** [157, p. 32], which for stationary pure states in the separable bipartite basis has the form (2.43)

$$C(\Upsilon) = 2\left|ab - cd\right|, \tag{2.44}$$

where a, b are the coefficients of the parallel separable basis states, and c, d are the coefficients of the antiparallel separable basis states.

This simple expression for concurrence (2.44) shows that the state is separable ($C = 0$) if, for example, any three of the separable-basis coefficients are zero, or if all the separable-basis coefficients have equal magnitude. The concurrence is maximized ($C = 1$) if

$$|a| = |b| = \frac{1}{\sqrt{2}} \text{ or } |c| = |d| = \frac{1}{\sqrt{2}}, \tag{2.45}$$

while values between zero and one indicate partial entanglement.

More generally, the concurrence of an arbitrary (and not necessarily stationary) pure state Υ is given by the magnitude of the Hermitian inner product (A.14) of a state with its time-reversed self:

$$C(\Upsilon) \equiv \left|\left\langle \Upsilon, \widetilde{\Upsilon} \right\rangle\right|$$

which is derived using APS and examined in detail in Appendix F.4.1.

2.3.6.5 Maximally entangled states

A state with the maximal measure of entanglement, i.e. having concurrence (2.44) $C = 1$, is called a **maximally entangled spin state**. By examining the general state (2.43) and the criteria for maximal entanglement (2.45), it can be determined that there are four unique (up to an overall phase[12]) maximally entangled states:

$$\begin{aligned} \Psi_{\pm} &= \tfrac{1}{\sqrt{2}} \left(\Upsilon_{\downarrow\uparrow} \pm \Upsilon_{\uparrow\downarrow} \right) \\ \Phi_{\pm} &= \tfrac{1}{\sqrt{2}} \left(\Upsilon_{\uparrow\uparrow} \pm \Upsilon_{\downarrow\downarrow} \right). \end{aligned} \tag{2.46}$$

The choice of state labels here adheres to the custom [43, 92, 124, 160] of using $\Psi_{\pm}$ to represent entanglement involving equal-magnitude linear superpositions of antiparallel separable basis states, and $\Phi_{\pm}$ to represent entanglement involving equal-magnitude linear superpositions of parallel separable basis states. The subscript $\pm$ indicates the relative phase between the terms in the superposition.

Each maximally entangled state can be interpreted as containing two *bits* of information [126, Ch. 4]: the *relative parity bit* describes whether the separable states involved are parallel or antiparallel, and the *relative phase bit* indicates whether the terms in the superposition have the same or opposite phase (i.e. a relative phase factor of ± 1).

The four maximally entangled states (2.46) are orthogonal, and can be used as a basis — the **Bell basis**[13] — for bipartite states. Any bipartite state, even separable states, may be expressed as a superposition of the four maximally entangled basis states:

$$\Upsilon = a\Phi_{+} + b\Phi_{-} + c\Psi_{+} + d\Psi_{-}. \tag{2.47}$$

Even though the components making up each entangled basis state involve only either parallel or antiparallel spins, predictions about correlations or anticorrelations are not straightforward, as will be seen next.

[12] Note that the overall phases of the Bell states as defined here are slightly different from the usual definitions of the Bell states. Usually, the first sub-state in the first term is chosen to have the same spin direction for all four basis states, although whether that state is spin-up or spin-down seems to depend on the notation: those using spin arrows tend to select $\uparrow$, while those using bit labels tend to select $|0\rangle$. Of course, any of these possible choices of global phase are valid, since overall global phases are insignificant unless compared to other overall global phases, and the defining of a basis set is for the purpose of comparing within the same description. The particular choice here, however, assists with describing a geometric interpretation of these states, and will also be shown to have an interesting relationship to projectors.

[13] Alternative names include the *Bell states*, *EPR pairs*, or *EPR states*, all of which are "in honour of several of the pioneers who first appreciated the novelty of entanglement" [117, p. 98]. "EPR" refers to Einstein, Podolsky, and Rosen, who proposed in 1935 that entanglement implied "that the quantum-mechanical description of physical reality given by wave functions is not complete" [64], with the implication that additional variables would complete the description. "Bell" refers to John S. Bell, who demonstrated in 1964 that a local hidden variable theory is incompatible with the statistical predictions of quantum mechanics [34].

2.3.6.6 Singlet state

A partially geometric interpretation of the maximally entangled basis states can be obtained through considering the effect of exchanging the spin states of the particles, by determining whether or not the expression changes sign when the order of factors is reversed in all the tensor products in an expression. A spin state that changes sign in such an exercise is said to be *antisymmetric under interchange*, and it has distinctly different properties than states that are symmetric.

Only one of the Bell basis states, Ψ_-, is antisymmetric under interchange. It has zero total spin, and thus it represents two spins that are directed oppositely to each other. It is invariant under a rotation of the entire state about any direction: i.e. $(R \otimes R)\,\Psi_- = \Psi_-$. This state is referred to as a **singlet state**, since it is the only one having this particular set of properties.

The singlet state is also called the **totally anticorrelated state**, because its spins are entangled to be oppositely directed no matter the measurement direction. The rotational invariance makes this state most useful for use in experiments, including the mathematical experimentation featured in this dissertation.

The form of the singlet density operator is worth noting [23, p. 408]:

$$\begin{aligned}
\rho_{\Psi-} &\equiv \Psi_- \Psi_-^\dagger \\
&= \frac{1}{2}\left(\mathbf{1}\otimes\mathbf{1} - \mathbf{e}_1\otimes\mathbf{e}_1\right)\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3 + \overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right] \\
&= \frac{1}{2}\left(\mathbf{1}\otimes\mathbf{1} - \mathbf{e}_1\otimes\mathbf{e}_1\right)\frac{1}{2}\left(\mathbf{1}\otimes\mathbf{1} - \mathbf{e}_3\otimes\mathbf{e}_3\right) \\
&= \frac{1}{4}\left(\mathbf{1}\otimes\mathbf{1} - \mathbf{e}_1\otimes\mathbf{e}_1 - \mathbf{e}_2\otimes\mathbf{e}_2 - \mathbf{e}_3\otimes\mathbf{e}_3\right)
\end{aligned}$$

where it was noted that $\mathrm{P}_n\otimes\overline{\mathrm{P}}_n + \overline{\mathrm{P}}_n\otimes\mathrm{P}_n = \frac{1}{2}\left(\mathbf{1}\otimes\mathbf{1} - \hat{\mathbf{n}}\otimes\hat{\mathbf{n}}\right)$ (B.16).

It is important to be able to recognize and interpret the other three basis states as well, since if a transformation of a singlet state is no longer a totally antisymmetric state, it must have some components that are symmetric.

2.3.6.7 Triplet states

The other three entangled basis states, Ψ_+ and $\Phi_\pm$, are symmetric under interchange. These states can be expressed in terms of the **triplet states** which have a total spin of magnitude one, since the spins of two spin-$\frac{1}{2}$ particles can maximally sum to one, but with net spin relative to a particular quantization direction (usually z) of $m_z = \pm 1$ or $m_z = 0$. The triplet states are *not* rotationally invariant: in fact, certain rotations or changes of basis can change a description from one representation of triplet state to another representation of triplet state.

The three triplet states corresponding to the $\mathbf{e}_3$ direction are:

- the state with both parts parallel and in the $\mathbf{e}_3$ direction, i.e. the double spin-up state, $\Upsilon_{\uparrow\uparrow}$;
- the state with both parts parallel and rotated from the $\mathbf{e}_3$ direction[14], i.e. the double spin-down state, $\Upsilon_{\downarrow\downarrow}$; and
- the positive-phase antiparallel state for $\mathbf{e}_3$, i.e. Ψ_+, which has correlations not obvious in its representation.

The triplet states are interrelated by **ladder operators**:

$$\begin{aligned} S_+ &= \frac{1}{\sqrt{2}}\left(\mathrm{P}_3\mathbf{e}_1 \otimes \mathbf{1} + \mathbf{1} \otimes \mathrm{P}_3\mathbf{e}_1\right) = S_-^\dagger \\ S_- &= \frac{1}{\sqrt{2}}\left(\mathbf{e}_1\mathrm{P}_3 \otimes \mathbf{1} + \mathbf{1} \otimes \mathbf{e}_1\mathrm{P}_3\right) = S_+^\dagger \end{aligned}$$

which change the state by either raising, S_+, or lowering, S_-, the expectation value for the z-direction measurement of spin, $m_z = 0,\pm 1$. Using these, the $m_z = 0$ state of Ψ_+ can be raised to the triplet state $\Upsilon_{\uparrow\uparrow}$ which has $m_z = +1$ (and raising further would give a zero result), or lowered to the triplet state $\Upsilon_{\downarrow\downarrow}$ which has $m_z = -1$ (and lowering further would give a zero result).

The $\Phi_\pm$ Bell states are *superpositions* of the parallel triplet states $\Upsilon_{\uparrow\uparrow}$ and $\Upsilon_{\downarrow\downarrow}$. However, due to the lack of rotational invariance of the correlated states, these are not generally as useful as starting points in experiments as the totally anticorrelated state Ψ_-. Hence, the main calculations in this dissertation will begin with the singlet state, as discussed in Section 3.3.2.

The form of the triplet density operators is worth noting [23, p. 409]:

$$\begin{aligned} \rho_{\Psi+} &\equiv \Psi_+\Psi_+^\dagger \\ &= \frac{1}{2}\left(\mathbf{1}\otimes\mathbf{1} + \mathbf{e}_1\otimes\mathbf{e}_1\right)\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3 + \overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right] \\ &= \frac{1}{2}\left(\mathbf{1}\otimes\mathbf{1} + \mathbf{e}_1\otimes\mathbf{e}_1\right)\frac{1}{2}\left(\mathbf{1}\otimes\mathbf{1} - \mathbf{e}_3\otimes\mathbf{e}_3\right) \\ &= \frac{1}{4}\left(\mathbf{1}\otimes\mathbf{1} + \mathbf{e}_1\otimes\mathbf{e}_1 + \mathbf{e}_2\otimes\mathbf{e}_2 - \mathbf{e}_3\otimes\mathbf{e}_3\right) \end{aligned}$$

$$\begin{aligned} \rho_{\Phi\pm} &\equiv \Phi_+\Phi_+^\dagger \\ &= \frac{1}{2}\left(\mathbf{1}\otimes\mathbf{1} \pm \mathbf{e}_1\otimes\mathbf{e}_1\right)\left[\mathrm{P}_3\otimes\mathrm{P}_3 + \overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right] \\ &= \frac{1}{2}\left(\mathbf{1}\otimes\mathbf{1} \pm \mathbf{e}_1\otimes\mathbf{e}_1\right)\frac{1}{2}\left(\mathbf{1}\otimes\mathbf{1} + \mathbf{e}_3\otimes\mathbf{e}_3\right) \\ &= \frac{1}{4}\left(\mathbf{1}\otimes\mathbf{1} \pm \mathbf{e}_1\otimes\mathbf{e}_1 \mp \mathbf{e}_2\otimes\mathbf{e}_2 + \mathbf{e}_3\otimes\mathbf{e}_3\right) \end{aligned}$$

[14] Note: spin down is defined as "rotated from" spin up, rather than merely in the opposite direction of spin up, in order to distinguish from opposite vector directions. A spin state is a spinor, not a vector, and as such it is described by relative rotations rather than directions.

where $\pm\mathbf{e}_1\mathbf{e}_3 = \mp i\mathbf{e}_2$. The relations $\mathrm{P}_n\otimes\overline{\mathrm{P}}_n+\overline{\mathrm{P}}_n\otimes\mathrm{P}_n = \frac{1}{2}\left(\mathbf{1}\otimes\mathbf{1}-\hat{\mathbf{n}}\otimes\hat{\mathbf{n}}\right)$ and $\mathrm{P}_n\otimes\mathrm{P}_n+\overline{\mathrm{P}}_n\otimes\overline{\mathrm{P}}_n = \frac{1}{2}\left(\mathbf{1}\otimes\mathbf{1}+\hat{\mathbf{n}}\otimes\hat{\mathbf{n}}\right)$ are both found in Appendix B.6.

Note that all three of the Bell-state density operators involving triplet states can be expressed in the form

$$\begin{aligned}\rho^{(\mathbf{e_k})} &\equiv \frac{1}{4}\left(\mathbf{1}\otimes\mathbf{1}+\mathbf{e}_1\otimes\mathbf{e}_1+\mathbf{e}_2\otimes\mathbf{e}_2+\mathbf{e}_3\otimes\mathbf{e}_3\right)-\frac{1}{2}\left(\mathbf{e}_k\otimes\mathbf{e}_k\right) \text{ for } k=1...3\\ \text{where} &\quad \rho^{(\mathbf{e}_1)}\equiv\rho_{\Phi-} \text{ and } \rho^{(\mathbf{e}_2)}\equiv\rho_{\Phi+} \text{ and } \rho^{(\mathbf{e}_3)}\equiv\rho_{\Psi+}\end{aligned}$$

which emphasizes the directional nature of these entangled states. The density matrix representation and $\rho^{(\mathbf{e_k})}$ notation also emphasize the similarities among these triplet-based entangled states, in particular how the Ψ_+ state is not so different from the $\Phi_\pm$ states.

2.4 How This Work Extends the APS Treatment of Entanglement

So far, APS has been used to discuss entanglement in terms of the states themselves, while the Dirac equation has suggested that the transformation amplitudes may be even more fundamental, and that the projection into spin components may be reserved for closer to the end of an investigation. Thus, although the above description of entanglement using APS has provided insight into the geometric nature of entangled states, this dissertation will set aside the treatment of states in Section 3.1 in favour of focusing on the transformations that obtain the entangled states from an initial separable state.

Next, although the combining of boosts (2.11) into bipartite transformation operators may seem like it should be a straightforward process, especially when the boosts are treated as separable, the interpretation of the combined boosts requires some care. Also, combinations of boosts can be examined for how they can be simplified in a way that assists with developing a geometric investigation of entanglement. This is all discussed in Section 3.2.

Finally, these two topics are combined in Section 3.3 to obtain entangling eigenspinors from the entangling operators and boost transformations. While none of the steps involved are drastically different from what has been demonstrated above, the assemblage of these steps has not yet been done until the present work. After these steps are completed in Chapter 3, an analysis and physical interpretation follows in Chapter 4.

Chapter 3

Entanglors, Bipartite Boosts, and Entangling Eigenspinors

This chapter begins by interpreting the typical description of spin entanglement in a new way. The new approach examined here is inspired by the fact that the Dirac equation (2.22) is satisfied by transformation operators, i.e. **eigenspinors** (2.7), even without explicitly applying these transformations to spin state projectors. The first step towards this new interpretation begins in Section 3.1 with factoring the entangled basis states into *entangling operators* or **entanglors**[1] that act on the bipartite spin-up state, so that the transformations themselves can be examined separately from any spin states. A geometric interpretation of these entanglors is offered, along with some analysis that justifies treating the entanglement independently from the spin states.

Section 3.2 examines transformations between inertial frames, i.e. **boosts** (2.5), that can be used to describe spin systems in relative motion. The key to combining boosts into a **bipartite** form is to treat both parts as being described from the same lab frame, since their rest frames may not coincide. Then, considering the intended eventual application of these boosts to a mathematical object that is transformed unilaterally, and whose rotational gauge freedom has aligned the coordinate system with the rest-frame spin direction in a particular way, certain simplified cases are suggested by the form of the bipartite boost.

Section 3.3 combines the previous two sections by applying bipartite boosts to the entangling transformation operators, obtaining **entangling eigenspinors**. Elements within these results are recognized as having the form of entanglors, and thus the entangling eigenspinors are expressed as linear superpositions of entanglors.

[1] The term "entanglor" was coined for this purpose during private communications between W. E. Baylis and his research group, around 2010.

Thus, this chapter outlines all the necessary calculational processes required for investigating the idea that entanglement can be represented as transformations, including a means of transforming the transformations relativistically. The calculations themselves are outlined and the method explained while it is demonstrated, but the interpretation of results is deferred until Chapter 4.

3.1 Entangling Operators

This section begins by describing entangled states where each of the constituent parts are described in its own rest frame. Although the parts may not share a rest frame, their relative velocities will not be taken into account until Section 3.2. While stationary entangled states have already been covered in existing APS treatments of entanglement [20, 23, 49] as described in Section 2.3.6, there are some geometrical interpretations that are offered here that have not been seen elsewhere.

Guiding the following treatment is the idea that the entanglement can be expressed as residing within *transformations of orientation* that can be — but do not necessarily need to be — applied to states. The interpretation is enacted in Section 3.1.1 by isolating the *entangling operators* or **entanglors** from the entangled basis states. These entanglors are then interpreted geometrically in Section 3.1.2, and relationships between these operators are analyzed to justify treating the entanglement independently from spin states.

3.1.1 Extracting the Entangling Operators

Recall that any general bipartite state can be expressed as a linear superposition of separable basis eigenstates (2.42), which are made up of all possible separable pairings of spin-up and spin-down states. Recall as well that the spinor representing the spin-down state (2.34) can be interpreted as a π-rotation about the $\mathbf{e}_2$ axis of the spin-up spinor (2.33), via the rotor $R_\pi = \mathbf{e}_1\mathbf{e}_3$, such that

$$\psi_\downarrow = R_\pi \psi_\uparrow = \mathbf{e}_1\mathbf{e}_3\mathrm{P}_3$$

and where $\psi_\uparrow \equiv \mathrm{P}_3$ projects any APS element onto the minimal left ideal (APS) P_3, giving a resulting matrix representation (B.9) that is equivalent to a traditional two-component spinor.

Note that although it is mathematically equivalent to represent the spin-down state in an abbreviated form, $\mathbf{e}_1\mathrm{P}_3$, the physically realizable transformation to achieve this is described by the application of the rotor, $\mathbf{e}_1\mathbf{e}_3\mathrm{P}_3$, and hence this is the form that lends itself best to the geometric analysis of this dissertation. However, since the abbreviated form will be encountered later, it is helpful to give it a name: henceforth, let it be referred to as the "flip" transformation, referring to the quantum computing "bit flip" change of state from $|0\rangle$ to $|1\rangle$, which is thus the same effect as the rotation, but without the emphasis on the rotation.

3.1.1.1 Factoring the Bell states

The entangled basis states (2.46) may be described as linear superpositions of separable combinations of spin-up and spin-down states (F.1), and can be factored into linear superpositions of bipartite rotors acting on the bipartite spin-up state $\Upsilon_{\uparrow\uparrow} \equiv \psi_\uparrow \otimes \psi_\uparrow$:

$$\begin{aligned} \Psi_\pm &= \frac{1}{\sqrt{2}} \left(R_\pi \otimes R_0 \pm R_0 \otimes R_\pi\right) \Upsilon_{\uparrow\uparrow} \\ &= \frac{1}{\sqrt{2}} \left(\mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{1} \pm \mathbf{1} \otimes \mathbf{e}_1\mathbf{e}_3\right) \left(\mathrm{P}_3 \otimes \mathrm{P}_3\right) \\ \Phi_\pm &= \frac{1}{\sqrt{2}} \left(R_0 \otimes R_0 \pm R_\pi \otimes R_\pi\right) \Upsilon_{\uparrow\uparrow} \\ &= \frac{1}{\sqrt{2}} \left(\mathbf{1} \otimes \mathbf{1} \pm \mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{e}_1\mathbf{e}_3\right) \left(\mathrm{P}_3 \otimes \mathrm{P}_3\right) \end{aligned} \tag{3.1}$$

where each term includes an *amplitude* that squares to a probability of $\frac{1}{2}$ for each bipartite rotation to occur, and where $R_0 \equiv \mathbf{1}$ is a zero-degree rotation. Since these transformations cannot be described as acting separably on the two parts of the $\Upsilon_{\uparrow\uparrow}$ state, the transformations themselves may be said to be entangled — although in order to emphasize that these are transformations, they will be described in this dissertation as *entangling*, rather than *entangled*. It is also worth mentioning that since the transformations and the states can be expressed as a product, the transformations may be considered separable from the states.

3.1.1.2 Labelling the entangling operators

It is useful now to assign labels to these bipartite rotor superpositions:

$$\boxed{\begin{aligned} \mathbf{R}_{\Psi\pm} &\equiv \frac{1}{\sqrt{2}} \left(R_\pi \otimes R_0 \pm R_0 \otimes R_\pi\right) = \frac{1}{\sqrt{2}} \left(\mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{1} \pm \mathbf{1} \otimes \mathbf{e}_1\mathbf{e}_3\right) \\ \mathbf{R}_{\Phi\pm} &\equiv \frac{1}{\sqrt{2}} \left(R_0 \otimes R_0 \pm R_\pi \otimes R_\pi\right) = \frac{1}{\sqrt{2}} \left(\mathbf{1} \otimes \mathbf{1} \pm \mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{e}_1\mathbf{e}_3\right) \end{aligned}} \tag{3.2}$$

and also to bestow a name upon them: **entanglors**, for **entangl**ing operat**ors**.

Note that the subscripts for the entanglors have been chosen as a reminder of what states the entanglors create when applied to the bipartite spin-up state. This simply serves as a reminder of the pattern the terms and factors within each one takes, since the labels clearly cannot indicate anything about the states that would result if these operators were applied to any bipartite state other than the double spin-up state.

Note also the use of a boldface $\mathbf{R}$ here, which indicates that these rotations are not simple single-system rotations. Similarly, a separable bipartite rotation can also be represented using boldface, as

$\mathbf{R}_{AB} \equiv R_A \otimes R_B$ for example, to emphasize rotations R_A and R_B as occurring in separate spaces, here arbitrarily labelled by A and B.

3.1.1.3 Alternative derivation of entanglors

The entanglors (3.2) may be obtained in an arguably more rigorous manner than the factoring approach described above. It has been noted [20, p. 141] that a full rotor R can be retrieved from its ideal projection $R\mathrm{P}_3$ by taking twice the even part: $2\langle R\mathrm{P}_3\rangle_+ = R$. To extend this idea to the bipartite case, the even part (A.11) must be taken separately in both spaces, which can be indicated by (ab)using a direct product symbol in the angle bracket subscript:

$$\begin{aligned} 4\langle\Psi_\pm\rangle_{+\otimes+} = 4\langle\mathbf{R}_{\Psi\pm}\Upsilon_{\uparrow\uparrow}\rangle_{+\otimes+} &\equiv \frac{1}{\sqrt{2}}\left(2\langle R_\pi \mathrm{P}_3\rangle_+ \otimes 2\langle R_0\mathrm{P}_3\rangle_+ \pm 2\langle R_0\mathrm{P}_3\rangle_+ \otimes 2\langle R_\pi\mathrm{P}_3\rangle_+\right) \\ &= \frac{1}{\sqrt{2}}\left(R_\pi \otimes R_0 \pm R_0 \otimes R_\pi\right) \\ &= \frac{1}{\sqrt{2}}\left(\mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{1} \pm \mathbf{1} \otimes \mathbf{e}_1\mathbf{e}_3\right) \;\equiv\; \mathbf{R}_{\Psi\pm} \end{aligned}$$

which shows that the full Ψ-entanglor $\mathbf{R}_{\Psi\pm}$ can be retrieved from four times the even part of the entangled projected states $\Psi_\pm \equiv \mathbf{R}_{\Psi\pm}\Upsilon_{\uparrow\uparrow}$. The full Ψ-entanglor $\mathbf{R}_{\Phi\pm}$ can similarly be retrieved from four times the even part of the entangled projected states $\Phi_\pm \equiv \mathbf{R}_{\Phi\pm}\Upsilon_{\uparrow\uparrow}$ as $\mathbf{R}_{\Phi\pm} \equiv 4\langle\Phi_\pm\rangle_{+\otimes+}$.

Note, however, that if the representation of spin down is as a "flipped" spin-up projector $\mathbf{e}_1\mathrm{P}_3$ rather than as the rotation $\mathbf{e}_1\mathbf{e}_3\mathrm{P}_3$, it cannot have the "flip" operator extracted from the projector in this way (the rotor is extracted instead). This is another reason — in addition to the physical unrealizability — that "flipping" is not used in the definition of the entanglor as a transformation operator.

Now that these so-called entanglors have been defined, what do they mean, and why are they useful or interesting?

3.1.2 Interpreting the Entanglors

In order to interpret the entanglors defined above, note that the single-subspace rotor $R_\pi = \mathbf{e}_1\mathbf{e}_3$ is described geometrically as invoking a π-rotation in the direction that would rotate $\mathbf{e}_3$ towards and past $\mathbf{e}_1$ to $-\mathbf{e}_3$; this may be defined as *clockwise* in the $\mathbf{e}_1\mathbf{e}_3$ plane. Then the inverse of this rotor is a *counterclockwise* rotation by π in the $\mathbf{e}_1\mathbf{e}_3$ plane, and is given by the bar conjugate $\overline{R}_\pi = -\mathbf{e}_1\mathbf{e}_3 = -R_\pi$ which in this case happens to be equal to the negation of R_π only because the rotor is represented by a simple bivector.

The entanglors, then, can be described in terms of these single-subspace π-rotations about $\mathbf{e}_2$ as follows:

- $\mathbf{R}_{\Psi+}$ is an equal superposition of rotors enacting: (a) **clockwise** π-rotation in the first subspace and no rotation in the second subspace; and (b) no rotation in the first subspace and **clockwise** π-rotation in the second subspace;
- $\mathbf{R}_{\Psi-}$ is an equal superposition of rotors enacting: (a) **clockwise** rotation in the first subspace and no rotation in the second subspace; and (b) no rotation in the first subspace and **counterclockwise** π-rotation in the second subspace;
- $\mathbf{R}_{\Phi+}$ is a superposition of rotors enacting: (a) no rotation in either subspace, and (b) π-rotations in both subspaces **in the same sense**;
- $\mathbf{R}_{\Phi-}$ is a superposition of rotors enacting: (a) no rotation in either subspace, and (b) π-rotations in the two subspaces **in opposite senses**.

Thus the phase factor of ± 1 between terms in each superposition indicates whether the relative rotation directions described by the terms are the same or opposite.

It is important to remember that the multiplicative factors for each of the terms here represent *amplitudes* relating to the probability of each of the rotations occurring. Until an appropriate probabilistic measurement is made, neither rotation takes place. Furthermore, amplitudes imply the possibility of interference with other amplitudes, if given an appropriate situation where interference may occur.

3.1.2.1 Interrelating the entanglors

Another way to interpret the entanglors (3.2) is to relate them to each other. To do so, consider the direction-inverting bipartite rotor $R_\pi = \mathbf{e}_1\mathbf{e}_3$, which can be included in two different single-subspace bipartite rotors as follows:

$$\begin{aligned} \mathbf{R}_{\pi,0} &\equiv R_\pi \otimes 1 = \mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{1} \\ \mathbf{R}_{0,\pi} &\equiv 1 \otimes R_\pi = \mathbf{1} \otimes \mathbf{e}_1\mathbf{e}_3. \end{aligned} \tag{3.3}$$

It can be easily reasoned that these have the effect of transforming a Ψ-entanglor to a Φ-entanglor, and vice versa.

However, when applying these rotors (3.3) to the entanglors (3.2) — unilaterally, since the entanglors

themselves are transformations — the *only* possible results are as follows:

$$\boxed{\begin{aligned}(R_\pi \otimes 1)\,\mathbf{R}_{\Phi+} &= +\mathbf{R}_{\Psi-}\\ (R_\pi \otimes 1)\,\mathbf{R}_{\Psi-} &= -\mathbf{R}_{\Phi+}\\ \\ (1 \otimes R_\pi)\,\mathbf{R}_{\Phi+} &= -\mathbf{R}_{\Psi-}\\ (1 \otimes R_\pi)\,\mathbf{R}_{\Psi-} &= +\mathbf{R}_{\Phi+}\end{aligned}} \qquad \boxed{\begin{aligned}(R_\pi \otimes 1)\,\mathbf{R}_{\Phi-} &= +\mathbf{R}_{\Psi+}\\ (R_\pi \otimes 1)\,\mathbf{R}_{\Psi+} &= -\mathbf{R}_{\Phi-}\\ \\ (1 \otimes R_\pi)\,\mathbf{R}_{\Phi-} &= +\mathbf{R}_{\Psi+}\\ (1 \otimes R_\pi)\,\mathbf{R}_{\Psi+} &= -\mathbf{R}_{\Phi-}\end{aligned}} \tag{3.4}$$

where the transformations are grouped to highlight the entanglors that interrelate. Specifically, it is seen that the single-subspace direction-inverting rotors (3.3) relate $\mathbf{R}_{\Psi+}$ and $\mathbf{R}_{\Phi-}$ to each other, and $\mathbf{R}_{\Psi-}$ and $\mathbf{R}_{\Phi+}$ to each other, with no other interrelationships possible.

So, the single-subspace direction-inverting rotors (3.3) not only change the relative directions, so Φ becomes Ψ and vice versa, but also change the relative phase, the $\pm$ between terms, and possibly the overall sign as well. The results can be abbreviated as follows:

$$\boxed{\begin{aligned}(R_\pi \otimes 1)\,\mathbf{R}_{\Phi\pm} &= +\mathbf{R}_{\Psi\mp}\\ (R_\pi \otimes 1)\,\mathbf{R}_{\Psi\pm} &= -\mathbf{R}_{\Phi\mp}\\ \\ (1 \otimes R_\pi)\,\mathbf{R}_{\Phi\pm} &= \mp\mathbf{R}_{\Psi\mp}\\ (1 \otimes R_\pi)\,\mathbf{R}_{\Psi\pm} &= \mp\mathbf{R}_{\Phi\mp}\end{aligned}}\ . \tag{3.5}$$

Note that the various signs relating the entanglors may vary for different choices of relative and global phases used to define the terms in the entanglors. What is interesting here is the fact that such interrelationships exist, and that the entanglors are divided into two groups that do not interrelate. This is true no matter the conventions applied in the defining of the basis Bell states or the entanglors that derive from them.

3.1.2.2 Verifying the restricted interrelationship

The restrictive interrelationship $\mathbf{R}_{\Psi\pm} \longleftrightarrow \mathbf{R}_{\Phi\mp}$ between entanglors is interesting enough to be worth verifying explicitly. To prove that there is no transformation that can completely change the entanglor to one having only either the opposite relative phase or the relative parity:

- Suppose there does exist a transformation of the form $A \otimes B + C \otimes D$ that directly relates the

two Ψ-entanglors by changing only the phase between the terms. Then:

$$\begin{aligned}
(A\otimes B + C\otimes D)\,\mathbf{R}_{\Psi+} &\stackrel{?}{=} \mathbf{R}_{\Psi-} \\
(A\otimes B + C\otimes D)\,(\mathbf{e}_1\mathbf{e}_3\otimes\mathbf{1} + \mathbf{1}\otimes\mathbf{e}_1\mathbf{e}_3) &\stackrel{?}{=} \mathbf{e}_1\mathbf{e}_3\otimes\mathbf{1} - \mathbf{1}\otimes\mathbf{e}_1\mathbf{e}_3 \\
(A\otimes B + C\otimes D)\,(\mathbf{e}_1\mathbf{e}_3\otimes\mathbf{1}) + (A\otimes B + C\otimes D)\,(\mathbf{1}\otimes\mathbf{e}_1\mathbf{e}_3) &\stackrel{?}{=} \mathbf{e}_1\mathbf{e}_3\otimes\mathbf{1} - \mathbf{1}\otimes\mathbf{e}_1\mathbf{e}_3\ .
\end{aligned}$$

  - ◇ For this to be true, $A\otimes B + C\otimes D$ must either:
    - ∗ leave $\mathbf{e}_1\mathbf{e}_3\otimes\mathbf{1}$ unchanged while changing $\mathbf{1}\otimes\mathbf{e}_1\mathbf{e}_3$ only by a sign; OR
    - ∗ it must transform both $\mathbf{e}_1\mathbf{e}_3\otimes\mathbf{1}$ to $-\mathbf{1}\otimes\mathbf{e}_1\mathbf{e}_3$ and $\mathbf{1}\otimes\mathbf{e}_1\mathbf{e}_3$ to $\mathbf{e}_1\mathbf{e}_3\otimes\mathbf{1}$.
  - ◇ Since there is no possible transformation that can accomplish either of these sets of simultaneous requirements, it must be *false* that there is a transformation that directly relates the two Ψ-entanglors.

- The proof that the pair of Φ-entanglors $\mathbf{R}_{\Phi\pm}$ cannot be interrelated directly is similar to the proof for the Ψ-entanglors, as seen above.

- Now suppose there exists a transformation of the form $A\otimes B + C\otimes D$ that directly relates the two same-phase entanglors $\mathbf{R}_{X+}$ by changing only the relative direction. Then:

$$\begin{aligned}
(A\otimes B + C\otimes D)\,\mathbf{R}_{\Psi+} &\stackrel{?}{=} \mathbf{R}_{\Phi+} \\
(A\otimes B + C\otimes D)\,(\mathbf{e}_1\mathbf{e}_3\otimes\mathbf{1} + \mathbf{1}\otimes\mathbf{e}_1\mathbf{e}_3) &\stackrel{?}{=} \mathbf{1}\otimes\mathbf{1} + \mathbf{e}_1\mathbf{e}_3\otimes\mathbf{e}_1\mathbf{e}_3 \\
(A\otimes B + C\otimes D)\,(\mathbf{e}_1\mathbf{e}_3\otimes\mathbf{1}) + (A\otimes B + C\otimes D)\,(\mathbf{1}\otimes\mathbf{e}_1\mathbf{e}_3) &\stackrel{?}{=} \mathbf{1}\otimes\mathbf{1} + \mathbf{e}_1\mathbf{e}_3\otimes\mathbf{e}_1\mathbf{e}_3\ .
\end{aligned}$$

  - ◇ For this to be true, $A\otimes B + C\otimes D$ must either:
    - ∗ rotate the first factor of $\mathbf{e}_1\mathbf{e}_3\otimes\mathbf{1}$ by $-\pi$, while simultaneously rotating the first factor of $\mathbf{1}\otimes\mathbf{e}_1\mathbf{e}_3$ by $+\pi$; OR
    - ∗ rotate the second factor of $\mathbf{e}_1\mathbf{e}_3\otimes\mathbf{1}$ by π, while simultaneously rotating the second factor of $\mathbf{1}\otimes\mathbf{e}_1\mathbf{e}_3$ by $-\pi$.
  - ◇ Since there is no possible transformation that can accomplish either of these sets of simultaneous requirements, it must be *false* that there is a transformation that directly relates the two same-phase entanglors $\mathbf{R}_{X+}$.

- The proof that the pair of opposite-phase entanglors $\mathbf{R}_{X-}$ cannot be interrelated directly is similar to the proof for the same-phase entanglors $\mathbf{R}_{X+}$, as seen above.

3.1.3 Entanglors and Entangled States

While the main premise of this dissertation is to model the entanglement as transformation amplitudes rather than the states, the intention is to eventually apply the entanglors to states in order

to interpret the results. Thus, it is useful to investigate some ways the results will be interpreted in terms of states, before complicating the situation with relativistic boosts.

3.1.3.1 Interrelating the Bell basis and separable basis

Now note that the single-subspace direction-inverting rotors (3.3) commute with the entanglors (3.2). This can be seen by noting that the only directed element in both sets of transformations is $\mathbf{e}_1\mathbf{e}_3$, which commutes with itself, and thus with every element of the rotors and entanglors in question. This means that the order of applying these two transformations can be reversed: instead of these rotors *acting on* the entanglors, they can *be acted upon* by the entanglors.

The full impact of this commutativity is revealed by investigating the effect of these transformations on the Bell states. Starting from the factored Bell basis states (3.1), the single-subspace direction-inverting rotors (3.3) can be used to re-express the entanglors, which can then be rearranged. The effect of this is as follows:

$$\begin{aligned}
\Phi_+ &= \mathbf{R}_{\Phi+}\Upsilon_{\uparrow\uparrow} &&= \mathbf{R}_{0,\pi}\mathbf{R}_{\Psi-}\Upsilon_{\uparrow\uparrow} &&= \mathbf{R}_{\Psi-}\mathbf{R}_{0,\pi}\Upsilon_{\uparrow\uparrow} &&= \mathbf{R}_{\Psi-}\Upsilon_{\uparrow\downarrow} \\
&&&= \overline{\mathbf{R}}_{\pi,0}\mathbf{R}_{\Psi-}\Upsilon_{\uparrow\uparrow} &&= \mathbf{R}_{\Psi-}\overline{\mathbf{R}}_{\pi,0}\Upsilon_{\uparrow\uparrow} &&= -\mathbf{R}_{\Psi-}\Upsilon_{\downarrow\uparrow} \\
\Psi_- &= \mathbf{R}_{\Psi-}\Upsilon_{\uparrow\uparrow} &&= \overline{\mathbf{R}}_{0,\pi}\mathbf{R}_{\Phi+}\Upsilon_{\uparrow\uparrow} &&= \mathbf{R}_{\Phi+}\overline{\mathbf{R}}_{0,\pi}\Upsilon_{\uparrow\uparrow} &&= -\mathbf{R}_{\Phi+}\Upsilon_{\uparrow\downarrow} \\
&&&= \mathbf{R}_{\pi,0}\mathbf{R}_{\Phi+}\Upsilon_{\uparrow\uparrow} &&= \mathbf{R}_{\Phi+}\mathbf{R}_{\pi,0}\Upsilon_{\uparrow\uparrow} &&= \mathbf{R}_{\Phi+}\Upsilon_{\downarrow\uparrow} \\
\Phi_- &= \mathbf{R}_{\Phi-}\Upsilon_{\uparrow\uparrow} &&= \overline{\mathbf{R}}_{0,\pi}\mathbf{R}_{\Psi+}\Upsilon_{\uparrow\uparrow} &&= \mathbf{R}_{\Psi+}\overline{\mathbf{R}}_{0,\pi}\Upsilon_{\uparrow\uparrow} &&= -\mathbf{R}_{\Psi+}\Upsilon_{\uparrow\downarrow} \\
&&&= \overline{\mathbf{R}}_{\pi,0}\mathbf{R}_{\Psi+}\Upsilon_{\uparrow\uparrow} &&= \mathbf{R}_{\Psi+}\overline{\mathbf{R}}_{\pi,0}\Upsilon_{\uparrow\uparrow} &&= -\mathbf{R}_{\Psi+}\Upsilon_{\downarrow\uparrow} \\
\Psi_+ &= \mathbf{R}_{\Psi+}\Upsilon_{\uparrow\uparrow} &&= \mathbf{R}_{0,\pi}\mathbf{R}_{\Phi-}\Upsilon_{\uparrow\uparrow} &&= \mathbf{R}_{\Phi-}\mathbf{R}_{0,\pi}\Upsilon_{\uparrow\uparrow} &&= \mathbf{R}_{\Phi-}\Upsilon_{\uparrow\downarrow} \\
&&&= \mathbf{R}_{\pi,0}\mathbf{R}_{\Phi-}\Upsilon_{\uparrow\uparrow} &&= \mathbf{R}_{\Phi-}\mathbf{R}_{\pi,0}\Upsilon_{\uparrow\uparrow} &&= \mathbf{R}_{\Phi-}\Upsilon_{\downarrow\uparrow}
\end{aligned}$$

which shows that the Bell states do not necessarily need to be expressed in terms of the double spin-up basis states, but may equivalently be expressed as entanglors acting on the antiparallel separable basis states $\Upsilon_{\uparrow\downarrow}$ or $\Upsilon_{\downarrow\uparrow}$.

Furthermore, each of the entanglors can be applied to a bipartite spin-down state, to complete the interrelating of entanglors and Bell basis states. Then the single-subspace direction-inverting rotors (3.3) can reveal equivalent representations in terms of antiparallel separable basis states, as

follows:

$$\begin{aligned}
\mathbf{R}_{\Psi+}\Upsilon_{\downarrow\downarrow} &= \mathbf{R}_{0,\pi}\mathbf{R}_{\Phi-}\Upsilon_{\downarrow\downarrow} = \mathbf{R}_{\Phi-}\mathbf{R}_{0,\pi}\Upsilon_{\downarrow\downarrow} = -\mathbf{R}_{\Phi-}\Upsilon_{\downarrow\uparrow} = -\Psi_+ \\
&= \mathbf{R}_{\pi,0}\mathbf{R}_{\Phi-}\Upsilon_{\downarrow\downarrow} = \mathbf{R}_{\Phi-}\mathbf{R}_{\pi,0}\Upsilon_{\downarrow\downarrow} = -\mathbf{R}_{\Phi-}\Upsilon_{\uparrow\downarrow} = -\Psi_+ \\
\mathbf{R}_{\Phi+}\Upsilon_{\downarrow\downarrow} &= \mathbf{R}_{0,\pi}\mathbf{R}_{\Psi-}\Upsilon_{\downarrow\downarrow} = \mathbf{R}_{\Psi-}\mathbf{R}_{0,\pi}\Upsilon_{\downarrow\downarrow} = -\mathbf{R}_{\Psi-}\Upsilon_{\downarrow\uparrow} = \Phi_+ \\
&= \overline{\mathbf{R}}_{\pi,0}\mathbf{R}_{\Psi-}\Upsilon_{\downarrow\downarrow} = \mathbf{R}_{\Psi-}\overline{\mathbf{R}}_{\pi,0}\Upsilon_{\downarrow\downarrow} = \mathbf{R}_{\Psi-}\Upsilon_{\uparrow\downarrow} = \Phi_+ \\
\mathbf{R}_{\Psi-}\Upsilon_{\downarrow\downarrow} &= \overline{\mathbf{R}}_{0,\pi}\mathbf{R}_{\Phi+}\Upsilon_{\downarrow\downarrow} = \mathbf{R}_{\Phi+}\overline{\mathbf{R}}_{0,\pi}\Upsilon_{\downarrow\downarrow} = \mathbf{R}_{\Phi+}\Upsilon_{\downarrow\uparrow} = \Psi_- \\
&= \mathbf{R}_{\pi,0}\mathbf{R}_{\Phi+}\Upsilon_{\downarrow\downarrow} = \mathbf{R}_{\Phi+}\mathbf{R}_{\pi,0}\Upsilon_{\downarrow\downarrow} = -\mathbf{R}_{\Phi+}\Upsilon_{\uparrow\downarrow} = \Psi_- \\
\mathbf{R}_{\Phi-}\Upsilon_{\downarrow\downarrow} &= \overline{\mathbf{R}}_{0,\pi}\mathbf{R}_{\Psi+}\Upsilon_{\downarrow\downarrow} = \mathbf{R}_{\Psi+}\overline{\mathbf{R}}_{0,\pi}\Upsilon_{\downarrow\downarrow} = \mathbf{R}_{\Psi+}\Upsilon_{\downarrow\uparrow} = -\Phi_- \\
&= \overline{\mathbf{R}}_{\pi,0}\mathbf{R}_{\Psi+}\Upsilon_{\downarrow\downarrow} = \mathbf{R}_{\Psi+}\overline{\mathbf{R}}_{\pi,0}\Upsilon_{\downarrow\downarrow} = \mathbf{R}_{\Psi+}\Upsilon_{\uparrow\downarrow} = -\Phi_-
\end{aligned}$$

where the single subspace relation $R_\pi\psi_\downarrow = -\psi_\uparrow$ was noted. The previous block of relations was also used to relate the results to the Bell basis states for the final conclusion in each line, thus wrapping up the demonstration of how all the bipartite basis states can be used with the entanglors to relate to the Bell basis states.

3.1.3.2 Summarizing the relationship between the Bell basis and separable basis

In summary, each of the Bell basis states (3.1) can be described in terms of entanglors (3.2) acting on each of the separable basis states (F.1), as follows:

$$\boxed{\begin{array}{rcrcrcrcr}
\Psi_+ &=& \mathbf{R}_{\Psi+}\Upsilon_{\uparrow\uparrow} &=& \mathbf{R}_{\Phi-}\Upsilon_{\uparrow\downarrow} &=& \mathbf{R}_{\Phi-}\Upsilon_{\downarrow\uparrow} &=& -\mathbf{R}_{\Psi+}\Upsilon_{\downarrow\downarrow} \\
\Phi_+ &=& \mathbf{R}_{\Phi+}\Upsilon_{\uparrow\uparrow} &=& \mathbf{R}_{\Psi-}\Upsilon_{\uparrow\downarrow} &=& -\mathbf{R}_{\Psi-}\Upsilon_{\downarrow\uparrow} &=& \mathbf{R}_{\Phi+}\Upsilon_{\downarrow\downarrow} \\
\Psi_- &=& \mathbf{R}_{\Psi-}\Upsilon_{\uparrow\uparrow} &=& -\mathbf{R}_{\Phi+}\Upsilon_{\uparrow\downarrow} &=& \mathbf{R}_{\Phi+}\Upsilon_{\downarrow\uparrow} &=& \mathbf{R}_{\Psi-}\Upsilon_{\downarrow\downarrow} \\
\Phi_- &=& \mathbf{R}_{\Phi-}\Upsilon_{\uparrow\uparrow} &=& -\mathbf{R}_{\Psi+}\Upsilon_{\uparrow\downarrow} &=& -\mathbf{R}_{\Psi+}\Upsilon_{\downarrow\uparrow} &=& -\mathbf{R}_{\Phi-}\Upsilon_{\downarrow\downarrow}
\end{array}}. \tag{3.6}$$

Most notably, this summary shows that each Bell basis state can be expressed in terms of each of the separable states but only two of the entanglors: the two that interrelate (3.5). Thus the Bell states and their corresponding entanglors form two distinct groups: those involving Ψ_- and Φ_+, and those involving Ψ_+ and Φ_-.

The point of the above exercise is to show that the entanglors relate the Bell states and the separable states in a fundamental way that goes beyond the specific states that are related. This supports the main idea of this dissertation, that it is reasonable to examine the entanglors independently from the states.

It is also worth noting that *any* single-subspace rotation acting directly on the separable basis states can be expressed in a form that commutes with the entanglor, since anything multiplying a projector can be expressed (2.38) as a complex linear combination of a scalar and the bivector $\mathbf{e}_1\mathbf{e}_3$. Thus any rotations of the initial states can be expressed equivalently as arbitrary single-subspace rotations in the $\mathbf{e}_1\mathbf{e}_3$ plane acting on the entanglor; however, for ease of calculations, it is most useful to transfer all possible single-subspace rotations to the rest frame as part of the process of simplification. This enables the single-subspace rotations to be considered gauge transformations (see Section 2.2.6).

Furthermore, it is worth noting that the Bell states themselves can be interrelated [38] by unilateral single-subspace π-rotations about any of the three basis directions:

- If the π-rotation is about $\mathbf{e}_2$, i.e. in the plane $-i\mathbf{e}_2 = \mathbf{e}_1\mathbf{e}_3$ as used above, the relationship between Bell states is $\Psi_\pm \leftrightarrow \Phi_\mp$ as seen above.
- If the π-rotation is about $\mathbf{e}_1$, the Bell states of opposite relative parity but same relative phase $\Psi_\pm \leftrightarrow \Phi_\pm$ are related.
- If the rotation is about $\mathbf{e}_3$, the Bell states of opposite relative phase but same relative parity are related: $\Phi_\pm \leftrightarrow \Phi_\mp$ and $\Psi_\pm \leftrightarrow \Psi_\mp$.

However, since the interpretation in the present work is based on the *entanglors* rather than the Bell states, the more restrictive relationship is noteworthy and will help with interpretation in Section 4.3.2.

3.2 Bipartite Boosts

Before discussing entanglement in a relativistic context, it is helpful to determine the **bipartite boost operators** that best serve the investigation of the geometry in question. This section begins by describing bipartite boosts in general, and then proceeds to describe specific simplified cases that are useful in the clarity they provide to the investigation of boosted entanglors that follows in Section 3.3.

3.2.1 General Bipartite Boost

The inertial reference frames of the two parts that make up a bipartite state can both be described relative to a common observer, in yet another inertial frame. This interpretation was developed to make sense of the application of a tensor product of boosts to a bipartite state; the details of this interpretation are discussed further in Section 3.2.2.

A pair of boosts joined by tensor product form a **bipartite boost operator**, i.e. $\mathbf{B}_{AB} \equiv B_A \otimes B_B$ — where, as in the case of $\mathbf{R}_{AB} \equiv R_A \otimes R_B$, the boldfaced symbol $\mathbf{B}$ serves to emphasize that this operator is not a mere single-subsystem boost. The boosts comprising $\mathbf{B}_{AB}$ are represented here as separable[2], so B_A describes the motion of what is described in the first subspace, and B_B describes the motion of what is described in the second subspace, both relative to the same lab frame.

In the present exploration of boosted entanglement, it is useful to express each boost B_X in a form (2.11) that explicitly emphasizes the scalar part, the vector part in direction $\hat{\mathbf{v}}_X$, and also the Lorentz factor $\gamma_X = \left(1 - v_X^2/c^2\right)^{-1/2}$:

$$B_X = \sqrt{\frac{\gamma_X + 1}{2}} + \hat{\mathbf{v}}_X \sqrt{\frac{\gamma_X - 1}{2}} \equiv (S_X + V_X \hat{\mathbf{v}}_X) \tag{3.7}$$

where S_X and V_X are used as shorthand for the scalar part and vector part of this expression. Using this, the general bipartite boost $B_A \otimes B_B$ expands as

$$\begin{aligned} B_A \otimes B_B &= (S_A + V_A \hat{\mathbf{v}}_A) \otimes (S_B + V_B \hat{\mathbf{v}}_B) \\ &= S_A S_B \,(\mathbf{1} \otimes \mathbf{1}) \;\; + V_A S_B \,(\hat{\mathbf{v}}_A \otimes \mathbf{1}) + \;\; S_A V_B \,(\mathbf{1} \otimes \hat{\mathbf{v}}_B) + V_A V_B \,(\hat{\mathbf{v}}_A \otimes \hat{\mathbf{v}}_B) \end{aligned} \tag{3.8}$$

where the coefficients are

$$\begin{aligned} S_A S_B &= \sqrt{\frac{\gamma_A + 1}{2}} \sqrt{\frac{\gamma_B + 1}{2}} = \frac{1}{2} \sqrt{(\gamma_A + 1)(\gamma_B + 1)} \\ V_A S_B &= \sqrt{\frac{\gamma_A - 1}{2}} \sqrt{\frac{\gamma_B + 1}{2}} = \frac{1}{2} \sqrt{(\gamma_A - 1)(\gamma_B + 1)} \\ S_A V_B &= \sqrt{\frac{\gamma_A + 1}{2}} \sqrt{\frac{\gamma_B - 1}{2}} = \frac{1}{2} \sqrt{(\gamma_A + 1)(\gamma_B - 1)} \\ \text{and } V_A V_B &= \sqrt{\frac{\gamma_A - 1}{2}} \sqrt{\frac{\gamma_B - 1}{2}} = \frac{1}{2} \sqrt{(\gamma_A - 1)(\gamma_B - 1)} \; . \end{aligned} \tag{3.9}$$

Specific cases of boosts enable simplification of these coefficients.

3.2.2 Changing Reference Frames

When discussing relative motion, it is often worth considering whether calculations can be simplified by changing to a different inertial reference frame. For example, one common simplification is to perform calculations in the rest frame of a system under examination, and then transform the result to the lab frame.

[2] Boosts that are themselves entangled are discussed as a potential future avenue of investigation, in Section 5.2.

3.2.2.1 Active versus passive transformations

In the bipartite case, the two separate parts may not have a common rest frame. However, each of the parts may be described by a separate boost relating its own rest frame to a *common observer*. The combination of these transformations, which establish the relative description of the subspaces as seen in a particular initial lab frame, may be considered an "**active transformation**" that is applied to the separable rest frames of the systems themselves. The transformation itself is separable,

$$\begin{aligned} \mathbf{B}_{\rm ini} &\equiv B_{{\rm ini}A} \otimes B_{{\rm ini}B} \\ &= (B_{{\rm ini}A} \otimes \mathbf{1})\,(\mathbf{1} \otimes B_{{\rm ini}B}) \ , \end{aligned} \tag{3.10}$$

however the two transformations are not independent: the choice of lab frame used to describe one of the subspaces must be the same lab frame from which the other is described. In a given situation, the relative velocities are generally known and can be incorporated into the model in a straightforward way.

Once the velocities of the pair of subsystems are described relative to a particular observer, it does not make sense to apply further boosts separably in the two subspaces. However, it is possible to transform a description of such a pair of subsystems from the initial lab frame to another lab frame. Since only the relative velocities and orientations are physically significant, a "**passive transformation**" of the lab frame may be applied by way of applying identical active transformations to both subsystem frames:

$$\mathbf{B}_{\rm lab} \equiv B_{\rm lab} \otimes B_{\rm lab} \ . \tag{3.11}$$

Such a transformation may be applied either: (a) to simplify a calculation describing a particular situation, or (b) to generalize a calculation that is initially performed simplistically.

Note that both the "active" and "passive" transformations described above are in fact applied in an active sense, to the systems themselves, and of course it is only relative motion that is physically significant. The names are given here to help distinguish the two possible ways of implementing the relative motion: the "active" case is in fact describing the relative motion between the subspaces, while the "passive" case is describing the relative motion between the lab frame and the system.

3.2.2.2 An active then a passive transformation

The initial bipartite boost, $\mathbf{B}_{\rm ini}$, relating the velocities between the parts of a system and its common observer, acts to its right on a system initially described in terms of the separable rest frames of the subsystems. The initial boost can then be transformed by the lab-frame boost $\mathbf{B}_{\rm lab}$ according to the multiplication rule for Kronecker products (E.2):

$$\mathbf{B}_{\rm lab}\mathbf{B}_{\rm ini} = (B_{\rm lab} \otimes B_{\rm lab})\,(B_{{\rm ini}A} \otimes B_{{\rm ini}B}) = B_{\rm lab}B_{{\rm ini}A} \otimes B_{\rm lab}B_{{\rm ini}B}$$

which acts rightward to transform the initial lab frame to another lab frame.

Note that if any pair of sequential boosts applied within a subsystem are not collinear, their effect is combined into an overall boost and a **Wigner rotation** that changes the orientation of the subsystems within their own rest frames, as described in Appendix C. In general, this means the combination of an active and passive transformation may be as follows:

$$\mathbf{B}_{\text{lab}}\mathbf{B}_{\text{ini}} = B_{\text{lab}}B_{\text{ini}A} \otimes B_{\text{lab}}B_{\text{ini}B} = B_A R_A \otimes B_B R_B \tag{3.12}$$

where R_A and R_B are the Wigner rotations in the rest frame of each subspace, and B_A and B_B are the overall boosts of each subsystem describing their inertial reference frames relative to a common observer in the new lab frame.

3.2.2.3 Justification for investigating the simplest cases

With an appropriate transformation of the lab frame, any complicated set of boosts can be simplified. Even if there are many boosts and many rotations, all the rotations can be brought together to act only in the rest frame of the subspace in question (C.2). Thus, the rest-frame orientation can be determined before the overall boost is applied.

Conversely, any simple boost can be investigated and used as an essential building block towards understanding the more complicated cases. That is the approach of this dissertation: the focus is on the simplest cases, in order to isolate the geometric effects without the complications present in more general cases. The relationship between the simplest cases and the more general is worth acknowledging, but only the simplest boost cases are examined in detail here. Generalizing this treatment is suggested as future work, and is discussed in Section 5.2.

3.2.3 Single-System Relative Boosts

A particularly simple bipartite boost is one that describes a transformation of only one of the subsystems. Taking into account the bipartite nature of the system, such a transformation describes both parts of the system relative to a lab frame which happens to be at rest relative to one of the subsystems.

3.2.3.1 Single-system boost, expanded

Expressions for a bipartite single-subsystem boost operator are obtained in a straightforward way, by applying a tensor product of the single-system boost (3.7) with the zero-boost operator, $\mathbf{1}$:

$$\begin{aligned} B_A \otimes \mathbf{1} \;&=\; (S_A + V_A \hat{\mathbf{v}}_A) \otimes \mathbf{1} \;=\; S_A\,(\mathbf{1}\otimes\mathbf{1}) \;+\; V_A\,(\hat{\mathbf{v}}_A \otimes \mathbf{1}) \\ \therefore B_A \otimes \mathbf{1} \;&=\; \sqrt{\frac{\gamma_A+1}{2}}\,(\mathbf{1}\otimes\mathbf{1}) \;+\; \sqrt{\frac{\gamma_A-1}{2}}\,(\hat{\mathbf{v}}_A \otimes \mathbf{1}) \end{aligned} \tag{3.13}$$

$$\begin{aligned} \mathbf{1} \otimes B_B \;&=\; \mathbf{1} \otimes (S_X + V_X \hat{\mathbf{v}}_X) \;=\; S_B\,(\mathbf{1}\otimes\mathbf{1}) \;+\; V_B\,(\mathbf{1} \otimes \hat{\mathbf{v}}_B) \\ \therefore \mathbf{1} \otimes B_B \;&=\; \sqrt{\frac{\gamma_B+1}{2}}\,(\mathbf{1}\otimes\mathbf{1}) \;+\; \sqrt{\frac{\gamma_B-1}{2}}\,(\mathbf{1} \otimes \hat{\mathbf{v}}_B)\ . \end{aligned}$$

These results are obviously simpler than the general bipartite boost $B_A \otimes B_B$ (3.8), since the scalar and vector parts of the non-boosted subsystem are one and zero respectively.

3.2.3.2 Transforming to and from a subsystem rest-frame

Any system with its constituent parts in relative motion can be described from the rest frame of one of its parts, by a passive transformation that converts its overall bipartite boost to the form $B_A \otimes \mathbf{1}$ or $\mathbf{1} \otimes B_B$ (with, perhaps, an additional rest-frame rotation). The transformation to such a form is achieved by taking the bar conjugate of one of the constituent boosts and applying it to both of its subspaces.

For example, if a system is initially described in terms of two different relative boosts, $\mathbf{B}_{\text{ini}} \equiv B_{\text{ini}A} \otimes B_{\text{ini}B}$, it can be transformed to the rest frame of the first subsystem by applying a boost $\overline{\mathbf{B}}_{\text{ini}A} = \overline{B}_{\text{ini}A} \otimes \overline{B}_{\text{ini}A}$ as follows:

$$\overline{\mathbf{B}}_{\text{ini}A}\mathbf{B}_{\text{ini}} = \left(\overline{B}_{\text{ini}A} \otimes \overline{B}_{\text{ini}A}\right)(B_{\text{ini}A} \otimes B_{\text{ini}B}) = \mathbf{1} \otimes \overline{B}_{\text{ini}A} B_{\text{ini}B}\ .$$

The effect of the pair of successive boosts in the second subsystem is equivalent to an overall boost B_Ω and possibly a rest-frame Wigner rotation (C.1) R_Ω, if $B_{\text{ini}A}$ and $B_{\text{ini}B}$ are not collinear:

$$\mathbf{1} \otimes \overline{B}_{\text{ini}A} B_{\text{ini}B} = \mathbf{1} \otimes B_\Omega R_\Omega = (\mathbf{1} \otimes B_\Omega)\,(\mathbf{1} \otimes R_\Omega)$$

where the Kronecker product factoring (E.2) serves to emphasize that the boost can be examined independently from the rotor that changes the orientation of the rest frame of the system.

Of course, it is also possible to use one of the rest frames as the initial lab frame, and deferring the option to transform to another more complicated reference frame until after some initial calculations

are performed. That is the perspective taken in this dissertation: the simplest case is used to examine the geometry of the entanglement, and the change of reference frame is simply acknowledged as an option that may be investigated further, if desired. However, in the context of the present work, the rest frame of one of the subsystems may not be the simplest case[3], which is why other simplifications of the bipartite boost are investigated next.

3.2.4 Equal-Velocity Boosts, Relative to Observer

Another simplification for the general bipartite boost $B_A \otimes B_B$ (3.8), which may be noticed from examining the form of its coefficients (3.9), is as follows. If the relative speeds of the two parts are equal with respect to the observer's frame, $v_A = v_B = v$, then the Lorentz factors are equal as well, $\gamma_A = \gamma_B = \gamma$, thus simplifying the coefficients for $B_{|A|} \otimes B_{|B|}$ to be

$$\begin{aligned} S_A S_B\Big|_{v_A=v_B} &= \frac{1}{2}\sqrt{(\gamma+1)(\gamma+1)} = \frac{1}{2}(\gamma+1) \\ V_A S_B\Big|_{v_A=v_B} = S_A V_B\Big|_{v_A=v_B} &= \frac{1}{2}\sqrt{(\gamma-1)(\gamma+1)} = \frac{1}{2}\sqrt{\gamma^2-1} = \frac{\gamma v}{2c} = \frac{1}{2}|\tilde{\mathbf{u}}| \\ \text{and } V_A V_B\Big|_{v_A=v_B} &= \frac{1}{2}\sqrt{(\gamma-1)(\gamma-1)} = \frac{1}{2}(\gamma-1) \end{aligned} \tag{3.14}$$

where it was noted that the unimodularity $u\overline{u} = 1$ of the proper velocity (2.10) gives:

$$\begin{aligned} 1 &= \gamma^2(1+\hat{\mathbf{v}}v/c)(1-\hat{\mathbf{v}}v/c) = \gamma^2\left(1-v^2/c^2\right) \\ &\therefore\ \gamma^2 v^2/c^2 = \gamma^2 - 1 \text{ and thus } \boxed{\sqrt{\gamma^2-1} = \gamma\frac{v}{c}} \\ &\text{since all quantities are positive and } \gamma \geq 1. \end{aligned} \tag{3.15}$$

Still, this only partially simplifies the calculation of the bipartite boost (3.8):

$$B_A \otimes B_{|B|=|A|} = \frac{1}{2}(\gamma+1)(\mathbf{1}\otimes\mathbf{1}) \ + \ \frac{\gamma v}{2c}(\hat{\mathbf{v}}_A \otimes \mathbf{1} + \mathbf{1}\otimes\hat{\mathbf{v}}_B) + \frac{1}{2}(\gamma-1)(\hat{\mathbf{v}}_A\otimes\hat{\mathbf{v}}_B) \tag{3.16}$$

since the vector directions remain distinct.

3.2.4.1 Orienting the coordinate system based on the boost directions

Before considering directions for the velocity, recall that the rotational gauge freedom of the rest frame allows for a choice of orientation of the coordinate system without loss of generality. The overall

[3] It's possible that the perspective from the rest frame of one of the subsystems may not simplify the description much at all. See Section 3.3.6 for the attempts at applying a single-subsystem boost to an entanglor.

rotational gauge freedom was first employed (2.16) in aligning $\mathbf{e}_3$ with the initial spin direction. The remaining rotational gauge freedom about that direction (2.17) can now exploited to orient the $\mathbf{e}_1$ direction in a way that assists the simplifications in Section 3.3.

To that end, since the entanglors (3.2) have been defined in terms of the bivector $\mathbf{e}_1\mathbf{e}_3$, it is convenient[4] to treat boosts in the same plane. Equivalently, the arbitrariness in orienting the plane of the entangling bivector can be removed by associating the $\mathbf{e}_1$ direction with the positive component of the boost perpendicular to the initial spin direction $\mathbf{e}_3$, provided of course that the boost is not aligned with the $\mathbf{e}_3$ direction.

Thus, it is reasonable to define the velocity in the first subspace, relative to the common observer, as having magnitude v in direction

$$\hat{\mathbf{v}} = \cos\delta\,\mathbf{e}_1 + \sin\delta\,\mathbf{e}_3 \text{ where } -\frac{\pi}{2} \leq \delta \leq \frac{\pi}{2}. \tag{3.17}$$

This restricted range for the angle takes into account the fact that the $\mathbf{e}_1$ direction can be selected so that its velocity component is positive, and that the $\mathbf{e}_3$ component of the boost may be either parallel or antiparallel to the initial spin direction.

Two specific relative velocities for the second subspace are described next.

3.2.4.2 Second boost direction the same as the first

First, consider a pair of subsystems that are at rest relative to each other, but in motion relative to the lab frame. The bipartite boost describing such a situation has identical boosts B_v applied to both subsections:

$$\mathbf{B}_{vv} \equiv B_v \otimes B_v = \frac{1}{2}(\gamma+1)(\mathbf{1}\otimes\mathbf{1}) \;+\; \frac{\gamma v}{2c}(\hat{\mathbf{v}}\otimes\mathbf{1} + \mathbf{1}\otimes\hat{\mathbf{v}}) + \frac{1}{2}(\gamma-1)(\hat{\mathbf{v}}\otimes\hat{\mathbf{v}}) \;. \tag{3.18}$$

Applying the details of the arbitrary boost direction $\hat{\mathbf{v}} = \cos\delta\,\mathbf{e}_1 + \sin\delta\,\mathbf{e}_3$ (3.17), the bipartite boost becomes

$$\begin{aligned}\mathbf{B}_{vv} = \frac{1}{2}(\gamma+1)(\mathbf{1}\otimes\mathbf{1}) + \frac{\gamma v}{2c}\Big\{\cos\delta\,(\mathbf{e}_1\otimes\mathbf{1} + \mathbf{1}\otimes\mathbf{e}_1) + \sin\delta\,(\mathbf{e}_3\otimes\mathbf{1} + \mathbf{1}\otimes\mathbf{e}_3)\Big\} \\ + \frac{1}{2}(\gamma-1)\Big\{\cos^2\delta\,(\mathbf{e}_1\otimes\mathbf{e}_1) + \sin^2\delta\,(\mathbf{e}_3\otimes\mathbf{e}_3) + \cos\delta\sin\delta\,(\mathbf{e}_1\otimes\mathbf{e}_3 + \mathbf{e}_3\otimes\mathbf{e}_1)\Big\}\;.\end{aligned} \tag{3.19}$$

[4] This convenience is most notable when compared to a boost with an $\mathbf{e}_2$ component, because such a case involves the unit imaginary i, which is identified in APS with the volume element $i \equiv \mathbf{e}_1\mathbf{e}_2\mathbf{e}_3$. While this is not necessarily a problem, it is useful to avoid unnecessary complexity: since the investigation here *can* be described within the $\mathbf{e}_1\mathbf{e}_3$ plane, why invoke the volume element? Avoiding this leaves room makes it easier for a later expansion to the Complexified APS (CAPS), which defines the volume element as an imaginary element I distinct from i.

3.2.4.3 Second boost direction opposite to first

Now consider an **EPR-like** situation, where equal speeds result from the conservation of linear momentum when equal-mass particles are emitted in opposite directions relative to the observer. The bipartite boost (3.16) that describes such a case can be described by assigning $B_A \equiv B_v$ and $B_B \equiv \overline{B}_v$, both relative to a common lab frame.

Explicitly, the bipartite boost with opposite directions is

$$\mathbf{B}_{v\bar{v}} \equiv B_v \otimes \overline{B}_v = \frac{1}{2}(\gamma+1)(\mathbf{1}\otimes\mathbf{1}) \;+\; \frac{\gamma v}{2c}(\hat{\mathbf{v}}\otimes\mathbf{1} - \mathbf{1}\otimes\hat{\mathbf{v}}) - \frac{1}{2}(\gamma-1)(\hat{\mathbf{v}}\otimes\hat{\mathbf{v}}) \tag{3.20}$$

and when its first boost direction is taken to be $\hat{\mathbf{v}} = \cos\delta\,\mathbf{e}_1 + \sin\delta\,\mathbf{e}_3$ (3.17) and second boost direction $-\hat{\mathbf{v}}$, this bipartite boost becomes

$$\begin{aligned}\mathbf{B}_{v\bar{v}} = \frac{1}{2}(\gamma+1)(\mathbf{1}\otimes\mathbf{1}) \;+\; \frac{\gamma v}{2c}\Big\{\cos\delta\,(\mathbf{e}_1\otimes\mathbf{1} - \mathbf{1}\otimes\mathbf{e}_1) + \sin\delta\,(\mathbf{e}_3\otimes\mathbf{1} - \mathbf{1}\otimes\mathbf{e}_3)\Big\} \\ - \frac{1}{2}(\gamma-1)\Big\{\cos^2\delta\,(\mathbf{e}_1\otimes\mathbf{e}_1) + \sin^2\delta\,(\mathbf{e}_3\otimes\mathbf{e}_3) + \cos\delta\sin\delta\,(\mathbf{e}_1\otimes\mathbf{e}_3 + \mathbf{e}_3\otimes\mathbf{e}_1)\Big\}\end{aligned} \tag{3.21}$$

which only differs from the $\mathbf{B}_{vv}$ case (3.19) by a few signs.

3.2.5 Other Boosts to Consider?

Other cases beyond the ones described so far are unlikely to be simple enough to clearly demonstrate any useful or interesting geometry. However, the examination provided by the present work may suggest some cases that have the potential to be interesting, or serve as a basis for comparing the more complicated cases.

One particular case that may be simple enough to be geometrically interesting is perpendicular equal-velocity cases. A proposal of how to investigate this idea is discussed briefly in Section 5.2.

Thus concludes the investigation of the bipartite boosts themselves. This dissertation is now in position to begin applying such boosts to the entanglors, so as to determine the effect of velocity transformations on entanglement.

3.3 Entangling Eigenspinors

Recall that the Dirac equation (2.20) is satisfied by an **eigenspinor** (2.7), which enacts the particular Lorentz transformation of a system from its rest frame to the lab frame from which it is being

described, including both a rotor R and a boost B.[5]

The **entangling eigenspinors** can be formed the same way, by combining the entanglors $\mathbf{R}_{X\pm}$ as introduced in Section 3.1 with the separable bipartite boosts $\mathbf{B}$ introduced in Section 3.2. However, it is important to be clear about the reasons for the particular order of operations involved in their combination; this is described in Section 3.3.1. There is also the matter of selecting which entanglors to explore, especially considering their interrelationship (3.4); this is discussed in Section 3.3.2. After this, Sections 3.3.3 through 3.3.6 present the entangling eigenspinors that are the focus of this dissertation.

3.3.1 Order of Operations

Recall that the single-system eigenspinor $\Lambda = BR$ (2.7) is defined as acting on a system in its rest frame. In particular, the rotor R sets the orientation of the system before the boost B introduces the information about the motion relating the lab frame and the rest frame.

It is also mathematically possible to describe an eigenspinor by way of a boost acting first and a rotation acting upon it, but the interpretation of such a sequence is less straightforward. Specifically, if attempting to describe the relationship between reference frames by a product $\Lambda = R'B'$, the rotor R' acting on the boost B' is effectively "correcting" the original boost direction, in addition to describing the relative orientation of the moving frame. Thus the alternative ordering drastically reduces the clarity of interpretation, since the direction of the boost B' may not correspond to the direction of the relative motion.[6]

The same reasoning for the ordering of boost and rotation factors applies to the bipartite case. That is, the relative rotation amplitudes of the entanglors (3.2) are most sensibly put in the rightmost position of the mathematical expression, in order for the boosts on their left to appropriately represent the relative motion between their reference frames. Just as there is no clear physical interpretation of the boost-in-rightmost-position B' in the single-system eigenspinor, there would be no clear physical interpretation of a separable bipartite boost applied on the right of an entanglor. (In attempting such a calculation, the superposition in the entangling operation makes it impossible to retain the separability of the boosts when attempting to transfer them through to the leftmost position, ala (C.2), to obtain information about the actual directions of relative motion.)

Finally, note that entanglement here is described as a relationship between the possibilities of a change of orientation, without any mention of an observer's frame of reference; thus it makes

[5] A more general eigenspinor may include information about a spatial reference-frame density $\rho_{ref}(x)$, as is done elsewhere [23, 25], but such a treatment is omitted here for the sake of focusing attention on the spins and entanglements thereof.

[6] Similarly, the Wigner rotation (C.1) is arguably most sensibly described as acting in the rest frame before the overall boost is applied. Explicit calculations (C.2) can even more clearly demonstrate the effect of reversing the order of the boost and rotor.

most sense for the entanglor to be the right-most operator in the model presented in this work. The observer's reference frame is introduced by the next operator to the left, the active bipartite boost (3.10), which describes the relative velocity of each part as seen from the lab frame.

A further overall transformation can be applied after that, as an additional leftmost operator, to change from one lab frame to another lab frame. However, in the name of clarity in the present work, additional transformations such as this will be omitted — except in acknowledging the possibility of applying such transformations.

Thus, the combination of a bipartite boost $\mathbf{B}$ and bipartite rotor $\mathbf{R}$ (whether an entanglor or not) are combined in the present work to form a **bipartite eigenspinor**:

$$\mathbf{BR} \equiv \mathbf{\Lambda}$$

which is represented in bold as a reminder that it is a bipartite operator. Note that if both the boost and rotor are separable, the eigenspinor is as well:

$$\mathbf{B}_{AB}\mathbf{R}_{AB} = \mathbf{\Lambda}_{AB}$$
$$(B_A \otimes B_B)\,(R_A \otimes R_B) = B_A R_A \otimes B_B R_B = \Lambda_A \otimes \Lambda_B$$

where the subscripts serve as a reminder that the operators acting on subsystem A may be different from those acting on subsystem B. The separable nature of all of the physical contributions to the separable bipartite eigenspinor $\mathbf{\Lambda}_{AB}$ make it a rather uninteresting example, except in its distinctness from the more interesting cases involving entanglement, which are examined below.

3.3.2 Which Entanglors?

Since the entanglors are interrelated (3.5) by single-subspace bipartite rotors (3.3), only a subset of the entanglors needs to be investigated in detail. Recall that $\mathbf{R}_{\Psi+}$ and $\mathbf{R}_{\Phi-}$ are so related, while $\mathbf{R}_{\Psi-}$ and $\mathbf{R}_{\Phi+}$ are as well. Thus, only one from each pair needs to be investigated relativistically, since the result of any calculation can be transformed by applying a simple rotation afterwards in order to obtain the other from the pair.

It is convenient to choose both the Ψ-entanglors, $\mathbf{R}_{\Psi\pm}$, for the purposes of this investigation.

One reason to select a pair of entanglors having the same primary subscript (here, Ψ) is to enable the combining of mathematical expressions when results differ only by a phase, by using the $\pm$ sign. This also makes it easier to compare the results between the cases, or highlight where the cases diverge.

A reason to select the Ψ-entanglors instead of the Φ-entanglors is because of the seemingly-closer relationship to the EPR-like situation. After all, conservation of angular momentum results in a

totally antisymmetric, rotationally-invariant, entangled basis state Ψ_-. However, it is important to remember that the present calculations are in terms of *transformations* rather than *states*. Still, the entanglors were derived and labelled according to the Bell states obtained when they are applied to the double spin-up state $\Upsilon_{\uparrow\uparrow}$; thus, there is a "closeness" in the description, even if only symbolically. The convenience of this choice, then, relates to the familiarity of the symbols and their interpretation.

With entanglors thus chosen and the choices justified, the calculations may begin.

3.3.3 General Boost of the Psi-Entanglors

First, consider a general, separable, bipartite boost $\mathbf{B}_{AB}$, applied to the Ψ-entanglors $\mathbf{R}_{\Psi\pm}$ (3.2):

$$\begin{aligned}
\mathbf{\Lambda}^{AB}_{\Psi\pm} \equiv \mathbf{B}_{AB}\mathbf{R}_{\Psi\pm} &= (B_A \otimes B_B)\,\frac{1}{\sqrt{2}}\,(R_\pi \otimes R_0 \pm R_0 \otimes R_\pi) \\
&= (B_A \otimes B_B)\,\frac{1}{\sqrt{2}}\,(\mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{1} \pm \mathbf{1} \otimes \mathbf{e}_1\mathbf{e}_3) \\
&= \frac{1}{\sqrt{2}}\,(B_A R_\pi \otimes B_B R_0 \pm B_A R_0 \otimes B_B R_\pi) \\
&= \frac{1}{\sqrt{2}}\,(\Lambda_{A\pi} \otimes \Lambda_{B0} \pm \Lambda_{A0} \otimes \Lambda_{B\pi})\,.
\end{aligned} \tag{3.22}$$

The result is a bipartite eigenspinor $\mathbf{\Lambda}^{AB}_{\Psi\pm}$ which has the form of a superposition of bipartite eigenspinor amplitudes of the form $\Lambda_{xy} \otimes \Lambda_{za}$. It is most useful to represent the entanglors $\mathbf{R}_{\Psi\pm}$ in terms of bivectors (3.22), because this is the form for which simplifications will be obvious in the calculations ahead.

The other representations of the bipartite eigenspinor $\mathbf{\Lambda}^{AB}_{\Psi\pm}$ shown above are for the sake of description, showing the form of the boost and entanglor as separate factors, then multiplied to form single-system products of the form $B_x R_y$, and then expressed as single-system eigenspinors of the form Λ_{xy}. Each representation of the systems with maximal entanglement that are considered here is a superposition of two terms with equal amplitudes of magnitude $\frac{1}{\sqrt{2}}$, and equal or opposite phase.

Note that there are no further simplifications obvious in $\mathbf{\Lambda}^{AB}_{\Psi\pm}$, since nothing has yet been specified about the boosts.

All the work up to this point was in preparation for the following sections, where entangling operators are subjected to explicit relativistic boosts: equal and parallel boosts in Section 3.3.4, equal and antiparallel boosts in Section 3.3.5, and bipartite single-subspace boosts in Section 3.3.6.

3.3.4 Equal Parallel Boost of Psi-Entanglors

As described in Section 3.2.4, the expression for the bipartite boost has a reasonably simplified form when both subsystems are described, relative to the lab frame that is used as common for both parts, as having equal velocity and direction (3.17):

$$\begin{aligned}\hat{\mathbf{v}} &= \cos\delta\,\mathbf{e}_1 + \sin\delta\,\mathbf{e}_3 \\ &= v_1\mathbf{e}_1 + v_3\mathbf{e}_3\end{aligned} \tag{3.23}$$

where the angle is constrained to $-\frac{\pi}{2} \leq \delta \leq \frac{\pi}{2}$, and the coefficients v_1, v_3 are introduced as shorthand. Using this, the bipartite boost $\mathbf{B}_{vv}$ (3.18) can be applied to the Ψ-entanglors $\mathbf{R}_{\Psi\pm}$ (3.2) to obtain the following expression for the entangling eigenspinors:

$$\begin{aligned}\boldsymbol{\Lambda}^{vv}_{\Psi\pm} &\equiv \mathbf{B}_{vv}\mathbf{R}_{\Psi\pm} \\ &= \frac{1}{2\sqrt{2}}\Big\{ \left[\mathbf{e}_1\otimes\mathbf{1} - \left(v_1^2 \mp v_3^2\right)(\mathbf{1}\otimes\mathbf{e}_1)\right][\mathbf{e}_3\otimes\mathbf{1}] \mp \left[\left(v_1^2 \mp v_3^2\right)(\mathbf{e}_1\otimes\mathbf{1}) - \mathbf{1}\otimes\mathbf{e}_1\right][\mathbf{1}\otimes\mathbf{e}_3]\Big\} \\ &\quad + \frac{\gamma}{2\sqrt{2}}\Big\{ \left[\mathbf{e}_1\otimes\mathbf{1} + \left(v_1^2 \mp v_3^2\right)(\mathbf{1}\otimes\mathbf{e}_1)\right][\mathbf{e}_3\otimes\mathbf{1}] \pm \left[\left(v_1^2 \mp v_3^2\right)(\mathbf{e}_1\otimes\mathbf{1}) + \mathbf{1}\otimes\mathbf{e}_1\right][\mathbf{1}\otimes\mathbf{e}_3]\Big\} \\ &\quad + \frac{\gamma v}{2\sqrt{2}c}\Big\{ v_1\left[\mathbf{1}\otimes\mathbf{1} + \mathbf{e}_1\otimes\mathbf{e}_1\right][\mathbf{e}_3\otimes\mathbf{1} \pm \mathbf{1}\otimes\mathbf{e}_3] - v_3\left[\mathbf{e}_1\otimes\mathbf{1} \pm \mathbf{1}\otimes\mathbf{e}_1\right][\mathbf{1}\otimes\mathbf{1} - \mathbf{e}_3\otimes\mathbf{e}_3]\Big\} \\ &\quad + \frac{v_1 v_3}{2\sqrt{2}}(\gamma-1)(\mathbf{e}_3\otimes\mathbf{e}_3 - \mathbf{e}_1\otimes\mathbf{e}_1)(1\pm 1)\end{aligned} \tag{3.24}$$

where the terms containing both $\mathbf{e}_1$ and $\mathbf{e}_3$ have been factored using the multiplication rule for Kronecker products (E.2). The factoring (and ordering, where there was a choice) will prove useful in the following steps.

As can be seen from the complicated expansion above, the same-phase Ψ-entanglor $\mathbf{R}_{\Psi+}$ and opposite-phase Ψ-entanglor $\mathbf{R}_{\Psi-}$ result in enough differences under a parallel boost that they are better off examined separately in order to determine how to simplify their description.

3.3.4.1 Equal parallel boost of opposite-phase Psi-entanglor (and method details)

It may be noticed from the entangling eigenspinor $\boldsymbol{\Lambda}^{vv}_{\Psi\pm}$ (3.24) that there are two major simplifications that result from using the lower sign, that is, by treating only the opposite-phase Ψ-entanglor $\mathbf{R}_{\Psi-}$. One simplification is that the term in the bottom line becomes zero; the other is that $v_1^2 \mp v_3^2$ becomes

$v_1^2 + v_3^2 = \cos^2\delta + \sin^2\delta = 1$. The entangling eigenspinor $\mathbf{\Lambda}^{vv}_{\Psi\pm}$ (3.24) therefore simplifies to

$$\begin{aligned}\mathbf{\Lambda}^{vv}_{\Psi-} \equiv \mathbf{B}_{vv}\mathbf{R}_{\Psi-} = {}& \frac{1}{2\sqrt{2}}\left[\mathbf{e}_1\otimes\mathbf{1} - \mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{e}_3\otimes\mathbf{1} + \mathbf{1}\otimes\mathbf{e}_3\right] \\ & + \frac{\gamma}{2\sqrt{2}}\left[\mathbf{e}_1\otimes\mathbf{1} + \mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{e}_3\otimes\mathbf{1} - \mathbf{1}\otimes\mathbf{e}_3\right] \\ & + \frac{\gamma v}{2\sqrt{2}c}\Big\{\cos\delta\left[\mathbf{1}\otimes\mathbf{1} + \mathbf{e}_1\otimes\mathbf{e}_1\right]\left[\mathbf{e}_3\otimes\mathbf{1} - \mathbf{1}\otimes\mathbf{e}_3\right] \\ & \qquad - \sin\delta\left[\mathbf{e}_1\otimes\mathbf{1} - \mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{1}\otimes\mathbf{1} - \mathbf{e}_3\otimes\mathbf{e}_3\right]\Big\}\end{aligned} \tag{3.25}$$

where the shorthand has been removed from the coefficients $v_1 = \cos\delta$ and $v_3 = \sin\delta$.

Note how the result above (3.25) has intriguingly similar factors appearing throughout, involving sums and differences of tensor products of unit vectors and the identity element $\mathbf{1}$. Interestingly, these may be realized as being equivalent to various sums and differences of **biprojectors** (B.16). However, even though the same forms appear in both $\mathbf{e}_1$ and $\mathbf{e}_3$ factors, there is another interpretation for the $\mathbf{e}_1$ factors that will be even more useful, and more obvious after the $\mathbf{e}_3$ factors are replaced by their "**biprojector equivalents**".

Making the abovementioned replacements of the $\mathbf{e}_3$ factors by their "biprojector equivalents" (B.16),

$$\begin{aligned}\mathbf{\Lambda}^{vv}_{\Psi-} \equiv \mathbf{B}_{vv}\mathbf{R}_{\Psi-} = {}& \frac{1}{\sqrt{2}}\left(\mathbf{e}_1\otimes\mathbf{1} - \mathbf{1}\otimes\mathbf{e}_1\right)\left[\mathrm{P}_3\otimes\mathrm{P}_3 - \overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right] \\ & + \frac{\gamma}{\sqrt{2}}\left(\mathbf{e}_1\otimes\mathbf{1} + \mathbf{1}\otimes\mathbf{e}_1\right)\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3 - \overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right] \\ & + \frac{\gamma v}{\sqrt{2}c}\Big\{\cos\delta\left(\mathbf{1}\otimes\mathbf{1} + \mathbf{e}_1\otimes\mathbf{e}_1\right)\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3 - \overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right] \\ & \qquad - \sin\delta\left(\mathbf{e}_1\otimes\mathbf{1} - \mathbf{1}\otimes\mathbf{e}_1\right)\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3 + \overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\Big\}\end{aligned}$$

it can now be more clearly seen that the $\mathbf{e}_1$ factors are reminiscent of the entanglors — especially since the adjacent biprojectors can "ungobble" factors of $\mathbf{e}_3$ via the Pacwoman property (B.7) to turn the reminiscence into reality, transforming the $\mathbf{e}_1$ factors from "spin flip" operators to π-rotation operators $\mathbf{e}_1\mathbf{e}_3$ instead.

Before applying the Pacwoman property (B.7) to simplify the entangling eigenspinor $\mathbf{\Lambda}^{vv}_{\Psi-}$ above, it is handy to label and name the $\mathbf{e}_1$ factors, since they arise in the rest of these calculations as well and will be handled the same way. Since each may be interpreted as a variant of an entanglor expressed in terms of an unphysical "spin flip" transformation, let their nickname be "**flip-tanglors**", and let them be denoted similarly to the entanglors $\mathbf{R}_{X\pm}$ (3.2):

$$\mathbf{F}_{\Psi\pm} = \frac{1}{\sqrt{2}}\left(\mathbf{e}_1\otimes\mathbf{1} \pm \mathbf{1}\otimes\mathbf{e}_1\right) \quad\text{and}\quad \mathbf{F}_{\Phi\pm} = \frac{1}{\sqrt{2}}\left(\mathbf{1}\otimes\mathbf{1} \pm \mathbf{e}_1\otimes\mathbf{e}_1\right). \tag{3.26}$$

Thus, the "flip-tanglors" $\mathbf{F}_{X\pm}$ are defined so that their application to a double spin-up projector $\Upsilon_{\uparrow\uparrow} = \mathrm{P}_3 \otimes \mathrm{P}_3$ allows the Pacwoman property (B.7) to transform them to the corresponding entanglors $\mathbf{R}_{X\pm}$ (3.2), where the correspondence is indicated by the same subscript.

Now, since these "flip-tanglors" arise adjacent to all possible biprojectors, it is useful to use the Pacwoman property (B.7) to obtain all the possible transformations, to help with simplifying the boosted eigenspinors:

$$\begin{array}{rclcrcl}
\mathbf{F}_{\Psi\pm}\left[\mathrm{P}_3 \otimes \mathrm{P}_3\right] &=& +\mathbf{R}_{\Psi\pm}\left[\mathrm{P}_3 \otimes \mathrm{P}_3\right] & \qquad & \mathbf{F}_{\Phi\pm}\left[\mathrm{P}_3 \otimes \mathrm{P}_3\right] &=& \mathbf{R}_{\Phi\pm}\left[\mathrm{P}_3 \otimes \mathrm{P}_3\right] \\
\mathbf{F}_{\Psi\pm}\left[\overline{\mathrm{P}}_3 \otimes \overline{\mathrm{P}}_3\right] &=& -\mathbf{R}_{\Psi\pm}\left[\overline{\mathrm{P}}_3 \otimes \overline{\mathrm{P}}_3\right] & & \mathbf{F}_{\Phi\pm}\left[\overline{\mathrm{P}}_3 \otimes \overline{\mathrm{P}}_3\right] &=& \mathbf{R}_{\Phi\pm}\left[\overline{\mathrm{P}}_3 \otimes \overline{\mathrm{P}}_3\right] \\
\mathbf{F}_{\Psi\pm}\left[\mathrm{P}_3 \otimes \overline{\mathrm{P}}_3\right] &=& +\mathbf{R}_{\Psi\mp}\left[\mathrm{P}_3 \otimes \overline{\mathrm{P}}_3\right] & & \mathbf{F}_{\Phi\pm}\left[\mathrm{P}_3 \otimes \overline{\mathrm{P}}_3\right] &=& \mathbf{R}_{\Phi\mp}\left[\mathrm{P}_3 \otimes \overline{\mathrm{P}}_3\right] \\
\mathbf{F}_{\Psi\pm}\left[\overline{\mathrm{P}}_3 \otimes \mathrm{P}_3\right] &=& -\mathbf{R}_{\Psi\mp}\left[\overline{\mathrm{P}}_3 \otimes \mathrm{P}_3\right] & & \mathbf{F}_{\Phi\pm}\left[\overline{\mathrm{P}}_3 \otimes \mathrm{P}_3\right] &=& \mathbf{R}_{\Phi\mp}\left[\overline{\mathrm{P}}_3 \otimes \mathrm{P}_3\right] .
\end{array} \tag{3.27}$$

Thus, in terms of the "flip-tanglors", the previous form of the entangling eigenspinor $\mathbf{\Lambda}^{vv}_{\Psi-}$ can be expressed instead as

$$\begin{aligned}
\mathbf{\Lambda}^{vv}_{\Psi-} \equiv \mathbf{B}_{vv}\mathbf{R}_{\Psi-} = \mathbf{F}_{\Psi-}&\left[\mathrm{P}_3 \otimes \mathrm{P}_3 - \overline{\mathrm{P}}_3 \otimes \overline{\mathrm{P}}_3\right] \\
&+ \gamma \mathbf{F}_{\Psi+}\left[\mathrm{P}_3 \otimes \overline{\mathrm{P}}_3 - \overline{\mathrm{P}}_3 \otimes \mathrm{P}_3\right] \\
&+ \frac{\gamma v}{c}\left\{\cos\delta\, \mathbf{F}_{\Phi+}\left[\mathrm{P}_3 \otimes \overline{\mathrm{P}}_3 - \overline{\mathrm{P}}_3 \otimes \mathrm{P}_3\right] - \sin\delta\, \mathbf{F}_{\Psi-}\left[\mathrm{P}_3 \otimes \overline{\mathrm{P}}_3 + \overline{\mathrm{P}}_3 \otimes \mathrm{P}_3\right]\right\}
\end{aligned}$$

and then the factors can be collected by biprojector so that the "flip-tanglors" (3.26) can be easily transformed (3.27) into physically-realizable entanglors instead:

$$\begin{aligned}
\mathbf{\Lambda}^{vv}_{\Psi-} \equiv \mathbf{B}_{vv}\mathbf{R}_{\Psi-} = \mathbf{F}_{\Psi-}&\left[\mathrm{P}_3 \otimes \mathrm{P}_3 - \overline{\mathrm{P}}_3 \otimes \overline{\mathrm{P}}_3\right] \\
&+ \gamma\left[\mathbf{F}_{\Psi+} + \frac{v}{c}\cos\delta\, \mathbf{F}_{\Phi+} - \frac{v}{c}\sin\delta\, \mathbf{F}_{\Psi-}\right]\left[\mathrm{P}_3 \otimes \overline{\mathrm{P}}_3\right] \\
&- \gamma\left[\mathbf{F}_{\Psi+} + \frac{v}{c}\cos\delta\, \mathbf{F}_{\Phi+} + \frac{v}{c}\sin\delta\, \mathbf{F}_{\Psi-}\right]\left[\overline{\mathrm{P}}_3 \otimes \mathrm{P}_3\right] \\
= \boxed{\begin{array}{l}
\mathbf{R}_{\Psi-}\left[\mathrm{P}_3 \otimes \mathrm{P}_3 + \overline{\mathrm{P}}_3 \otimes \overline{\mathrm{P}}_3\right] \\
+ \gamma\left[\mathbf{R}_{\Psi-} + \frac{v}{c}\cos\delta\, \mathbf{R}_{\Phi-} - \frac{v}{c}\sin\delta\, \mathbf{R}_{\Psi+}\right]\left[\mathrm{P}_3 \otimes \overline{\mathrm{P}}_3\right] \\
+ \gamma\left[\mathbf{R}_{\Psi-} - \frac{v}{c}\cos\delta\, \mathbf{R}_{\Phi-} + \frac{v}{c}\sin\delta\, \mathbf{R}_{\Psi+}\right]\left[\overline{\mathrm{P}}_3 \otimes \mathrm{P}_3\right]
\end{array}} .
\end{aligned} \tag{3.28}$$

For the sake of completeness[7], it is worth evaluating the entangling eigenspinor $\mathbf{\Lambda}^{vv}_{\Psi-}$ (3.28) for some special cases. In particular, consider velocity directions that are aligned with the basis directions $\mathbf{e}_1$ or $\pm\mathbf{e}_3$, recalling that $\mathbf{e}_1$ is aligned with the component of the boost and thus its coefficient is defined as positive.

For the $\hat{\mathbf{v}} = \mathbf{e}_1$ case, the coefficients are $v_1 = \cos\delta = 1$ and $v_3 = \sin\delta = 0$, and the entangling

[7]The "completeness" here refers to the fact that later examinations of other entangling eigenspinors will require explicit examples such as this in order to obtain simplifications at all.

eigenspinor $\mathbf{\Lambda}^{vv}_{\Psi-}$ (3.28) becomes

$$\begin{aligned}\mathbf{\Lambda}^{1,1}_{\Psi-} &\equiv \mathbf{B}_{1,1}\mathbf{R}_{\Psi-} \\ &= \boxed{\begin{array}{l}\mathbf{R}_{\Psi-}\left[\mathrm{P}_3\otimes\mathrm{P}_3+\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right] \\ +\,\gamma\left\{\left[\mathbf{R}_{\Psi-}+\dfrac{v}{c}\mathbf{R}_{\Phi-}\right]\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3\right]+\left[\mathbf{R}_{\Psi-}-\dfrac{v}{c}\mathbf{R}_{\Phi-}\right]\left[\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\right\}\end{array}}\,.\end{aligned} \tag{3.29}$$

For the $\hat{\mathbf{v}}=\pm\mathbf{e}_3$ case, the coefficients are $v_1=\cos\delta=0$, $v_3=\sin\delta=\pm1$, and the entangling eigenspinor $\mathbf{\Lambda}^{vv}_{\Psi-}$ (3.28) becomes

$$\begin{aligned}\mathbf{\Lambda}^{\pm3,\pm3}_{\Psi-} &\equiv \mathbf{B}_{\pm3,\pm3}\mathbf{R}_{\Psi-} \\ &= \boxed{\begin{array}{l}\mathbf{R}_{\Psi-}\left[\mathrm{P}_3\otimes\mathrm{P}_3+\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right] \\ +\,\gamma\left\{\left[\mathbf{R}_{\Psi-}\mp\dfrac{v}{c}\mathbf{R}_{\Psi+}\right]\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3\right]+\left[\mathbf{R}_{\Psi-}\pm\dfrac{v}{c}\mathbf{R}_{\Psi+}\right]\left[\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\right\}\end{array}}\,.\end{aligned} \tag{3.30}$$

3.3.4.2 Equal parallel boost of same-phase Psi-entanglor

Next, consider the upper sign in $\mathbf{\Lambda}^{vv}_{\Psi\pm}$ (3.24), i.e. the same-phase Ψ-entanglor case. The general result for arbitrary direction $\hat{\mathbf{v}}=v_1\mathbf{e}_1+v_3\mathbf{e}_3$ does not simplify much:

$$\begin{aligned}\mathbf{\Lambda}^{vv}_{\Psi+} &\equiv \mathbf{B}_{vv}\mathbf{R}_{\Psi+} \\ &= \frac{1}{2\sqrt{2}}\left\{\left[\mathbf{e}_1\otimes\mathbf{1}-\left(v_1^2-v_3^2\right)\left(\mathbf{1}\otimes\mathbf{e}_1\right)\right]\left[\mathbf{e}_3\otimes\mathbf{1}\right]-\left[\left(v_1^2-v_3^2\right)\left(\mathbf{e}_1\otimes\mathbf{1}\right)-\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{1}\otimes\mathbf{e}_3\right]\right\} \\ &\quad+\frac{\gamma}{2\sqrt{2}}\left\{\left[\mathbf{e}_1\otimes\mathbf{1}+\left(v_1^2-v_3^2\right)\left(\mathbf{1}\otimes\mathbf{e}_1\right)\right]\left[\mathbf{e}_3\otimes\mathbf{1}\right]+\left[\left(v_1^2-v_3^2\right)\left(\mathbf{e}_1\otimes\mathbf{1}\right)+\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{1}\otimes\mathbf{e}_3\right]\right\} \\ &\quad+\frac{\gamma v}{2\sqrt{2}c}\left\{v_1\left[\mathbf{1}\otimes\mathbf{1}+\mathbf{e}_1\otimes\mathbf{e}_1\right]\left[\mathbf{e}_3\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_3\right]-v_3\left[\mathbf{e}_1\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{1}\otimes\mathbf{1}-\mathbf{e}_3\otimes\mathbf{e}_3\right]\right\} \\ &\quad+\frac{v_1v_3}{\sqrt{2}}\left(\gamma-1\right)\left(\mathbf{e}_3\otimes\mathbf{e}_3-\mathbf{e}_1\otimes\mathbf{e}_1\right)\,,\end{aligned} \tag{3.31}$$

not even if the shorthand is removed from the coefficients $v_1=\cos\delta$ and $v_3=\sin\delta$ to give $v_1^2-v_3^2=\cos^2\delta-\sin^2\delta=\cos2\delta$ and $v_1v_3=\cos\delta\sin\delta=\frac{1}{2}\sin2\delta$. Thus, the same-phase Ψ-entanglor case $\mathbf{\Lambda}^{vv}_{\Psi+}$ (3.31) is drastically different from the opposite-phase Ψ-entanglor case $\mathbf{\Lambda}^{vv}_{\Psi-}$ (3.25).

Since there are no further obvious simplifications for the same-phase Ψ-entanglor case $\mathbf{\Lambda}^{vv}_{\Psi+}$ (3.31) using the general velocity direction $\hat{\mathbf{v}}$, the next simplifications to consider are the special cases where the velocity directions align with the basis directions $\mathbf{e}_1$ or $\pm\mathbf{e}_3$. Then, at least, it is possible to compare results with the similar-case results $\mathbf{\Lambda}^{1,1}_{\Psi-}$ (3.29) and $\mathbf{\Lambda}^{\pm3,\pm3}_{\Psi-}$ (3.30) involving the opposite-phase Ψ-entanglor $\mathbf{R}_{\Psi-}$

For the $\hat{\mathbf{v}} = \mathbf{e}_1$ case, the coefficients are $v_1 = \cos\delta = 1$ and $v_3 = \sin\delta = 0$, and the entangling eigenspinor $\mathbf{\Lambda}^{vv}_{\Psi+}$ (3.31) simplifies to

$$\begin{aligned}\mathbf{\Lambda}^{1,1}_{\Psi+} \equiv \mathbf{B}_{1,1}\mathbf{R}_{\Psi+} = \frac{1}{2\sqrt{2}}\left[\mathbf{e}_1\otimes\mathbf{1}-\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{e}_3\otimes\mathbf{1}-\mathbf{1}\otimes\mathbf{e}_3\right]\\ + \frac{\gamma}{2\sqrt{2}}\left[\mathbf{e}_1\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{e}_3\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_3\right]\\ + \frac{\gamma v}{2\sqrt{2}c}\left[\mathbf{1}\otimes\mathbf{1}+\mathbf{e}_1\otimes\mathbf{e}_1\right]\left[\mathbf{e}_3\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_3\right]\ .\end{aligned}$$

Replacing the $\mathbf{e}_3$ factors by their "biprojector equivalents" (B.16), and the $\mathbf{e}_1$ factors with the "flip-tanglors" (3.26), and collecting in terms of like biprojectors, this becomes

$$\begin{aligned}\mathbf{\Lambda}^{1,1}_{\Psi+} &\equiv \mathbf{B}_{1,1}\mathbf{R}_{\Psi+}\\ &= \mathbf{F}_{\Psi-}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3-\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\\ &\quad+\gamma\left\{\mathbf{F}_{\Psi+}\left[\mathrm{P}_3\otimes\mathrm{P}_3-\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]+\frac{v}{c}\mathbf{F}_{\Phi+}\left[\mathrm{P}_3\otimes\mathrm{P}_3-\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\right\}\\ &= \mathbf{F}_{\Psi-}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3-\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\\ &\quad+\gamma\left\{\left[\mathbf{F}_{\Psi+}+\frac{v}{c}\mathbf{F}_{\Phi+}\right]\left[\mathrm{P}_3\otimes\mathrm{P}_3\right]-\left[\mathbf{F}_{\Psi+}+\frac{v}{c}\mathbf{F}_{\Phi+}\right]\left[\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\right\}\end{aligned}$$

where the "flip-tanglors" can be transformed (3.27) into physically-realizable entanglors:

$$\begin{aligned}\mathbf{\Lambda}^{1,1}_{\Psi+} &\equiv \mathbf{B}_{1,1}\mathbf{R}_{\Psi+}\\ &= \boxed{\begin{array}{l}\mathbf{R}_{\Psi+}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3+\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\\ \quad+\gamma\left\{\left[\mathbf{R}_{\Psi+}+\frac{v}{c}\mathbf{R}_{\Phi+}\right]\left[\mathrm{P}_3\otimes\mathrm{P}_3\right]+\left[\mathbf{R}_{\Psi+}-\frac{v}{c}\mathbf{R}_{\Phi+}\right]\left[\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\right\}\end{array}}\end{aligned}\tag{3.32}$$

and thus this special case for the entangling eigenspinor $\mathbf{\Lambda}^{vv}_{\Psi+}$ (3.31) is in a form comparable to that of $\mathbf{\Lambda}^{1,1}_{\Psi-}$ (3.29).

For the $\hat{\mathbf{v}} = \pm\mathbf{e}_3$ case, the coefficients are $v_1 = \cos\delta = 0$, $v_3 = \sin\delta = \pm 1$, and the entangling eigenspinor $\mathbf{\Lambda}^{vv}_{\Psi+}$ (3.31) simplifies to

$$\begin{aligned}\mathbf{\Lambda}^{\pm3,\pm3}_{\Psi+} \equiv \mathbf{B}_{\pm3,\pm3}\mathbf{R}_{\Psi+} = \frac{1}{2\sqrt{2}}\left[\mathbf{e}_1\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{e}_3\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_3\right]\\ + \frac{\gamma}{2\sqrt{2}}\left[\mathbf{e}_1\otimes\mathbf{1}-\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{e}_3\otimes\mathbf{1}-\mathbf{1}\otimes\mathbf{e}_3\right]\\ \mp \frac{\gamma v}{2\sqrt{2}c}\left[\mathbf{e}_1\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{1}\otimes\mathbf{1}-\mathbf{e}_3\otimes\mathbf{e}_3\right]\ .\end{aligned}$$

Replacing the $\mathbf{e}_3$ factors by their "biprojector equivalents" (B.16), and the $\mathbf{e}_1$ factors with the "flip-

tanglors" (3.26), this becomes

$$\begin{aligned}\boldsymbol{\Lambda}_{\Psi+}^{\pm 3,\pm 3} &\equiv \mathbf{B}_{\pm 3,\pm 3}\mathbf{R}_{\Psi+} \\ &= \mathbf{F}_{\Psi+}\left[\mathrm{P}_3\otimes\mathrm{P}_3-\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right] \\ &\quad+\gamma\left\{\mathbf{F}_{\Psi-}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3-\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\mp\frac{v}{c}\mathbf{F}_{\Psi+}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3+\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\right\} \\ &= \mathbf{F}_{\Psi+}\left[\mathrm{P}_3\otimes\mathrm{P}_3-\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right] \\ &\quad+\gamma\left\{\left[\mathbf{F}_{\Psi-}\mp\frac{v}{c}\mathbf{F}_{\Psi+}\right]\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3\right]-\left[\mathbf{F}_{\Psi-}\pm\frac{v}{c}\mathbf{F}_{\Psi+}\right]\left[\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\right\}\end{aligned}$$

where the "flip-tanglors" can be transformed (3.27) into physically-realizable entanglors:

$$\begin{aligned}\boldsymbol{\Lambda}_{\Psi+}^{\pm 3,\pm 3} &\equiv \mathbf{B}_{\pm 3,\pm 3}\mathbf{R}_{\Psi+} \\ &= \boxed{\begin{array}{l}\mathbf{R}_{\Psi+}\left[\mathrm{P}_3\otimes\mathrm{P}_3+\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right] \\ \quad+\gamma\left\{\left[\mathbf{R}_{\Psi+}\mp\frac{v}{c}\mathbf{R}_{\Psi-}\right]\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3\right]+\left[\mathbf{R}_{\Psi+}\pm\frac{v}{c}\mathbf{R}_{\Psi-}\right]\left[\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\right\}\end{array}}\end{aligned} \tag{3.33}$$

and thus this special case for the entangling eigenspinor $\boldsymbol{\Lambda}_{\Psi+}^{vv}$ (3.31) is in a form comparable to that of $\boldsymbol{\Lambda}_{\Psi-}^{\pm 3,\pm 3}$ (3.30).

3.3.5 Equal Antiparallel Boosts of Psi-Entanglors

As described in Section 3.2.4, the expression for the bipartite boost has a reasonably simplified form when both subsystems have equal velocity and opposite direction relative to the lab frame that is used as common for both parts. In the following, let the velocity associated with the first boost be in direction $\hat{\mathbf{v}} = v_1\mathbf{e}_1 + v_3\mathbf{e}_3$ (3.23), just as it was in Section 3.3.4 — in fact, all of the calculations here follow the method outlined in that section — and let the velocity associated with the second boost be in direction $-\hat{\mathbf{v}}$.

Using this, the bipartite boost $\mathbf{B}_{v\bar{v}}$ (3.20) can be applied to the Ψ-entanglors $\mathbf{R}_{\Psi\pm}$ (3.2) to obtain

the following expression for the entangling eigenspinors:

$$\begin{aligned}\boldsymbol{\Lambda}_{\Psi\pm}^{v\bar{v}} &\equiv \mathbf{B}_{v\bar{v}}\mathbf{R}_{\Psi\pm} \\ &= \frac{1}{2\sqrt{2}}\left(\left[\mathbf{e}_1\otimes\mathbf{1}+\left(v_1^2\mp v_3^2\right)(\mathbf{1}\otimes\mathbf{e}_1)\right][\mathbf{e}_3\otimes\mathbf{1}]\pm\left[\left(v_1^2\mp v_3^2\right)(\mathbf{e}_1\otimes\mathbf{1})+\mathbf{1}\otimes\mathbf{e}_1\right][\mathbf{1}\otimes\mathbf{e}_3]\right) \\ &\quad+\frac{\gamma}{2\sqrt{2}}\left\{\left(\left[\mathbf{e}_1\otimes\mathbf{1}-\left(v_1^2\mp v_3^2\right)(\mathbf{1}\otimes\mathbf{e}_1)\right][\mathbf{e}_3\otimes\mathbf{1}]\mp\left[\left(v_1^2\mp v_3^2\right)(\mathbf{e}_1\otimes\mathbf{1})-\mathbf{1}\otimes\mathbf{e}_1\right][\mathbf{1}\otimes\mathbf{e}_3]\right)\right\} \\ &\quad+\frac{\gamma v}{2\sqrt{2}c}\left\{v_1\left[\mathbf{1}\otimes\mathbf{1}-\mathbf{e}_1\otimes\mathbf{e}_1\right][\mathbf{e}_3\otimes\mathbf{1}\mp\mathbf{1}\otimes\mathbf{e}_3]-v_3\left[\mathbf{e}_1\otimes\mathbf{1}\mp\mathbf{1}\otimes\mathbf{e}_1\right][\mathbf{1}\otimes\mathbf{1}+\mathbf{e}_3\otimes\mathbf{e}_3]\right\} \\ &\quad-\frac{v_1v_3}{2\sqrt{2}}(\gamma-1)(\mathbf{e}_3\otimes\mathbf{e}_3-\mathbf{e}_1\otimes\mathbf{e}_1)(1\pm1)\end{aligned} \tag{3.34}$$

where, similarly to the $\boldsymbol{\Lambda}_{\Psi\pm}^{vv}$ (3.24) case, the terms containing both $\mathbf{e}_1$ and $\mathbf{e}_3$ have been factored using the multiplication rule for Kronecker products E.3.

Also evident again is that the same-phase Ψ-entanglor $\mathbf{R}_{\Psi+}$ and opposite-phase Ψ-entanglor $\mathbf{R}_{\Psi-}$ have enough differences under an antiparallel boost that they are better off examined separately in order to determine how to simplify their description.

3.3.5.1 Equal antiparallel boost of opposite-phase Psi-entanglor

The same two major simplifications that were made available by selecting the opposite-phase Ψ-entanglor case for $\boldsymbol{\Lambda}_{\Psi\pm}^{vv}$ (3.24) are available in $\boldsymbol{\Lambda}_{\Psi\pm}^{v\bar{v}}$ (3.34) as well. One of the simplifications is that the term in the bottom line becomes zero; the other is that $v_1^2\mp v_3^2$ becomes $v_1^2+v_3^2=\cos^2\delta+\sin^2\delta=1$. The entangling eigenspinor $\boldsymbol{\Lambda}_{\Psi\pm}^{v\bar{v}}$ (3.34) therefore simplifies to

$$\begin{aligned}\boldsymbol{\Lambda}_{\Psi-}^{v\bar{v}} \equiv \mathbf{B}_{v\bar{v}}\mathbf{R}_{\Psi-} &= \frac{1}{2\sqrt{2}}\left[\mathbf{e}_1\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_1\right][\mathbf{e}_3\otimes\mathbf{1}-\mathbf{1}\otimes\mathbf{e}_3] \\ &\quad+\frac{\gamma}{2\sqrt{2}}\left[\mathbf{e}_1\otimes\mathbf{1}-\mathbf{1}\otimes\mathbf{e}_1\right][\mathbf{e}_3\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_3] \\ &\quad+\frac{\gamma v}{2\sqrt{2}c}\left\{\cos\delta\left[\mathbf{1}\otimes\mathbf{1}-\mathbf{e}_1\otimes\mathbf{e}_1\right][\mathbf{e}_3\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_3]\right. \\ &\qquad\left.-\sin\delta\left[\mathbf{e}_1\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_1\right][\mathbf{1}\otimes\mathbf{1}+\mathbf{e}_3\otimes\mathbf{e}_3]\right\}\end{aligned} \tag{3.35}$$

where, just as in the $\boldsymbol{\Lambda}_{\Psi\pm}^{vv}$ (3.24) case, the shorthand has been removed from the coefficients $v_1=\cos\delta$ and $v_3=\sin\delta$, and the $\mathbf{e}_3$ factors can be replaced by "biprojector equivalents" (B.16), as follows:

$$\begin{aligned}\mathbf{\Lambda}_{\Psi-}^{v\bar{v}} \equiv \mathbf{B}_{v\bar{v}}\mathbf{R}_{\Psi-} &= \frac{1}{\sqrt{2}}\left[\mathbf{e}_1\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3-\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\\ &+\frac{\gamma}{\sqrt{2}}\left[\mathbf{e}_1\otimes\mathbf{1}-\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathrm{P}_3\otimes\mathrm{P}_3-\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\\ &+\frac{\gamma v}{\sqrt{2}c}\Bigg\{\cos\delta\left[\mathbf{1}\otimes\mathbf{1}-\mathbf{e}_1\otimes\mathbf{e}_1\right]\left[\mathrm{P}_3\otimes\mathrm{P}_3-\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\\ &\qquad-\sin\delta\left[\mathbf{e}_1\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathrm{P}_3\otimes\mathrm{P}_3+\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\Bigg\}.\end{aligned}$$

As before, for the sake of conciseness, the "flip-tanglors" (3.26) can replace the factors containing $\mathbf{e}_1$, the factors can be collected by biprojector, and the "flip-tanglors" can be transformed (3.27) into physically-realizable entanglors:

$$\begin{aligned}\mathbf{\Lambda}_{\Psi-}^{v\bar{v}} \equiv \mathbf{B}_{v\bar{v}}\mathbf{R}_{\Psi-} &= \mathbf{F}_{\Psi+}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3-\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\\ &+\gamma\mathbf{F}_{\Psi-}\left[\mathrm{P}_3\otimes\mathrm{P}_3-\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\\ &+\frac{\gamma v}{c}\Bigg\{\cos\delta\,\mathbf{F}_{\Phi-}\left[\mathrm{P}_3\otimes\mathrm{P}_3-\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]-\sin\delta\,\mathbf{F}_{\Psi+}\left[\mathrm{P}_3\otimes\mathrm{P}_3+\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\Bigg\}\end{aligned}$$

$$\begin{aligned}&=\mathbf{F}_{\Psi+}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3-\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\\ &+\gamma\left[\mathbf{F}_{\Psi-}+\frac{v}{c}\cos\delta\,\mathbf{F}_{\Phi-}-\frac{v}{c}\sin\delta\,\mathbf{F}_{\Psi+}\right]\left[\mathrm{P}_3\otimes\mathrm{P}_3\right]\\ &-\gamma\left[\mathbf{F}_{\Psi-}+\frac{v}{c}\cos\delta\,\mathbf{F}_{\Phi-}-\frac{v}{c}\sin\delta\,\mathbf{F}_{\Psi+}\right]\left[\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\end{aligned}$$

$$=\boxed{\begin{array}{l}\mathbf{R}_{\Psi-}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3+\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\\ +\gamma\left[\mathbf{R}_{\Psi-}+\frac{v}{c}\cos\delta\,\mathbf{R}_{\Phi-}-\frac{v}{c}\sin\delta\,\mathbf{R}_{\Psi+}\right]\left[\mathrm{P}_3\otimes\mathrm{P}_3\right]\\ +\gamma\left[\mathbf{R}_{\Psi-}-\frac{v}{c}\cos\delta\,\mathbf{R}_{\Phi-}+\frac{v}{c}\sin\delta\,\mathbf{R}_{\Psi+}\right]\left[\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\end{array}}. \tag{3.36}$$

For the sake of completeness, it is worth evaluating the entangling eigenspinor $\mathbf{\Lambda}_{\Psi-}^{v\bar{v}}$ (3.36) for some special cases. In particular, consider velocity directions that are aligned with the basis directions $\mathbf{e}_1$ or $\pm\mathbf{e}_3$, recalling that $\mathbf{e}_1$ is aligned with the component of the boost and thus its coefficient is defined as positive, and whichever sign the $\mathbf{e}_3$ component has for the first subspace, the second subspace has the opposite.

For the $\hat{\mathbf{v}}=\mathbf{e}_1$ case, the coefficients are $v_1=\cos\delta=1$ and $v_3=\sin\delta=0$, and the entangling

eigenspinor $\mathbf{\Lambda}^{v,\bar{v}}_{\Psi-}$ (3.36) becomes

$$\begin{aligned}\mathbf{\Lambda}^{1,-1}_{\Psi-} &\equiv \mathbf{B}_{1,-1}\mathbf{R}_{\Psi-} \\ &= \boxed{\begin{array}{l}\mathbf{R}_{\Psi-}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3+\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right] \\ +\,\gamma\left\{\left[\mathbf{R}_{\Psi-}+\dfrac{v}{c}\mathbf{R}_{\Phi-}\right]\left[\mathrm{P}_3\otimes\mathrm{P}_3\right]+\left[\mathbf{R}_{\Psi-}-\dfrac{v}{c}\mathbf{R}_{\Phi-}\right]\left[\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\right\}\end{array}}\end{aligned} \tag{3.37}$$

and thus this special case is in a form comparable to that of $\mathbf{\Lambda}^{1,1}_{\Psi-}$ (3.29) and $\mathbf{\Lambda}^{1,1}_{\Psi+}$ (3.32).

For the $\hat{\mathbf{v}} = \pm\mathbf{e}_3$ case, the coefficients are $v_1 = \cos\delta = 0$, $v_3 = \sin\delta = \pm 1$, and the entangling eigenspinor $\mathbf{\Lambda}^{v,\bar{v}}_{\Psi-}$ (3.36) becomes

$$\begin{aligned}\mathbf{\Lambda}^{\pm 3,\mp 3}_{\Psi-} &\equiv \mathbf{B}_{\pm 3,\mp 3}\mathbf{R}_{\Psi-} \\ &= \boxed{\begin{array}{l}\mathbf{R}_{\Psi-}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3+\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right] \\ +\,\gamma\left\{\left[\mathbf{R}_{\Psi-}\mp\dfrac{v}{c}\mathbf{R}_{\Psi+}\right]\left[\mathrm{P}_3\otimes\mathrm{P}_3\right]+\left[\mathbf{R}_{\Psi-}\pm\dfrac{v}{c}\mathbf{R}_{\Psi+}\right]\left[\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\right\}\end{array}}\end{aligned} \tag{3.38}$$

and thus this special case is in a form comparable to that of $\mathbf{\Lambda}^{\pm 3,\pm 3}_{\Psi-}$ (3.30) and $\mathbf{\Lambda}^{\pm 3,\pm 3}_{\Psi+}$ (3.33).

3.3.5.2 Equal antiparallel boost of same-phase Psi-entanglor

Next, consider the upper sign in $\mathbf{\Lambda}^{v\bar{v}}_{\Psi\pm}$ (3.34), i.e. the sign that selects the same-phase Ψ-entanglor case from the antiparallel boosted case. As was found in the $\mathbf{\Lambda}^{vv}_{\Psi+}$ (3.31) case, the general result in this case for arbitrary direction $\hat{\mathbf{v}} = v_1\mathbf{e}_1 + v_3\mathbf{e}_3$ does not simplify much:

$$\begin{aligned}\mathbf{\Lambda}^{v\bar{v}}_{\Psi+} &\equiv \mathbf{B}_{v,\bar{v}}\mathbf{R}_{\Psi+} \\ &= \frac{1}{2\sqrt{2}}\left\{\left[\mathbf{e}_1\otimes\mathbf{1}+\left(v_1^2-v_3^2\right)\left(\mathbf{1}\otimes\mathbf{e}_1\right)\right]\left[\mathbf{e}_3\otimes\mathbf{1}\right]+\left[\left(v_1^2-v_3^2\right)\left(\mathbf{e}_1\otimes\mathbf{1}\right)+\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{1}\otimes\mathbf{e}_3\right]\right\} \\ &+ \frac{\gamma}{2\sqrt{2}}\left\{\left[\mathbf{e}_1\otimes\mathbf{1}-\left(v_1^2-v_3^2\right)\left(\mathbf{1}\otimes\mathbf{e}_1\right)\right]\left[\mathbf{e}_3\otimes\mathbf{1}\right]-\left[\left(v_1^2-v_3^2\right)\left(\mathbf{e}_1\otimes\mathbf{1}\right)-\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{1}\otimes\mathbf{e}_3\right]\right\} \\ &+ \frac{\gamma v}{2\sqrt{2}c}\left\{v_1\left[\mathbf{1}\otimes\mathbf{1}-\mathbf{e}_1\otimes\mathbf{e}_1\right]\left[\mathbf{e}_3\otimes\mathbf{1}-\mathbf{1}\otimes\mathbf{e}_3\right]-v_3\left[\mathbf{e}_1\otimes\mathbf{1}-\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{1}\otimes\mathbf{1}+\mathbf{e}_3\otimes\mathbf{e}_3\right]\right\} \\ &- \frac{v_1v_3}{\sqrt{2}}\left(\gamma-1\right)\left(\mathbf{e}_3\otimes\mathbf{e}_3-\mathbf{e}_1\otimes\mathbf{e}_1\right)\end{aligned} \tag{3.39}$$

not even if the shorthand is removed from the coefficients $v_1 = \cos\delta$ and $v_3 = \sin\delta$ to give $v_1^2 - v_3^2 = \cos^2\delta - \sin^2\delta = \cos 2\delta$ and $v_1v_3 = \cos\delta\sin\delta = \frac{1}{2}\sin 2\delta$. Thus, just as in the comparison between the entanglors for the parallel boosts, the same-phase Ψ-entanglor $\mathbf{\Lambda}^{v\bar{v}}_{\Psi+}$ (3.39) is drastically different from the opposite-phase Ψ-entanglor $\mathbf{\Lambda}^{v\bar{v}}_{\Psi-}$ (3.35).

Again, since there are no further obvious simplifications for the same-phase Ψ-entanglor $\mathbf{\Lambda}^{v\bar{v}}_{\Psi+}$ (3.39) using the general velocity direction $\hat{\mathbf{v}}$, the next simplifications to consider are the special cases where the velocity directions align with the basis directions $\mathbf{e}_1$ or $\pm\mathbf{e}_3$. Then, at least, it is possible to compare these results with the similar-case results $\mathbf{\Lambda}^{1,-1}_{\Psi-}$ (3.37) and $\mathbf{\Lambda}^{\pm 3,\mp 3}_{\Psi-}$ (3.38) involving the opposite-phase Ψ-entanglor $\mathbf{R}_{\Psi-}$.

For the $\hat{\mathbf{v}} = \mathbf{e}_1$ case, the coefficients are $v_1 = \cos\delta = 1$ and $v_3 = \sin\delta = 0$, and the entangling eigenspinor $\mathbf{\Lambda}^{v\bar{v}}_{\Psi+}$ (3.39) simplifies to

$$\begin{aligned}\mathbf{\Lambda}^{1,-1}_{\Psi+} \equiv \mathbf{B}_{1,-1}\mathbf{R}_{\Psi+} = {}& \frac{1}{2\sqrt{2}}\left[\mathbf{e}_1\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{e}_3\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_3\right]\\ &+\frac{\gamma}{2\sqrt{2}}\left[\mathbf{e}_1\otimes\mathbf{1}-\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{e}_3\otimes\mathbf{1}-\mathbf{1}\otimes\mathbf{e}_3\right]\\ &+\frac{\gamma v}{2\sqrt{2}c}\left[\mathbf{1}\otimes\mathbf{1}-\mathbf{e}_1\otimes\mathbf{e}_1\right]\left[\mathbf{e}_3\otimes\mathbf{1}-\mathbf{1}\otimes\mathbf{e}_3\right]\end{aligned}$$

Replacing the $\mathbf{e}_3$ factors by their "biprojector equivalents" (B.16), and the $\mathbf{e}_1$ factors with the "flip-tanglors" (3.26), and collecting in terms of like biprojector factors, this becomes

$$\begin{aligned}\mathbf{\Lambda}^{1,-1}_{\Psi+} &\equiv \mathbf{B}_{1,-1}\mathbf{R}_{\Psi+}\\ &= \mathbf{F}_{\Psi+}\left[\mathrm{P}_3\otimes\mathrm{P}_3-\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\\ &\quad+\gamma\left\{\mathbf{F}_{\Psi-}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3-\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]+\frac{v}{c}\mathbf{F}_{\Phi-}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3-\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\right\}\\ &= \mathbf{F}_{\Psi+}\left[\mathrm{P}_3\otimes\mathrm{P}_3-\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\\ &\quad+\gamma\left\{\left[\mathbf{F}_{\Psi-}+\frac{v}{c}\mathbf{F}_{\Phi-}\right]\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3\right]-\left[\mathbf{F}_{\Psi-}+\frac{v}{c}\mathbf{F}_{\Phi-}\right]\left[\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\right\}\end{aligned}$$

where the "flip-tanglors" can be transformed (3.27) into physically-realizable entanglors:

$$\begin{aligned}\mathbf{\Lambda}^{1,-1}_{\Psi+} &\equiv \mathbf{B}_{1,-1}\mathbf{R}_{\Psi+}\\ &= \boxed{\begin{array}{l}\mathbf{R}_{\Psi+}\left[\mathrm{P}_3\otimes\mathrm{P}_3+\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\\ \quad+\gamma\left\{\left[\mathbf{R}_{\Psi+}+\frac{v}{c}\mathbf{R}_{\Phi+}\right]\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3\right]+\left[\mathbf{R}_{\Psi+}-\frac{v}{c}\mathbf{R}_{\Phi+}\right]\left[\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\right\}\end{array}}\end{aligned} \tag{3.40}$$

and thus this special case for the entangling eigenspinor $\mathbf{\Lambda}^{v\bar{v}}_{\Psi+}$ (3.39) is in a form comparable to that of $\mathbf{\Lambda}^{1,1}_{\Psi-}$ (3.29), $\mathbf{\Lambda}^{1,1}_{\Psi+}$ (3.32), and $\mathbf{\Lambda}^{1,-1}_{\Psi-}$ (3.37).

For the $\hat{\mathbf{v}} = \pm\mathbf{e}_3$ case, the coefficients are $v_1 = \cos\delta = 0$, $v_3 = \sin\delta = \pm 1$, and the entangling

eigenspinor $\mathbf{\Lambda}^{v\bar{v}}_{\Psi+}$ (3.39) simplifies to

$$\begin{aligned}\mathbf{\Lambda}^{\pm 3,\mp 3}_{\Psi+} \equiv \mathbf{B}_{\pm 3,\mp 3}\mathbf{R}_{\Psi+} &= \frac{1}{2\sqrt{2}}\left[\mathbf{e}_1\otimes\mathbf{1}-\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{e}_3\otimes\mathbf{1}-\mathbf{1}\otimes\mathbf{e}_3\right]\\ &+\frac{\gamma}{2\sqrt{2}}\left[\mathbf{e}_1\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{e}_3\otimes\mathbf{1}+\mathbf{1}\otimes\mathbf{e}_3\right]\\ &\mp\frac{\gamma v}{2\sqrt{2}c}\left[\mathbf{e}_1\otimes\mathbf{1}-\mathbf{1}\otimes\mathbf{e}_1\right]\left[\mathbf{1}\otimes\mathbf{1}+\mathbf{e}_3\otimes\mathbf{e}_3\right]\end{aligned}$$

Replacing the $\mathbf{e}_3$ factors by their "biprojector equivalents" (B.16), and the $\mathbf{e}_1$ factors with the "flip-tanglors" (3.26), this becomes

$$\begin{aligned}\mathbf{\Lambda}^{\pm 3,\mp 3}_{\Psi+} &\equiv \mathbf{B}_{\pm 3,\mp 3}\mathbf{R}_{\Psi+}\\ &= \mathbf{F}_{\Psi-}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3-\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\\ &\quad+\gamma\left\{\mathbf{F}_{\Psi+}\left[\mathrm{P}_3\otimes\mathrm{P}_3-\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\mp\frac{v}{c}\mathbf{F}_{\Psi-}\left[\mathrm{P}_3\otimes\mathrm{P}_3+\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\right\}\\ &\quad\mathbf{F}_{\Psi-}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3-\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\\ &\quad+\gamma\left\{\left[\mathbf{F}_{\Psi+}\mp\frac{v}{c}\mathbf{F}_{\Psi-}\right]\left[\mathrm{P}_3\otimes\mathrm{P}_3\right]-\left[\mathbf{F}_{\Psi+}\pm\frac{v}{c}\mathbf{F}_{\Psi-}\right]\left[\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\right\}\end{aligned}$$

where the "flip-tanglors" can be transformed (3.27) into physically-realizable entanglors:

$$\begin{aligned}\mathbf{\Lambda}^{\pm 3,\mp 3}_{\Psi+} &\equiv \mathbf{B}_{\pm 3,\mp 3}\mathbf{R}_{\Psi+}\\ &= \boxed{\begin{array}{l}\mathbf{R}_{\Psi+}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3+\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\\ \quad+\gamma\left\{\left[\mathbf{R}_{\Psi+}\mp\frac{v}{c}\mathbf{R}_{\Psi-}\right]\left[\mathrm{P}_3\otimes\mathrm{P}_3\right]+\left[\mathbf{R}_{\Psi+}\pm\frac{v}{c}\mathbf{R}_{\Psi-}\right]\left[\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\right\}\end{array}}\end{aligned} \tag{3.41}$$

and thus this special case for the entangling eigenspinor $\mathbf{\Lambda}^{v\bar{v}}_{\Psi+}$ (3.39) is in a form comparable to that of $\mathbf{\Lambda}^{\pm 3,\pm 3}_{\Psi-}$ (3.30), $\mathbf{\Lambda}^{\pm 3,\pm 3}_{\Psi+}$ (3.33), and $\mathbf{\Lambda}^{\pm 3,\mp 3}_{\Psi-}$ (3.38).

3.3.6 Single-Subspace Boost of Psi-Entanglors

As described in Section 3.2.2, it is possible to choose a lab frame that coincides with the rest frame of one subsystem. This is equivalent to applying a boost only to the subsystem not coinciding with the lab frame.

As noted previously (3.13), a single-subspace bipartite boost has the form $1\otimes B$ or $B\otimes 1$. Using the simple boost $B=\sqrt{\frac{\gamma+1}{2}}+\hat{\mathbf{v}}\sqrt{\frac{\gamma-1}{2}}$ (3.7) in specific arbitrary direction $\hat{\mathbf{v}}=\cos\delta\,\mathbf{e}_1+\sin\delta\,\mathbf{e}_3$ (3.17),

the single-subspace bipartite boosts affect the Ψ-entanglors (3.2) as follows:

$$(1 \otimes B)\,\mathbf{R}_{\Psi\pm} = \sqrt{\frac{\gamma+1}{2}}\mathbf{R}_{\Psi\pm} + \frac{1}{2}\sqrt{\gamma-1}\Big\{\cos\delta\,[\mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{e}_1 \pm \mathbf{1} \otimes \mathbf{e}_3] + \sin\delta\,[\mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{e}_3 \mp \mathbf{1} \otimes \mathbf{e}_1]\Big\}$$

and

$$(B \otimes 1)\,\mathbf{R}_{\Psi\pm} = \sqrt{\frac{\gamma+1}{2}}\mathbf{R}_{\Psi\pm} + \frac{1}{2}\sqrt{\gamma-1}\Big\{\cos\delta\,[\mathbf{e}_3 \otimes \mathbf{1} \pm \mathbf{e}_1 \otimes \mathbf{e}_1\mathbf{e}_3] - \sin\delta\,[\mathbf{e}_1 \otimes \mathbf{1} \mp \mathbf{e}_3 \otimes \mathbf{e}_1\mathbf{e}_3]\Big\}.$$

Notice that there are no obvious simplifications in either case above. First, there is no way to group the $\mathbf{e}_3$ factors into forms having "biprojector equivalents", or $\mathbf{e}_1$ factors into "flip-tanglors", which were both helpful steps in re-expressing all the previous cases. Furthermore, there is no way within these expressions to separate the parts unmodified by the boost from those parts that do involve relativity, because the Lorentz factor γ only occurs here under square roots.

If these boosts were to be applied in a bilateral way, such as when transforming a vector or bivector, more simplifications would be possible. However, since this dissertation is interpreting the entanglement as residing within operators, which are transformed unilaterally, these single-subspace bipartite boosts are not helpful in the investigation at hand.

3.4 Bipartite Dirac Equation

One important question to ask at this point is whether the superpositions of separable eigenspinors derived in Section 3.3 satisfy a bipartite form of the classical[8] Dirac equation (2.19).

To begin to answer this, first recall that the linearity of the Dirac equation means that not only simple eigenspinors Λ but also *real linear superpositions* of eigenspinors are possible solutions. Thus, if it can be shown here that a bipartite form of the classical Dirac equation is a real linear equation and is satisfied by a *separable* eigenspinor $\mathbf{\Lambda}_{AB} = \Lambda_A \otimes \Lambda_B$, then real linear superpositions of such solutions, including solutions involving entanglement, would also satisfy it.

Now, recall that the classical Dirac equation (2.19) was derived from the relationship between the momentum p and proper velocity u of an elementary system described by a single eigenspinor. For

[8]The classical Dirac equation is the simplest form that is expected to sufficiently describe the dynamics of the systems in question. Extending this description to quantum form is one of the extensions to this approach that is discussed in Section 5.2.

the bipartite case, consider two elementary systems of masses m_A and m_B, momenta p_A and p_B, and proper velocities $u_A = \Lambda_A \Lambda_A^\dagger$ and $u_B = \Lambda_B \Lambda_B^\dagger$, joined into a separable direct product:

$$\begin{aligned} p_A \otimes p_B &= m_A u_A \otimes m_B u_B \\ &= m_A \Lambda_A \Lambda_A^\dagger \otimes m_B \Lambda_B \Lambda_B^\dagger \end{aligned}$$

where $c = 1$. Now, using the unimodularity of both eigenspinors, $\Lambda_X \bar{\Lambda}_X = 1 = \Lambda_X^\dagger \bar{\Lambda}_X^\dagger$, and using the multiplication rule for the Kronecker product (E.2), the expression above can be rearranged to

$$\begin{aligned} (p_A \otimes p_B)\left(\bar{\Lambda}_A^\dagger \otimes \bar{\Lambda}_B^\dagger\right) &= \left(m_A \Lambda_A \Lambda_A^\dagger \otimes m_B \Lambda_B \Lambda_B^\dagger\right)\left(\bar{\Lambda}_A^\dagger \otimes \bar{\Lambda}_B^\dagger\right) \\ (p_A \otimes p_B)\left(\bar{\Lambda}_A^\dagger \otimes \bar{\Lambda}_B^\dagger\right) &= (m_A \mathbf{1} \otimes m_B \mathbf{1})(\Lambda_A \otimes \Lambda_B) \end{aligned} \tag{3.42}$$

which may be deemed the **bipartite classical Dirac equation**. It may then be reasoned that the typical quantum replacement of the momentum by its operator equivalent $p\bar{\Psi}^\dagger \to i\hbar\partial\bar{\Psi}^\dagger \mathbf{e}_3$ (2.21) would lead to an expression of (3.42) in quantum form as

$$-\hbar^2 (\partial \otimes \partial)\left(\bar{\Lambda}_A^\dagger \otimes \bar{\Lambda}_B^\dagger\right)(\mathbf{e}_3 \otimes \mathbf{e}_3) = (m_A \mathbf{1} \otimes m_B \mathbf{1})(\Lambda_A \otimes \Lambda_B) \tag{3.43}$$

which is proposed here to be a **bipartite quantum Dirac equation**.

Note that the equation above is, by design, satisfied by a tensor product of eigenspinors $\Lambda_A \otimes \Lambda_B$. The real linear nature of the bipartite Dirac equation follows from the real linear nature of its single-space predecessor, since nothing about this derivation introduced any changes to its reality or linearity. Thus, the solutions for this equation can indeed take the form of real linear superpositions of tensor products of eigenspinors, including — most importantly to the interpretation presented in is dissertation — the entangling and boosting transformations derived in Section 3.3.

3.5 Up to Here, and What is Next

This chapter of the dissertation began by setting up all the details that were required to make the calculations according to the interpretation that has guided this work. To that end, Section 3.1 showed how entanglement can be represented by transformations instead of the states, and Section 3.2 showed the form of the applicable bipartite relativistic velocity transformations. Section 3.3 applied the relativistic transformations to the entangling transformations, and Section 3.4 verified that the results of the calculations were suitable solutions to a bipartite Dirac equation.

The next step is to analyze the results from these calculations and discuss their interpretation, which will be done in Chapter 4.

Chapter 4

Analysis of the Interpretation

This chapter analyzes the interpretation set forth in this dissertation by way of examining the calculation results from the previous chapter, and by performing a few additional calculations to examine the results more clearly.

Section 4.1 begins by summarizing the approach that was taken in Chapter 3, highlighting the facets of the interpretation that were applied throughout. Then, Section 4.2 summarizes and analyzes the results of the calculations from Section 3.3. Next, Section 4.3 attempts to reconcile the idea of entanglement as a transformation by applying the operators to spin states, and then attempting to quantify the entanglement in the transformed states by calculating the concurrence. Finally, Section 4.4 discusses the problems inherent in the approach attempted in Section 4.3, highlighting how these problems support the interpretation of entanglement as a relationship between reference frames, rather than as a property of quantum states.

4.1 Connecting the Interpretation and the Calculations

Before discussing the results, it is worth briefly reviewing how the interpretation led to the calculations, and then how the calculations implemented the interpretation. Certain key features of the approach are emphasized, as they were the realizations that solidified the interpretation of entanglement as a relationship between reference frames.

4.1.1 How the Interpretation Led to the Calculations

With the primary focus of this dissertation being an interpretation of spin entanglement, it was necessary to come up with an investigation to test the idea of modelling the information as being

contained within entangling operators (3.2), rather than entangled spin states (2.46). The ideas leading to the calculations are outlined below.

4.1.1.1 All information is present in the eigenspinor

The first key idea behind the interpretation of entanglement as contained within the operators rather than the states is based on the definition of eigenspinors (2.7) as transformation operators that describe the relative orientation and relative velocity of a rest frame relative to a lab reference frame. This builds on the idea that eigenspinors satisfy the classical Dirac equation (2.20) even without explicitly applying spin projectors.

Spin information *may* be extracted by an application of such projectors, but all the information about the possible spin states is present in the full eigenspinor, so there is *no need* to apply the projectors until a particular analysis is desired. The present interpretation extends this idea to the bipartite case: that all information about the possibilities for a pair of spin states is present in a bipartite eigenspinor. Hence, the present interpretation encourages holding off on applying the spin projectors until a particular analysis is desired.

4.1.1.2 Superpositions lead to probabilistic wavefunctions

Another key to the present approach is the fact that real linear superpositions of solutions to the Dirac equation are also solutions. The eigenspinors that are involved in such a linear superposition each represent a relationship between the rest frame of the system in question and reference frames of different velocities and/or orientations.

The linear superposition can be interpreted as indicating the *possibilities* that may describe a situation. Each coefficient can be treated as a *transformation amplitude*, which squares to obtain the probability that a particular relationship is the one that will bring about an eventual measurement result. Thus, a solution that is a linear superposition of transformation operators can be associated with a *probabilistic wavefunction* when it is eventually applied to a particular spin state projector.

The bipartite case has an additional complication: a subsystem can be rotated relative to the other, and a superposition of such rotations can be described as *relative rotation amplitudes* between the subsystems. This is how entanglement is interpreted in the present work. Again, the amplitudes may be squared to determine the probabilities of each relative rotation taking place. The result is that it is uncertain how the reference frames are described relatively to some observer, not until a measurement takes place, but it is certain how they are described relatively to each other.

4.1.1.3 The observer must be taken into account at each step

Care is necessary in considering the observer when applying boost operators, because velocities must be given relative to a particular reference frame. The entanglement information given in the previous step does not complete the description of the system, unless it is meant to describe a bipartite system at rest from the perspective of an observer.

The systems of interest in the present work involve nonzero relative velocities. Such relative velocities may be between the parts of the subsystem relative to an observer "at rest" between them, or between the system and an observer, so that the parts of the subsystem are at rest relative to each other (or some combination of these cases, beyond the scope of the present work). Note that the relative velocity between the reference frames of the subsystems is part of the description of a *situation*, rather than a change of *perspective*, where the latter is something that can be applied after the fact and even changed in order to get information about a different point of view.

Altogether, the abovementioned considerations led to the calculations of entangling eigenspinors which are used in the present work to explore the interpretation of entanglement in terms of information relating reference frames, rather than as state information. The details of how these calculations were implemented are summarized in Section 4.1.2, and then the results are examined in Section 4.2.

4.1.2 How the Calculations Implemented the Interpretation

Before discussing the calculation results in Section 4.2, it is worth reviewing the steps that were taken in the calculations. That way, the results can be appreciated with regard to how they were obtained.

4.1.2.1 Preparations for the main calculation

In order to examine the entanglement as relative information — or as a *transformation* — rather than a property of a state, entangled states were interpreted in Section 3.1 as being the result of a superposition of relative rotations applied to a given state. The transformation operator was extracted from the description of the entangled state, and deemed an **entanglor**. As a superposition of relative rotations, it was treated as acting in the rest-frame of each part of the bipartite system in question. With the states extracted from the treatment, the entanglor relates only the reference frames where the states reside.

Then, **separable boosts** were investigated in Section 3.2. Their application was interpreted as introducing a reference frame from which both the subspaces are described, taking into account

their relative motion. The most useful reference frames were ones from which both subspaces are observed as having equal velocity relative to the observer.

At this point, the dissertation was ready for the main calculation, involving applying the boosts to the entanglors. This was done in Section 3.3, and the details of how it was done are outlined below.

4.1.2.2 The starting point for the main calculations

The mathematical approach in this dissertation has been guided by the desire to highlight geometrical understanding of entanglement by examining it relativistically. In order to do this, it is useful to investigate similarities and differences between calculations made for similar situations. A particular subset of the possible calculations was selected in order to keep the investigation simple enough to provide clarity, while leaving as much generalized as possible.

Specifically, the restrictions applied in this work are as follows:

- The boosts are separable, of equal magnitude (3.16), and are either parallel or antiparallel to each other, aligned with an arbitrary direction $\hat{\mathbf{v}} = \cos\delta\,\mathbf{e}_1 + \sin\delta\,\mathbf{e}_3$ (3.17).
- The only entanglors addressed are the ones of opposite relative parity, i.e. the Ψ-entanglors, denoted $\mathbf{R}_{\Psi\pm}$ (3.2), which can transform parallel separable spin states $\Upsilon_{\uparrow\uparrow}$ or $\Upsilon_{\downarrow\downarrow}$ (F.1) to antiparallel entangled states $\Psi_\pm$ (3.1) or antiparallel separable spin states to parallel entangled states (3.6). (The entanglors of the same relative parity, $\mathbf{R}_{\Phi\pm}$, can be related to these ones $\mathbf{R}_{\Psi\pm}$ by a single-subspace direction-inverting rotation (3.3).)

These restrictions formed the starting point for the main calculations of entangling eigenspinors.

4.1.2.3 Separately investigating the Psi-entanglors

However, even with the abovementioned restrictions, the calculation results were too general and could not be easily compared to one another. It was discovered that the Ψ-entanglors differ enough from each other that they need to be separated from each other in order to investigate their underlying geometry further.

One of the main differences between the two entanglors was in how much they could be simplified: the general results from boosting the $\mathbf{R}_{\Psi+}$ entanglor to get the eigenspinors $\mathbf{\Lambda}^{v,\pm v}_{\Psi+}$ (3.31)(3.39) could not be simplified anywhere near as much as was possible for the results of boosting the $\mathbf{R}_{\Psi-}$ entanglor to get the eigenspinors $\mathbf{\Lambda}^{v,\pm v}_{\Psi-}$ (3.28)(3.36). This is related to the rotational invariance of the Ψ_- Bell state from which the $\mathbf{R}_{\Psi-}$ entanglor was derived. Since rotational invariance does not apply to the Ψ_+ Bell state from which the $\mathbf{R}_{\Psi+}$ entanglor was derived, the arbitrary boost direction in the latter case left too many unknowns to enable simplification.

4.1.2.4 Investigating specific boost directions

In order to reduce the unknowns that were hindering a geometric analysis of the effect of the boosts applied to the two Ψ-entanglors, specific boost directions were selected for further calculations. In particular, the directions investigated were exclusively perpendicular, parallel, or antiparallel to the $\mathbf{e}_3$ direction which itself had been oriented to coincide with the initial spin direction via the rotational gauge invariance (2.16) of the rest frame.

With these additional restrictions to the boost directions, the results of the specific various boosts applied to the $\mathbf{R}_{\Psi+}$ entanglor could be simplified as much as was possible when the more general boosts were applied to the $\mathbf{R}_{\Psi-}$ entanglor. The specific boosts could then be applied to both entanglors in order to obtain a set of calculation results that could be compared to one another.

4.1.2.5 The insights towards obtaining the final form

The most-simplified final form for each of the entangling eigenspinors was obtained by factoring the $\mathbf{e}_1$ and $\mathbf{e}_3$ parts, and noting that the $\mathbf{e}_3$ parts could be identified as sums and differences of $\pm\mathbf{e}_3$ **biprojectors** (B.16). These biprojectors, which are products of P_3 and/or $\overline{\mathrm{P}}_3$ projectors, allowed for a reinterpretation of the $\mathbf{e}_1$ factors multiplying them, which ended up with each term expressed as an *entanglor* multiplying a biprojector.

Thus the final results — which are summarized in Section 4.2 — are each in a form consisting entirely of entanglors multiplying biprojectors, with some factors including the Lorentz factor γ. The significance and effect of each aspect of this form will be investigated in Section 4.2.

4.2 Analysis of Results

The results analyzed in this section are the entangling eigenspinors from Sections 3.3.4 and 3.3.5 obtained from boosting both the Ψ-entanglors in specific boost directions. These results are first summarized, and then various features of the results are analyzed in detail.

4.2.1 Summary of Results

Before proceeding, recall how the definition of the "relatively arbitrary" boost direction $\hat{\mathbf{v}} = \cos\delta\,\mathbf{e}_1 + \sin\delta\,\mathbf{e}_3$ (3.17) took advantage of the rotational gauge freedom of the rest frame to orient the coordinate system according to specific physical properties under consideration. In particular, the $\mathbf{e}_3$ direction had already been selected to align with some initial rest-frame spin direction — even

though spin states themselves are not yet being addressed directly — and the $\mathbf{e}_1$ direction was then selected to align with the positive perpendicular-to-$\mathbf{e}_3$ component of the first-subsystem boost[1].

The abovementioned physical associations of the two basis directions $\mathbf{e}_1$ and $\mathbf{e}_3$ are significant in the analysis of the calculation results. Thus, the results from Sections 3.3.4 and 3.3.5 are grouped together in the summary below according to their boost direction(s). Of all the possible groupings of results, this grouping best emphasizes the similarities as well as highlighting the geometric differences between the entangling eigenspinors.

To that end, each group of four results below involves a first-subspace boost in the same direction, either $\mathbf{e}_1$ or $\pm\mathbf{e}_3$. The first of each pair within the group has its second-subspace boost in the same direction as the first, and the second of each pair has the opposite. Both options within each pair are applied to the same Ψ-entanglor. The other pair within the group describes the other Ψ-entanglor.

Recall that the bipartite eigenspinor symbols are labelled in a way that indicates the boosts applied and the entanglor to which they are applied: $\mathbf{\Lambda}_c^{a,b}$ has superscripts indicating that its boosts are in the $\mathbf{e}_a$ and $\mathbf{e}_b$ directions (where $\mathbf{e}_{-a} \equiv -\mathbf{e}_a$), and that the boosts are applied to entanglor $\mathbf{R}_c$. Explicit symbols for the boost $\mathbf{B}_{a,b}$ and entanglor $\mathbf{R}_c$ are also shown for additional clarity.

4.2.1.1 Boosts in e1 direction(s)

The $\mathbf{e}_1$ boosts applied to the $\mathbf{R}_{\Psi-}$ eigenspinor give $\mathbf{\Lambda}_{\Psi-}^{1,1}$ (3.29) and $\mathbf{\Lambda}_{\Psi-}^{1,-1}$ (3.37):

$$\begin{aligned}
\mathbf{\Lambda}_{\Psi-}^{1,1} \equiv \mathbf{B}_{1,1}\mathbf{R}_{\Psi-} = {} & \boxed{\begin{array}{l} \mathbf{R}_{\Psi-}\left[\mathrm{P}_3 \otimes \mathrm{P}_3 + \overline{\mathrm{P}}_3 \otimes \overline{\mathrm{P}}_3\right] \\ + \gamma\left\{ \left[\mathbf{R}_{\Psi-} + \frac{v}{c}\mathbf{R}_{\Phi-}\right]\left[\mathrm{P}_3 \otimes \overline{\mathrm{P}}_3\right] + \left[\mathbf{R}_{\Psi-} - \frac{v}{c}\mathbf{R}_{\Phi-}\right]\left[\overline{\mathrm{P}}_3 \otimes \mathrm{P}_3\right] \right\} \end{array}} \\
\mathbf{\Lambda}_{\Psi-}^{1,-1} \equiv \mathbf{B}_{1,-1}\mathbf{R}_{\Psi-} = {} & \boxed{\begin{array}{l} \mathbf{R}_{\Psi-}\left[\mathrm{P}_3 \otimes \overline{\mathrm{P}}_3 + \overline{\mathrm{P}}_3 \otimes \mathrm{P}_3\right] \\ + \gamma\left\{ \left[\mathbf{R}_{\Psi-} + \frac{v}{c}\mathbf{R}_{\Phi-}\right]\left[\mathrm{P}_3 \otimes \mathrm{P}_3\right] + \left[\mathbf{R}_{\Psi-} - \frac{v}{c}\mathbf{R}_{\Phi-}\right]\left[\overline{\mathrm{P}}_3 \otimes \overline{\mathrm{P}}_3\right] \right\} \end{array}}
\end{aligned} \tag{4.1}$$

[1] If the boosts are aligned with the initial spin direction $\mathbf{e}_3$, then the $\mathbf{e}_1$ direction is not oriented in any specific way because directionality can only be defined in a relative sense. The arbitrary direction for $\mathbf{e}_1$ does not pose any greater problem than existed for the stationary case; the direction affects the global phase of the state, and the unknowability would be resolved by the asymmetric nature of any situation where the global phase would matter.

The $\mathbf{e}_1$ boosts applied to the $\mathbf{R}_{\Psi+}$ eigenspinor give $\mathbf{\Lambda}^{1,1}_{\Psi+}$ (3.32) and $\mathbf{\Lambda}^{1,-1}_{\Psi+}$ (3.40):

$$\mathbf{\Lambda}^{1,1}_{\Psi+} \equiv \mathbf{B}_{1,1}\mathbf{R}_{\Psi+} = \boxed{\begin{array}{l}\mathbf{R}_{\Psi+}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3+\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\\ +\gamma\left\{\left[\mathbf{R}_{\Psi+}+\frac{v}{c}\mathbf{R}_{\Phi+}\right]\left[\mathrm{P}_3\otimes\mathrm{P}_3\right]+\left[\mathbf{R}_{\Psi+}-\frac{v}{c}\mathbf{R}_{\Phi+}\right]\left[\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\right\}\end{array}}$$

$$\mathbf{\Lambda}^{1,-1}_{\Psi+} \equiv \mathbf{B}_{1,-1}\mathbf{R}_{\Psi+} = \boxed{\begin{array}{l}\mathbf{R}_{\Psi+}\left[\mathrm{P}_3\otimes\mathrm{P}_3+\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\\ +\gamma\left\{\left[\mathbf{R}_{\Psi+}+\frac{v}{c}\mathbf{R}_{\Phi+}\right]\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3\right]+\left[\mathbf{R}_{\Psi+}-\frac{v}{c}\mathbf{R}_{\Phi+}\right]\left[\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\right\}\end{array}} \tag{4.2}$$

4.2.1.2 Boosts in e3 directions

The $\pm\mathbf{e}_3$ boosts applied to the $\mathbf{R}_{\Psi-}$ eigenspinor give $\mathbf{\Lambda}^{\pm3,\pm3}_{\Psi-}$ (3.30) and $\mathbf{\Lambda}^{\pm3,\mp3}_{\Psi-}$ (3.38):

$$\mathbf{\Lambda}^{\pm3,\pm3}_{\Psi-} \equiv \mathbf{B}_{\pm3,\pm3}\mathbf{R}_{\Psi-} = \boxed{\begin{array}{l}\mathbf{R}_{\Psi-}\left[\mathrm{P}_3\otimes\mathrm{P}_3+\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\\ +\gamma\left\{\left[\mathbf{R}_{\Psi-}\mp\frac{v}{c}\mathbf{R}_{\Psi+}\right]\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3\right]+\left[\mathbf{R}_{\Psi-}\pm\frac{v}{c}\mathbf{R}_{\Psi+}\right]\left[\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\right\}\end{array}}$$

$$\mathbf{\Lambda}^{\pm3,\mp3}_{\Psi-} \equiv \mathbf{B}_{\pm3,\mp3}\mathbf{R}_{\Psi-} = \boxed{\begin{array}{l}\mathbf{R}_{\Psi-}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3+\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\\ +\gamma\left\{\left[\mathbf{R}_{\Psi-}\mp\frac{v}{c}\mathbf{R}_{\Psi+}\right]\left[\mathrm{P}_3\otimes\mathrm{P}_3\right]+\left[\mathbf{R}_{\Psi-}\pm\frac{v}{c}\mathbf{R}_{\Psi+}\right]\left[\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\right\}\end{array}} \tag{4.3}$$

The $\pm\mathbf{e}_3$ boosts applied to the $\mathbf{R}_{\Psi+}$ eigenspinor give $\mathbf{\Lambda}^{\pm3,\pm3}_{\Psi+}$ (3.33) and $\mathbf{\Lambda}^{\pm3,\mp3}_{\Psi+}$ (3.41):

$$\mathbf{\Lambda}^{\pm3,\pm3}_{\Psi+} \equiv \mathbf{B}_{\pm3,\pm3}\mathbf{R}_{\Psi+} = \boxed{\begin{array}{l}\mathbf{R}_{\Psi+}\left[\mathrm{P}_3\otimes\mathrm{P}_3+\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\\ +\gamma\left\{\left[\mathbf{R}_{\Psi+}\mp\frac{v}{c}\mathbf{R}_{\Psi-}\right]\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3\right]+\left[\mathbf{R}_{\Psi+}\pm\frac{v}{c}\mathbf{R}_{\Psi-}\right]\left[\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\right\}\end{array}}$$

$$\mathbf{\Lambda}^{\pm3,\mp3}_{\Psi+} \equiv \mathbf{B}_{\pm3,\mp3}\mathbf{R}_{\Psi+} = \boxed{\begin{array}{l}\mathbf{R}_{\Psi+}\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3+\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\\ +\gamma\left\{\left[\mathbf{R}_{\Psi+}\mp\frac{v}{c}\mathbf{R}_{\Psi-}\right]\left[\mathrm{P}_3\otimes\mathrm{P}_3\right]+\left[\mathbf{R}_{\Psi+}\pm\frac{v}{c}\mathbf{R}_{\Psi-}\right]\left[\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\right\}\end{array}} \tag{4.4}$$

Note that for these $\mathbf{e}_3$ cases, the first-subspace boost could be in the same direction as the spin or the opposite. Hence, both possible arrangements of directions are included in these results using the $\pm\mathbf{e}_3$ direction for the first-subspace boost and either $\pm\mathbf{e}_3$ or $\mp\mathbf{e}_3$ for the second. Contrast this with the $\mathbf{e}_1$ cases (4.1)4.2, where $+\mathbf{e}_1$ was defined by the first-subspace boost direction.

4.2.1.3 General form of the results

All the entangling eigenspinors (4.1)–(4.4) summarized above have the same general form:

$$\boldsymbol{\Lambda}^{a,b}_{\Psi\pm} = \mathbf{B}_{a,b}\mathbf{R}_{\Psi\pm} = \boxed{\begin{array}{l} \mathbf{R}_{\Psi\pm}\Big[\text{the sum of two biprojectors, both either parallel or antiparallel}\Big] \\ \quad + \gamma\left\{\left[\mathbf{R}_{\Psi\pm}\begin{pmatrix}+\\ \text{or}\\ -\end{pmatrix}\frac{v}{c}\mathbf{R}_{\text{other}}\right]\Big[\text{the remaining biprojector of form } \mathrm{P}_3\otimes Q\Big]\right. \\ \quad \left. + \left[\mathbf{R}_{\Psi\pm}\begin{pmatrix}-\\ \text{or}\\ +\end{pmatrix}\frac{v}{c}\mathbf{R}_{\text{other}}\right]\Big[\text{the remaining biprojector of form } \overline{\mathrm{P}}_3\otimes \overline{Q}\Big]\right\} \end{array}} \tag{4.5}$$

where the original entanglor that is boosted in each case is represented in this general result by $\mathbf{R}_{\Psi\pm}$, while $\mathbf{R}_{\text{other}}$ represents some other entanglor (to be discussed below). The sign uncertainty denoted here by $\begin{pmatrix}+\\ \text{or}\\ -\end{pmatrix}$ is distinct from the $\pm$ sign indicating the relative phase of the boosted entanglor. The "remaining" biprojectors are those not yet appearing in the expression, so that each of the results above include all four biprojectors.

4.2.2 Biprojectors in the Results

One feature that stands out about the general form of the results is the presence of **biprojectors** (B.15). As described in Section 4.1.2, sums and differences of biprojectors were discovered to be useful replacements for the factors containing $\mathbf{e}_3$ (B.16). Recognition of this replacement further enabled valuable interpretations of the adjacent factors of $\mathbf{e}_1$ as entanglors (3.27), thus allowing great simplification of the form of the results.

There is much more value to be revealed about the role of these biprojectors within the results. This section introduces some of the valuable features of biprojectors in detail, while others are discussed further in Section 4.3, when these transformation operators are applied to spin states.

4.2.2.1 Complete set of biprojectors

First, note that all four of the complementary biprojectors (B.15) are present in each expression. This is represented in the expression for the general result (4.5) by indicating that the third and fourth biprojectors are ones that have not yet appeared in the expression. If the first and second biprojectors are parallel, the third and fourth are antiparallel, and thus it must be that $Q = \overline{\mathrm{P}}_3$. If the first and second biprojectors are antiparallel, the third and fourth are parallel, and thus it must be that $Q = \mathrm{P}_3$.

The presence of all four complementary biprojectors (B.15) makes it easy to see that the unboosted case, with $v = 0$ and $\gamma = 1$, gives for each case

$$\mathbf{\Lambda}^{0,0}_{\Psi\pm} \;=\; \mathbf{B}_0\mathbf{R}_{\Psi\pm} \;=\; \mathbf{R}_{\Psi\pm}\left[\mathrm{P}_3\otimes\mathrm{P}_3 + \overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3 + \mathrm{P}_3\otimes\overline{\mathrm{P}}_3 + \overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right] \;=\; \mathbf{R}_{\Psi\pm},$$

thus leaving the original entanglor $\mathbf{R}_{\Psi\pm}$ unchanged, as it should. Although this case may be trivial, it begins to highlight the importance of all four complementary biprojectors being present in the results.

4.2.2.2 Utility of the biprojectors

Now recall that one of the many roles played by a set of complementary projectors $\mathrm{P}_3, \overline{\mathrm{P}}_3$ is by making use of how their sum, $\mathrm{P}_3 + \overline{\mathrm{P}}_3 = 1$, acting upon a general spin state, can be used to express the state as a linear superposition (2.38) of the standard basis states $\psi_\uparrow = \mathrm{P}_3$ and $\psi_\downarrow = \mathbf{e}_1\mathrm{P}_3$. The case here is similar, because the sum of all four complementary biprojectors gives a bipartite completeness relation (B.15) which can act upon a general bipartite state to express it as a linear superposition of the separable bipartite basis states (2.43) $\Upsilon_{ij} \equiv \psi_i \otimes \psi_j$ where $i, j \in \{\uparrow, \downarrow\}$.

The entangling transformation differs from the completeness relation since it is not an identity (except in the limiting case), but the presence of all four biprojectors is suggestive in how it may be applied to a general state. Specifically, each biprojector (B.15) only gives a nonzero result for one of the separable basis states Υ_{ij}:

$$\begin{aligned}
\left[\mathrm{P}_3\otimes\mathrm{P}_3\right]\Upsilon_{\uparrow\uparrow} &= \left[\mathrm{P}_3\otimes\mathrm{P}_3\right](\mathrm{P}_3\otimes\mathrm{P}_3) &&= \Upsilon_{\uparrow\uparrow}\\
\left[\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right]\Upsilon_{\downarrow\downarrow} &= \left[\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right](\mathbf{e}_1\mathrm{P}_3\otimes\mathbf{e}_1\mathrm{P}_3) &&= \Upsilon_{\downarrow\downarrow}\\
\left[\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\Upsilon_{\downarrow\uparrow} &= \left[\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right](\mathbf{e}_1\mathrm{P}_3\otimes\mathrm{P}_3) &&= \Upsilon_{\downarrow\uparrow}\\
\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3\right]\Upsilon_{\uparrow\downarrow} &= \left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3\right](\mathrm{P}_3\otimes\mathbf{e}_1\mathrm{P}_3) &&= \Upsilon_{\uparrow\downarrow}
\end{aligned} \tag{4.6}$$

where $\mathbf{e}_1\mathrm{P}_3 = \overline{\mathrm{P}}_3\mathbf{e}_1$ (B.8) and $\mathrm{P}_3^2 = \mathrm{P}_3$ (B.5) were used to obtain these results, and $\mathrm{P}_3\overline{\mathrm{P}}_3 = 0$ (B.3) gives the twelve zero-valued cases not shown explicitly here.

Using the above, suppose there exists a general transformation operator $\mathbf{T}$, which contains sub-operators A, B, C, D multiplying each of the four biprojectors. This transformation operator may be applied to a general state Υ expressed in terms of the separable bipartite basis states Υ_{ij} in the following simple way:

$$\begin{aligned}
\mathbf{T}\Upsilon &= \left(A\left[\mathrm{P}_3\otimes\mathrm{P}_3\right] + B\left[\overline{\mathrm{P}}_3\otimes\overline{\mathrm{P}}_3\right] + C\left[\mathrm{P}_3\otimes\overline{\mathrm{P}}_3\right] + D\left[\overline{\mathrm{P}}_3\otimes\mathrm{P}_3\right]\right)\left(a\Upsilon_{\uparrow\uparrow} + b\Upsilon_{\downarrow\downarrow} + c\Upsilon_{\downarrow\uparrow} + d\Upsilon_{\uparrow\downarrow}\right)\\
&= aA\Upsilon_{\uparrow\uparrow} + bB\Upsilon_{\downarrow\downarrow} + cC\Upsilon_{\downarrow\uparrow} + dD\Upsilon_{\uparrow\downarrow}\,.
\end{aligned} \tag{4.7}$$

Therefore, the importance of having all four biprojectors in a transformation is solidified: it ensures that a transformation can be applied to a spin state and obtain another spin state as a result.

4.2.3 Relativity in the Results

Next, note that the general result (4.5) shows that there are always two biprojector terms that involve no relativity, since they are not multiplied by the Lorentz factor γ, while the other two biprojector terms do involve relativity. Which biprojector terms involve relativity and which do not depend on both the entanglor and the boost direction.

4.2.3.1 The nonrelativistic part of the results

The nonrelativistic terms of the entangling eigenspinor contain only the original entanglor, with no boost. This is equivalent to saying that there is part of the entanglor that is unaffected by the applied boost. The existence of such cases unaffected by relativistic transformations may be applicable in the field of quantum communication, to appropriately orient entangled signal beams so that the messages are not distorted.

The two biprojectors in the nonrelativistic terms of the entangling eigenspinors are either both parallel or both antiparallel:

- in $\mathbf{\Lambda}^{1,1}_{\Psi-}$, $\mathbf{\Lambda}^{1,-1}_{\Psi+}$, $\mathbf{\Lambda}^{\pm3,\pm3}_{\Psi-}$, and $\mathbf{\Lambda}^{\pm3,\pm3}_{\Psi+}$, it is the *parallel* biprojectors that are nonrelativistic; and
- in $\mathbf{\Lambda}^{1,-1}_{\Psi-}$, $\mathbf{\Lambda}^{1,1}_{\Psi+}$, $\mathbf{\Lambda}^{\pm3,\mp3}_{\Psi-}$, and $\mathbf{\Lambda}^{\pm3,\mp3}_{\Psi+}$, it is the *antiparallel* biprojectors that are nonrelativistic.

Overall, it may be noticed that there is no sweeping generalization that can be made based on which eigenspinor is applied, which directions are involved, or whether the boosts are parallel or antiparallel. Thus, all that can be concluded here is that the nonrelativistic parts of the resulting eigenspinors for the two entanglors do depend on the geometry of the boosts.

4.2.3.2 The relativistic part of the results

The biprojectors in the relativistic terms of each result are parallel if the biprojectors in the nonrelativistic part were antiparallel, and vice versa. This is indicated in the general result (4.5) by an unknown projector Q that is present in both the "remaining biprojectors", i.e. not those appearing in the nonrelativistic part, and which is P_3or $\overline{\mathrm{P}}_3$ depending on whether the remaining biprojector of the form $\mathrm{P}_3 \otimes Q$ must be parallel or antiparallel.

As may be seen in the general result (4.5), the factor multiplying each relativistically-affected biprojector has two terms: one involving the original entanglor , and the other involving another entanglor (to be discussed next) that is multiplied by an additional factor of $\frac{v}{c}$. The sign for the speed factor depends on direction of the boost, and also whether the adjacent biprojector has its first factor P_3 or $\overline{\mathrm{P}}_3$ (notably, *not* depending on the projector Q).

- For the term containing the biprojector of the form $\mathrm{P}_3 \otimes Q$, the sign of the speed part is positive for first-subspace boost direction $\mathbf{e}_1$ or $-\mathbf{e}_3$, and negative for first-subspace boost direction $+\mathbf{e}_3$.
- For the term containing the biprojector of the form $\overline{\mathrm{P}}_3 \otimes \overline{Q}$, the sign of the speed part is opposite of that in the $\mathrm{P}_3 \otimes Q$ case.

This generalization further reiterates the trend in the results where the differences between results depend on the entanglors that were boosted, and the direction of the boost, since which biprojectors arise are dependent on both of these things.

4.2.4 Entanglors in the Results

Another noteworthy element present in the general form of the results (4.5) is the entanglor $\mathbf{R}_{\text{other}}$ which is not the one that was originally boosted $\mathbf{R}_{\Psi\pm}$. In each case, this "other" entanglor is the only other one that arises within the result, and which one it is depends on which entanglor $\mathbf{R}_{\Psi\pm}$ was originally boosted, and in which direction.

4.2.4.1 Which entanglors arise, and in what circumstances?

Comparing the results for the boosts in the $\mathbf{e}_1$ direction (4.1)(4.2), it is seen that this "other" entanglor $\mathbf{R}_{\text{other}}$ that arises always has the *same relative phase* as the original entanglor, but the *opposite relative parity*. That is, since the boosts have been applied in the present work only to the Ψ-entanglors, the "other" entanglor that arises from the $\mathbf{e}_1$ boosts is the Φ-entanglor with the same sign as the Ψ-entanglor that was boosted. Thus, the $\mathbf{e}_1$ boosts effectively shift an entanglor $\mathbf{R}_{\Psi\pm}$ so that it becomes a linear superposition that includes the entanglor $\mathbf{R}_{\Phi\pm}$ as well.

Comparing the results for the boosts in the $\mathbf{e}_3$ direction (4.3)(4.4), it is seen that the "other" entanglor $\mathbf{R}_{\text{other}}$ that arises always has the *same relative parity* as the original entanglor, but the *opposite relative phase*. That is, since the boosts have been applied in the present work only to the Ψ-entanglors, the "other" entanglor that arises from the $\mathbf{e}_3$ boosts is the other Ψ-entanglor, with the opposite sign. Thus, the $\mathbf{e}_3$ boosts effectively shift an entanglor $\mathbf{R}_{\Psi\pm}$ so that it becomes a linear superposition that includes the entanglor $\mathbf{R}_{\Psi\mp}$ as well.

4.2.4.2 How one entanglor can be shifted towards the other three

Now recall the single-subspace direction-inverting rotors (3.3), and their effect (3.5) of interrelating the entanglors: these changed an entanglor $\mathbf{R}_{X\pm}$ to another that was different in both its relative

parity and its relative phase, say, $\mathbf{R}_{Z\mp}$ where $Z \neq X$. It was also shown that it is not possible to completely transform an entanglor $\mathbf{R}_{X\pm}$ so that it becomes the entanglor $\mathbf{R}_{Z\pm}$ having opposite relative parity but same relative phase, or the entanglor $\mathbf{R}_{X\mp}$ having opposite relative phase but same relative parity.

The impossibility of completely transforming an entanglor to another that differs only in relative phase or only in relative parity also applies to the velocity transformation. As can be seen from the general result (4.5), a boost can at most shift the entanglor $\mathbf{R}_{\Psi\pm}$ to have some *component* of either entanglor $\mathbf{R}_{\Phi\pm}$ or $\mathbf{R}_{\Psi\mp}$, since the terms containing the Lorentz factor also include the original entanglor. That is, the limiting value of the part of the result containing the other entanglor $\mathbf{R}_{\text{other}}$ can be loosely expressed in the form

$$\lim_{v\to c} \Lambda^{a,b}_{\Psi\pm} = \gamma \left\{ \left[\mathbf{R}_{\Psi\pm} \begin{pmatrix} + \\ \text{or} \\ - \end{pmatrix} \mathbf{R}_{\text{other}} \right] [\mathrm{P}_3 \otimes Q] + \left[\mathbf{R}_{\Psi\pm} \begin{pmatrix} - \\ \text{or} \\ + \end{pmatrix} \mathbf{R}_{\text{other}} \right] [\overline{\mathrm{P}}_3 \otimes \overline{Q}] \right\} \tag{4.8}$$

where since $\gamma \to \infty$ it was noted that $\gamma \gg 1$ and so the first term of the general result (4.5) was dropped (but γ itself was left in place). Thus, even as $v \to c$, the magnitude of the $\mathbf{R}_{\text{other}}$ term will never exceed the magnitude of the term containing the original entanglor $\mathbf{R}_{\Psi\pm}$.

Altogether, there are three transformations that can be applied to either Ψ-entanglor to shift it "towards" the other three entanglors in the sense that it can become a linear superposition having that entanglor as a component:

- a single-subspace rotation in the $\mathbf{e}_1\mathbf{e}_3$ plane can shift an entanglor $\mathbf{R}_{\Psi\pm}$ towards $\mathbf{R}_{\Phi\mp}$;
- a bipartite boost in the $\pm\mathbf{e}_1$ direction(s) shifts an entanglor $\mathbf{R}_{\Psi\pm}$ towards $\mathbf{R}_{\Phi\pm}$;
- a bipartite boost in the $\pm\mathbf{e}_3$ direction(s) shifts an entanglor $\mathbf{R}_{\Psi\pm}$ towards $\mathbf{R}_{\Psi\mp}$,

where only the first of these three can be fully shifted from one entanglor to the other.

4.3 Application of Results to States

The results of combining specific separable boosts with entanglors, as summarized above, have the form of *transformation operators* that contain all the information about the entanglement and relative velocities of reference frames. However, in order to compare to results involving spin and relativity in the literature, these transformation operators must be applied to states. As will be demonstrated below, care is required in determining appropriate applications of these transformations.

4.3.1 Which Spin States to Transform?

A completely arbitrary state can be expressed as a linear superposition of either the separable or entangled basis states, and there may be any degree of entanglement in the state. The first of the sections below explain why such a completely arbitrary state is not appropriate for the application of the entangling eigenspinors.

Following this, the specific case of an arbitrarily-rotated separable state is investigated — but even that turns out to be excessively complicated, at least for the purpose of investigating the geometric effect of relativistic boosts. It turns out that the most relativistically-illuminating cases are some of the simplest ones; these are then transformed in Section 4.3.2.

4.3.1.1 Why not transform completely arbitrary states?

First, recall from Section 3.2.2 that a pair of separable boosts may be interpreted as a *description* of the motions, relative to some common observer, of the two subspaces in which some bipartite system can eventually be described. After the first application of boosts relating the individual rest frames to the frames in relative motion, the only further boosts that may be applied are ones that have identical effects on both subspaces: this is equivalent to boosting only the observer or the entire system relative to the observer. The part involving the relative boost information simply completes the *setting up of the mathematical model* of the subspaces that will eventually be used to describe a particular bipartite system.

Similarly, the entanglor can be interpreted as a *description* of relative orientations of the subspaces that will eventually contain the parts of a bipartite spin system. If such relative information already exists within the bipartite spin state, then it makes no sense to apply more such information; in fact, attempting to do so tends to give nonsensical mathematical results in addition to the interpretational difficulties. Thus, ***an entanglor can only be applied to a separable bipartite state***, and the same is true of the boosted entanglor, i.e. entangling eigenspinor.

So, each of the entangling eigenspinors investigated in this dissertation can be said to provide superpositions of relative information that connects the information in a pair of separable subspaces to the information in a non-separable bipartite space. Interpreted as transformation operators, the entangling eigenspinors may be applied to arbitrarily rotated separable states — but then it is just as reasonable to include the separable arbitrary rotations in the eigenspinor, and have it and act on simpler basis states such as $\Upsilon_{\uparrow\uparrow}$ or $\Upsilon_{\uparrow\downarrow}$. This idea will be shown to be justifiable for other reasons as well; but first, it is worth investigating a most general separable state.

4.3.1.2 A general separable rest-frame bipartite state

As noted above, the entangling eigenspinor can only be applied to a state that is separable. It is also worth mentioning that the separable parts should both be described in their own rest frames, since the purpose of the eigenspinor is to introduce both the relative entanglement and relative boost information to the state.

Thus, it is reasonable to consider a particular form for the bipartite state Υ_{AB} which can assure its separability: its coefficients can be obtained via a separable bipartite rotor $\mathbf{R}_{AB}$ applied to the separable spin-up basis state $\Upsilon_{\uparrow\uparrow}$. Equivalently, such a state is described by a pair of spin-up states independently rotated by simple rotors, which can be simplified before being combined into a tensor product. Note that this state remains normalized, $|\Upsilon_{AB}| = 1$, since it is not boosted.

Let the simple rotations R_A and R_B relating the spin directions to the initial spin direction $\mathbf{e}_3$ be given in terms of Euler angles (2.36), as

$$\begin{aligned} R_A &\equiv \exp\left(-i\mathbf{e}_3\frac{\alpha}{2}\right)\exp\left(-i\mathbf{e}_2\frac{\varepsilon}{2}\right)\exp\left(-i\mathbf{e}_3\frac{\eta}{2}\right) \\ R_B &\equiv \exp\left(-i\mathbf{e}_3\frac{\beta}{2}\right)\exp\left(-i\mathbf{e}_2\frac{\delta}{2}\right)\exp\left(-i\mathbf{e}_3\frac{\zeta}{2}\right), \end{aligned}$$

where the angles are all real. The spin states in Euler form can be expressed as linear combinations (2.39) of the spin-up $\psi_\uparrow$ and spin-down $\psi_\downarrow$ states, as

$$\begin{aligned} \psi_A &= e^{-i\eta/2}\left[e^{-i\alpha/2}\cos\frac{\varepsilon}{2}\psi_\uparrow + e^{i\alpha/2}\sin\frac{\varepsilon}{2}\psi_\downarrow\right] \\ \psi_B &= e^{-i\zeta/2}\left[e^{-i\beta/2}\cos\frac{\delta}{2}\psi_\uparrow + e^{i\beta/2}\sin\frac{\delta}{2}\psi_\downarrow\right]. \end{aligned}$$

Thus, the general separable state Υ_{AB} combining the two simple rotors above can be expressed as

$$\begin{aligned} \Upsilon_{AB} \equiv \mathbf{R}_{AB}\Upsilon_{\uparrow\uparrow} &= (R_A \otimes R_B)\,\Upsilon_{\uparrow\uparrow} \\ &= R_A\mathrm{P}_3 \otimes R_B\mathrm{P}_3 = \psi_A \otimes \psi_B \\ &= c_{\uparrow\uparrow}\Upsilon_{\uparrow\uparrow} + c_{\downarrow\downarrow}\Upsilon_{\downarrow\downarrow} + c_{\downarrow\uparrow}\Upsilon_{\downarrow\uparrow} + c_{\uparrow\downarrow}\Upsilon_{\uparrow\downarrow} \end{aligned} \tag{4.9}$$

where

$$\begin{aligned} c_{\uparrow\uparrow} &= e^{-i\eta/2}e^{-i\zeta/2}e^{-i\alpha/2}e^{-i\beta/2}\cos\frac{\varepsilon}{2}\cos\frac{\delta}{2} \\ c_{\downarrow\downarrow} &= e^{-i\eta/2}e^{-i\zeta/2}e^{i\alpha/2}e^{i\beta/2}\sin\frac{\varepsilon}{2}\sin\frac{\delta}{2} \\ c_{\downarrow\uparrow} &= e^{-i\eta/2}e^{-i\zeta/2}e^{i\alpha/2}e^{-i\beta/2}\sin\frac{\varepsilon}{2}\cos\frac{\delta}{2} \\ c_{\uparrow\downarrow} &= e^{-i\eta/2}e^{-i\zeta/2}e^{-i\alpha/2}e^{i\beta/2}\cos\frac{\varepsilon}{2}\sin\frac{\delta}{2} \end{aligned} \tag{4.10}$$

are the criteria that ensure the arbitrary state is separable.

4.3.1.3 Checking the concurrence of an arbitrary separable state

The concurrence (F.9) of this general state, which is neither entangled nor boosted, is calculated as follows:

$$\begin{aligned}
C\left(\Upsilon_{AB}\right) &= \frac{2}{\left|\Upsilon_{AB}\right|}\left|c_{\uparrow\uparrow}c_{\downarrow\downarrow} - c_{\downarrow\uparrow}c_{\uparrow\downarrow}\right| \\
&= 2\Bigg|\left[e^{-i\eta/2}e^{-i\zeta/2}e^{-i\alpha/2}e^{-i\beta/2}\cos\frac{\varepsilon}{2}\cos\frac{\delta}{2}\right]\left[e^{-i\eta/2}e^{-i\zeta/2}e^{i\alpha/2}e^{i\beta/2}\sin\frac{\varepsilon}{2}\sin\frac{\delta}{2}\right] \\
&\quad - \left[e^{-i\eta/2}e^{-i\zeta/2}e^{i\alpha/2}e^{-i\beta/2}\sin\frac{\varepsilon}{2}\cos\frac{\delta}{2}\right]\left[e^{-i\eta/2}e^{-i\zeta/2}e^{-i\alpha/2}e^{i\beta/2}\cos\frac{\varepsilon}{2}\sin\frac{\delta}{2}\right]\Bigg)\Bigg| \\
&= 2\left|e^{-i\eta/2}e^{-i\zeta/2}\left[\cos\frac{\delta}{2}\sin\frac{\delta}{2}\cos\frac{\varepsilon}{2}\sin\frac{\varepsilon}{2} - \cos\frac{\delta}{2}\sin\frac{\delta}{2}\cos\frac{\varepsilon}{2}\sin\frac{\varepsilon}{2}\right]\right| \\
&= 0.
\end{aligned}$$

Of course, its concurrence is zero, as it must be since the state was created by separable rotations from a separable state.

Note that the individual global phase factors $e^{-i\eta/2}$ and $e^{-i\zeta/2}$ from the separable parts end up as an overall global phase factor $e^{-i\eta/2}e^{-i\zeta/2}$ in bipartite form. Furthermore, this overall global phase factor does not contribute to the concurrence, since it is a part of every term and its absolute value is $\left|e^{-i\eta/2}e^{-i\zeta/2}\right| = 1$. Thus, this factor will be dropped in the analysis below.

4.3.1.4 Selecting even more specific separable bipartite states to investigate

It is still interesting to consider as general a state as possible, while still adhering to the separability criteria (4.10). This section considers the effects of the entangling eigenspinors (4.5) as applied to such a general separable state, for two cases: one where the relativistic boost has no effect, and one where the boost affects the state in a "maximal" sense.

Recall from the general form of the entangling eigenspinors that each result contains a pair of biprojectors for which no relativistic effects occur. Thus, if the transformation could be applied to a completely general state, no relativistic effects would occur — only entangling effects — in the terms containing the separable bipartite basis states Υ_{ij} that are left unchanged 4.6 by those biprojectors (4.6). If it is of interest to consider cases that are entirely nonrelativistic, the eigenspinors can be applied to the separable basis states individually, but most emphatically not as a linear superposition of these states: the initial state must not have any pre-existing entanglement, as explained above.

Now, since the purpose of this dissertation is to examine the *relativistic* effect of the entangling operators, it is particularly valuable to consider a general separable state (4.9) with coefficients of

zero for the parts that are invariant under the boost being investigated are zero. For example, if the sum of the parallel biprojectors $\mathrm{P}_3 \otimes \mathrm{P}_3 + \overline{\mathrm{P}}_3 \otimes \overline{\mathrm{P}}_3$ indicates the part of the transformation that is nonrelativistic, the general state that is "maximally" affected by the relativistic part of the transformation will have zeroes that correspond to these biprojectors, such that the general state (4.9) is simplified by $c_{\uparrow\uparrow} = c_{\downarrow\downarrow} = 0$. There are two ways to obtain a separable state with this condition: either (a) $\cos\frac{\epsilon}{2} = 0$ and $\sin\frac{\delta}{2} = 0$, or (b) $\cos\frac{\delta}{2} = 0$ and $\sin\frac{\epsilon}{2} = 0$. According to the separability criteria, both of these conditions render one of the other coefficients zero as well: case (a) gives $c_{\uparrow\downarrow} = 0$, while case (b) gives $c_{\downarrow\uparrow} = 0$. Thus, the only separable states that do not include the parallel basis states are the antiparallel basis states $\Upsilon_{\downarrow\uparrow}$ and $\Upsilon_{\uparrow\downarrow}$ — specifically, not linear superpositions of these, since such cases would have some degree of entanglement.

Similarly, for the cases where the sum of the antiparallel biprojectors $\mathrm{P}_3 \otimes \overline{\mathrm{P}}_3 + \overline{\mathrm{P}}_3 \otimes \mathrm{P}_3$ indicates the part of the transformation that is nonrelativistic, the only separable states worth investigating for relativistic purposes are the parallel basis states $\Upsilon_{\uparrow\uparrow}$ and $\Upsilon_{\downarrow\downarrow}$, each considered separately.

Thus, despite the attempt to examine the transformation operators most generally by applying them to as arbitrary a state as possible, it has been reasoned that the next appropriate step is to apply the entangling eigenspinors, separately, to only the separable basis states.

4.3.2 Applying the Transformation Operators to States

As found above, each of the entangling eigenspinors only gives relativistically-interesting results when applied to either the pair of parallel basis states, $\Upsilon_{\uparrow\uparrow}$ or $\Upsilon_{\downarrow\downarrow}$, or the antiparallel ones, $\Upsilon_{\downarrow\uparrow}$ or $\Upsilon_{\uparrow\downarrow}$. In any case, one of these pairs gives fully nonrelativistic results, while the other gives "maximally" relativistic results — and, which has which type of result depends on both the entanglor and the boost direction.

The work below is split into investigations of either the parallel or antiparallel separable basis states. In each case, the relativistic results are shown first, since the purpose of the present work is to investigate relativistic spin entanglement. The nonrelativistic results are also given, to complete the collection of results.

4.3.2.1 Entangling transformations of parallel basis states

The entangling eigenspinors that give relativistic results only when applied to *parallel* basis states are $\mathbf{\Lambda}^{1,-1}_{\Psi-}$ (4.1), $\mathbf{\Lambda}^{1,1}_{\Psi+}$ (4.2), $\mathbf{\Lambda}^{\pm3,\mp3}_{\Psi-}$ (4.3), and $\mathbf{\Lambda}^{\pm3,\mp3}_{\Psi+}$ (4.4). That is, the transformation operators investigated here are those with boost direction perpendicular to the initial spin direction and the boost directions and spin phases are either both the same (+ sign in subscript, and same signs for

both superscripts) or both opposite (− sign in subscript, and opposite signs for both superscripts); or, the boosts are antiparallel and aligned with the initial spin direction.

In each case below, the applicable result from above has already had all its biprojectors simplified away (4.6), leaving only the basis state that is the one being transformed. Then the entanglors and basis states are simplified (3.6) to obtain the Bell states.

The results of the relativistic entangling transformations of parallel separable basis states are:

$$\mathbf{\Lambda}^{1,-1}_{\Psi_-}\Upsilon_{\uparrow\uparrow} = \gamma\left[\mathbf{R}_{\Psi_-} + \frac{v}{c}\mathbf{R}_{\Phi_-}\right]\Upsilon_{\uparrow\uparrow} = \gamma\left[\ \ \Psi_- + \frac{v}{c}\Phi_-\right] \tag{4.11}$$

$$\mathbf{\Lambda}^{1,-1}_{\Psi_-}\Upsilon_{\downarrow\downarrow} = \gamma\left[\mathbf{R}_{\Psi_-} - \frac{v}{c}\mathbf{R}_{\Phi_-}\right]\Upsilon_{\downarrow\downarrow} = \gamma\left[\ \ \Psi_- + \frac{v}{c}\Phi_-\right] \tag{4.12}$$

$$\mathbf{\Lambda}^{1,1}_{\Psi_+}\Upsilon_{\uparrow\uparrow} = \gamma\left[\mathbf{R}_{\Psi_+} + \frac{v}{c}\mathbf{R}_{\Phi_+}\right]\Upsilon_{\uparrow\uparrow} = \gamma\left[\ \ \Psi_+ + \frac{v}{c}\Phi_+\right] \tag{4.13}$$

$$\mathbf{\Lambda}^{1,1}_{\Psi_+}\Upsilon_{\downarrow\downarrow} = \gamma\left[\mathbf{R}_{\Psi_+} - \frac{v}{c}\mathbf{R}_{\Phi_+}\right]\Upsilon_{\downarrow\downarrow} = \gamma\left[-\Psi_+ - \frac{v}{c}\Phi_+\right] \tag{4.14}$$

$$\mathbf{\Lambda}^{\pm 3,\mp 3}_{\Psi_-}\Upsilon_{\uparrow\uparrow} = \gamma\left[\mathbf{R}_{\Psi_-} \mp \frac{v}{c}\mathbf{R}_{\Psi_+}\right]\Upsilon_{\uparrow\uparrow} = \gamma\left[\ \ \Psi_- \mp \frac{v}{c}\Psi_+\right] \tag{4.15}$$

$$\mathbf{\Lambda}^{\pm 3,\mp 3}_{\Psi_-}\Upsilon_{\downarrow\downarrow} = \gamma\left[\mathbf{R}_{\Psi_-} \pm \frac{v}{c}\mathbf{R}_{\Psi_+}\right]\Upsilon_{\downarrow\downarrow} = \gamma\left[\ \ \Psi_- \mp \frac{v}{c}\Psi_+\right] \tag{4.16}$$

$$\mathbf{\Lambda}^{\pm 3,\mp 3}_{\Psi_+}\Upsilon_{\uparrow\uparrow} = \gamma\left[\mathbf{R}_{\Psi_+} \mp \frac{v}{c}\mathbf{R}_{\Psi_-}\right]\Upsilon_{\uparrow\uparrow} = \gamma\left[\ \ \Psi_+ \mp \frac{v}{c}\Psi_-\right] \tag{4.17}$$

$$\mathbf{\Lambda}^{\pm 3,\mp 3}_{\Psi_+}\Upsilon_{\downarrow\downarrow} = \gamma\left[\mathbf{R}_{\Psi_+} \pm \frac{v}{c}\mathbf{R}_{\Psi_-}\right]\Upsilon_{\downarrow\downarrow} = \gamma\left[-\Psi_+ \pm \frac{v}{c}\Psi_-\right]. \tag{4.18}$$

It is interesting to note two pairs of equal results, (4.11) & (4.12) and (4.15) & (4.16), all of which involve the Ψ_- entanglor:

$$\begin{aligned}\mathbf{\Lambda}^{1,-1}_{\Psi_-}\Upsilon_{\uparrow\uparrow} = \mathbf{\Lambda}^{1,-1}_{\Psi_-}\Upsilon_{\downarrow\downarrow} \ &= \gamma\left[\Psi_- + \frac{v}{c}\Phi_-\right] \\ \mathbf{\Lambda}^{\pm 3,\mp 3}_{\Psi_-}\Upsilon_{\uparrow\uparrow} = \mathbf{\Lambda}^{\pm 3,\mp 3}_{\Psi_-}\Upsilon_{\downarrow\downarrow} &= \gamma\left[\Psi_- \mp \frac{v}{c}\Psi_+\right]\end{aligned} \tag{4.19}$$

and two pairs of results that are negations of each other, (4.13) & (4.14) and (4.17) & (4.18), all of which involve the Ψ_+ entanglor:

$$\begin{aligned}\mathbf{\Lambda}^{1,1}_{\Psi_+}\Upsilon_{\uparrow\uparrow} = -\mathbf{\Lambda}^{1,1}_{\Psi_+}\Upsilon_{\downarrow\downarrow} \ &= \gamma\left[\Psi_+ + \frac{v}{c}\Phi_+\right] \\ \mathbf{\Lambda}^{\pm 3,\mp 3}_{\Psi_+}\Upsilon_{\uparrow\uparrow} = -\mathbf{\Lambda}^{\pm 3,\mp 3}_{\Psi_+}\Upsilon_{\downarrow\downarrow} &= \gamma\left[\Psi_+ \mp \frac{v}{c}\Psi_-\right].\end{aligned} \tag{4.20}$$

In all four pairings above, the result of transforming the separable spin-down basis state is, up to a sign, equal to the result of applying the *same* transformation to the separable spin-up basis state.

For completeness, here are the nonrelativistic entangling transformations of parallel separable basis

states:

$$\begin{aligned} \boldsymbol{\Lambda}^{1,1}_{\Psi-}\Upsilon_{\uparrow\uparrow} = \quad \boldsymbol{\Lambda}^{1,1}_{\Psi-}\Upsilon_{\downarrow\downarrow} = \quad \boldsymbol{\Lambda}^{\pm3,\pm3}_{\Psi-}\Upsilon_{\uparrow\uparrow} = \quad \boldsymbol{\Lambda}^{\pm3,\pm3}_{\Psi-}\Upsilon_{\downarrow\downarrow} = \quad \Psi_- \\ \boldsymbol{\Lambda}^{1,-1}_{\Psi+}\Upsilon_{\uparrow\uparrow} = -\boldsymbol{\Lambda}^{1,-1}_{\Psi+}\Upsilon_{\downarrow\downarrow} = \quad \boldsymbol{\Lambda}^{\pm3,\pm3}_{\Psi+}\Upsilon_{\uparrow\uparrow} = -\boldsymbol{\Lambda}^{\pm3,\pm3}_{\Psi+}\Upsilon_{\downarrow\downarrow} = \quad \Psi_+. \end{aligned} \tag{4.21}$$

It is seen that these transformations simply give the antiparallel entangled basis states $\Psi_\pm$, since the relativistic parts of these bipartite eigenspinors have antiparallel biprojectors that multiply with the parallel separable states to give zero.

4.3.2.2 Entangling transformations of antiparallel basis states

The entangling eigenspinors that give relativistic results only when applied to *antiparallel* basis states are $\boldsymbol{\Lambda}^{1,1}_{\Psi-}$ (4.1), $\boldsymbol{\Lambda}^{1,-1}_{\Psi+}$ (4.2), $\boldsymbol{\Lambda}^{\pm3,\pm3}_{\Psi-}$ (4.3), and $\boldsymbol{\Lambda}^{\pm3,\pm3}_{\Psi+}$ (4.4). That is, the transformation operators investigated here are those with boost direction perpendicular to the initial spin direction and either the boost directions or spin phases antiparallel (so for the phase subscript of $-$ or $+$, the signs in the superscript are the same or opposite, respectively); or, the boosts either both parallel or both antiparallel to the initial spin direction.

The same simplifications are applied here as were applied in the cases above. The results of the relativistic entangling transformations of antiparallel separable basis states are:

$$\boldsymbol{\Lambda}^{1,1}_{\Psi-}\Upsilon_{\uparrow\downarrow} = \gamma\left[\mathbf{R}_{\Psi-} + \frac{v}{c}\mathbf{R}_{\Phi-}\right]\Upsilon_{\uparrow\downarrow} = \gamma\left[\quad \Phi_+ + \frac{v}{c}\Psi_+\right] \tag{4.22}$$

$$\boldsymbol{\Lambda}^{1,1}_{\Psi-}\Upsilon_{\downarrow\uparrow} = \gamma\left[\mathbf{R}_{\Psi-} - \frac{v}{c}\mathbf{R}_{\Phi-}\right]\Upsilon_{\downarrow\uparrow} = \gamma\left[-\Phi_+ - \frac{v}{c}\Psi_+\right] \tag{4.23}$$

$$\boldsymbol{\Lambda}^{1,-1}_{\Psi+}\Upsilon_{\uparrow\downarrow} = \gamma\left[\mathbf{R}_{\Psi+} + \frac{v}{c}\mathbf{R}_{\Phi+}\right]\Upsilon_{\uparrow\downarrow} = \gamma\left[-\Phi_- - \frac{v}{c}\Psi_-\right] \tag{4.24}$$

$$\boldsymbol{\Lambda}^{1,-1}_{\Psi+}\Upsilon_{\downarrow\uparrow} = \gamma\left[\mathbf{R}_{\Psi+} - \frac{v}{c}\mathbf{R}_{\Phi+}\right]\Upsilon_{\downarrow\uparrow} = \gamma\left[-\Phi_- - \frac{v}{c}\Psi_-\right] \tag{4.25}$$

$$\boldsymbol{\Lambda}^{\pm3,\pm3}_{\Psi-}\Upsilon_{\uparrow\downarrow} = \gamma\left[\mathbf{R}_{\Psi-} \mp \frac{v}{c}\mathbf{R}_{\Psi+}\right]\Upsilon_{\uparrow\downarrow} = \gamma\left[\quad \Phi_+ \pm \frac{v}{c}\Phi_-\right] \tag{4.26}$$

$$\boldsymbol{\Lambda}^{\pm3,\pm3}_{\Psi-}\Upsilon_{\downarrow\uparrow} = \gamma\left[\mathbf{R}_{\Psi-} \pm \frac{v}{c}\mathbf{R}_{\Psi+}\right]\Upsilon_{\downarrow\uparrow} = \gamma\left[-\Phi_+ \mp \frac{v}{c}\Phi_-\right] \tag{4.27}$$

$$\boldsymbol{\Lambda}^{\pm3,\pm3}_{\Psi+}\Upsilon_{\uparrow\downarrow} = \gamma\left[\mathbf{R}_{\Psi+} \mp \frac{v}{c}\mathbf{R}_{\Psi-}\right]\Upsilon_{\uparrow\downarrow} = \gamma\left[-\Phi_- \mp \frac{v}{c}\Phi_+\right] \tag{4.28}$$

$$\boldsymbol{\Lambda}^{\pm3,\pm3}_{\Psi+}\Upsilon_{\downarrow\uparrow} = \gamma\left[\mathbf{R}_{\Psi+} \pm \frac{v}{c}\mathbf{R}_{\Psi-}\right]\Upsilon_{\downarrow\uparrow} = \gamma\left[-\Phi_- \mp \frac{v}{c}\Phi_+\right]. \tag{4.29}$$

It is interesting to note two pairs of equal results, (4.24) & (4.25) and (4.28) & (4.29), all of which

involve the Ψ_+ entanglor:

$$\begin{aligned} \boldsymbol{\Lambda}^{1,-1}_{\Psi_+}\Upsilon_{\uparrow\downarrow} = \boldsymbol{\Lambda}^{1,-1}_{\Psi_+}\Upsilon_{\downarrow\uparrow} &= \gamma\left[-\Phi_- - \frac{v}{c}\Psi_-\right] \\ \boldsymbol{\Lambda}^{\pm 3,\pm 3}_{\Psi_+}\Upsilon_{\uparrow\downarrow} = \boldsymbol{\Lambda}^{\pm 3,\pm 3}_{\Psi_+}\Upsilon_{\downarrow\uparrow} &= \gamma\left[-\Phi_- \mp \frac{v}{c}\Phi_+\right] \end{aligned} \tag{4.30}$$

and two pairs of results that are negations of each other, (4.22) & (4.23) and (4.26) & (4.27), all of which involve the Ψ_- entanglor:

$$\begin{aligned} \boldsymbol{\Lambda}^{1,1}_{\Psi_-}\Upsilon_{\uparrow\downarrow} = -\boldsymbol{\Lambda}^{1,1}_{\Psi_-}\Upsilon_{\downarrow\uparrow} &= \gamma\left[\Phi_+ + \frac{v}{c}\Psi_+\right] \\ \boldsymbol{\Lambda}^{\pm 3,\pm 3}_{\Psi_-}\Upsilon_{\uparrow\downarrow} = -\boldsymbol{\Lambda}^{\pm 3,\pm 3}_{\Psi_-}\Upsilon_{\downarrow\uparrow} &= \gamma\left[\Phi_+ \pm \frac{v}{c}\Phi_-\right]. \end{aligned} \tag{4.31}$$

In all four pairings above, the result of transforming the separable up-down basis state is, up to a sign, equal to the result of applying the *same* transformation to the separable down-up basis state.

For completeness, here are the nonrelativistic entangling transformations of antiparallel separable basis states:

$$\begin{aligned} \boldsymbol{\Lambda}^{1,-1}_{\Psi_-}\Upsilon_{\uparrow\downarrow} = -\boldsymbol{\Lambda}^{1,-1}_{\Psi_-}\Upsilon_{\downarrow\uparrow} &= \quad \boldsymbol{\Lambda}^{\pm 3,\mp 3}_{\Psi_-}\Upsilon_{\uparrow\downarrow} = -\boldsymbol{\Lambda}^{\pm 3,\mp 3}_{\Psi_-}\Upsilon_{\downarrow\uparrow} = \quad \Phi_+ \\ -\boldsymbol{\Lambda}^{1,1}_{\Psi_+}\Upsilon_{\uparrow\downarrow} = -\boldsymbol{\Lambda}^{1,1}_{\Psi_+}\Upsilon_{\downarrow\uparrow} &= -\boldsymbol{\Lambda}^{\pm 3,\mp 3}_{\Psi_+}\Upsilon_{\uparrow\downarrow} = -\boldsymbol{\Lambda}^{\pm 3,\mp 3}_{\Psi_+}\Upsilon_{\downarrow\uparrow} = \quad \Phi_-. \end{aligned} \tag{4.32}$$

It is seen that these transformations simply give the parallel entangled basis states $\Phi_\pm$, since the relativistic parts of these bipartite eigenspinors have parallel biprojectors that multiply with the antiparallel separable states to give zero.

4.3.2.3 From transformations of basis states to other entangling eigenspinors

The resulting transformed basis states (4.11)–(4.18) and (4.22)–(4.29) from the sections above, along with the general form of the entangling eigenspinors involving the Ψ-entanglors, can help infer expressions for the entangling eigenspinors involving the Φ-entanglors. The method outlined below makes ample use of the interrelationships between entanglors (3.5) and Bell basis states (3.6), as well as the assumption that the form of the Φ-entangling eigenspinors $\boldsymbol{\Lambda}^{a,b}_{\Phi_\pm} \equiv \mathbf{B}_{a,b}\mathbf{R}_{\Phi\pm}$ is similar to the general result (4.5) for the Ψ-entangling eigenspinors $\boldsymbol{\Lambda}^{a,b}_{\Psi_\pm}$.

To begin, note that the interrelationship $\mathbf{R}_{\Phi+}\Upsilon_{\uparrow\uparrow} = \mathbf{R}_{\Psi-}\Upsilon_{\uparrow\downarrow}$ (3.6) as well as the result (4.22) from the previous section give the effect of the entangling eigenspinor $\boldsymbol{\Lambda}^{1,1}_{\Phi_+}$ on the double spin-up basis state:

$$\begin{aligned} \boldsymbol{\Lambda}^{1,1}_{\Phi_+}\Upsilon_{\uparrow\uparrow} &= \boldsymbol{\Lambda}^{1,1}_{\Psi_-}\Upsilon_{\uparrow\downarrow} \\ &= \gamma\left[\Phi_+ + \frac{v}{c}\Psi_+\right] \\ &= \gamma\left[\mathbf{R}_{\Phi+} + \frac{v}{c}\mathbf{R}_{\Psi+}\right]\Upsilon_{\uparrow\uparrow}. \end{aligned}$$

This result gives information about one portion of the entangling eigenspinor $\mathbf{\Lambda}^{1,1}_{\Phi+}$: the portion containing the biprojector $\mathrm{P}_3 \otimes \mathrm{P}_3$. The other portions may be obtained similarly:

$$\begin{aligned}
\mathbf{\Lambda}^{1,1}_{\Phi+}\Upsilon_{\downarrow\downarrow} &= \quad \mathbf{\Lambda}^{1,1}_{\Phi+}\Upsilon_{\uparrow\uparrow} = \gamma\left[\mathbf{R}_{\Phi+} + \frac{v}{c}\mathbf{R}_{\Psi+}\right]\Upsilon_{\uparrow\uparrow} \\
\mathbf{\Lambda}^{1,1}_{\Phi+}\Upsilon_{\uparrow\downarrow} &= -\mathbf{\Lambda}^{1,1}_{\Psi-}\Upsilon_{\uparrow\uparrow} = -\mathbf{R}_{\Psi-}\Upsilon_{\uparrow\uparrow} = \mathbf{R}_{\Phi+}\Upsilon_{\uparrow\downarrow} \\
\mathbf{\Lambda}^{1,1}_{\Phi+}\Upsilon_{\downarrow\uparrow} &= \quad \mathbf{\Lambda}^{1,1}_{\Psi-}\Upsilon_{\uparrow\uparrow} = \quad \mathbf{R}_{\Psi-}\Upsilon_{\uparrow\uparrow} = \mathbf{R}_{\Phi+}\Upsilon_{\downarrow\uparrow}
\end{aligned}$$

where the latter two lines give information about the nonrelativistic portions of the entangling eigenspinor $\mathbf{\Lambda}^{1,1}_{\Phi+}$.

Combining the above results gives an entangling eigenspinor $\mathbf{\Lambda}^{1,1}_{\Phi+}$ of

$$\mathbf{\Lambda}^{1,1}_{\Phi+} \equiv \mathbf{B}_{a,b}\mathbf{R}_{\Phi\pm} = \boxed{\begin{aligned}&\mathbf{R}_{\Phi+}\left(\overline{\mathrm{P}}_3 \otimes \mathrm{P}_3 + \mathrm{P}_3 \otimes \overline{\mathrm{P}}_3\right) \\ &+ \gamma\left\{\left[\mathbf{R}_{\Phi+} + \frac{v}{c}\mathbf{R}_{\Psi+}\right]\mathrm{P}_3 \otimes \mathrm{P}_3 + \left[\mathbf{R}_{\Phi+} - \frac{v}{c}\mathbf{R}_{\Psi+}\right]\overline{\mathrm{P}}_3 \otimes \overline{\mathrm{P}}_3\right\}\end{aligned}}$$

which is unsurprisingly similar to the general result (4.5) for the Ψ-entangling eigenspinors $\mathbf{\Lambda}^{a,b}_{\Psi\pm}$. What is noteworthy is that this demonstrates how any entangling eigenspinor of the form $\mathbf{\Lambda}^{a,b}_{\Phi\pm}$ may be determined using the results obtained from the entangling eigenspinors of the form $\mathbf{\Lambda}^{a,b}_{\Psi\mp}$. The rest of the Φ-entangling eigenspinors $\mathbf{\Lambda}^{a,b}_{\Phi\pm}$ will not all be determined here; it suffices to know that it is possible.

Note that the interrelationship between the Ψ-entangling eigenspinors $\mathbf{\Lambda}^{a,b}_{\Psi\pm}$ and the Φ-entangling eigenspinors $\mathbf{\Lambda}^{a,b}_{\Phi\pm}$ can be understood by remembering that the separable basis states can be related to each other by single-subspace direction-inverting rotors (3.3). When these rotors are "shifted" to apply to the entanglor instead of the basis state, the states themselves become less important to consider, since the transformation operator or **full eigenspinor** contains all of the *relative information* within the system and between the system and the specific observer's frame.

The full eigenspinor will be discussed further in Section 4.4. Before moving on to that, it is valuable to further examine the boosted entangled states that were determined above, in the interest of working towards determining the significance of such calculations.

4.3.3 Concurrence of boosted entangled states

In all of the boosted entangled states calculated above, the resulting linear combination is expressed in terms of two entangled basis states, each of which is a member of a different "entanglement group".[2]

[2] As described in Appendix F.4.2, the entangled basis states can be divided into two groups based on the contributions of their coefficients to the formula for concurrence: "Group I" consists of Φ_+ and Ψ_- basis states, and "Group II" consists of Φ_- and Ψ_+ basis states. A state made up of a real linear combination involving members of only one group is maximally entangled.

The entangled basis state that is not multiplied by v is, in each case, in the same "group" as the entanglor that was applied to the initial separable basis state, while the one that was multiplied by v is in the other "group" — thus, the amount of entanglement in the boosted states as calculated above is expected to be *reduced* from the maximal entanglement applied by just the entanglor.

To verify the reduction in concurrence in the states as calculated above, consider the concurrence formula (F.11) that includes the normalization factor, since boosts effectively modify the magnitude of the state (D.1) and it is important to keep state expressions normalized so that the coefficients can represent probability amplitudes. Note that each of the state expressions under consideration here have coefficients of the same magnitudes. Thus, the same expression for concurrence applies for all the boosted entangled states as calculated above:

$$\begin{aligned} C\left(\mathbf{\Lambda}^{a,b}_{\Psi\pm}\Upsilon_{ij}\right) &= \frac{1}{\left|\mathbf{\Lambda}^{a,b}_{\Psi\pm}\Upsilon_{ij}\right|}\left|\gamma^2 - \gamma^2\frac{v^2}{c^2}\right| \\ &= \frac{1}{\left|\mathbf{\Lambda}^{a,b}_{\Psi\pm}\Upsilon_{ij}\right|}\gamma^2\left[1-\frac{v^2}{c^2}\right] \\ &= \frac{1}{\left|\mathbf{\Lambda}^{a,b}_{\Psi\pm}\Upsilon_{ij}\right|} \end{aligned}$$

where it was noted (3.15) that $\gamma^2\left(1-v^2/c^2\right)=1$.

It is interesting to consider that if the concurrence formula developed in Appendix F.4 did *not* take into account the magnitude of the boosted state, it would appear that the amount of entanglement remains constant. This contradicts the interpretation of the formula for the concurrence (F.11), which clearly shows that a linear superposition of entangled basis states from different "entanglement groups" should *reduce* the measure of entanglement. However, as will be discussed in Section 4.4, the boosted entangled state expression may be missing something important, in which case the normalization — and thus, also this calculation of concurrence — would need to be re-examined (this is strongly recommended as future work, in Section 5.2).

To complete the present (albeit simplistic) calculation, note that the magnitudes of all the boosted entangled states as calculated above are identical, because the absolute values of coefficients are the same in each. Thus, the Hilbert-space magnitude of each expression is

$$\begin{aligned} \left|\mathbf{\Lambda}^{a,b}_{\Psi\pm}\Upsilon_{ij}\right| &= \sqrt{|\gamma|^2+\left|\frac{\gamma v}{c}\right|^2} \\ &= \sqrt{\gamma^2\left(1+\frac{v^2}{c^2}\right)} \\ &= \sqrt{\frac{c^2+v^2}{c^2-v^2}}, \end{aligned}$$

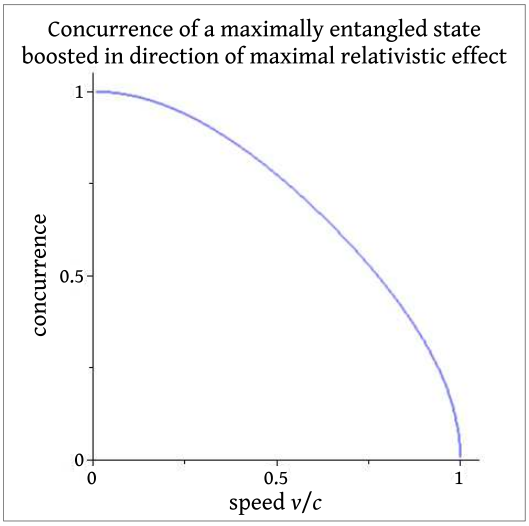


Figure 4.1: The concurrence (4.33) of a state that is entangled via an unboosted entanglor has the maximal value of $C = 1$. If the entanglor is boosted in a direction that has maximal relativistic effect on the particular state, the concurrence reduces to $C = 0$ as the speed v approaches c according to the curve shown here.

and therefore, the concurrence of any of the boosted entangled states as calculated above is

$$\boxed{C\left(\mathbf{\Lambda}^{a,b}_{\Psi\pm}\Upsilon_{ij}\right) = \sqrt{\frac{c^2 - v^2}{c^2 + v^2}}}. \tag{4.33}$$

This result for the concurrence as a function of speed v/c is plotted in Figure 4.1.

Note that the concurrence is equal to 1 if there is no boost (i.e. $v = 0$), which is as it should be. After all, in such a case, the eigenspinor contains only the entanglor, $\mathbf{R}_{\Psi\pm}$, which acts upon the separable state Υ_{ij} to obtain one of the maximally entangled Bell basis states, for which $C = 1$.

Note as well that this calculated concurrence is equal to 0 for a maximal boost (i.e. as the relative speed $v \to c$). This agrees with the previous realization (4.8) that the magnitude of the term affected by $\mathbf{R}_{\text{other}}$, the "other" entanglor that arose due to the boost, will never exceed the magnitude of the term affected by the original entanglor $\mathbf{R}_{\Psi\pm}$. For an equal linear superposition such as this, involving two entanglors of different "entanglement groups" (as described in Footnote 2 on page 103), the concurrence would be zero, as shown in Appendix F.4.2.

4.4 Connecting Calculated Results to their Interpretation

The interpretation of entanglement as being encoded within transformations rather than states has come full circle, with the transformations being applied to the quantum states from which they were originally separated for the purpose of examining this interpretation. What has been the point, then, if not to keep the entanglors as independent entities? It was to highlight the advantages of treating the entanglement as encoded within the entanglors, instead of encoded in the states.

As is discussed below, there are problems with the way the transformations were applied to states in the work above. This also means that there are problems with the way many existing treatments of relativistic entanglement naïvely apply boosts to spin states. The interpretation presented in the present work demonstrates how these naïve approaches fall short of reality.

4.4.1 Comparing to the Single-Particle Dirac Equation

To compare the difference between the boosted entanglors as operators and boosted entangled states, it is helpful to revisit the Dirac equation (2.20). Recall that the single-system eigenspinors (2.7) completely satisfy the free-particle Dirac equation. The connection to spin-$\frac{1}{2}$ states and spin measurement probability amplitudes is only made after the eigenspinors are applied to projectors (2.26) — and this was not only involving the projector P_3 representing spin-up, but also its complement $\overline{\mathrm{P}}_3$. The complementary projections were then transformed and manipulated in a way that obtained information about the particles of opposite chirality to the first case, expressed in the same minimal left ideal, which made it possible to compare to existing results (namely the Weyl bispinor (2.30), though a simple transformation relates these to the more typical Dirac bispinor (2.31)).

In Section 4.3.2, the eigenspinors representing the relativistic entangling transformations were applied to bipartite spin states, but only those involving the spin-up projectors P_3 and rotations of it. The calculations attempted to quantify the entanglement using concurrence (4.33), but the P_3-based spin basis states only extracted part of the available information, and thus the concurrence and normalization measures demonstrated cannot be considered entirely valid. These calculations must be modified in order to include information that describes particles of opposite chirality.

The full description of relativistically boosted entanglement must include some information from both possible chiralities. To appropriately interpret this, a description of annihilation and creation operators using APS [49] could be applied in a second quantization of the present approach. However, even without going beyond the scope of the present work, there is value in acknowledging the omission of the opposite-chirality case. Specifically, the erroneous calculations of Section 4.3.2 have value in how they demonstrate *why* one cannot simply apply a Lorentz transformation to a spin state in order to investigate relativity: there is information lost when applying the full eigenspinor to any specific separable spin basis state, since three of the four biprojectors multiply such a state will obtain zero.

Even more importantly, the calculations presented here can provide geometric insight towards a comprehensive understanding of the nature of spin. In particular, it is interesting that projectors serve not only to represent the spin-$\frac{1}{2}$ nature of elementary fermions, but also to filter part of the eigenspinor in such a way that the remainder only arises in relativistic situations. The question now is how to assemble the parts in a more satisfyingly geometric manner, rather than merely "stacked" as a bispinor (Section 5.2 mentions a complexified extension to the APS used in this dissertation, which may be promising in this regard).

Ultimately, it may be concluded that **this work highlights the information that is missing if an entangled spin state is naïvely transformed by the simple application of a Lorentz boost**. Thus, in the other works (which were mentioned in Section 1.2.3) that use overly simplistic models for relativistic entanglement, the resulting reduction in entanglement that has been reported may be a consequence of the modelling method rather than an indication of what happens in boosted entangled states.

Overall, the present work is more than merely a proof-of-concept for the idea of treating the entanglor as a transformation. Rather, it demonstrates the *necessity* of such an approach, or at least of an approach that ensures all the relativistic information is taken into account (such as by using bispinors, or the full entangling eigenspinors presented here).

4.4.2 The Physical Meaning of the Interpretation

Now that it has been deemed *necessary* to interpret entanglement in a way that ensures all the relativistic information is taken into account, such as is done in the present work, what does it mean to describe entanglement as residing within a transformation operator rather than within a state?

To compare the physical interpretations of "entanglement as a transformation" and "entanglement as a state", note that both still describe probabilistic information about possible measurement results, but as a transformation it may be described in a more *atemporal* way. Specifically, "entanglement as a transformation" can describe equally well the probabilities for a transformation *yet to take place*, and also the probabilities of which transformation *has taken place* but with the resulting information not yet available to the observer.

Although the present work is restricted to describing the entangling transformation as a superposition of rotations of a spin state, the general idea behind the interpretation may also be applied to more general situations involving superpositions. For example, the state of Schrodinger's cat need not be described as "half-alive and half-dead", because it may instead be interpreted by the observer as "alive until proven otherwise" — after all, it is only the transition, the *probability of death happening*, that is in superposition. If the death has taken place and the observer is unaware, the superposition does not change because there is no need to distinguish between the probability of death happening and the probability of the death having happened, not until actively interacting with the system to check the measurement result.

By interpreting entanglement (or any other superposition) as a collection of possible transformations, "weirdness" no longer needs to be attributed to quantum states themselves. Instead, it is the "probabilistic happenings" which are in superposition and can be discussed in terms of the probabilities of being yet to happen or even of having happened already. The present work argues that such an interpretation involves markedly less "weirdness" than superpositions of states of physical systems.

Furthermore, this interpretation bypasses the idea of "spooky action at a distance" (which, of course, cannot truly be "action" since that violates causality); instead, the collapse of the wavefunction simply represents the revelation about which relative rotations ended up describing the particles in question, which is information that can only be obtained *after* the measurement takes place. (The measurement operator is not discussed in the present work but in Section 5.2 it is strongly recommended for future investigation.)

All in all, each of the entangling eigenspinors investigated in this dissertation can be said to provide relative information that connects the information in a pair of separable subspaces to the information in the entangled and relatively boosted bipartite space. The superpositions, as in many standard interpretations, describe how the information is conditional or even mutually exclusive, and the information contained within can be interpreted probabilistically if the expression can be normalized. The difference here is that the probabilities describe the *odds that each of transformations has taken or will take place*, rather than probabilities "weirdly" applying to the states directly.

Chapter 5

The Aftermath

To conclude this work, this final chapter[1] summarizes what was accomplished here, describes some directions that the work can (and should!) be extended in the future, and then offers a few final words on the subject.

5.1 Review of This Dissertation

Section 5.1.1 begins this chapter by describing the interpretation that is proposed in this work along with a very brief review of the way it was investigated. Following that, Section 5.1.2 summarizes how the present work examined a simplistic approach to spin transformations and found problems from such an approach, and in what way the proposed interpretation is a step towards remedying this problem. Finally, in Section 5.1.3, the conclusions from Chapter 4 are restated, and some implications of the interpretation is discussed in the context of the related literature.

5.1.1 The Interpretation and its Investigation

This dissertation proposed (in Chapter 3) an interpretation of spin entanglement as information encoded within the relationship between reference frames, rather than as a property of a bipartite state. The interpretation was treated mathematically using the Algebra of Physical Space (introduced in Chapter 2, with supplemental details primarily in Appendices A and B). Specifically, the geometric description of the proposed interpretation of entanglement is as a superposition of relative

[1] This chapter is facetiously entitled "The Aftermath" only in reference to the fact that the mathematical parts of this work are complete.

rotations of reference frames, i.e. an **entanglor** (3.2), *without* including the particles themselves in the description.

Since the algebra used here provides an easy connection to relativity, the proposed interpretation of entanglement was then subjected to relativistic boost transformations (which themselves are explored in Section 3.2). The resulting operators — **entangling eigenspinors**, calculated in Section 3.3 — were reasoned in Section 3.4 to satisfy a proposed form for a bipartite Dirac equation (3.43), in the same way that simple eigenspinors (2.7) satisfy the usual Dirac equation (2.22), *without* having to include the projectors that represent the spin states. The last portion of the previous statement is what inspired the interpretation that is proposed in this dissertation: that the transformations contain valuable information *independently* from the entities being transformed. Specifically, since the Dirac equation is satisfied by any real linear superposition of its solutions, valuable information exists in the *relationships* between possible relatively rotated inertial frames — and this is what is proposed here to be the mechanism explaining **entanglement**.

The results of calculating the entangling eigenspinors were analyzed in Section 4.2. Unfortunately, the similarities between the different cases did not suggest a straightforward generalization, other than a blanket statement about how the effect of the boost on the entanglor depended on which boost it was and which entanglor. There was, however, some value in taking a "misstep" forward for the sake of further analyzing the results.

5.1.2 The Problem with Simplistic Spin Transformations

The general form found for the entangling eigenspinors (4.5) suggests that they be applied as transformation operators acting on states, as explained in Section 4.2.2. In order to most effectively investigate relativistic entanglement, it was determined in Section 4.3.1 that the most informative calculations would be those involving the separable basis states — mostly because the entangling eigenspinor must be applied to a separable state, since the entanglement is modelled within the transformation rather than existing within the state itself.

These simplistic spin calculations, which were done in Section 4.3, effectively mirror a common simple representation of relativistic entanglement in the literature (as described in Section 1.2.3.2): as relativistic boosts applied to entangled states — and in some works, the state kets even include momentum (1.1), ignoring the relativistic effect on the spin. However, there is a problem with even this geometrically-inspired yet simple attempt at describing relativistic entanglement: much of the information in the full entangling eigenspinor gets zeroed away by the projectors that are used to represent the spin states (4.6). The approaches that naïvely apply relativistic boosts to entangled states do not encounter this step of "discarding" information, and so the problem is not typically recognized.

The simplified results that are obtained in the geometric-yet-naïve approach in Section 4.3.2 still do have physical significance, although only as *part* of the complete picture. What is missing in this treatment is the opposite-chirality features of the Dirac bispinor, as described in Section 4.4. Once the missing parts are examined separately, their geometry is suspected to illuminate a way to better represent the full eigenspinor or an alternative combination of its parts.

5.1.3 Conclusions and Implications

As discussed in Section 4.4.1, one of the primary features of the present work is that it has revealed the geometric nature of the "missing" information that results from transforming spins in too simplistic a manner. The next step will be to complete the description using all four equations that can be obtained from the bipartite Dirac equation (3.42). The geometric analysis of all these parts is sure to reveal something interesting about the nature of spin and entanglement, in a way that adheres beautifully to the nature of spacetime that is modelled so well in APS (as described in Section 2.2). Some ideas of how this could be approached are discussed in Section 5.2.

It is worth noting that there is existing research [2, 54, 56, 74, 75, 115] that does, in fact, use the Dirac equation appropriately in modelling entangled states, but there are many more frequently-referenced articles (as described in Section 1.2.3.2) that do not. However, even in the cases where the Dirac equation has been applied appropriately, the work focuses on the particle/antiparticle states, which is quite unlike the present approach. Here, the APS Dirac equation (2.22) does not require explicit representations of particles or antiparticles, but only their Lorentz transformations or real linear superpositions thereof.

It is surely possible to relate the present work to existing second-quantized approaches to entanglement via Dirac fields, by using APS to model the creation and annihilation of particles and antiparticles [49, Ch. 6], but there is more at stake here than merely interconnecting existing descriptions. Quantum theory is rife with "weirdness" that should be able to be resolved by appropriate geometric models that provide a clear picture of the underlying mechanisms or explanations, and the importance of the present work is that it sets forth a proposal for the way of explaining entanglement, rather than merely reaffirming or refining mathematical relationships.

Furthermore, the present work can be extended to suggest that the quantum "weirdness" involved in *any* superposition should be thought of as applying to the possible *occurrences* rather than *states*. This notion was discussed in detail in Section 4.4.2, but it is important to highlight this idea here because of the popular debates [108] about whether quantum wavefunctions should be thought of as epistemic (delineating what is knowable or not knowable) or ontic (representing some property that exists in reality). Treating a superposition such as entanglement as a property of a bipartite state is a feature of the ontic approach, while the present work is epistemic instead. For example, it is not knowable which occurence entangled a pair of particles to have opposite spins, only that it

can be described as a superposition of rotating the first to be opposite to the second or rotating the second to be opposite the first.

Overall, of utmost importance in the present work is that the interpretation supports the idea that it is not the *state* that is fundamental, but the *processes* that occur — although this too is not a completely new idea [68]. However, the literature search did not reveal any works that highlight the pitfalls of treating the state as fundamental or ontic, as it has been discussed here.

5.2 Future Directions

The present work builds a foundation for much future progress. The most logical aspects to examine next are the ones that follow up on the directions that were started in the present work; several of these are in progress, and others are intended to be included in future work.

There are also several directions this work could be extended, either in terms of the scope or the mathematical framework applied.

5.2.1 Next Logical Extensions of the Present Work

First and foremost, the full bipartite eigenspinors need to be examined in a way that utilizes the entire completeness relation of biprojectors (B.15). This would enable an examination of bipartite versions of the Weyl bispinors (2.30), which can be used to express states in their chiral parts. It would then likely be straightforward to convert these to bipartite versions of the Dirac bispinors (2.31), which describe states in particle and antiparticle parts. However, it is very possible that the extension to covering the full bispinor forms of solutions would benefit from using the complexified Algebra of Physical Space, CAPS [31], instead of merely APS as was done here, but the same overall approach would apply.

One way or another, it will be important to ensure a proper normalization of boosted states that takes into account the effect of the full eigenspinor, in the name of preserving the interpretation of the coefficients as probability amplitudes. Furthermore, it would be valuable to calculate the concurrence (or some other measure of entanglement) in a way that takes into account all the information present in the full eigenspinor, in order to determine whether the entanglement depends on the reference frame from which it is described. If the entanglement is frame-dependent, then it would be interesting to compare perspectives between multiple observers, such as the pair of observers typically referred to as Alice and Bob who each hold one part of an entangled pair. The interpretation of entanglement as a relationship between inertial frames may highlight nuances between multiple inertial frames.

Also important would be to further generalize the approach presented here. In particular, the restrictions and limitations applied in the examination of the interpretation served to simplify the geometry for the sake of calculation, but they do not in general imply that the interpretation is only valid in such restricted cases. Thus, the present work serves as a "proof of concept" for the idea of treating entanglement as a transformation instead of a state. Further analysis of the geometry involved in this interpretation will surely be fruitful

It would also be interesting to see how this treatment of entanglement as a transformation contributes insight into the "measurement problem". This would require including electromagnetic fields instead of treating only free particles, since measuring devices for spin states typically involve magnetic fields.

5.2.2 Further Extensions of the Present Work

There are other extensions to the present work that would be interesting to examine geometrically, farther in the future. Some extensions, such as multiparticle states and general relativity, are fairly obvious in why they would be interesting. The literature also often mentions the idea of "entangled momenta" [77], particularly when the momentum states under investigation are not pure states, that is, not the plane waves that were investigated here. Further investigation of this subject, in light of the present work, could prove quite illuminating.

Finally, there is much discussion in the literature about an appropriate relativistic spin operator [9, 75, 96, 149], which would benefit from an investigation using APS. Specifically, the Pauli-Lubański spin, which arises frequently in discussions of possible relativistic spin operators, is easily understood in APS to be the result of a relativistic boost applied to the rest-frame spin direction $\mathbf{e}_3$. Such an investigation could lead to further clarity in the discussion of spin measurement in relativistic situations, which may then be related to the present work through the eigenspinors involved in both topics.

5.3 Final Words

The work presented in this dissertation addresses only a small part of the big picture that is quantum mechanics. However, the interpretation it presents highlights a perspective that is subtly different from the existing points of view, with lofty consequences: it may be closer than most traditional approaches to describing the reality behind quantum processes. As such, it may help with explaining and teaching quantum physics without having to rely so much on accepting "weirdness" as a quantum fact.

Appendix A

Notation, Conventions, and Useful Relations

The notational conventions of this dissertation are summarized here for convenience. Many are inspired by the conventions used in the Algebra of Physical Space (APS) [16, 25]. Some are adapted for use here.

A.1 Lettering and Letters

A.1.1 Roman Letters: Lowercase

- Boldfaced lowercase roman letters (e.g. $\mathbf{x}$) indicate **spatial vectors**.
  - ◇ The standard **basis unit vectors** (describing any observer's own rest frame) are denoted $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$ (representing the x, y, and z axes respectively) and the **basis scalar element** may be denoted $\mathbf{e}_0 \equiv 1$ or may be omitted for brevity.
  - ◇ Boldface lowercase roman letters with hats (e.g. $\hat{\mathbf{x}}$) represent **unit vectors** other than the basis vectors.
- Italicized lowercase roman letters (e.g. x) usually indicate **general elements of APS**, which may range from simple scalars to paravectors (see Section 2.1.1), unless otherwise stated. The following symbols have specific meanings throughout:
  - ◇ the **unit imaginary** i , **speed of light** c, or the **reduced Planck constant** $\hbar$ are examples of specific scalarlike quantities;

- ◇ the **time** t, and **mass** m, are scalar variables;
- ◇ **Euler's constant** e is occasionally used as an abbreviation for the exponential function exp, especially when its argument is purely scalar;
- ◇ if there are numeric superscripts (e.g. x^0, x^2)[1], such symbols may indicate the **scalarlike coefficients** of basis vectors with the corresponding subscripts (e.g. $x^0\mathbf{e}_0 + x^2\mathbf{e}_2$) — however, if there is any risk of confusion with an exponent, the quantity raised to an exponent will be enclosed in brackets (e.g. $\left(x^0\right)^2$);
- ◇ as a subscript or superscript, a roman letter (often j or k) represents an **index** $\in \{1, 2, 3\}$. If the index is repeated in the same term, the **Einstein summation convention** is to be interpreted; thus

$$x^k\mathbf{e}_k \equiv \sum x^k\mathbf{e}_k = x^1\mathbf{e}_1 + x^2\mathbf{e}_2 + x^3\mathbf{e}_3 \tag{A.1}$$

  i.e. $x^k\mathbf{e}_k$ is to be interpreted using $k = 1 \ldots 3$ to include all three components of the vector.

A.1.2 Roman Letters: Uppercase

- Uppercase roman letters:
  - ◇ If italicized, they represent particularly useful APS elements that may be linear combinations of scalarlike and and vectorlike parts.
    - ∗ Examples: R, B, and L, represent a spatial **rotor**, a **boost** operator, and a **Lorentz rotor**, respectively.
    - ∗ Another example: S and V are used to represent the scalarlike scalar and vector coefficients of a particular expression that would be confusing if represented by lowercase letters.
  - ◇ If in boldface, $\mathbf{R}$ and $\mathbf{B}$ represent **bipartite** versions of rotors and boosts.
  - ◇ A letter P that is neither bold nor italicized represents a **projector** (a special case of a paravector). (See Appendix B for details.)

A.1.3 Greek Letters

- Lowercase Greek letters (e.g. θ, ϕ, χ, α) often indicate angles, but some have specific other meanings that will be specified as necessary.

[1] Superscripts are used here to indicate that they transform differently than subscripted elements under a change of basis. The way basis vectors are transformed is deemed a "*covariant* transformation", while vector coefficients transform in the opposite sense, so their transformation is deemed "*contravariant*". Attention to this detail is not strictly required for following through this thesis, but the superscripted indices used here for demonstration warranted an explanation.

◇ Notable Greek letters with specific meanings include π as the ratio of circumference to diameter, τ as proper time, ψ as a quantum state paravector/wavefunction, ρ as a state density operator.

◇ A σ with an index indicates one of the **Pauli matrices**:

$$\sigma_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \tag{A.2}$$

or σ_k where $k \in \{1, 2, 3\}$. The 2×2 identity matrix $I_2 \equiv \sigma_0$ may be included in the set by using a Greek index: σ_μ where $\mu \in \{0, 1, 2, 3\}$.

◇ γ without an index is the **Lorentz factor**: $\gamma = \left(1 - v^2/c^2\right)^{-1/2}$.

◇ γ_X with an uppercase subscript represents the Lorentz factor corresponding to a particular speed v_X that needs to be distinguished from another, so $X \in \{A, B, \ldots\}$.

◇ γ^μ with a superscripted index (numeric, or lowercase roman or Greek to represent $1 \ldots 3$ and $0 \ldots 3$ respectively) indicates one of the **gamma matrices** used in the Dirac equation. In the Dirac-Pauli ("standard") and Weyl ("chiral") representations, the gamma matrices are given by[118, p. 46, p. 86]:

$$\begin{aligned} \text{standard rep:} \quad \gamma^0 &\equiv \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \gamma^k \equiv \begin{pmatrix} 0 & \sigma_k \\ -\sigma_k & 0 \end{pmatrix} \\ \text{Weyl rep:} \quad \gamma^0 &\equiv \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \gamma^k \equiv \begin{pmatrix} 0 & \sigma_k \\ -\sigma_k & 0 \end{pmatrix} \end{aligned} \tag{A.3}$$

with each element in each matrix as a 2×2 matrix, and the σ_k as the Pauli spin matrices.

◇ an **epsilon** with three indices, ϵ_{jkl}, is the fully antisymmetric **Levi-Civita symbol**:

$$\epsilon_{jkl} = \begin{cases} 0 & \text{if any two indices are equal,} \\ +1 & \text{if indices are some cyclic permutation of sequential} \\ -1 & \text{if indices are an anticyclic permutation of sequential} \end{cases} \tag{A.4}$$

◇ as a superscript or subscript, a Greek letter (often μ or ν) represents an **index** $\in \{0, 1, 2, 3\}$. If the index is repeated in the same term, the **Einstein summation convention** is to be interpreted; thus

$$x^\mu \mathbf{e}_\mu \equiv \sum x^\mu \mathbf{e}_\mu = x^0 \mathbf{e}_0 + x^1 \mathbf{e}_1 + x^2 \mathbf{e}_2 + x^3 \mathbf{e}_3 \tag{A.5}$$

i.e. $x^\mu \mathbf{e}_\mu$ is to be interpreted using $\mu = 0 \ldots 3$ to include the scalar part along with all three components of the vector part.

- Uppercase Greek letters generally indicate states, such as $\Psi_\pm$ and $\Phi_\pm$ as the entangled basis states, and Υ_{jk} where $j, k \in \{\uparrow, \downarrow\}$ as the separable basis states.

- ◇ Exception: the capital Greek letter Λ represents an **eigenspinor**, which is a specific Lorentz rotor that describes the relationship between the observed system's rest frame and a specific lab frame. The bolded case, $\mathbf{\Lambda}$, represents a **bipartite** eigenspinor.

A.2 Involutions & Hodge Dual

An **involution** is a mapping that is equal to its inverse.

- A superscripted asterisk following a known *scalarlike* element (e.g. x^*) represents **complex conjugation**, so for $x = a + ib$ with real a and b, the complex conjugate is $x^* = a - ib$. Note that this conjugation does not apply to general elements of APS.
- Given an arbitrary element $x = x^0 + x^k \mathbf{e}_k$ or product of elements p and q ,
  - ◇ $\bar{x}$ indicates **Clifford conjugation**, also called **bar conjugation**, such that

$$\bar{x} = x^0 - x^k \mathbf{e}_k \qquad \text{and} \qquad \overline{pq} = \bar{q}\bar{p} \tag{A.6}$$

  - ◇ $x^\dagger$ indicates **Hermitian conjugation**, also called **reversion** or **dagger conjugation**, such that

$$x^\dagger = x^{0*} + x^{k*} \mathbf{e}_k \qquad \text{and} \qquad (pq)^\dagger = q^\dagger p^\dagger \ . \tag{A.7}$$

    Note that the basis unit vectors $\mathbf{e}_k$ are unchanged by this conjugation, because they are *Hermitian.*

  - ◇ The combination of the two conjugations above is generally referred to as the **bar-dagger conjugate**, and its effect is a **spatial inversion** or **parity inversion**. Notably, this conjugation applied to a product,

$$\overline{(pq)}^\dagger = \bar{p}^\dagger \bar{q}^\dagger \ , \tag{A.8}$$

    does not reverse the order of factors. The order of application of the bar and dagger conjugate does not matter.

- Given a **projector** (See Appendix B) $\mathrm{P}_j = \frac{1}{2}(1 + \mathbf{e}_j)$ of basis vector $\mathbf{e}_j$ or $\mathrm{P}_n = \frac{1}{2}(1 + \hat{\mathbf{n}})$ of arbitrary direction $\hat{\mathbf{n}}$, the bar conjugate may be denoted $\overline{\mathrm{P}}_j = \mathrm{P}_{-j} = \frac{1}{2}(1 - \mathbf{e}_j)$ or $\overline{\mathrm{P}}_n = \mathrm{P}_{-n} = \frac{1}{2}(1 - \hat{\mathbf{n}})$ respectively.

A.2.1 Applications of Involutions

- Various subscripts to **angled brackets** indicate the use of involutions of APS to select or extract certain parts of an element or expression. For example, for a general APS element x,
  - ◇ the **scalarlike** and **vectorlike** parts are given by subscripts S and V, and are defined as

$$\langle x\rangle_S = \frac{1}{2}\left(x + \overline{x}\right) \text{ and } \langle x\rangle_V = \frac{1}{2}\left(x - \overline{x}\right) \tag{A.9}$$

    respectively, so that $x = \langle x\rangle_S + \langle x\rangle_V$,
  - ◇ the **real** and **imaginary** parts are given by subscripts $\Re$ and $\Im$, and are defined as

$$\langle x\rangle_\Re = \frac{1}{2}\left(x + x^\dagger\right) \text{ and } \langle x\rangle_\Im = \frac{1}{2}\left(x - x^\dagger\right) \tag{A.10}$$

    respectively, so that $x = \langle x\rangle_\Re + \langle x\rangle_\Im$ (where the unit imaginary i is included in $\langle x\rangle_\Im$),
  - ◇ the **even** and **odd** parts are given by subscripts $+$ and $-$, and are defined as

$$\langle x\rangle_+ = \frac{1}{2}\left(x + \overline{x}^\dagger\right) \text{ and } \langle x\rangle_- = \frac{1}{2}\left(x - \overline{x}^\dagger\right) \tag{A.11}$$

    respectively, so that $x = \langle x\rangle_+ + \langle x\rangle_-$.
  - ◇ Furthermore, a general element may be split into real scalars $\Re S$, real vectors $\Re V$, pseudovectors $\Im V$, and pseudoscalars $\Im S$, via a combination of subscripts:

$$x = \langle x\rangle_{\Re S} + \langle x\rangle_{\Re V} + \langle x\rangle_{\Im V} + \langle x\rangle_{\Im S} \ .$$

    Each of these parts may also be selected by the **vector grade**:

$$\begin{aligned}\langle x\rangle_0 &= \langle x\rangle_{\Re S}\\ \langle x\rangle_1 &= \langle x\rangle_{\Re V}\\ \langle x\rangle_2 &= \langle x\rangle_{\Im V}\\ \langle x\rangle_3 &= \langle x\rangle_{\Im S}\end{aligned}$$

    where, for example, the imaginary scalar part is numbered as vector grade *three* because it selects *tri*vectors.
    - ∗ The even elements of APS have vector grades 0 and 2 (scalar and bivector), while odd elements have vector grades 1 and 3 (vector and trivector).
- Certain products involve the bar (A.6) and dagger (A.7) conjugations:
  - ◇ The **Lorentzian inner product** of any pair of elements u and v may be denoted

$$\langle u, v\rangle_L \equiv 2\langle u\bar{v}\rangle_S = 2\langle v\bar{u}\rangle_S = 2\langle \bar{v}u\rangle_S = 2\langle \bar{u}v\rangle_S \tag{A.12}$$

where the result is a Lorentz-invariant scalar (see Section 2.2.4).

- ∗ If the Lorentzian inner product of two real paravectors is zero, they are **spacetime orthogonal**:

$$\langle u\bar{v}\rangle_S = 0 \quad \Longrightarrow \quad u \perp v. \tag{A.13}$$

◇ The **Hermitian inner product** of any pair of elements u and v may be denoted

$$\langle u, v\rangle_H \equiv 2\left\langle u^\dagger v\right\rangle_S = 2\left\langle v^\dagger u\right\rangle_S^* \tag{A.14}$$

where the result is a complex scalar that is invariant under a unitary transformation of its factors. The convention of conjugating the first of the pair of factors is based on the form of Dirac's bra-ket notation, where the bra is the leftmost factor and is the Hermitian conjugate of its corresponding ket.

- ∗ If the Hermitian inner product of two eigenstates of a Hermitian operator is zero, they are **Hilbert-space orthogonal**.

A.2.2 The Hodge Dual

The **Hodge dual** is a mapping that, when it is applied twice, the result is the identity mapping but only up to a sign. Thus it generally takes four applications to return the element to its original value. There are different sign conventions, but one possible definition (which has been called the *Clifford-Hodge dual* [23, p. 398]) is:

$$\star X \quad = \quad -iX. \tag{A.15}$$

This mapping converts a vector to a bivector and vice versa, with the sign determined by the handedness of the space and order of factors in the expressions being related. For example, applying the Hodge dual four times to one of the APS basis elements, for example, gives

$$\begin{aligned}
\star\mathbf{e}_1 &= -i\mathbf{e}_1 = -\mathbf{e}_2\mathbf{e}_3 \\
\star\star\,\mathbf{e}_1 &= (-i)^2\,\mathbf{e}_1 = -\mathbf{e}_1 \\
\star\star\star\mathbf{e}_1 &= (-i)^3\,\mathbf{e}_1 = \mathbf{e}_2\mathbf{e}_3 \\
\star\star\star\star\,\mathbf{e}_1 &= (-i)^4\,\mathbf{e}_1 = \mathbf{e}_1
\end{aligned}$$

A.3 Direct/Tensor Product (see also Appendix E)

- The $\otimes$ symbol generally indicates a direct product of vector spaces, or a tensor product of vector elements. (See Section E.1 for details.)

- This symbol is also used in subscripts of angled brackets, to indicate selection/extraction from specific subspaces. For example:
  - $\langle a \otimes b \rangle_{S\otimes 1} = \langle a \rangle_S b$ takes the scalar part (A.9) only in the first subspace;
  - $\langle a \otimes b \rangle_{S\otimes S} = \langle a \rangle_S \langle b \rangle_S$ takes the scalar part (A.9) in both subspaces;
  - $\langle a \otimes b \rangle_{+\otimes +} = \langle a \rangle_+ \otimes \langle b \rangle_+$ takes the even part (A.11) in both subspaces.
- Note that the selection/extraction from specific multipartite subspaces is often a part of an expression requiring a factor of 2^n where n is the number of subspaces.
  - For example, the Hermitian inner product (A.14) of a pair of bipartite states Υ_1 and Υ_2 has the form
$$2^2 \left\langle \Upsilon_1^\dagger \Upsilon_2 \right\rangle_{S\otimes S} . \tag{A.16}$$

A.4 Derivatives

- The **del** or **nabla** symbol represents the usual gradient operator:
$$\nabla \equiv \mathbf{e}_k \partial_k \tag{A.17}$$
with spatial coordinate derivatives
$$\partial_k \equiv \frac{\partial}{\partial x^k} \tag{A.18}$$
where the x^k represent the coefficients of the position vector $\mathbf{x} \equiv x^k \mathbf{e}_k$. Note that superscripted components transform like vector components do, while superscripted components transform contravariantly.
- As a standalone symbol, ∂ represents the **APS derivative**:
$$\partial \equiv \mathbf{e}_\mu \partial^\mu \quad = \quad \frac{1}{c}\partial_0 - \nabla \tag{A.19}$$
where the negative sign is from $\partial^k = -\partial_k$, which is due to the metric of the space, and the derivative of the time component is
$$\partial^0 = \partial_0 = \frac{1}{c}\partial^t = \frac{1}{c}\partial_t = \frac{1}{c}\frac{\partial}{\partial t} \tag{A.20}$$
where the unit scalar element $\mathbf{e}_0 \equiv 1$ is not shown explicitly.

Appendix B

Projectors

Projectors are particularly useful in the discussion of spin and entanglement. Their definition, properties, and various applications are outlined below.

B.1 Definition of a Projector in APS

Before investigating its properties in Appendix B.2, it may be helpful to define a **projector** as a particular APS element, having a real scalar and real vector part, each of magnitude $\frac{1}{2}$. It can be denoted as follows:

- If the vector part of the projector is in basis vector direction $\mathbf{e}_j$, the projector is denoted

$$\mathrm{P}_j = \frac{1}{2}\left(1 + \mathbf{e}_j\right) \tag{B.1}$$

  and its complementary projector (B.4) is denoted using the bar conjugate (A.6):

$$\overline{\mathrm{P}}_j = \mathrm{P}_{-j} = \frac{1}{2}\left(1 - \mathbf{e}_j\right) \ .$$

- If the vector part of the projector is in any other arbitrary direction $\hat{\mathbf{n}}$, the projector is denoted

$$\mathrm{P}_n = \frac{1}{2}\left(1 + \hat{\mathbf{n}}\right) \tag{B.2}$$

  and its complementary projector is denoted

$$\overline{\mathrm{P}}_n = \mathrm{P}_{-n} = \frac{1}{2}\left(1 - \hat{\mathbf{n}}\right) \ .$$

However, a **projector** is more typically defined in terms of its properties as a real (A.10), null (B.3), idempotent (B.5) element.

B.2 Properties of Projectors

$$\text{null norm:} \quad \mathrm{P}_n\overline{\mathrm{P}}_n = 0 \tag{B.3}$$

$$\text{complementarity:} \quad \mathrm{P}_n + \overline{\mathrm{P}}_n = 1 \tag{B.4}$$

$$\text{idempotence:} \quad \mathrm{P}_n^2 = \mathrm{P}_n \tag{B.5}$$

$$\text{reality:} \quad \mathrm{P}_n^\dagger = \mathrm{P}_n \tag{B.6}$$

$$\text{Pacwoman property:} \quad \hat{\mathbf{n}}\mathrm{P}_n = \mathrm{P}_n = \mathrm{P}_n\hat{\mathbf{n}} \tag{B.7}$$

$$\text{"perpendicular inversion":} \quad \hat{\mathbf{x}}\mathrm{P}_n = \overline{\mathrm{P}}_n\hat{\mathbf{x}}, \text{ iff } \hat{\mathbf{x}} \perp \hat{\mathbf{n}} \tag{B.8}$$

Note that the name "Pacwoman property" [16] is based on the Pac-Man video game [95], and refers to the way a projector can "gobble up" the unit vector matching its vector part. A projector can also "ungobble" such a unit vector. The ubiquitous utility of this property should not be underestimated as a result of its jocular appellation.

B.3 Spin, Matrix Representations, and Projectors

Although there are infinitely many possibilities that are valid matrix representations of the elements of APS, the Pauli matrix representation is particularly useful because of its connection to traditional representations of spin.

In particular, with the $\mathbf{e}_3$ axis typically oriented to align with the spin direction, the P_3 projector acting on any element of APS results in an entity whose matrix representation contains a two-component spinor as its first column, and an ignorable column of zeros as its second. Similarly, the $\overline{\mathrm{P}}_3$ projector extracts the second column from the matrix representation of an APS element.

To see this, take an arbitrary APS element $X = s\mathbf{e}_0 + v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + v_3\mathbf{e}_3$ expressed in Pauli matrix representation:

$$X \sim \begin{pmatrix} s + v_3 & v_1 - iv_2 \\ v_1 + iv_2 & s - v_3 \end{pmatrix}$$

where the complex coefficients $s,\ v_k$ for $k = 1...3$ are so labelled in order to facilitate a particular

conjugation later. The $\mathrm{P}_{\pm 3}$ projectors are seen to have the following effect on X:

$$X\mathrm{P}_3 \sim \begin{pmatrix} s+v_3 & v_1-iv_2 \\ v_1+iv_2 & s-v_3 \end{pmatrix} \frac{1}{2}\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \right) = \begin{pmatrix} s+v_3 & 0 \\ v_1+iv_2 & 0 \end{pmatrix} \quad \text{(B.9)}$$

$$X\overline{\mathrm{P}}_3 \sim \begin{pmatrix} s+v_3 & v_1-iv_2 \\ v_1+iv_2 & s-v_3 \end{pmatrix} \frac{1}{2}\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \right) = \begin{pmatrix} 0 & v_1-iv_2 \\ 0 & s-v_3 \end{pmatrix} \quad \text{(B.10)}$$

i.e. each of the projectors is indeed "projecting" one column of the matrix representation.

The information content in the part projected by $\overline{\mathrm{P}}_3$ can be transformed by taking the **bar-dagger conjugate** (A.8) $\overline{\left(X\overline{\mathrm{P}}_3\right)}^{\dagger} = \bar{X}^{\dagger}\mathrm{P}_3$ where $\bar{X}^{\dagger} = s^*\mathbf{e}_0 - v_1^*\mathbf{e}_1 - v_2^*\mathbf{e}_2 - v_3^*\mathbf{e}_3$, which has a Pauli matrix representation of

$$\bar{X}^{\dagger}\mathrm{P}_3 \sim \begin{pmatrix} s^*-v_3^* & -v_1^*+iv_2^* \\ -v_1^*-iv_2^* & s^*+v_3^* \end{pmatrix} \mathrm{P}_3 = \begin{pmatrix} s^*-v_3^* & 0 \\ -v_1^*-iv_2^* & 0 \end{pmatrix} = \begin{pmatrix} (s-v_3)^* & 0 \\ -(v_1-iv_2)^* & 0 \end{pmatrix} \quad \text{(B.11)}$$

and thus the information is now represented in the same **minimal left ideal** (defined below) as the information in $X\mathrm{P}_3$, so that each of the projected parts can be compared to a two-component spinor simply by ignoring the second column of zeros.

B.4 Minimal Left Ideals

APS contains certain sub-algebras, known as **minimal left ideals** of APS, that are important to this dissertation. An **ideal** is a sub-algebra whose elements sum to other elements within the sub-algebra, and multiplication (perhaps in only one direction) by any other element gives an element within the same ideal — this property is known as *closure*. If the closure rule applies only to multiplication by an element on the left of the element of the ideal, the ideal is a **left ideal**. A **minimal** ideal does not contain any other ideal within it.

In this dissertation, the minimal left ideals of APS that are encountered are obtained by applying a projector, generally either P_3 or $\overline{\mathrm{P}}_3$, leftward onto an expression in APS. The Pacwoman property (B.7) is generally invoked to re-express the results in a form that has a valuable geometric interpretation.

B.4.1 Examining a Minimal Left Ideal

Consider an arbitrary APS element with possibly-complex coefficients $x_\mu \equiv a_\mu + ib_\mu$ with a_μ, b_μ real:

$$\begin{aligned} x &= x_0 + x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + x_3\mathbf{e}_3 \\ &= (a_0 + ib_0) + (a_1 + ib_1)\,\mathbf{e}_1 + (a_2 + ib_2)\,\mathbf{e}_2 + (a_3 + ib_3)\,\mathbf{e}_3 \\ &= a_0 + ib_0 + a_1\mathbf{e}_1 + ib_1\mathbf{e}_1 + a_2\mathbf{e}_2 + ib_2\mathbf{e}_2 + a_3\mathbf{e}_3 + ib_3\mathbf{e}_3 \\ &= a_0 + a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3 + b_1\mathbf{e}_2\mathbf{e}_3 + b_2\mathbf{e}_3\mathbf{e}_1 + b_3\mathbf{e}_1\mathbf{e}_2 + b_0\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3. \end{aligned}$$

Multiply the arbitrary element x by P_3, acting leftward, to obtain an arbitrary element in the "primary" minimal left ideal used in this dissertation:

$$\begin{aligned} x\mathrm{P}_3 &= \{a_0 + a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3 + b_1\mathbf{e}_2\mathbf{e}_3 + b_2\mathbf{e}_3\mathbf{e}_1 + b_3\mathbf{e}_1\mathbf{e}_2 + b_0\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3\}\,\mathrm{P}_3 \\ &= \{a_0 + a_1\mathbf{e}_1\mathbf{e}_3 + a_2\mathbf{e}_2\mathbf{e}_3 + a_3 + b_1\mathbf{e}_2\mathbf{e}_3 + b_2\mathbf{e}_3\mathbf{e}_1 + b_3\mathbf{e}_1\mathbf{e}_2 + b_0\mathbf{e}_1\mathbf{e}_2\}\,\mathrm{P}_3 \\ &= \{[a_0 + a_3] + ([b_1 + a_2]\,\mathbf{e}_2\mathbf{e}_3 + [b_2 - a_1]\,\mathbf{e}_3\mathbf{e}_1 + [b_3 + b_0]\,\mathbf{e}_1\mathbf{e}_2)\}\,\mathrm{P}_3 \\ &= \{[a_0 + a_3] + i\,([b_1 + a_2]\,\mathbf{e}_1 + [b_2 - a_1]\,\mathbf{e}_2 + [b_3 + b_0]\,\mathbf{e}_3)\}\,\mathrm{P}_3 \\ &\equiv \{\epsilon_0 + i\,(\epsilon_1\mathbf{e}_1 + \epsilon_2\mathbf{e}_2 + \epsilon_3\mathbf{e}_3)\}\,\mathrm{P}_3. \end{aligned}$$

Here, it can be seen that an arbitrary element of the minimal left ideal denoted [30] $(\mathrm{APS})\,\mathrm{P}_3$ can be expressed as an *even element* — involving only scalars and bivectors — multiplied by the projector of the ideal. Note that the coefficients ϵ_μ of the resulting even expression are all real.

Even elements are useful because they can be expressed as proportional to a rotor, since a rotor $R = \exp\left(-i\hat{\mathbf{r}}\frac{\theta}{2}\right) = \cos\frac{\theta}{2} - i\hat{\mathbf{r}}\sin\frac{\theta}{2}$ consists of a real scalar part $\cos\frac{\theta}{2}$ and a bivector part $-i\hat{\mathbf{r}}\sin\frac{\theta}{2}$. Since a rotor has magnitude 1, the factor of proportionality relating the rotor to the even element is the magnitude of the even element; so, for the $x\mathrm{P}_3$ case above, it is

$$\begin{aligned} k &= \sqrt{\epsilon_0^2 + \epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2} \\ &= \sqrt{[a_0 + a_3]^2 + [b_1 + a_2]^2 + [b_2 - a_1]^2 + [b_3 + b_0]^2} \\ &= \sqrt{a_0^2 + a_1^2 + a_2^2 + a_3^2 + b_0^2 + b_1^2 + b_2^2 + b_3^2 + 2a_0a_3 + 2b_1a_2 - 2a_1b_2 + 2b_3b_0}. \end{aligned}$$

The rotor R is then found to be the even element divided by k. Equivalently, the original expression can be expressed in terms of a rotor in the minimal left ideal $(\mathrm{APS})\,\mathrm{P}_3$ as

$$x\mathrm{P}_3 = kR\mathrm{P}_3 \tag{B.12}$$

where the rotor itself can be expressed as $R = \exp\left(-i\hat{\mathbf{r}}\frac{\theta}{2}\right) = \cos\frac{\theta}{2} - i\hat{\mathbf{r}}\sin\frac{\theta}{2}$ with

$$\begin{aligned}\theta &= 2\arccos\left(\frac{1}{k}\left[a_0 + a_3\right]\right)\\ \hat{\mathbf{r}} &= -\frac{1}{k\sin\frac{\theta}{2}}\left(\left[b_1 + a_2\right]\mathbf{e}_1 + \left[b_2 - a_1\right]\mathbf{e}_2 + \left[b_3 + b_0\right]\mathbf{e}_3\right).\end{aligned}$$

B.4.2 The Complementary Minimal Left Ideal

The complementary projector $\overline{\mathrm{P}}_3 = \mathrm{P}_{-3}$ has a similar effect as P_3 does on an arbitrary APS element x:

$$\begin{aligned}x\overline{\mathrm{P}}_3 &= \left\{a_0 + a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3 + b_1\mathbf{e}_2\mathbf{e}_3 + b_2\mathbf{e}_3\mathbf{e}_1 + b_3\mathbf{e}_1\mathbf{e}_2 + b_0\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3\right\}\overline{\mathrm{P}}_3\\ &= \left\{a_0 - a_1\mathbf{e}_1\mathbf{e}_3 - a_2\mathbf{e}_2\mathbf{e}_3 - a_3 + b_1\mathbf{e}_2\mathbf{e}_3 + b_2\mathbf{e}_3\mathbf{e}_1 + b_3\mathbf{e}_1\mathbf{e}_2 - b_0\mathbf{e}_1\mathbf{e}_2\right\}\overline{\mathrm{P}}_3\\ &= \left\{\left[a_0 - a_3\right] + \left(\left[b_1 - a_2\right]\mathbf{e}_2\mathbf{e}_3 + \left[b_2 + a_1\right]\mathbf{e}_3\mathbf{e}_1 + \left[b_3 - b_0\right]\mathbf{e}_1\mathbf{e}_2\right)\right\}\overline{\mathrm{P}}_3\\ &= \left\{\left[a_0 - a_3\right] + i\left(\left[b_1 - a_2\right]\mathbf{e}_1 + \left[b_2 + a_1\right]\mathbf{e}_2 + \left[b_3 - b_0\right]\mathbf{e}_3\right)\right\}\overline{\mathrm{P}}_3\\ &= \left\{\varepsilon_0 + i\left(\varepsilon_1\mathbf{e}_1 + \varepsilon_2\mathbf{e}_2 + \varepsilon_3\mathbf{e}_3\right)\right\}\overline{\mathrm{P}}_3\ ,\end{aligned}$$

that is, the minimal left ideal (APS) $\overline{\mathrm{P}}_3$ also consists of only even elements multiplied by the projector of the ideal, but with even elements that are different from the previous case: $\epsilon_\mu \neq \varepsilon_\mu$.

An arbitrary element of this complementary minimal left ideal $x\overline{\mathrm{P}}_3$ can be expressed in terms of a rotor as

$$x\overline{\mathrm{P}}_3 = k'R'\overline{\mathrm{P}}_3\ , \tag{B.13}$$

but the scalar dilation factor k' is different from the $x\mathrm{P}_3$ case:

$$\begin{aligned}k' &= \sqrt{\varepsilon_0^2 + \varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2}\\ &= \sqrt{\left[a_0 - a_3\right]^2 + \left[b_1 - a_2\right]^2 + \left[b_2 + a_1\right]^2 + \left[b_3 - b_0\right]^2}\\ &= \sqrt{a_0^2 + a_1^2 + a_2^2 + a_3^2 + b_0^2 + b_1^2 + b_2^2 + b_3^2 - 2a_0a_3 - 2b_1a_2 + 2a_1b_2 - 2b_3b_0}\end{aligned}$$

and so the rotor $R' = \exp\left(-i\hat{\mathbf{r}}'\frac{\phi'}{2}\right) = \cos\frac{\phi'}{2} - i\hat{\mathbf{r}}'\sin\frac{\phi'}{2}$ is different as well:

$$\begin{aligned}\phi' &= 2\arccos\left(\frac{1}{k'}\left[a_0 - a_3\right]\right)\\ \hat{\mathbf{r}}' &= -\frac{1}{k'\sin\frac{\phi}{2}}\left(\left[b_1 - a_2\right]\mathbf{e}_1 + \left[b_2 + a_1\right]\mathbf{e}_2 + \left[b_3 - b_0\right]\mathbf{e}_3\right)\ .\end{aligned}$$

B.4.3 Comparing Information From Complementary Ideals

The information content of an arbitrary element x can be split into complementary minimal left ideals by multiplying the element with the complementarity relation (B.4) $1 = \mathrm{P}_3 + \overline{\mathrm{P}}_3$, obtaining:

$$\begin{aligned} x &= x\left(\mathrm{P}_3 + \overline{\mathrm{P}}_3\right) \\ &= x\mathrm{P}_3 + x\overline{\mathrm{P}}_3 \\ &= kR\mathrm{P}_3 + k'R'\overline{\mathrm{P}}_3 \ . \end{aligned}$$

However, the information content in the two parts cannot be compared to one another by way of simple rotations and scaling factors, as would be possible when comparing elements within the same minimal left ideal.

It is possible to transform the information in the second term by taking the bar-dagger conjugate (A.8) of it:

$$\begin{aligned} \overline{\left(x\overline{\mathrm{P}}_3\right)}^{\dagger} &= \bar{x}^{\dagger}\mathrm{P}_3 \\ &= \overline{\{a_0 + ib_0 + a_1\mathbf{e}_1 + ib_1\mathbf{e}_1 + a_2\mathbf{e}_2 + ib_2\mathbf{e}_2 + a_3\mathbf{e}_3 + ib_3\mathbf{e}_3\}}^{\dagger}\mathrm{P}_3 \\ &= \{a_0 - ib_0 - a_1\mathbf{e}_1 + ib_1\mathbf{e}_1 - a_2\mathbf{e}_2 + ib_2\mathbf{e}_2 - a_3\mathbf{e}_3 + ib_3\mathbf{e}_3\}\,\mathrm{P}_3 \\ &= \{a_0 - a_1\mathbf{e}_1 - a_2\mathbf{e}_2 - a_3\mathbf{e}_3 + b_1\mathbf{e}_2\mathbf{e}_3 + b_2\mathbf{e}_3\mathbf{e}_1 + b_3\mathbf{e}_1\mathbf{e}_2 - b_0\mathbf{e}_1\mathbf{e}_2\mathbf{e}_3\}\,\mathrm{P}_3 \\ &= \{a_0 - a_1\mathbf{e}_1\mathbf{e}_3 - a_2\mathbf{e}_2\mathbf{e}_3 - a_3 + b_1\mathbf{e}_2\mathbf{e}_3 + b_2\mathbf{e}_3\mathbf{e}_1 + b_3\mathbf{e}_1\mathbf{e}_2 - b_0\mathbf{e}_1\mathbf{e}_2\}\,\mathrm{P}_3 \\ &= \{[a_0 - a_3] + ([b_1 - a_2]\,\mathbf{e}_2\mathbf{e}_3 + [b_2 + a_1]\,\mathbf{e}_3\mathbf{e}_1 + [b_3 - b_0]\,\mathbf{e}_1\mathbf{e}_2)\}\,\mathrm{P}_3 \\ &= \{[a_0 - a_3] + i\,([b_1 - a_2]\,\mathbf{e}_1 + [b_2 + a_1]\,\mathbf{e}_2 + [b_3 - b_0]\,\mathbf{e}_3)\}\,\mathrm{P}_3 \\ &= \{\varepsilon_0 + i\,(\varepsilon_1\mathbf{e}_1 + \varepsilon_2\mathbf{e}_2 + \varepsilon_3\mathbf{e}_3)\}\,\mathrm{P}_3 \end{aligned}$$

which involves the same even element as was found for $x\overline{\mathrm{P}}_3$, but it is now expressed in the (APS) P_3 minimal left ideal instead. This expression as a dilated rotor has the form

$$\bar{x}^{\dagger}\mathrm{P}_3 = k'R'\mathrm{P}_3 \tag{B.14}$$

where R' and k' are same ones as found for $x\overline{\mathrm{P}}_3$.

Now that the two terms are in the same minimal left ideal, the rotors R and R' and the scaling factors k and k' can be compared to one another by rotations and dilations.

B.5 Interpretations of Projectors

It may seem to be a point of confusion that there are so many interpretations for projectors in the APS model. For example, P_3 may represent a state spinor $\psi_{\uparrow}$ or a state density operator $\rho_{\uparrow}$, or one

of a pair of complementary projectors used to re-express a spin state (2.38) in terms of spin-up and spin-down states. Note that it should be clear from the context how a projector is to be interpreted: it may be seen as undergoing a transformation via an operator that is acting either unilaterally (if the projector is a spinor) or bilaterally (if the projector is a density operator), or the projector itself may be behaving as an operator on a spinor state to re-express it in a specific way.

The confusion regarding the projectors may be slightly alleviated (or, at least, justified) by considering the idempotence of the projector: $\mathrm{P}_3\mathrm{P}_3 = \mathrm{P}_3$. This relation may be considered an eigenvalue equation as follows: the leftmost P_3 in the expression is treated as an operator (which is traditionally represented as a matrix), acting on the adjacent P_3 which is seen to be an eigenstate (and traditionally represented as a column spinor) of the operator P_3, with a corresponding eigenvalue of magnitude one.

This dissertation strives to ensure clarity of the interpretations of projectors based on the wording nearby, and also by using alternative symbols such as $\psi_\uparrow$ and $\rho_\uparrow$ to indicate the role of the projector, instead of just using the projector itself.

B.6 Biprojectors

A pair of projectors joined together by the tensor product symbol may be called a **biprojector**. This term was coined for use in this dissertation as shorthand for **bi**partite **projector**.

B.6.1 Complementarity of Biprojectors

In a direct product space, the complementarity of projectors gives a sum of four tensor products of projectors that span the space:

$$\begin{aligned} \mathbf{1} \otimes \mathbf{1} &= \left(\mathrm{P}_n + \overline{\mathrm{P}}_n\right) \otimes \left(\mathrm{P}_n + \overline{\mathrm{P}}_n\right) \\ &= \mathrm{P}_n \otimes \mathrm{P}_n + \overline{\mathrm{P}}_n \otimes \overline{\mathrm{P}}_n + \mathrm{P}_n \otimes \overline{\mathrm{P}}_n + \overline{\mathrm{P}}_n \otimes \mathrm{P}_n \end{aligned} \tag{B.15}$$

where each term is a **primitive idempotent** of the two-part space, which means they sum to one and multiply to zero and cannot be decomposed into smaller idempotents. In this work, such terms are deemed **biprojectors** since they are projectors for the bipartite space investigated here.

B.6.2 Sums and Differences of Biprojectors

The biprojectors may be usefully multiplied out into explicit scalars and unit vectors:

$$\begin{aligned}
\mathrm{P}_n \otimes \mathrm{P}_n &= \frac{1}{4}\left(\mathbf{1}\otimes\mathbf{1} + \mathbf{1}\otimes\hat{\mathbf{n}} + \hat{\mathbf{n}}\otimes\mathbf{1} + \hat{\mathbf{n}}\otimes\hat{\mathbf{n}}\right) \\
\mathrm{P}_n \otimes \overline{\mathrm{P}}_n &= \frac{1}{4}\left(\mathbf{1}\otimes\mathbf{1} - \mathbf{1}\otimes\hat{\mathbf{n}} + \hat{\mathbf{n}}\otimes\mathbf{1} - \hat{\mathbf{n}}\otimes\hat{\mathbf{n}}\right) \\
\overline{\mathrm{P}}_n \otimes \overline{\mathrm{P}}_n &= \frac{1}{4}\left(\mathbf{1}\otimes\mathbf{1} - \mathbf{1}\otimes\hat{\mathbf{n}} - \hat{\mathbf{n}}\otimes\mathbf{1} + \hat{\mathbf{n}}\otimes\hat{\mathbf{n}}\right) \\
\overline{\mathrm{P}}_n \otimes \mathrm{P}_n &= \frac{1}{4}\left(\mathbf{1}\otimes\mathbf{1} + \mathbf{1}\otimes\hat{\mathbf{n}} - \hat{\mathbf{n}}\otimes\mathbf{1} - \hat{\mathbf{n}}\otimes\hat{\mathbf{n}}\right).
\end{aligned}$$

These are even more useful when grouped into similar sums and differences, where the similarity is based on whether the constituent biprojectors are parallel or antiparallel:

$$\begin{aligned}
\mathrm{P}_n \otimes \mathrm{P}_n + \overline{\mathrm{P}}_n \otimes \overline{\mathrm{P}}_n &= \frac{1}{2}\left(\mathbf{1}\otimes\mathbf{1} + \hat{\mathbf{n}}\otimes\hat{\mathbf{n}}\right) \\
\mathrm{P}_n \otimes \overline{\mathrm{P}}_n + \overline{\mathrm{P}}_n \otimes \mathrm{P}_n &= \frac{1}{2}\left(\mathbf{1}\otimes\mathbf{1} - \hat{\mathbf{n}}\otimes\hat{\mathbf{n}}\right) \\
\mathrm{P}_n \otimes \mathrm{P}_n - \overline{\mathrm{P}}_n \otimes \overline{\mathrm{P}}_n &= \frac{1}{2}\left(\hat{\mathbf{n}}\otimes\mathbf{1} + \mathbf{1}\otimes\hat{\mathbf{n}}\right) \\
\mathrm{P}_n \otimes \overline{\mathrm{P}}_n - \overline{\mathrm{P}}_n \otimes \mathrm{P}_n &= \frac{1}{2}\left(\hat{\mathbf{n}}\otimes\mathbf{1} - \mathbf{1}\otimes\hat{\mathbf{n}}\right).
\end{aligned} \tag{B.16}$$

The expressions on the right-hand-side above arise naturally in several calculations in this dissertation, and in replacing them with the biprojectors, greater insight is obtained.

The other possible sums of biprojectors simply obtain the single-subspace projectors:

$$\begin{aligned}
\mathrm{P}_n \otimes \mathrm{P}_n + \mathrm{P}_n \otimes \overline{\mathrm{P}}_n &= \mathrm{P}_n \otimes \left(\mathrm{P}_n + \overline{\mathrm{P}}_n\right) = \mathrm{P}_n \otimes \mathbf{1} \\
\mathrm{P}_n \otimes \mathrm{P}_n + \overline{\mathrm{P}}_n \otimes \mathrm{P}_n &= \left(\mathrm{P}_n + \overline{\mathrm{P}}_n\right) \otimes \mathrm{P}_n = \mathbf{1} \otimes \mathrm{P}_n \\
\overline{\mathrm{P}}_n \otimes \mathrm{P}_n + \overline{\mathrm{P}}_n \otimes \overline{\mathrm{P}}_n &= \overline{\mathrm{P}}_n \otimes \left(\mathrm{P}_n + \overline{\mathrm{P}}_n\right) = \overline{\mathrm{P}}_n \otimes \mathbf{1} \\
\mathrm{P}_n \otimes \overline{\mathrm{P}}_n + \overline{\mathrm{P}}_n \otimes \overline{\mathrm{P}}_n &= \left(\mathrm{P}_n + \overline{\mathrm{P}}_n\right) \otimes \overline{\mathrm{P}}_n = \mathbf{1} \otimes \overline{\mathrm{P}}_n
\end{aligned}$$

while the other possible differences of the biprojectors are

$$\begin{aligned}
\mathrm{P}_n \otimes \mathrm{P}_n - \mathrm{P}_n \otimes \overline{\mathrm{P}}_n &= \mathrm{P}_n \otimes \left(\mathrm{P}_n - \overline{\mathrm{P}}_n\right) = \mathrm{P}_n \otimes \hat{\mathbf{n}} \\
\mathrm{P}_n \otimes \mathrm{P}_n - \overline{\mathrm{P}}_n \otimes \mathrm{P}_n &= \left(\mathrm{P}_n - \overline{\mathrm{P}}_n\right) \otimes \mathrm{P}_n = \hat{\mathbf{n}} \otimes \mathrm{P}_n \\
\overline{\mathrm{P}}_n \otimes \mathrm{P}_n - \overline{\mathrm{P}}_n \otimes \overline{\mathrm{P}}_n &= \overline{\mathrm{P}}_n \otimes \left(\mathrm{P}_n - \overline{\mathrm{P}}_n\right) = \overline{\mathrm{P}}_n \otimes \hat{\mathbf{n}} \\
\mathrm{P}_n \otimes \overline{\mathrm{P}}_n - \overline{\mathrm{P}}_n \otimes \overline{\mathrm{P}}_n &= \left(\mathrm{P}_n - \overline{\mathrm{P}}_n\right) \otimes \overline{\mathrm{P}}_n = \hat{\mathbf{n}} \otimes \overline{\mathrm{P}}_n .
\end{aligned}$$

which have dubious usefulness but are included here for the sake of completeness.

Appendix C

Wigner Rotation

The Algebra of Physical Space (APS) can provide geometric clarity to the interpretation and derivation of a Wigner rotation [155]. It can also, in Section C.5, be applied to explain a rotation that is occasionally confused with the Wigner rotation.

C.1 Definition of Wigner Rotation

When combining non-collinear boosts B_A and B_B, the resulting transformation can be described by the product

$$B_B B_A = B_\Omega R_\Omega \tag{C.1}$$

where B_Ω is an overall boost, and R_Ω is known as a **Wigner rotation**, which describes an adjustment to the orientation within the reference frame of the system to which the combination of boosts is applied.

Note that every boost in APS is described relatively to the reference frame of whatever it is acting upon, applied rightward. Thus, suppose B_A is described relatively to some rest frame, with the resulting description given by velocity $\mathbf{v}_A$. Then the velocity $\mathbf{v}_B$ involved in the definition of B_B is described relatively to the reference frame moving with velocity $\mathbf{v}_A$.

C.2 Alternative Interpretation of Wigner Rotation

Some references — including the one [4] that provided valuable insight in the development of the present derivation — express the product of two boosts (C.1) as a rotation multiplying a boost, in that order. In this form, since operators are applied via rightward multiplication, the boost operation acts first on the rest frame, followed by Wigner rotation acting on the boost. This ordering can introduce confusion into the interpretation, as described below.

The two ways to factor a product of boosts into a rotation and boost are related by noting the unitarity of the rotor: $R_\Omega^\dagger R_\Omega = R_\Omega R_\Omega^\dagger = 1$, which is verifiable by noting that every rotor is even, $R^\dagger = \bar{R}$, and unimodular, $R\bar{R} = 1$. Thus, the result from the product of boosts (C.1) can be rearranged as follows:

$$\begin{aligned} B_B B_A &= B_\Omega R_\Omega \\ &= R_\Omega \left(R_\Omega^\dagger B_\Omega R_\Omega \right) \\ &\equiv R_\Omega B_{\text{alt}} \end{aligned} \tag{C.2}$$

where $B_{\text{alt}} \equiv R_\Omega^\dagger B_\Omega R_\Omega$ is a boost direction that is rotated from the one used in the product form preferred here, $B_\Omega R_\Omega$. The application of the Wigner rotation can then be interpreted in this alternative case as "correcting" the boost direction after it is applied to the object system.

Both possible orderings of the boost and rotation are valid, since the overall effective orientation and relative velocity transformations are the same in both cases. However, for the sake of providing a clear geometric interpretation of not only the results but the intermediate steps, it is more clear to present the boost factor in a way that does not need "correcting".

One point about the above comparison is helpful in the derivation below: since R_Ω can be applied bilaterally to B_{alt} to turn it to B_Ω, and since the boost obtained from a combination of boosts must be coplanar with them, then the Wigner rotation must also be within the same plane.

C.3 Details of Overall Boost and Wigner Rotation

The products in the definition of the Wigner rotation (C.1) will be expanded here, in order to derive expressions for the rapidity w and direction $\hat{\mathbf{w}}$ of the overall boost B_Ω, as well as the Wigner rotation angle Ω.

C.3.1 Setting up for Deriving Details

To be clear about what is being calculated below, this section first outlines how the combining boosts and resulting boost and rotation are denoted, and how they relate to one another. Then, the products that are to be equated are expanded so that the relationships between them can be examined in the section that follows this one.

C.3.1.1 Denoting the boosts and resulting Wigner rotation

To begin, suppose the two combining boosts are given by $B_A = \cosh\frac{a}{2} + \hat{\mathbf{a}}\sinh\frac{a}{2}$ and $B_B = \cosh\frac{b}{2} + \hat{\mathbf{b}}\sinh\frac{b}{2}$, where a, b are the rapidities in the $\hat{\mathbf{a}}$, $\hat{\mathbf{b}}$ directions. The resulting boost, $B_\Omega = \cosh\frac{w}{2} + \hat{\mathbf{w}}\sinh\frac{w}{2}$, is thus in the plane of $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$. Since the three boosts must be coplanar, it is convenient to define their directions in terms of perpendicular directions. Thus we can let $\hat{\mathbf{c}} \perp \hat{\mathbf{a}}$, such that:

- $\hat{\mathbf{b}} = \hat{\mathbf{a}}\cos\beta + \hat{\mathbf{c}}\sin\beta$ describes the direction of the second boost as a counterclockwise rotation from $\hat{\mathbf{a}}$, $0 \le \beta \le 2\pi$, and
- $\hat{\mathbf{w}} = \hat{\mathbf{a}}\cos\omega + \hat{\mathbf{c}}\sin\omega$ describes the direction of the resultant boost as a counterclockwise rotation from $\hat{\mathbf{a}}$, with $0 \le \omega \le 2\pi$.

Thus, in terms of $\hat{\mathbf{a}}$ and $\hat{\mathbf{c}}$, the three boosts are:

$$\begin{aligned} B_A &\equiv \cosh\frac{a}{2} + \hat{\mathbf{a}}\sinh\frac{a}{2} \\ B_B &\equiv \cosh\frac{b}{2} + [\hat{\mathbf{a}}\cos\beta + \hat{\mathbf{c}}\sin\beta]\sinh\frac{b}{2} \\ B_\Omega &\equiv \cosh\frac{w}{2} + [\hat{\mathbf{a}}\cos\omega + \hat{\mathbf{c}}\sin\omega]\sinh\frac{w}{2}\ . \end{aligned}$$

Since the rotation is also in the same plane, it can be represented by

$$R_\Omega = \cos\frac{\Omega}{2} - \hat{\mathbf{a}}\hat{\mathbf{c}}\sin\frac{\Omega}{2}$$

where $0 \le \Omega \le 2\pi$ describes the counterclockwise angle that the rest-frame orientation is changed before the overall boost B_Ω is applied in order to obtain the equivalent effect as the product $B_B B_A$.

C.3.1.2 Expanding the Wigner rotation products

Using the definitions above, the left-hand side of (C.1) expands to:

$$\begin{aligned} B_B B_A &= \left(\cosh\frac{b}{2} + [\hat{\mathbf{a}}\cos\beta + \hat{\mathbf{c}}\sin\beta]\sinh\frac{b}{2}\right)\left(\cosh\frac{a}{2} + \hat{\mathbf{a}}\sinh\frac{a}{2}\right) \\ &= \cosh\frac{b}{2}\cosh\frac{a}{2} + \cos\beta\sinh\frac{b}{2}\sinh\frac{a}{2} + \hat{\mathbf{c}}\left(\sin\beta\sinh\frac{b}{2}\cosh\frac{a}{2}\right) \\ &\quad + \hat{\mathbf{a}}\left(\cos\beta\sinh\frac{b}{2}\cosh\frac{a}{2} + \cosh\frac{b}{2}\sinh\frac{a}{2}\right) - \hat{\mathbf{a}}\hat{\mathbf{c}}\left(\sin\beta\sinh\frac{b}{2}\sinh\frac{a}{2}\right) \end{aligned}$$

and the right-hand side of (C.1) expands to:

$$\begin{aligned} B_\Omega R_\Omega &= \left(\cosh\frac{w}{2} + [\hat{\mathbf{a}}\cos\omega + \hat{\mathbf{c}}\sin\omega]\sinh\frac{w}{2}\right)\left(\cos\frac{\Omega}{2} - \hat{\mathbf{a}}\hat{\mathbf{c}}\sin\frac{\Omega}{2}\right) \\ &= \cosh\frac{w}{2}\cos\frac{\Omega}{2} + \hat{\mathbf{c}}\left(\sin\omega\sinh\frac{w}{2}\cos\frac{\Omega}{2} - \cos\omega\sinh\frac{w}{2}\sin\frac{\Omega}{2}\right) \\ &\quad + \hat{\mathbf{a}}\left(\cos\omega\sinh\frac{w}{2}\cos\frac{\Omega}{2} + \sin\omega\sinh\frac{w}{2}\sin\frac{\Omega}{2}\right) - \hat{\mathbf{a}}\hat{\mathbf{c}}\cosh\frac{w}{2}\sin\frac{\Omega}{2} \end{aligned}$$

C.3.2 Extracting the Details

Using the products above, it is possible to calculate the Wigner rotation angle Ω, the resultant rapidity w, and the resultant boost angle ω describing the direction $\hat{\mathbf{w}}$ of the overall boost B_Ω relative to the initial boost direction $\hat{\mathbf{a}}$.

C.3.2.1 Finding the Wigner rotation angle

The scalar and bivector parts can be separately equated:

$$\begin{aligned} \text{scalar:}\quad & \cosh\frac{w}{2}\cos\frac{\Omega}{2} = \cosh\frac{b}{2}\cosh\frac{a}{2} + \cos\beta\sinh\frac{b}{2}\sinh\frac{a}{2} \\ \text{bivector } \hat{\mathbf{a}}\hat{\mathbf{c}}\text{:}\quad & \cosh\frac{w}{2}\sin\frac{\Omega}{2} = \sin\beta\sinh\frac{b}{2}\sinh\frac{a}{2}. \end{aligned}$$

and divided to obtain an expression for Ω in terms of the rapidities a, b, and the angle β relating them:

$$\boxed{\tan\frac{\Omega}{2} = \frac{\sin\beta\sinh\frac{b}{2}\sinh\frac{a}{2}}{\cosh\frac{b}{2}\cosh\frac{a}{2} + \cos\beta\sinh\frac{b}{2}\sinh\frac{a}{2}}} \tag{C.3}$$

C.3.2.2 Finding the resultant rapidity

To determine the parameters for the overall boost B_Ω, the easiest method [4] is to consider the two rapidities a and b, with interior angle between $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ of $\delta = \pi - \beta$, as two sides of a *hyperbolic triangle* as in Figure C.1. Then the third side can be obtained using the *hyperbolic rule of cosines* and $\cos\delta = \cos(\pi - \beta) = -\cos\beta$:

$$\boxed{\cosh w = \cosh a \cosh b + \sinh a \sinh b \cos\beta} \tag{C.4}$$

and thus the **resultant rapidity** w is now expressed entirely in terms of the rapidities a, b, and the angle β relating them.

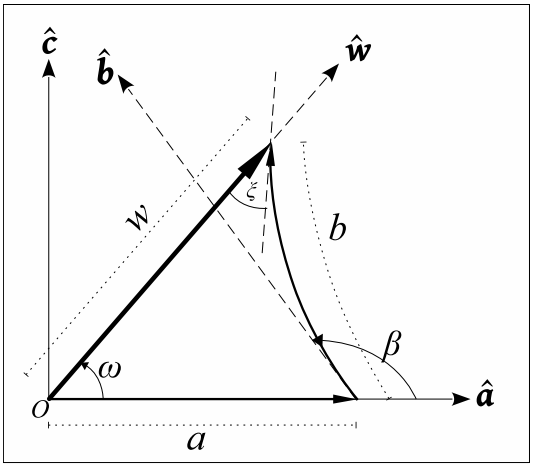


Figure C.1: A hyperbolic triangle in the $\hat{\mathbf{a}}\hat{\mathbf{c}}$ plane formed from a rapidity a in the direction $\hat{\mathbf{a}}$ followed by a rapidity b in the direction $\hat{\mathbf{b}} = \hat{\mathbf{a}}\cos\beta + \hat{\mathbf{c}}\sin\beta$, and their resultant rapidity w in direction $\hat{\mathbf{w}} = \hat{\mathbf{a}}\cos\omega + \hat{\mathbf{c}}\sin\omega$. The curved b rapidity vector is indicative of the hyperbolic space in which rapidities combine; straight lines are geodesics through the origin, and all geodesics are perpendicular to the infinite-rapidity circular boundary of the Poincaré disk model of hyperbolic space [129].

C.3.2.3 Finding the resultant boost angle

The resultant angle ω describing the direction $\hat{\mathbf{w}}$ relative to $\hat{\mathbf{a}}$ can be obtained from Figure C.1 using the *hyperbolic rule of sines* and $\sin\delta = \sin(\pi - \beta) = \sin\beta$:

$$\frac{\sin\omega}{\sinh b} = \frac{\sin\delta}{\sinh w} \rightarrow \sin\omega = \frac{\sin\beta \sinh b}{\sinh w}$$

as well as the cosine rule again:

$$\cosh b = \cosh a \cosh w - \sinh a \sinh w \cos\omega$$
$$\rightarrow \cos\omega = \frac{\cosh b - \cosh a \cosh w}{\sinh a \sinh w}$$

where $\sin\omega$ and $\cos\omega$ can be divided as a first step towards eliminating w from the expression, and then (C.4) can be substituted in to eliminate the rest of the terms containing w, giving (after some manipulation):

$$\boxed{\tan\omega = -\frac{\sin\beta\sinh b}{\cosh b\sinh a + \sinh b\cosh a\cos\beta}} \tag{C.5}$$

which is an expression for the **resultant boost angle** ω in terms of the rapidities a, b, and the angle β relating them.

C.4 Relating These Results to Other Works

The result for the rapidity (C.4) agrees with the result from the work [4] that was the inspiration for this method, but the results for the rotation angle (C.3) and boost angle (C.5) each differ by a sign. This may be explained by the fact that the source material used the opposite order for the boost and rotation; the result is that the two boost angles differ by the Wigner angle. It also may be explained as an error: another work [129] obtained the same result with the same sign as found here, and claimed it agreed with the result from the first-quoted work [4], despite the difference in sign.

The Wigner rotation may also be expressed [53] in terms of the Lorentz factors, γ, of the constituent boosts. Results from the present work can be compared to those approaches by noting the following hyperbolic trig functions of the rapidity x:

$$\begin{aligned}\tanh x &= \frac{v}{c}, \text{ which defines the rapidity, } x\\ \cosh x &= \gamma = \left(1 - v_x^2/c^2\right)^{-1/2}\\ \sinh x &= \frac{\gamma v}{c}\end{aligned} \tag{C.6}$$

— which, in fact, is also interesting to consider in light of the calculation results summarized and analyzed in Section 4.2, since it means the coefficients of results may be expressed as hyperbolic trigonometric functions of the rapidity.

C.5 Effect of Boost on Spin? NOT Wigner Rotation (1 of 2)

There are many examples in the literature [3, 77, 134, 133, 144, 148] where the term "Wigner rotation" is used to mean something different than it does above. Specifically, the Wigner rotation is often given as the explanation of the effective rotation of a spin direction due to a boost — yes, only one boost, which is why this is not truly a Wigner rotation.

Several of the early investigations of boosted spin states involved a combination of boosts applied, and thus did result in a Wigner rotation. But not *every* boost acting on a rest-frame spin state invokes a Wigner rotation, as has been implied even in some oft-cited works [70, 77, 148]. The potential for confusion due to the inappropriate usage of the term "Wigner rotation" is concerning.

In Appendix D, the Algebra of Physical Space is used to examine the geometry of the effect of a boost on a spin state in a way that further distinguishes the effect of a boost on a spin state from a true Wigner rotation, as emphasized in Section D.4.

Appendix D

Relativistic Boosts of Spin States

Prior to the present work, it has been recognized that spin is best described relativistically using the Dirac equation (2.20). In APS, this is done [20, 23] by taking minimal left ideals (2.27) & (2.28) and reformulating the latter to be in the same minimal left ideal as the former, and then combining the separable solutions of the two projected equations into bispinors of either Weyl (2.30) or Dirac (2.31) form. The minimal left ideal is taken by using the projector (B.1) that represents a particular spin-up state (2.33).

The solutions to the Dirac equation are seen in APS to correspond to Lorentz transformations, involving rest-frame rotations (2.6) and relativistic boosts (2.5). The rest-frame rotations can be thought of as rotating the spin-up state projector (2.37) to any other linear superposition of spin up and spin down (2.39). Thus, it is of great interest to examine the effect of applying relativistic boosts to the spin up, spin down, and arbitrary state representations in APS, even if this is only part of the whole picture provided by the Dirac equation.

D.1 Effect of Boost on Spin-Up State

To begin examining the effect of a boost operator on a spin state, consider a rest-frame spin-up state spinor $\psi_\uparrow \equiv \mathrm{P}_3$ (2.33) under a boost transformation (2.5) in an arbitrary direction $\hat{\mathbf{v}} = \cos\delta\,\mathbf{e}_1 + \sin\delta\,\mathbf{e}_3$ (3.17), with $-\frac{\pi}{2} \leq \delta \leq \frac{\pi}{2}$. Note that this boost direction has a component in the initial spin direction $\mathbf{e}_3$ and a component perpendicular to the spin direction, where this

perpendicular direction may (without loss of generality) deemed the $+\mathbf{e}_1$ direction, thus keeping the representation of the arbitrary boost direction as simple as possible.

Using the above considerations, the effect of the boost on the spin-up state spinor may be expressed as follows (making use of the Pacwoman property (B.7) in the form $\mathrm{P}_3 = \mathbf{e}_3\mathrm{P}_3$):

$$\begin{aligned} B\psi_\uparrow &= \left(\cosh\frac{w}{2} + \hat{\mathbf{v}}\sinh\frac{w}{2}\right)\mathrm{P}_3 \\ &= \left(\cosh\frac{w}{2} + \sin\delta\sinh\frac{w}{2} + \mathbf{e}_1\mathbf{e}_3\cos\delta\sinh\frac{w}{2}\right)\mathrm{P}_3 \\ &\equiv (X_0 - i\mathbf{e}_2 X_2)\,\mathrm{P}_3 \;\equiv\; kR\psi_\uparrow \qquad \text{(D.1)} \\ &= X_0\mathrm{P}_3 + X_2\mathbf{e}_1\mathrm{P}_3 \;=\; X_0\psi_\uparrow + X_2\psi_\downarrow \qquad \text{(D.2)} \end{aligned}$$

where it was noted that an APS element consisting only of scalar ($X_0 \equiv \cosh\frac{w}{2} + \sin\delta\sinh\frac{w}{2}$) and bivector ($-i\mathbf{e}_2X_2 \equiv \mathbf{e}_1\mathbf{e}_3\cos\delta\sinh\frac{w}{2}$) elements is an *even* element, which can be equated (B.12) to a dilated rotation operator, here denoted kR. Furthermore,

$$\begin{aligned} \text{the dilation factor is } k &= \sqrt{X_0^2 + X_2^2} \\ &= \sqrt{\left(\cosh\frac{w}{2} + \sin\delta\sinh\frac{w}{2}\right)^2 + \cos^2\delta\sinh^2\frac{w}{2}} \\ &= \sqrt{\cosh^2\frac{w}{2} - 2\sin\delta\cosh\frac{w}{2}\sinh\frac{w}{2} + \sin^2\delta\sinh^2\frac{w}{2} + \cos^2\delta\sinh^2\frac{w}{2}} \\ &= \sqrt{\cosh w + \sin\delta\sinh w} \qquad \text{(D.3)} \\ &= \sqrt{\gamma\,(1 + \sin\delta v/c)} \qquad \text{via (C.6)} \\ \text{and the rotor is } R &= \exp\left(-i\hat{\mathbf{r}}\frac{\alpha}{2}\right) = \cos\frac{\alpha}{2} - i\hat{\mathbf{r}}\sin\frac{\alpha}{2} \\ \text{with rotation axis } \hat{\mathbf{r}} &= \mathbf{e}_2 \\ \text{and rotation angle } \alpha &= 2\arccos\left(X_0/k\right) \;=\; 2\arccos\left(\frac{1}{k}\left[\cosh\frac{w}{2} + \sin\delta\sinh\frac{w}{2}\right]\right) \\ &= 2\arcsin\left(X_2/k\right) \;=\; 2\arcsin\left(\frac{1}{k}\cos\delta\sinh\frac{w}{2}\right). \end{aligned}$$

If the dilated rotated spin state found above may be associated with a state in a Hilbert space, the dilation factor k may be factored out to leave a normalized state with coefficients that can be interpreted as probability amplitudes.

D.2 Effect of Boost on Spin-Down State

The calculation above (D.1) described the effect of a boost operator acting on a rest-frame spin-up state. The next step is to calculate the effect of a boost on a spin-down state, to compare to the

previous calculation and also to complete the preparations for describing the effect of a boost on a general spin state in the following subsection.

Consider a rest-frame spin-down state spinor $\psi_\downarrow = \mathbf{e}_1\mathrm{P}_3$ (2.34) under the same boost transformation as above. The effect of the boost on the spin-down state spinor may be expressed as follows (making use of the Pacwoman property (B.7) in the form $\overline{\mathrm{P}}_3 = -\mathbf{e}_3\overline{\mathrm{P}}_3$):

$$\begin{aligned} B\psi_\downarrow &= \left(\cosh\frac{w}{2} + \hat{\mathbf{v}}\sinh\frac{w}{2}\right)\mathbf{e}_1\mathrm{P}_3 \\ &= \left(\cosh\frac{w}{2} - \sin\delta\sinh\frac{w}{2} - \mathbf{e}_1\mathbf{e}_3\cos\delta\sinh\frac{w}{2}\right)\mathbf{e}_1\mathrm{P}_3 \\ &\equiv (Y_0 - i\mathbf{e}_2 Y_2)\,\mathbf{e}_1\mathrm{P}_3 \quad \equiv \; k'R'\psi_\downarrow \qquad \text{(D.4)} \\ &= Y_0\mathbf{e}_1\mathrm{P}_3 - Y_2\mathrm{P}_3 \quad = \; -Y_2\psi_\uparrow + Y_0\psi_\downarrow \qquad \text{(D.5)} \end{aligned}$$

where it was noted again that an APS element consisting only of scalar ($Y_0 \equiv \cosh\frac{w}{2} - \sin\delta\sinh\frac{w}{2}$) and bivector ($-i\mathbf{e}_2 Y_2 \equiv -\mathbf{e}_1\mathbf{e}_3\cos\delta\sinh\frac{w}{2}$) elements is an *even* element, which can be equated (B.12) to a dilated rotation operator $k'R'$ distinct from kR in the previous calculation. In the present case,

$$\begin{aligned} \text{the dilation factor is } k' &= \sqrt{Y_0^2 + Y_2^2} \\ &= \sqrt{\left(\cosh\frac{w}{2} - \sin\delta\sinh\frac{w}{2}\right)^2 + \cos^2\delta\sinh^2\frac{w}{2}} \\ &= \sqrt{\cosh^2\frac{w}{2} - 2\sin\delta\cosh\frac{w}{2}\sinh\frac{w}{2} + \sin^2\delta\sinh^2\frac{w}{2} + \cos^2\delta\sinh^2\frac{w}{2}} \\ &= \sqrt{\cosh w - \sin\delta\sinh w} \qquad \text{(D.6)} \\ &= \sqrt{\gamma\,(1 - \sin\delta v/c)} \qquad \text{via (C.6)} \\ \text{and the rotor is } R' &= \exp\left(-i\hat{\mathbf{r}}\frac{\beta}{2}\right) = \cos\frac{\beta}{2} - i\hat{\mathbf{r}}\sin\frac{\beta}{2} \\ \text{with rotation axis } \hat{\mathbf{r}} &= \mathbf{e}_2 \\ \text{and rotation angle } \beta &= 2\arccos\left(Y_0/k'\right) \;=\; 2\arccos\left(\frac{1}{k'}\left[\cosh\frac{w}{2} - \sin\delta\sinh\frac{w}{2}\right]\right) \\ &= 2\arcsin\left(Y_2/k'\right) \;=\; 2\arcsin\left(-\frac{1}{k'}\cos\delta\sinh\frac{w}{2}\right) \end{aligned}$$

where it is noteworthy that the dilation factors in this and the previous case are not equal, $k' \neq k$. This means the rotation angles are also not of equal magnitude, although they can both be expressed in a form that emphasizes their similarities and differences:

$$\begin{aligned} \alpha &= 2\arcsin\left(\frac{1}{k}\cos\delta\sinh\frac{w}{2}\right) \quad = 2\arcsin\left(\frac{\cos\delta\sinh\frac{w}{2}}{\sqrt{\cosh w + \sin\delta\sinh w}}\right) \\ \beta &= 2\arcsin\left(-\frac{1}{k'}\cos\delta\sinh\frac{w}{2}\right) = -2\arcsin\left(\frac{\cos\delta\sinh\frac{w}{2}}{\sqrt{\cosh w - \sin\delta\sinh w}}\right). \end{aligned}$$

The rotation axis can be treated as the same in both cases: $\hat{\mathbf{r}} = \mathbf{e}_2$, with the signs of α and β

indicating rotations in opposite directions about that axis, since from the definition of the initial boost direction (3.17) it is given that $0 \leq \cos\delta \leq 1$.

With unequal dilation factors, and unequal rotation angles, the calculations above imply that a boost has a different effect on each of the spin basis states. However, this need not pose much of a problem in calculating the effect of a boost on an initial arbitrary state.

D.3 Effect of Boost on Arbitrary State

Before explicitly calculating the effect of a boost operator acting on an arbitrary initial state, note that if the resulting state can be normalized in the manner demonstrated above, its normalization factor n would be somewhere between the dilation factors for the spin-up k and spin-down k' case. Furthermore, the boost direction (3.17) is allowed to make an angle $-\frac{\pi}{2} \leq \delta \leq \frac{\pi}{2}$ with the $\mathbf{e}_1$ direction, such that $\delta = \pm\frac{\pi}{2}$ aligns the boost with the $\pm\mathbf{e}_3$ directions. If the boost has a component in the $+\mathbf{e}_3$ direction, the dilation factor for the boosted spin-up state is found to be larger than that of the spin-down state, so:

$$k' \leq n \leq k,$$

with the inequalities inverted if the boost has a component in the $-\mathbf{e}_3$ direction instead. If $\sin\delta = 0$, the boost direction is aligned with the $+\mathbf{e}_1$ direction and the dilation factors are equal: $k = k'$.

Now, consider an arbitrarily rotated state given by Euler rotations (2.37), which can be equivalently represented as a linear superposition of spin-up and spin-down states (2.39):

$$\begin{aligned} R_{\phi\theta\chi}\psi_\uparrow &= \exp\left(-i\mathbf{e}_3\frac{\phi}{2}\right)\exp\left(-i\mathbf{e}_2\frac{\theta}{2}\right)\exp\left(-i\frac{\chi}{2}\right)\psi_\uparrow \\ &= e^{-i\chi/2}\left[e^{-i\phi/2}\cos\frac{\theta}{2}\psi_\uparrow + e^{i\phi/2}\sin\frac{\theta}{2}\psi_\downarrow\right] \end{aligned}$$

where the overall phase factor χ may be interpreted as describing a rotation about the basis spin-up direction $\mathbf{e}_3$; θ describes a rotation away from spin up towards or past $+\mathbf{e}_1$ (which is defined as aligning with the component of the boost direction perpendicular to the initial spin direction); and ϕ describes a rotation away from $\mathbf{e}_1$ towards $\mathbf{e}_2$ (where the direction of $\mathbf{e}_2$ is defined as completing a right-handed coordinate system).

A boost transformation can be applied to the arbitrarily rotated state by making use of the linear superposition forms of the boost transformations of spin up (D.2) and spin down (D.5):

$$\begin{aligned} BR_{\phi\theta\chi}\psi_\uparrow &= Be^{-i\chi/2}\left[e^{-i\phi/2}\cos\frac{\theta}{2}\psi_\uparrow + e^{i\phi/2}\sin\frac{\theta}{2}\psi_\downarrow\right] \\ &= e^{-i\chi/2}\left[e^{-i\phi/2}\cos\frac{\theta}{2}B\psi_\uparrow + e^{i\phi/2}\sin\frac{\theta}{2}B\psi_\downarrow\right] \end{aligned}$$

$$
\begin{aligned}
&= e^{-i\chi/2}\left[e^{-i\phi/2}\cos\frac{\theta}{2}\left(X_0\psi_\uparrow + X_2\psi_\downarrow\right) + e^{i\phi/2}\sin\frac{\theta}{2}\left(-Y_2\psi_\uparrow + Y_0\psi_\downarrow\right)\right] \\
&= e^{-i\chi/2}\Bigg[\left(e^{-i\phi/2}\cos\frac{\theta}{2}X_0 - e^{i\phi/2}\sin\frac{\theta}{2}Y_2\right)\psi_\uparrow \\
&\qquad + \left(e^{-i\phi/2}\cos\frac{\theta}{2}X_2 + e^{i\phi/2}\sin\frac{\theta}{2}Y_0\right)\psi_\downarrow\Bigg] \\
&= e^{-i\chi/2}\Bigg[\left(e^{-i\phi/2}\cos\frac{\theta}{2}\left[\cosh\frac{w}{2} + \sin\delta\sinh\frac{w}{2}\right] + e^{i\phi/2}\sin\frac{\theta}{2}\cos\delta\sinh\frac{w}{2}\right)\psi_\uparrow \\
&\qquad + \left(e^{-i\phi/2}\cos\frac{\theta}{2}\cos\delta\sinh\frac{w}{2} + e^{i\phi/2}\sin\frac{\theta}{2}\left[\cosh\frac{w}{2} - \sin\delta\sinh\frac{w}{2}\right]\right)\psi_\downarrow\Bigg] \\
&\equiv c_\uparrow\psi_\uparrow + c_\downarrow\psi_\downarrow
\end{aligned}
$$

where $c_\uparrow$ and $c_\downarrow$ denote the coefficients for the spin-up and spin-down components respectively, although these would not be probability amplitudes if the magnitude of the state is not one.

D.3.1 Normalization of an Arbitrary Spin State

If the expression above is to be treated as a state in Hilbert space, it can be normalized so that the coefficients of each basis state give probability amplitudes for each state. As a first step to calculating the normalization factor n, note that the square magnitudes of the two coefficients above are found to be

$$
\begin{aligned}
|c_\uparrow|^2 &= c_\uparrow c_\uparrow^* \\
&= e^{-i\chi/2}e^{i\chi/2}\Bigg\{\cos^2\frac{\theta}{2}\left[\cosh\frac{w}{2} + \sin\delta\sinh\frac{w}{2}\right]^2 + \sin^2\frac{\theta}{2}\cos^2\delta\sinh^2\frac{w}{2} \\
&\qquad + \left(e^{-i\phi} + e^{i\phi}\right)\cos\frac{\theta}{2}\sin\frac{\theta}{2}\cos\delta\sinh\frac{w}{2}\left[\cosh\frac{w}{2} + \sin\delta\sinh\frac{w}{2}\right]\Bigg\} \\
&= \cos^2\frac{\theta}{2}\cosh^2\frac{w}{2} + \left\{\cos^2\frac{\theta}{2}\sin^2\delta + \sin^2\frac{\theta}{2}\cos^2\delta + \frac{1}{2}\cos\phi\sin\theta\sin 2\delta\right\}\sinh^2\frac{w}{2} \\
&\qquad + \left\{\cos^2\frac{\theta}{2}\sin\delta + \frac{1}{2}\cos\phi\sin\theta\cos\delta\right\}\sinh w
\end{aligned}
$$

and

$$
\begin{aligned}
|c_\downarrow|^2 &= c_\downarrow c_\downarrow^* \\
&= e^{-i\chi/2}e^{i\chi/2}\Bigg\{\cos^2\frac{\theta}{2}\cos^2\delta\sinh^2\frac{w}{2} + \sin^2\frac{\theta}{2}\left[\cosh\frac{w}{2} - \sin\delta\sinh\frac{w}{2}\right]^2 \\
&\qquad + \left(e^{-i\phi} + e^{i\phi}\right)\cos\frac{\theta}{2}\sin\frac{\theta}{2}\cos\delta\sinh\frac{w}{2}\left[\cosh\frac{w}{2} - \sin\delta\sinh\frac{w}{2}\right]\Bigg\}
\end{aligned}
$$

$$= \sin^2\frac{\theta}{2}\cosh^2\frac{w}{2} + \left\{\cos^2\frac{\theta}{2}\cos^2\delta + \sin^2\frac{\theta}{2}\sin^2\delta - \frac{1}{2}\cos\phi\sin\theta\sin 2\delta\right\}\sinh^2\frac{w}{2}$$
$$+ \left\{-\sin^2\frac{\theta}{2}\sin\delta + \frac{1}{2}\cos\phi\sin\theta\cos\delta\right\}\sinh w.$$

From these, the normalization factor n for the boosted arbitrary state can be determined:

$$n^2 \;=\; |c_\uparrow|^2 + |c_\downarrow|^2 = \left(\cos^2\frac{\theta}{2} + \sin^2\frac{\theta}{2}\right)\cosh^2\frac{w}{2} + \left(\cos^2\frac{\theta}{2} + \sin^2\frac{\theta}{2}\right)\left(\sin^2\delta + \cos^2\delta\right)\sinh^2\frac{w}{2}$$
$$+ \left\{\left[\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}\right]\sin\delta + \frac{1}{2}\left[\sin\theta + \sin\theta\right]\cos\phi\cos\delta\right\}\sinh w$$
$$\therefore \boxed{n \;=\; \sqrt{\cosh w + \{\cos\theta\sin\delta + \sin\theta\cos\phi\cos\delta\}\sinh w}}$$

which is, of course, equal to 1 if there is no boost, since $w = 0$ gives $\cosh w = 1$ and $\sinh w = 0$.

The expression above agrees with the previously-obtained normalization factors k and k' for spin-up (when $\theta = 0$) and spin-down (when $\theta = \pi$). There is, however, an additional term in this more general case, which only applies when θ is not a multiple of π and thus the spin is not simply up or down; this additional term includes a factor dependent on the initial spin's azimuthal angle ϕ about the $\mathbf{e}_3$ axis. Note that $\phi = 0$ when the component of the spin direction perpendicular to the $\mathbf{e}_3$ axis is in the $+\mathbf{e}_1$ direction — i.e., in the same direction as the boost component that is perpendicular to $\mathbf{e}_3$. It may also be noted that the normalization factor can be expressed as

$$n \;=\; \sqrt{\gamma\,(1 + \{\cos\theta\sin\delta + \sin\theta\cos\phi\cos\delta\}\,v/c)}, \tag{D.7}$$

by making use of the definition (C.6) of the rapidity as the hyperbolic tangent of the speed v/c. Note that this proposed normalization factor for the boosted arbitrary state involves all the possible geometric factors that could reasonably modify the effect of the relativistic velocity transformation that is applied.

Overall, then, the effect of the arbitrary boost on an arbitrarily-rotated initial spin state has been calculated here to have the form of the following normalized linear superposition of spin-up and spin-down:

$$\frac{1}{n}\psi_{B,\,\mathrm{arb}} = \frac{1}{n}BR_{\phi\theta\chi}\psi_\uparrow \quad = \quad \frac{c_\uparrow}{n}\psi_\uparrow + \frac{c_\downarrow}{n}\psi_\downarrow$$
$$= e^{-i\chi/2}\left[\frac{e^{-i\phi/2}\cos\frac{\theta}{2}\left[\cosh\frac{w}{2} + \sin\delta\sinh\frac{w}{2}\right] + e^{i\phi/2}\sin\frac{\theta}{2}\cos\delta\sinh\frac{w}{2}}{\sqrt{\cosh w + \{\cos\theta\sin\delta + \sin\theta\cos\phi\cos\delta\}\sinh w}}\psi_\uparrow\right.$$
$$\left. + \;\frac{e^{-i\phi/2}\cos\frac{\theta}{2}\cos\delta\sinh\frac{w}{2} + e^{i\phi/2}\sin\frac{\theta}{2}\left[\cosh\frac{w}{2} - \sin\delta\sinh\frac{w}{2}\right]}{\sqrt{\cosh w + \{\cos\theta\sin\delta + \sin\theta\cos\phi\cos\delta\}\sinh w}}\psi_\downarrow\right].$$

D.3.2 Interpretation of Boosted Arbitrary Spin State

It is important to note that the non-unitary nature of this transformation (which is what gives rise to the need for a new normalization factor) is not a problem for the probabilistic interpretation, because *the boost is not modifying the spin state*. Instead, the boost is merely shifting the perspective of the reference frame of the observer, which is included implicitly in the mathematical representation of a spin state that is described from its own rest frame. The description of the boosted case explicitly includes a different observer than the rest-frame case, and it is not unreasonable to require normalization for a state that newly takes into account additional parts. Once an inertial reference frame is selected and the resulting situation is described by a normalized wavefunction, the usual rules of probability conservation would apply when time-dependent unitary transformations are applied to the state.

If the boosted arbitrary state can be normalized in the method demonstrated above, it is also possible to determine an expression for the effective rotation of the arbitrary state, by making use again of the results from the spin-up and spin-down cases:

$$\begin{aligned}
\frac{1}{n}BR_{\phi\theta\chi}\psi_\uparrow &= \frac{1}{n}e^{-i\chi/2}\left[e^{-i\phi/2}\cos\frac{\theta}{2}B\psi_\uparrow + e^{i\phi/2}\sin\frac{\theta}{2}B\psi_\downarrow\right] \\
&= \frac{1}{n}e^{-i\chi/2}\left[e^{-i\phi/2}\cos\frac{\theta}{2}\left(X_0\mathrm{P}_3 + X_2\mathbf{e}_1\mathrm{P}_3\right) + e^{i\phi/2}\sin\frac{\theta}{2}\left(Y_0\mathbf{e}_1\mathrm{P}_3 - Y_2\mathrm{P}_3\right)\right] \\
&= \frac{1}{n}e^{-i\chi/2}\left[e^{-i\phi/2}\cos\frac{\theta}{2}\left(X_0 + X_2\mathbf{e}_1\right) + e^{i\phi/2}\sin\frac{\theta}{2}\left(Y_0\mathbf{e}_1 - Y_2\right)\right]\mathrm{P}_3 \\
&= \frac{1}{n}e^{-i\chi/2}\Bigg[\left\{e^{-i\phi/2}\cos\frac{\theta}{2}X_0 - e^{i\phi/2}\sin\frac{\theta}{2}Y_2\right\} \\
&\qquad + \left\{e^{-i\phi/2}\cos\frac{\theta}{2}X_2 + e^{i\phi/2}\sin\frac{\theta}{2}Y_0\right\}\mathbf{e}_1\mathbf{e}_3\Bigg]\mathrm{P}_3 \\
&\equiv e^{-i\chi/2}\left[Z_0 - i\mathbf{e}_2 Z_2\right]\mathrm{P}_3 \quad \equiv \quad \kappa\mathcal{R}\mathrm{P}_3
\end{aligned}$$

where the overall phase factor is kept out of the rotor for simplicity, and, again, it was noted that an APS element consisting only of scalar ($Z_0 \equiv \left(e^{-i\phi/2}\cos\frac{\theta}{2}X_0 - e^{i\phi/2}\sin\frac{\theta}{2}Y_2\right)/n$) and bivector ($-i\mathbf{e}_2 Z_2 \equiv \left(\mathbf{e}_1\mathbf{e}_3\left\{e^{-i\phi/2}\cos\frac{\theta}{2}X_2 + e^{i\phi/2}\sin\frac{\theta}{2}Y_0\right\}/n\right)$) elements is an even element, which can be equated (B.12) to a dilated rotation operator $\kappa\mathcal{R}$ (which is denoted in a stylized form here solely to distinguish it from both kR and $k'R'$ for the spin-up and spin-down cases, respectively). Here, since the normalization factor was included in the initial expression, the even element would be of unit length and thus the dilation factor is $\kappa = 1$. Then,

$$\text{the rotor is } \mathcal{R} = \exp\left(-i\hat{\mathbf{r}}\frac{\eta}{2}\right) = \cos\frac{\eta}{2} - i\hat{\mathbf{r}}\sin\frac{\eta}{2}$$

with rotation axis $\hat{\mathbf{r}} = \mathbf{e}_2$

and rotation angle $\eta = 2\arccos(Z_0)$

$$= 2\arccos\left[\frac{e^{-i\phi/2}\cos\frac{\theta}{2}\left[\cosh\frac{w}{2}+\sin\delta\sinh\frac{w}{2}\right]+e^{i\phi/2}\sin\frac{\theta}{2}\left[\cos\delta\sinh\frac{w}{2}\right]}{\sqrt{\cosh w+\{\cos\theta\sin\delta+\sin\theta\cos\phi\cos\delta\}\sinh w}}\right]$$

$$= 2\arcsin(Z_2)$$

$$= 2\arcsin\left[\frac{e^{-i\phi/2}\cos\frac{\theta}{2}\left[\cos\delta\sinh\frac{w}{2}\right]+e^{i\phi/2}\sin\frac{\theta}{2}\left[\cosh\frac{w}{2}-\sin\delta\sinh\frac{w}{2}\right]}{\sqrt{\cosh w+\{\cos\theta\sin\delta+\sin\theta\cos\phi\cos\delta\}\sinh w}}\right]$$

which describes the effective rotation that results from the application of an arbitrary boost to an arbitrarily-rotated spin state.

D.4 Effect of Boost on Spin? NOT Wigner Rotation (2 of 2)

As was shown above, any spin state boosted from rest to a moving frame is, effectively, rotated in the perspective of the relative frame of the observer. However, this rotation is not to be confused with a Wigner rotation (C.1), as is described in Appendix C.5. The main way the two rotations are distinguished is that a Wigner rotation involves two non-collinear boosts, whereas the above rotation occurs even when there is only one boost.

Another important feature of the effect of a relativistic boost on a spin state, which serves to further distinguish it from a Wigner rotation, is the fact that there are more components in the Dirac or Weyl bispinor than is generally represented in the rest-frame spin state. Thus, when simply applying a simple boost to a rest-frame spin state, the effective rotation that results is not the entire picture. To see what else needs to be considered, recall that a boost acting on an arbitrary state can also be expressed as a boost and rotor acting on a spin-up state:

$$B(a\mathrm{P}_3 + b\mathbf{e}_1\mathrm{P}_3) = BR\mathrm{P}_3$$

and that the rotor and boost together give an eigenspinor $\Lambda = BR$ (2.7) which, itself, satisfies the Dirac equation (2.19). The product of this eigenspinor with the projector P_3 gives just part of the information available in the full eigenspinor. The rest of the information can be obtained by applying the complementary projector $\overline{\mathrm{P}}_3$:

$$\Lambda\overline{\mathrm{P}}_3 = BR\overline{\mathrm{P}}_3$$

which can then be bar-dagger conjugated to change this expression to the same minimal left ideal as the previous case:

$$\overline{\left(BR\overline{\mathrm{P}}_3\right)}^{\dagger} = \overline{B}R\mathrm{P}_3$$

such that this along with $BR\mathrm{P}_3$ can be combined together to form the Weyl bispinor (2.30).

Of emphasis in the calculation above is that a boost B and its bar conjugate $\overline{B}$ must *both* be examined when describing a spin using the Dirac equation. Fortunately, the bar-conjugated boost $\overline{B}$ has the same form as the original boost B but its vector part is in the opposite direction; thus, it is straightforward to use the above calculations to obtain the results for the conjugated case. For a conjugated boost, which is in direction $-\hat{\mathbf{v}} = -\cos\delta\,\mathbf{e}_1 - \sin\delta\,\mathbf{e}_3$, the normalization factor is

$$n_c \;=\; \sqrt{\gamma\,(1 - \{\cos\theta\sin\delta + \sin\theta\cos\phi\cos\delta\}\,v/c)},$$

which may be compared to the normalization factor (D.7) for the original boosted case.

The present dissertation only makes explicit use of some of the ideas presented here, specifically during the normalization process that was incorporated into the calculation of concurrence (F.10). The problems highlighted in Section 4.4 of this dissertation are a direct result of the difference between examining boosts applied only to one projector, and applying to both projectors and interpreting both together, as is done in relating the Dirac equation to the bispinor form often used for its solutions.

Appendix E

Tensor Products

This appendix clarifies some details about tensor products, so that their use in this dissertation is clear.

E.1 Direct, Tensor, or Kronecker Product?

The symbol $\otimes$ is associated with several names having related but distinct meanings:

- If it is applied to *vector spaces*, it means the **direct product** of those spaces. For example, (APS) $\otimes$ (APS) is the direct product of two copies of the APS vector space.
- If it is applied to *linear maps* describing transformations between vector spaces, it is considered a **tensor product**.
- If it is applied to *matrices*, it is generally known as the **Kronecker product** of the matrices, but this is also considered a special case of the **tensor product** so it may be called that as well.

In this dissertation, the phrase "tensor product" is generally used interchangeably with "Kronecker product", since the subject matter here deals mostly with transformations (i.e. rotations and boosts) which may be represented as matrices. Hence, both names are technically appropriate.

It is important to note that the parts before and after the $\otimes$ symbol indicate **independent** vector spaces. However, their combination may have an interpretation that is more than the sum of its parts (see, for example, Section 3.2.2 regarding the interpretation of bipartite boost transformations).

E.2 Definition of the Kronecker Product

Let $A = [a_{jk}]$ be an $m \times n$ matrix, and B be any other rectangular matrix, say, of dimension $p \times q$. Then the **Kronecker product** of A and B is

$$A \otimes B = \begin{bmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{bmatrix} \tag{E.1}$$

and the result is of dimension $mp \times nq$.

The Kronecker or tensor product is bilinear (linear in each parameter), associative, but not commutative. It is generally possible to expand a tensor product into a linear combination of tensor products of basis elements.

E.3 Multiplication Rule for Kronecker Products

Let A, B, C, D be matrices such that A is $m \times n$, B is $p \times q$, C is $n \times r$, and D is $q \times s$. Then

$$(A \otimes B)(C \otimes D) = AC \otimes BD \tag{E.2}$$

where the matrix sizes are constrained so that standard matrix multiplication is possible.

Proof, using $A = [a_{jk}]$ and $C = [c_{kl}]$ for $j = 1..m$, $k = 1..n$, $l = 1..r$:

$$\begin{aligned} (A \otimes B)(C \otimes D) &= \begin{bmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{bmatrix} \begin{bmatrix} c_{11}D & \cdots & c_{1r}D \\ \vdots & \ddots & \vdots \\ c_{n1}D & \cdots & c_{nr}D \end{bmatrix} \\ &= \begin{bmatrix} a_{11}c_{11}BD + a_{1n}c_{n1}BD & \cdots & a_{11}c_{1r}BD + a_{1n}c_{nr}BD \\ \vdots & \ddots & \vdots \\ a_{m1}c_{11}BD + a_{mn}c_{n1}BD & \cdots & a_{m1}c_{1r}BD + a_{mn}c_{nr}BD \end{bmatrix} \end{aligned}$$

$$
\begin{aligned}
AC \otimes BD &= \begin{bmatrix} a_{11}c_{11} + a_{1n}c_{n1} & \cdots & a_{11}c_{1r} + a_{1n}c_{nr} \\ \vdots & \ddots & \vdots \\ a_{m1}c_{11} + a_{mn}c_{n1} & \cdots & a_{m1}c_{1r} + a_{mn}c_{nr} \end{bmatrix} \otimes BD \\
&= \begin{bmatrix} a_{11}c_{11}BD + a_{1n}c_{n1}BD & \cdots & a_{11}c_{1r}BD + a_{1n}c_{nr}BD \\ \vdots & \ddots & \vdots \\ a_{m1}c_{11}BD + a_{mn}c_{n1}BD & \cdots & a_{m1}c_{1r}BD + a_{mn}c_{nr}BD \end{bmatrix} \\
&= (A \otimes B)(C \otimes D),
\end{aligned}
$$

Q.E.D.

E.4 Other Properties of Kronecker Product

The following properties of Kronecker products [78, Table 2] are also useful. Let A, B, C be matrices (or APS elements that may be represented by matrices that do not necessarily commute) and a be any scalarlike element (or an APS element that commutes with all others). Then:

$$A \otimes (aB) = a\,(A \otimes B) \qquad \text{scalars commute} \tag{E.3}$$

$$\left.\begin{aligned} (A+B) \otimes C &= A \otimes C + B \otimes C \\ A \otimes (B+C) &= A \otimes B + A \otimes C \end{aligned}\right\} \text{sums are distributive} \tag{E.4}$$

$$A \otimes (B \otimes C) = (A \otimes B) \otimes C \qquad \text{product is associative} \tag{E.5}$$

$$(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger \qquad \text{transpose is distributive} \tag{E.6}$$

$$A \otimes B = U_1\,(B \otimes A)\,U_2 \qquad \begin{cases} \text{permutation matrices exist;} \\ \text{in general, } A \otimes B \neq B \otimes A\ . \end{cases} \tag{E.7}$$

Appendix F

Bipartite Spin States: Separable & Entangled

Any quantum state may be expressed as various possible linear superpositions of basis states. Each basis state can be represented as a linear superposition in some other basis, which gives a way to re-express any state in terms of any other basis states.

After initially defining the various bipartite basis states in Section F.1, transformations between them are given in Section F.2. Then, using the assumption that all of the above states are meant to represent the same state, the coefficients of each representation are related to one another in Section F.3.

Finally, in Section F.4, a measure of entanglement known as **concurrence** is investigated in the separable bipartite basis, following a derivation given in the literature but performed using APS. Then the resulting expression for concurrence is transformed to the entangled basis, since it is more useful in that form in this dissertation. This work is newly presented within this dissertation.

F.1 Bipartite Spin Basis State Definitions

The **separable bipartite basis states** (2.43) are defined in APS as follows:

$$\begin{aligned}
\Upsilon_{\uparrow\uparrow} &\equiv \mathrm{P}_3 \otimes \mathrm{P}_3 \\
\Upsilon_{\downarrow\downarrow} &\equiv \mathbf{e}_1\mathrm{P}_3 \otimes \mathbf{e}_1\mathrm{P}_3 \\
\Upsilon_{\downarrow\uparrow} &\equiv \mathbf{e}_1\mathrm{P}_3 \otimes \mathrm{P}_3 \\
\Upsilon_{\uparrow\downarrow} &\equiv \mathrm{P}_3 \otimes \mathbf{e}_1\mathrm{P}_3
\end{aligned} \tag{F.1}$$

The **entangled basis states** (2.46) are defined, with relative and global phases particular to this dissertation (see Footnote 12 on page 47), as follows:

$$\begin{aligned}
\Phi_+ &\equiv \frac{1}{\sqrt{2}}\left[\Upsilon_{\uparrow\uparrow} + \Upsilon_{\downarrow\downarrow}\right] = \frac{1}{\sqrt{2}}\left[\mathrm{P}_3 \otimes \mathrm{P}_3 + \mathbf{e}_1\mathrm{P}_3 \otimes \mathbf{e}_1\mathrm{P}_3\right] \equiv \mathbf{R}_{\Phi+}\Upsilon_{\uparrow\uparrow} \\
\Phi_- &\equiv \frac{1}{\sqrt{2}}\left[\Upsilon_{\uparrow\uparrow} - \Upsilon_{\downarrow\downarrow}\right] = \frac{1}{\sqrt{2}}\left[\mathrm{P}_3 \otimes \mathrm{P}_3 - \mathbf{e}_1\mathrm{P}_3 \otimes \mathbf{e}_1\mathrm{P}_3\right] \equiv \mathbf{R}_{\Phi-}\Upsilon_{\uparrow\uparrow} \\
\Psi_+ &\equiv \frac{1}{\sqrt{2}}\left[\Upsilon_{\downarrow\uparrow} + \Upsilon_{\uparrow\downarrow}\right] = \frac{1}{\sqrt{2}}\left[\mathbf{e}_1\mathrm{P}_3 \otimes \mathrm{P}_3 + \mathrm{P}_3 \otimes \mathbf{e}_1\mathrm{P}_3\right] \equiv \mathbf{R}_{\Psi+}\Upsilon_{\uparrow\uparrow} \\
\Psi_- &\equiv \frac{1}{\sqrt{2}}\left[\Upsilon_{\downarrow\uparrow} - \Upsilon_{\uparrow\downarrow}\right] = \frac{1}{\sqrt{2}}\left[\mathbf{e}_1\mathrm{P}_3 \otimes \mathrm{P}_3 - \mathrm{P}_3 \otimes \mathbf{e}_1\mathrm{P}_3\right] \equiv \mathbf{R}_{\Psi-}\Upsilon_{\uparrow\uparrow}
\end{aligned} \tag{F.2}$$

The separable bipartite basis states can be expressed in terms of the entangled basis states as follows:

$$\begin{aligned}
\Upsilon_{\uparrow\uparrow} &= \frac{1}{\sqrt{2}}\left[\Phi_+ + \Phi_-\right] \\
\Upsilon_{\downarrow\downarrow} &= \frac{1}{\sqrt{2}}\left[\Phi_+ - \Phi_-\right] \\
\Upsilon_{\downarrow\uparrow} &= \frac{1}{\sqrt{2}}\left[\Psi_+ + \Psi_-\right] \\
\Upsilon_{\uparrow\downarrow} &= \frac{1}{\sqrt{2}}\left[\Psi_+ - \Psi_-\right]
\end{aligned} \tag{F.3}$$

The other basis encountered in this dissertation (albeit briefly, in subsections 2.3.6.6 and 2.3.6.7) is one made up of **the singlet state and the three triplet states**. These basis states were not given symbols of their own, and indeed, not much emphasis was placed on their collective identity as a basis. Their representation in terms of other basis states is presented here for interest's sake, in Section F.2.3.

F.2 Transforming Between Bipartite Spin Bases

In each subsection below, an initial general expression is given in terms of one basis, and then re-expressed in terms of the others discussed in this dissertation. The coefficients for the initial representation have subscripts that clearly indicate the associated basis states, for the purpose of ensuring that the re-expressed coefficients in terms of the new basis are most clear in how they relate back to the initial representation.

Each representation has its uses, but since the present work is a study of entanglement, the entangled basis states are the most applicable here. However, the ability to recognize the relationships to the other bases is useful.

Notation notes: The fully expanded APS form of each superposition shown below is highlighted with a box, both to highlight it and to isolate the initial representation above the box from the alternative representations below the box. A transition step is shown between representations, where such seems useful or interesting. Where brackets are required, square brackets are used around basis states, while round brackets are used around coefficients; where brackets are not required but helpful for interpretation, braces are used to group terms, and square brackets are used to highlight basis states.

F.2.1 From Separable to Entangled Basis

Using the definitions from Section F.1, a general state initially expressed in terms of the separable bipartite basis states $\{\Upsilon_{\uparrow\uparrow}, \Upsilon_{\downarrow\downarrow}, \Upsilon_{\downarrow\uparrow}, \Upsilon_{\uparrow\downarrow}\}$ can be expressed in terms of the entangled basis states $\{\Phi_+, \Phi_-, \Psi_+, \Psi_-\}$ as follows:

$$
\begin{aligned}
\Upsilon &= a_{\uparrow\uparrow}\left[\Upsilon_{\uparrow\uparrow}\right] + a_{\downarrow\downarrow}\left[\Upsilon_{\downarrow\downarrow}\right] + a_{\downarrow\uparrow}\left[\Upsilon_{\downarrow\uparrow}\right] + a_{\uparrow\downarrow}\left[\Upsilon_{\uparrow\downarrow}\right] \\
&= \boxed{a_{\uparrow\uparrow}\left[\mathrm{P}_3 \otimes \mathrm{P}_3\right] + a_{\downarrow\downarrow}\left[\mathbf{e}_1\mathrm{P}_3 \otimes \mathbf{e}_1\mathrm{P}_3\right] + a_{\downarrow\uparrow}\left[\mathbf{e}_1\mathrm{P}_3 \otimes \mathrm{P}_3\right] + a_{\uparrow\downarrow}\left[\mathrm{P}_3 \otimes \mathbf{e}_1\mathrm{P}_3\right]} \\
&= a_{\uparrow\uparrow}\left[\frac{\Phi_+ + \Phi_-}{\sqrt{2}}\right] + a_{\downarrow\downarrow}\left[\frac{\Phi_+ - \Phi_-}{\sqrt{2}}\right] + a_{\downarrow\uparrow}\left[\frac{\Psi_+ + \Psi_-}{\sqrt{2}}\right] + a_{\uparrow\downarrow}\left[\frac{\Psi_+ - \Psi_-}{\sqrt{2}}\right] \\
&= \left(\frac{a_{\uparrow\uparrow} + a_{\downarrow\downarrow}}{\sqrt{2}}\right)\Phi_+ + \left(\frac{a_{\uparrow\uparrow} - a_{\downarrow\downarrow}}{\sqrt{2}}\right)\Phi_- + \left(\frac{a_{\downarrow\uparrow} + a_{\uparrow\downarrow}}{\sqrt{2}}\right)\Psi_+ + \left(\frac{a_{\downarrow\uparrow} - a_{\uparrow\downarrow}}{\sqrt{2}}\right)\Psi_-
\end{aligned}
$$

F.2.2 From Entangled to Separable Basis

Using the definitions from Section F.1, a general state initially expressed in terms of the entangled basis states $\{\Phi_+, \Phi_-, \Psi_+, \Psi_-\}$ can be expressed instead in terms of the separable bipartite basis states $\{\Upsilon_{\uparrow\uparrow}, \Upsilon_{\downarrow\downarrow}, \Upsilon_{\downarrow\uparrow}, \Upsilon_{\uparrow\downarrow}\}$ as follows:

$$
\begin{aligned}
\Upsilon &= b_{\Phi+}\left[\Phi_+\right] + b_{\Phi-}\left[\Phi_-\right] + b_{\Psi+}\left[\Psi_+\right] + b_{\Psi-}\left[\Psi_-\right] \\
&= b_{\Phi+}\left[\frac{\Upsilon_{\uparrow\uparrow} + \Upsilon_{\downarrow\downarrow}}{\sqrt{2}}\right] + b_{\Phi-}\left[\frac{\Upsilon_{\uparrow\uparrow} - \Upsilon_{\downarrow\downarrow}}{\sqrt{2}}\right] + b_{\Psi+}\left[\frac{\Upsilon_{\downarrow\uparrow} + \Upsilon_{\uparrow\downarrow}}{\sqrt{2}}\right] + b_{\Psi-}\left[\frac{\Upsilon_{\downarrow\uparrow} - \Upsilon_{\uparrow\downarrow}}{\sqrt{2}}\right] \\
&= \boxed{\begin{aligned} &b_{\Phi+}\left[\frac{\mathrm{P}_3 \otimes \mathrm{P}_3 + \mathbf{e}_1\mathrm{P}_3 \otimes \mathbf{e}_1\mathrm{P}_3}{\sqrt{2}}\right] + b_{\Phi-}\left[\frac{\mathrm{P}_3 \otimes \mathrm{P}_3 - \mathbf{e}_1\mathrm{P}_3 \otimes \mathbf{e}_1\mathrm{P}_3}{\sqrt{2}}\right] \\ &+ b_{\Psi+}\left[\frac{\mathbf{e}_1\mathrm{P}_3 \otimes \mathrm{P}_3 + \mathrm{P}_3 \otimes \mathbf{e}_1\mathrm{P}_3}{\sqrt{2}}\right] + b_{\Psi-}\left[\frac{\mathbf{e}_1\mathrm{P}_3 \otimes \mathrm{P}_3 - \mathrm{P}_3 \otimes \mathbf{e}_1\mathrm{P}_3}{\sqrt{2}}\right] \end{aligned}} \\
&= \left(\frac{b_{\Phi+} + b_{\Phi-}}{\sqrt{2}}\right)\Upsilon_{\uparrow\uparrow} + \left(\frac{b_{\Phi+} - b_{\Phi-}}{\sqrt{2}}\right)\Upsilon_{\downarrow\downarrow} + \left(\frac{b_{\Psi+} + b_{\Psi-}}{\sqrt{2}}\right)\Upsilon_{\downarrow\uparrow} + \left(\frac{b_{\Psi+} - b_{\Psi-}}{\sqrt{2}}\right)\Upsilon_{\uparrow\downarrow}
\end{aligned}
$$

F.2.3 Basis of Singlet and Triplet States, in Terms of the Others

An arbitrary general state expressed in terms of the singlet and triplet states would have the following forms, if expressed in terms of the separable bipartite $\{\Upsilon_{\uparrow\uparrow}, \Upsilon_{\downarrow\downarrow}, \Upsilon_{\downarrow\uparrow}, \Upsilon_{\uparrow\downarrow}\}$ and entangled $\{\Phi_+, \Phi_-, \Psi_+, \Psi_-\}$ basis states:

$$\begin{aligned}
\Upsilon &= s_{0,0}\left[\frac{\Upsilon_{\downarrow\uparrow} - \Upsilon_{\uparrow\downarrow}}{\sqrt{2}}\right] \;+\; \left\{t_{1,1}\left[\Upsilon_{\uparrow\uparrow}\right] + t_{1,0}\left[\frac{\Upsilon_{\downarrow\uparrow} + \Upsilon_{\uparrow\downarrow}}{\sqrt{2}}\right] + t_{1,-1}\left[\Upsilon_{\downarrow\downarrow}\right]\right\} \\
&= s_{0,0}\left[\Psi_-\right] \;+\; \left\{t_{1,1}\left[\Upsilon_{\uparrow\uparrow}\right] \;+\; t_{1,0}\left[\Psi_+\right] \;+\; t_{1,-1}\left[\Upsilon_{\downarrow\downarrow}\right]\right\} \\
&= s_{0,0}\left[\Psi_-\right] \;+\; t_{1,1}\left[\frac{\Phi_+ + \Phi_-}{\sqrt{2}}\right] + t_{1,0}\left[\Psi_+\right] + t_{1,-1}\left[\frac{\Phi_+ - \Phi_-}{\sqrt{2}}\right] \\
&= s_{0,0}\left[\Psi_-\right] \;+\; \left(\frac{t_{1,1} + t_{1,-1}}{\sqrt{2}}\right)\left[\Phi_+\right] + \left(\frac{t_{1,1} - t_{1,-1}}{\sqrt{2}}\right)\left[\Phi_-\right] + t_{1,0}\left[\Psi_+\right] \\
&= t_{1,1}\left[\Upsilon_{\uparrow\uparrow}\right] \;+\; t_{1,-1}\left[\Upsilon_{\downarrow\downarrow}\right] \;+\; \left(\frac{t_{1,0} + s_{0,0}}{\sqrt{2}}\right)\left[\Upsilon_{\downarrow\uparrow}\right] + \left(\frac{t_{1,0} - s_{0,0}}{\sqrt{2}}\right)\left[\Upsilon_{\uparrow\downarrow}\right]
\end{aligned} \tag{F.4}$$

where the coefficient $s_{0,0}$ is for the singlet state having $s = 0$ and $m = 0$, and $t_{1,k}$ is for the triplet state with $s = 1$ and $m = k$.

As can be seen above, the general state expressed in terms of singlet and triplet states is most simply expressed (F.4) using the symbols for the two parallel separable bipartite basis states, $\Upsilon_{\uparrow\uparrow}$ and $\Upsilon_{\downarrow\downarrow}$, and two entangled basis states, Ψ_- and Ψ_+. However, it is interesting to see how the coefficients relate when expressed fully in terms of one basis or the other.

F.3 Relating the Coefficients in the Various Bases

If all of the above representations are to represent the same general state Υ, the following relations between the coefficients must hold:

$$\begin{aligned}
a_{\uparrow\uparrow} &= \frac{1}{\sqrt{2}}\left(b_{\Phi+} + b_{\Phi-}\right) = t_{1,1} \\
a_{\downarrow\downarrow} &= \frac{1}{\sqrt{2}}\left(b_{\Phi+} - b_{\Phi-}\right) = t_{1,-1} \\
a_{\downarrow\uparrow} &= \frac{1}{\sqrt{2}}\left(b_{\Psi+} + b_{\Psi-}\right) = \frac{1}{\sqrt{2}}\left(t_{1,0} + s_{0,0}\right) \\
a_{\uparrow\downarrow} &= \frac{1}{\sqrt{2}}\left(b_{\Psi+} - b_{\Psi-}\right) = \frac{1}{\sqrt{2}}\left(t_{1,0} - s_{0,0}\right)
\end{aligned} \tag{F.5}$$

$$
\begin{aligned}
b_{\Phi+} &= \frac{1}{\sqrt{2}}\left(a_{\uparrow\uparrow} + a_{\downarrow\downarrow}\right) &&= \frac{1}{\sqrt{2}}\left(t_{1,1} + t_{1,-1}\right) \\
b_{\Phi-} &= \frac{1}{\sqrt{2}}\left(a_{\uparrow\uparrow} - a_{\downarrow\downarrow}\right) &&= \frac{1}{\sqrt{2}}\left(t_{1,1} - t_{1,-1}\right) \\
b_{\Psi+} &= \frac{1}{\sqrt{2}}\left(a_{\downarrow\uparrow} + a_{\uparrow\downarrow}\right) &&= t_{1,0} \\
b_{\Psi-} &= \frac{1}{\sqrt{2}}\left(a_{\downarrow\uparrow} - a_{\uparrow\downarrow}\right) &&= s_{0,0}
\end{aligned}
$$

F.4 Concurrence: Derivation and Re-expression

In an arbitrary bipartite linear combination state, it may be of interest to have some idea how much entanglement is described by the state, though perhaps not necessarily in a resource-based or information-theoretical sense. A simple means of quantifying entanglement that is appropriate for use in the present work is the **concurrence** [156, 157].

In the original paper [156] introducing this means of quantifying entanglement, the concurrence of an arbitrary pure state Υ is defined as the magnitude of the inner product

$$
C\left(\Upsilon\right) \equiv \left|\left\langle \Upsilon \middle| \widetilde{\Upsilon} \right\rangle\right| \tag{F.6}
$$

where $\left|\widetilde{\Upsilon}\right\rangle = \sigma_2 \otimes \sigma_2 \left|\Upsilon^*\right\rangle$ indicates a "spin flip" transformation of the given state $\left|\Upsilon\right\rangle$. This transformation is also described there as a **time reversal**, and it is enacted by complex conjugation of the state in the separable basis, $\left|\Upsilon^*\right\rangle$, and then the application of the Pauli spin operator σ_2 on both spins.

Deriving the concurrence in APS clarifies its geometric interpretation, and also ensures that it is applied appropriately within this dissertation.

F.4.1 Derivation of Concurrence in the Separable Basis

In APS, the discrete symmetry transformation of time reversal is described [31, p. 552] as being applicable to an *eigenspinor* satisfying the Dirac equation (2.20), rather than being applicable to a state obtained from the application of the eigenspinor to a projector. The effect of the time-reversal or **T** transformation on an eigenspinor takes the form $\Psi \to i\overline{\Psi}^{\dagger}\hat{\mathbf{n}}$ for $\hat{\mathbf{n}} \perp \mathbf{e}_3$, where for the present purposes it is reasonable to take $i\hat{\mathbf{n}} \equiv i\mathbf{e}_2 = \mathbf{e}_1\mathbf{e}_3$.

Thus, the time reversal of a simple eigenspinor $\Lambda = BR$ is given by

$$
\Lambda \to \widetilde{\Lambda} = \overline{\Lambda}^{\dagger}\mathbf{e}_1\mathbf{e}_3, \tag{F.7}
$$

where the over-tilde symbol is nonstandard but useful here to relate to the notation (F.6) used for the "spin-flip" in the original derivation. Since conjugation and multiplication are linear operations, it is straightforward to apply time reversal to an arbitrary linear superposition of eigenspinors.

F.4.1.1 Preparation of steps for deriving concurrence

To apply the time reversal transformation (F.7) to an arbitrary bipartite spin state expressed as a linear superposition of the separable bipartite basis states Υ_{ij}, where $i, j \in \{\uparrow, \downarrow\}$, the transformation and projector parts need to be factored:

$$\begin{aligned}
\Upsilon &= a_{\uparrow\uparrow}\Upsilon_{\uparrow\uparrow} + a_{\downarrow\downarrow}\Upsilon_{\downarrow\downarrow} + a_{\downarrow\uparrow}\Upsilon_{\downarrow\uparrow} + a_{\uparrow\downarrow}\Upsilon_{\uparrow\downarrow} \\
&= a_{\uparrow\uparrow}\left[\mathrm{P}_3 \otimes \mathrm{P}_3\right] + a_{\downarrow\downarrow}\left[\mathbf{e}_1\mathrm{P}_3 \otimes \mathbf{e}_1\mathrm{P}_3\right] + a_{\downarrow\uparrow}\left[\mathbf{e}_1\mathrm{P}_3 \otimes \mathrm{P}_3\right] + a_{\uparrow\downarrow}\left[\mathrm{P}_3 \otimes \mathbf{e}_1\mathrm{P}_3\right] \\
&= \Big\{a_{\uparrow\uparrow}\left[\mathbf{1} \otimes \mathbf{1}\right] + a_{\downarrow\downarrow}\left[\mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{e}_1\mathbf{e}_3\right] + a_{\downarrow\uparrow}\left[\mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{1}\right] + a_{\uparrow\downarrow}\left[\mathbf{1} \otimes \mathbf{e}_1\mathbf{e}_3\right]\Big\}\left[\mathrm{P}_3 \otimes \mathrm{P}_3\right] \\
&\equiv \mathbf{\Lambda}_\Upsilon\left[\mathrm{P}_3 \otimes \mathrm{P}_3\right]
\end{aligned}$$

where the Pacwoman property (B.7) enables the factors of $\mathbf{e}_1$ to be interpreted instead as π-rotations, $R_\pi \equiv \mathbf{e}_1\mathbf{e}_3$, which enables their identification as eigenspinors $\Lambda = BR$ with $B = 1$.

Note that if the arbitrary spin state above does include some information about relative velocities (i.e. there have been boosts applied), then an overall multiplying factor would need to be taken into account. This is because projectors render the effect of each boost as a dilated rotation. Suppose for now that the above state is not boosted, or that the effects of any boosts have been factored out and omitted from the description, so that the squares of the coefficients sum to $|a_{\uparrow\uparrow}|^2 + |a_{\downarrow\downarrow}|^2 + |a_{\downarrow\uparrow}|^2 + |a_{\uparrow\downarrow}|^2 = 1$.

Now, the time-inverted eigenspinor can be found, noting that $\overline{\mathbf{e}_1\mathbf{e}_3}^\dagger = \mathbf{e}_1\mathbf{e}_3$:

$$\begin{aligned}
\widetilde{\mathbf{\Lambda}}_\Upsilon &= \overline{\Big\{a_{\uparrow\uparrow}\left[\mathbf{1} \otimes \mathbf{1}\right] + a_{\downarrow\downarrow}\left[\mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{e}_1\mathbf{e}_3\right] + a_{\downarrow\uparrow}\left[\mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{1}\right] + a_{\uparrow\downarrow}\left[\mathbf{1} \otimes \mathbf{e}_1\mathbf{e}_3\right]\Big\}}^\dagger \left(\mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{e}_1\mathbf{e}_3\right) \\
&= \Big\{a^*_{\uparrow\uparrow}\left[\mathbf{1} \otimes \mathbf{1}\right] + a^*_{\downarrow\downarrow}\left[\mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{e}_1\mathbf{e}_3\right] + a^*_{\downarrow\uparrow}\left[\mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{1}\right] + a^*_{\uparrow\downarrow}\left[\mathbf{1} \otimes \mathbf{e}_1\mathbf{e}_3\right]\Big\}\left(\mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{e}_1\mathbf{e}_3\right) \\
&= a^*_{\uparrow\uparrow}\left[\mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{e}_1\mathbf{e}_3\right] + a^*_{\downarrow\downarrow}\left[(-\mathbf{1}) \otimes (-\mathbf{1})\right] + a^*_{\downarrow\uparrow}\left[(-\mathbf{1}) \otimes \mathbf{e}_1\mathbf{e}_3\right] + a^*_{\uparrow\downarrow}\left[\mathbf{e}_1\mathbf{e}_3 \otimes (-\mathbf{1})\right] \\
&= a^*_{\uparrow\uparrow}\left[\mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{e}_1\mathbf{e}_3\right] + a^*_{\downarrow\downarrow}\left[\mathbf{1} \otimes \mathbf{1}\right] - a^*_{\downarrow\uparrow}\left[\mathbf{1} \otimes \mathbf{e}_1\mathbf{e}_3\right] - a^*_{\uparrow\downarrow}\left[\mathbf{e}_1\mathbf{e}_3 \otimes \mathbf{1}\right]
\end{aligned}$$

which means the time-inverted arbitrary bipartite spin state is

$$\begin{aligned}
\widetilde{\Upsilon} &\equiv \widetilde{\mathbf{\Lambda}}_\Upsilon\left[\mathrm{P}_3 \otimes \mathrm{P}_3\right] \\
&= a^*_{\uparrow\uparrow}\left[\mathbf{e}_1\mathrm{P}_3 \otimes \mathbf{e}_1\mathrm{P}_3\right] + a^*_{\downarrow\downarrow}\left[\mathrm{P}_3 \otimes \mathrm{P}_3\right] - a^*_{\downarrow\uparrow}\left[\mathrm{P}_3 \otimes \mathbf{e}_1\mathrm{P}_3\right] - a^*_{\uparrow\downarrow}\left[\mathbf{e}_1\mathrm{P}_3 \otimes \mathrm{P}_3\right] \\
&= a^*_{\uparrow\uparrow}\Upsilon_{\downarrow\downarrow} + a^*_{\downarrow\downarrow}\Upsilon_{\uparrow\uparrow} - a^*_{\downarrow\uparrow}\Upsilon_{\uparrow\downarrow} - a^*_{\uparrow\downarrow}\Upsilon_{\downarrow\uparrow}.
\end{aligned}$$

Recall that the concurrence is defined (F.6) as the magnitude of the (Hermitian) inner product between the state Υ and its time inversion $\widetilde{\Upsilon}$, which in APS is given (A.16) as:

$$\begin{aligned} C(\Upsilon) \equiv \left|\left\langle \Upsilon, \widetilde{\Upsilon} \right\rangle\right| &= \left\langle \Upsilon, \widetilde{\Upsilon} \right\rangle \left\langle \widetilde{\Upsilon}, \Upsilon \right\rangle = \left|\left\langle \widetilde{\Upsilon}, \Upsilon \right\rangle\right| \\ &= 2^2 \left|\left\langle \Upsilon^\dagger \widetilde{\Upsilon} \right\rangle_{S\otimes S}\right| = 2^2 \left|\left\langle \widetilde{\Upsilon}^\dagger \Upsilon \right\rangle_{S\otimes S}\right| \end{aligned} \tag{F.8}$$

where, because the absolute value is taken, the ordering of factors may be chosen according to which is easiest or most clear. In this case, it is helpful to take the form involving the dagger conjugate (A.7) of the time-inverted state $\widetilde{\Upsilon}$, so the complex conjugation of the coefficients does not clutter up the representation:

$$\begin{aligned} \widetilde{\Upsilon}^\dagger &= a_{\uparrow\uparrow} \left[\mathbf{e}_1 \mathrm{P}_3 \otimes \mathbf{e}_1 \mathrm{P}_3\right]^\dagger + a_{\downarrow\downarrow} \left[\mathrm{P}_3 \otimes \mathrm{P}_3\right]^\dagger - a_{\downarrow\uparrow} \left[\mathrm{P}_3 \otimes \mathbf{e}_1 \mathrm{P}_3\right]^\dagger - a_{\uparrow\downarrow} \left[\mathbf{e}_1 \mathrm{P}_3 \otimes \mathrm{P}_3\right]^\dagger \\ &= a_{\uparrow\uparrow} \left[\mathrm{P}_3 \mathbf{e}_1 \otimes \mathrm{P}_3 \mathbf{e}_1\right] + a_{\downarrow\downarrow} \left[\mathrm{P}_3 \otimes \mathrm{P}_3\right] - a_{\downarrow\uparrow} \left[\mathrm{P}_3 \otimes \mathrm{P}_3 \mathbf{e}_1\right] - a_{\uparrow\downarrow} \left[\mathrm{P}_3 \mathbf{e}_1 \otimes \mathrm{P}_3\right] \\ &= a_{\uparrow\uparrow} \left[\mathbf{e}_1 \overline{\mathrm{P}}_3 \otimes \mathbf{e}_1 \overline{\mathrm{P}}_3\right] + a_{\downarrow\downarrow} \left[\mathrm{P}_3 \otimes \mathrm{P}_3\right] - a_{\downarrow\uparrow} \left[\mathrm{P}_3 \otimes \mathbf{e}_1 \overline{\mathrm{P}}_3\right] - a_{\uparrow\downarrow} \left[\mathbf{e}_1 \overline{\mathrm{P}}_3 \otimes \mathrm{P}_3\right] \end{aligned}$$

where it was also helpful to rearrange the projectors (B.8) so that they are in the rightmost position, to prepare them for right-multiplying with the state Υ.

Similarly, it is helpful to rearrange the expression for the initial state Υ so its projectors are in the leftmost position:

$$\begin{aligned} \Upsilon &= a_{\uparrow\uparrow} \left[\mathrm{P}_3 \otimes \mathrm{P}_3\right] + a_{\downarrow\downarrow} \left[\mathbf{e}_1 \mathrm{P}_3 \otimes \mathbf{e}_1 \mathrm{P}_3\right] + a_{\downarrow\uparrow} \left[\mathbf{e}_1 \mathrm{P}_3 \otimes \mathrm{P}_3\right] + a_{\uparrow\downarrow} \left[\mathrm{P}_3 \otimes \mathbf{e}_1 \mathrm{P}_3\right] \\ &= a_{\uparrow\uparrow} \left[\mathrm{P}_3 \otimes \mathrm{P}_3\right] + a_{\downarrow\downarrow} \left[\overline{\mathrm{P}}_3 \mathbf{e}_1 \otimes \overline{\mathrm{P}}_3 \mathbf{e}_1\right] + a_{\downarrow\uparrow} \left[\overline{\mathrm{P}}_3 \mathbf{e}_1 \otimes \mathrm{P}_3\right] + a_{\uparrow\downarrow} \left[\mathrm{P}_3 \otimes \overline{\mathrm{P}}_3 \mathbf{e}_1\right] \end{aligned}$$

which prepares them for left-multiplying with the conjugated time-reversed state $\widetilde{\Upsilon}^\dagger$.

F.4.1.2 Application of steps for deriving concurrence

Now the final multiplication within the concurrence formula can be performed. Because of the null property (B.3) of projectors, there are only four nonzero terms that result:

$$\begin{aligned} \widetilde{\Upsilon}^\dagger \Upsilon &= a_{\uparrow\uparrow} a_{\downarrow\downarrow} \left[\mathbf{e}_1 \overline{\mathrm{P}}_3 \mathbf{e}_1 \otimes \mathbf{e}_1 \overline{\mathrm{P}}_3 \mathbf{e}_1\right] + a_{\downarrow\downarrow} a_{\uparrow\uparrow} \left[\mathrm{P}_3 \otimes \mathrm{P}_3\right] \\ &\quad - a_{\downarrow\uparrow} a_{\uparrow\downarrow} \left[\mathrm{P}_3 \otimes \mathbf{e}_1 \overline{\mathrm{P}}_3 \mathbf{e}_1\right] - a_{\uparrow\downarrow} a_{\downarrow\uparrow} \left[\mathbf{e}_1 \overline{\mathrm{P}}_3 \mathbf{e}_1 \otimes \mathrm{P}_3\right] \\ &= \left\{a_{\uparrow\uparrow} a_{\downarrow\downarrow} + a_{\downarrow\downarrow} a_{\uparrow\uparrow} - a_{\downarrow\uparrow} a_{\uparrow\downarrow} - a_{\uparrow\downarrow} a_{\downarrow\uparrow}\right\} \left[\mathrm{P}_3 \otimes \mathrm{P}_3\right] \\ &= 2 \left\{a_{\uparrow\uparrow} a_{\downarrow\downarrow} - a_{\downarrow\uparrow} a_{\uparrow\downarrow}\right\} \left[\mathrm{P}_3 \otimes \mathrm{P}_3\right]. \end{aligned}$$

Now this can be inputted into the APS form of the definition (F.8) for the concurrence, using the fact that the scalar part of the spin-up biprojector is $\langle \mathrm{P}_3 \otimes \mathrm{P}_3 \rangle_{S\otimes S} = \frac{1}{4}$:

$$\begin{aligned} C(\Upsilon) &= 2^2 \left| \left\langle \Upsilon^\dagger \tilde{\Upsilon} \right\rangle_{S\otimes S} \right| \\ &= 4 \left| \left\langle 2 \left\{ a_{\uparrow\uparrow} a_{\downarrow\downarrow} - a_{\downarrow\uparrow} a_{\uparrow\downarrow} \right\} [\mathrm{P}_3 \otimes \mathrm{P}_3] \right\rangle_{S\otimes S} \right| \\ &= 2 \left| a_{\uparrow\uparrow} a_{\downarrow\downarrow} - a_{\downarrow\uparrow} a_{\uparrow\downarrow} \right| \\ &\boxed{\therefore C(\Upsilon) = 2 \left| a_{\uparrow\uparrow} a_{\downarrow\downarrow} - a_{\downarrow\uparrow} a_{\uparrow\downarrow} \right|} \end{aligned} \tag{F.9}$$

which is an appropriate calculation for the **concurrence of a normalized state in the separable bipartite basis**.

This simple form for the concurrence C shows that the state is separable ($C = 0$) if, for example, any three of the coefficients are zero, or if all the coefficients have equal magnitude. A state is maximally entangled ($C = 1$) if $|a_{\uparrow\uparrow}| = |a_{\downarrow\downarrow}| = 2^{-1/2}$ or $|a_{\downarrow\uparrow}| = |a_{\uparrow\downarrow}| = 2^{-1/2}$. Values between 0 and 1 indicate partial entanglement.

Note that the simple form given above (F.9) does not take into account the possibility that the transformation relating the arbitrary state to the initial spin-up bipartite state $\Upsilon_{\uparrow\uparrow}$ may include boosts. A boost, when applied to a projector, can be factored into a rotor and a dilation factor which is greater than or equal to one, since the effect of a boost is to introduce the Lorentz factor $\gamma \geq 1$ into the expression. However, according to the definition of a state as a ray in Hilbert space, any scalar factor multiplying a state does not change the state. Thus the state itself can be usefully normalized, and any multiplicative factors treated as extrinsic to the state. If the concurrence of a state is treated as intrinsic to the state, any multiplicative factor that arises due to a transformation of a state must be factored out in order to calculate the true concurrence of the state.

Thus, a slightly more general form for the concurrence of an unnormalized state $\tilde{\Upsilon}$ is as follows:

$$\boxed{\therefore C(\tilde{\Upsilon}) = \frac{2}{|\tilde{\Upsilon}|} \left| a_{\uparrow\uparrow} a_{\downarrow\downarrow} - a_{\downarrow\uparrow} a_{\uparrow\downarrow} \right|} \tag{F.10}$$

F.4.2 Concurrence in the Entangled Basis

For a measure of entanglement more useful to this dissertation, the concurrence of a general pure state Υ in the separable bipartite basis (F.10) can be easily re-expressed (F.5) in terms of the

coefficients of the entangled basis as follows:

$$\begin{aligned} C(\Upsilon) &= \frac{2}{|\Upsilon|}\left|a_{\uparrow\uparrow}a_{\downarrow\downarrow} - a_{\downarrow\uparrow}a_{\uparrow\downarrow}\right| \\ &= \frac{2}{|\Upsilon|}\left|\frac{1}{\sqrt{2}}\left(b_{\Phi+}+b_{\Phi-}\right)\frac{1}{\sqrt{2}}\left(b_{\Phi+}-b_{\Phi-}\right) - \frac{1}{\sqrt{2}}\left(b_{\Psi+}+b_{\Psi-}\right)\frac{1}{\sqrt{2}}\left(b_{\Psi+}-b_{\Psi-}\right)\right| \\ &= \frac{1}{|\Upsilon|}\left|\left(b_{\Phi+}^2 - b_{\Phi-}^2\right) - \left(b_{\Psi+}^2 - b_{\Psi-}^2\right)\right]\right| \\ &\boxed{\therefore C(\Upsilon) = \frac{1}{|\Upsilon|}\left|\left\{b_{\Phi+}^2 + b_{\Psi-}^2\right\} - \left\{b_{\Phi-}^2 + b_{\Psi+}^2\right\}\right|} \end{aligned} \tag{F.11}$$

where the normalization factor $|\Upsilon|$ was included in case the given state was not of magnitude 1.

In this new expression for concurrence, it is seen that the contributions from the coefficients for the Φ_+ and Ψ_- states are weighted against the contributions from the Φ_- and Ψ_+ states. Entanglement is then maximal if

$$b_{\Phi+}^2 + b_{\Psi-}^2 = 1 \quad \text{or} \quad b_{\Phi-}^2 + b_{\Psi+}^2 = 1. \tag{F.12}$$

A real linear combination state involving only the elements of one or the other of the groups is maximally entangled.

An equal linear combination of one element from each of the two groups can be shown to have a concurrence of zero. For example,

$$\begin{aligned} \alpha\left(\Phi_+ + \Psi_+\right) &= \alpha\left(\frac{1}{\sqrt{2}}\left[\psi_\uparrow \otimes \psi_\uparrow + \psi_\downarrow \otimes \psi_\downarrow\right] + \frac{1}{\sqrt{2}}\left[\psi_\downarrow \otimes \psi_\uparrow + \psi_\uparrow \otimes \psi_\downarrow\right]\right) \\ &= \frac{\alpha}{\sqrt{2}}\left(\left[\psi_\uparrow + \psi_\downarrow\right] \otimes \left[\psi_\uparrow + \psi_\downarrow\right]\right) \end{aligned}$$

is seen to be separable, and thus has no entanglement.

Notice that the pairs of entangled-basis coefficients that sum to maximal entanglement have the same subscripts as the pairs of entanglors interrelated by single-subspace rotations (3.5). This confirms that single-subspace rotations (any such rotations, if they are applied in the rest frame) do not change the amount of correlation.

This also demonstrates that the Bell states and their corresponding entanglors form two complementary groups: "Group I" involving Φ_+ and Ψ_- entanglors or states, and "Group II" involving Φ_- and Ψ_+ entanglors or states. Each of these groups has a distinct type of entanglement, which can be classified by recalling that each entangled state encodes two bits of information: the relative parity bit may be thought of as having value 1 for the Φ case or 0 for the Ψ case, while the relative phase bit may be thought of as having value 1 for the + sign and 0 for the − sign. In "Group I" entanglement, both bits have the same value, while in "Group II" entanglement the bits have opposite values.

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